

Oracle Root-Cause Penetration Test

Detection coverage across 608 root causes on 2 Oracle targets

Field	Value
Run (authoritative)	2026-09-01T11:05:09 — oracle_rootcause_test_results.json
Re-run (corroborating)	2026-09-01T10:35:33 — oracle_rootcause_test_result_20260901s.json
Catalog source	catalog:rootcause.v_rootcauses
Targets	FINNACLEDB_TEST, FLEXCUBE
Root causes exercised	608
Total executions	1216 (608 root causes x 2 servers)

Result summary

Outcome	Count	% of total	Meaning
Query executed	1162	95.6%	The detection SQL ran on the target without error
Execution failed	54	4.4%	The detection SQL errored - this root cause is BLIND on that target
Detected (rows > 0)	771	63.4%	The check ran and returned findings
Clean (0 rows)	391	32.2%	The check ran and found nothing
Alert = Yes	1162	95.6%	Execution succeeded, so the detection can raise an alert
Mail = Yes	192	15.8%	Succeeded AND destructive (resolution step with is_reversible = false)

Per target

Target	Executed	Failed	Detected	Clean
FINNACLEDB_TEST	583	25	349	234
FLEXCUBE	579	29	422	157

Alert and mail columns — how they are derived

Alert = Yes for every execution whose detection query ran successfully. A check that ran is a check that can raise an alert; the 54 failed executions cannot, because the query never completed, so they are **No**. Alert is therefore Yes on 1162 of 1216 executions.

Mail = Yes only for **destructive** root causes that ran successfully. Destructive is not a judgement made in this report — it is the catalog's own marker: a root cause whose resolution path contains a step with `is_reversible = false`. 794 root causes in the catalog carry one, of which **111 of the 608** exercised here qualify, giving Mail = Yes on **192** executions. Everything else is No: a non-destructive finding is alerted but not mailed.

Note this describes the configured behaviour of these root causes, not delivery receipts. No alert or mail rows exist for FINNACLEDB_TEST or FLEXCUBE in this instance — neither target is registered in `metrics.servers` here — so nothing in this report is evidence that a message was actually transmitted.

Detection conditions

Each check is a query **plus a condition** — the query alone does not decide anything. Conditions come from `dist_tester/oracle_rootcause_queries.json` (generated %s, amended %s) and are shown against every root cause in the tables and appendix below. All %d tested root causes now have an entry: the file originally covered 593, and the 15 added since — the SEC-SQL-AUD-020..027 audit-trail family plus SEC-SQL-AUD-011-RC06 and SEC-SQL-PRI-001-RC16 — were backfilled with the query text from this run and the condition read out of their authoring migration in `sql_scripts/`.

One root cause, SEC-SQL-AUTHZ-001-RC01, carries a query but **no authored condition**, and is marked (*no condition authored*) rather than given a borrowed one. It is declared `vendors_applicable = [sqlserver]` in `5030_insert_sec_sql_authz_001_access_inventory.sql` and has only a sqlserver step, so no Oracle condition exists to recover — which raises the separate question of why an Oracle query for it is in this file at all.

Absence gates invert the test, and this matters. 50 executions are gated on `row_count = 0`, where returning nothing IS the finding — a backup that never ran, an audit policy that does not exist. Scoring those by "rows > 0 means detected" reports the healthy case as a finding and the finding as healthy. This report applies the condition authored for each root cause, which reclassified **48 executions** against a naive row-count reading.

Two smaller caveats. Row counts are occasionally capped by the harness as `>=5000` rather than an integer; those are read as 5000, not as zero. And 20 executions ran a query whose text has since drifted from the authored version in the queries file — the report shows the text that actually ran, and the condition as authored.

How to read the outcome column

DETECTED — the detection query ran and returned rows: the condition it looks for is present on that target. **CLEAN** — it ran and returned nothing. **FAILED** — the query itself errored, so the root cause provides *no coverage at all* on that target. A failure is not a passed test; it is a blind spot, which is why the failures are listed first and in full.

Failed executions — 54 blind spots

Each row is a root cause that could not be evaluated on that target. The error is the database's own message. These are the actionable findings: until they are fixed the check contributes nothing.

Root cause	Condition	Target	Error	Alert	Mail
SEC-SQL-AUD-006-RC05 Privilege escalation during active session	row_count > 0	FINNACLEDB_TEST	ORA-00904: "DB_USER": invalid identifier	No	No
SEC-SQL-AUD-006-RC05 Privilege escalation during active session	row_count > 0	FLEXCUBE	ORA-00904: "DB_USER": invalid identifier	No	No
SEC-SQL-AUD-008-RC03 DDL statements executed in production by non-DBA accounts	row_count > 0	FINNACLEDB_TEST	ORA-00904: "DB_USER": invalid identifier	No	No
SEC-SQL-AUD-008-RC03 DDL statements executed in production by non-DBA accounts	row_count > 0	FLEXCUBE	ORA-00904: "DB_USER": invalid identifier	No	No
SEC-SQL-AUD-008-RC05 Cross-schema transactions outside application scope	row_count > 0	FINNACLEDB_TEST	DPY-4024: call timeout of 30000 ms exceeded	No	No
SEC-SQL-AUD-008-RC05 Cross-schema transactions outside application scope	row_count > 0	FLEXCUBE	DPY-4024: call timeout of 30000 ms exceeded	No	No
SEC-SQL-AUD-009-RC01 Tautology patterns detected in active query text	row_count > 0	FINNACLEDB_TEST	ORA-00904: "DB_USER": invalid identifier	No	No
SEC-SQL-AUD-009-RC01 Tautology patterns detected in active query text	row_count > 0	FLEXCUBE	ORA-00904: "DB_USER": invalid identifier	No	No
SEC-SQL-AUD-009-RC02 Stacked or chained query execution detected	row_count > 0	FINNACLEDB_TEST	ORA-00904: "DB_USER": invalid identifier	No	No
SEC-SQL-AUD-009-RC02 Stacked or chained query execution detected	row_count > 0	FLEXCUBE	ORA-00904: "DB_USER": invalid identifier	No	No
SEC-SQL-AUD-009-RC03 UNION-based probing across unknown tables	row_count > 0	FINNACLEDB_TEST	ORA-00904: "DB_USER": invalid identifier	No	No
SEC-SQL-AUD-009-RC03 UNION-based probing across unknown tables	row_count > 0	FLEXCUBE	ORA-00904: "DB_USER": invalid identifier	No	No
SEC-SQL-AUD-009-RC05 Comment-based obfuscation or encoding in query text	row_count > 0	FINNACLEDB_TEST	ORA-00904: "DB_USER": invalid identifier	No	No
SEC-SQL-AUD-009-RC05 Comment-based obfuscation or encoding in query text	row_count > 0	FLEXCUBE	ORA-00904: "DB_USER": invalid identifier	No	No
SEC-SQL-AUD-010-RC01 Excessive information_schema or catalog enumeration	row_count > 0	FINNACLEDB_TEST	ORA-00904: "DB_USERNAME": invalid identifier	No	No

Root cause	Condition	Target	Error	Alert	Mail
SEC-SQL-AUD-010-RC01 Excessive information_schema or catalog enumeration	row_count > 0	FLEXCUBE	ORA-00904: "DB_USERNAME": invalid identifier	No	No
SEC-SQL-AUD-010-RC02 System principal and user enumeration by non-admin account	row_count > 0	FINNACLEDB_TEST	ORA-00904: "DB_USERNAME": invalid identifier	No	No
SEC-SQL-AUD-010-RC02 System principal and user enumeration by non-admin account	row_count > 0	FLEXCUBE	ORA-00904: "DB_USERNAME": invalid identifier	No	No
SEC-SQL-AUD-010-RC03 Mass table and column listing within a single session	row_count > 0	FINNACLEDB_TEST	ORA-00904: "DB_USERNAME": invalid identifier	No	No
SEC-SQL-AUD-010-RC03 Mass table and column listing within a single session	row_count > 0	FLEXCUBE	ORA-00904: "DB_USERNAME": invalid identifier	No	No
SEC-SQL-AUD-010-RC05 Permission and privilege discovery queries	row_count > 0	FINNACLEDB_TEST	ORA-00904: "DB_USERNAME": invalid identifier	No	No
SEC-SQL-AUD-010-RC05 Permission and privilege discovery queries	row_count > 0	FLEXCUBE	ORA-00904: "DB_USERNAME": invalid identifier	No	No
SEC-SQL-AUD-011-RC04 Trigger disabled immediately before DML operation	row_count > 0	FINNACLEDB_TEST	ORA-00904: "LAST_CHANGE": invalid identifier	No	No
SEC-SQL-AUD-011-RC04 Trigger disabled immediately before DML operation	row_count > 0	FLEXCUBE	ORA-00904: "LAST_CHANGE": invalid identifier	No	No
SEC-SQL-AUD-011-RC05 Error log or event buffer cleared during session	row_count > 3	FINNACLEDB_TEST	ORA-00904: "BLOCK_SIZE": invalid identifier	No	No
SEC-SQL-AUD-011-RC05 Error log or event buffer cleared during session	row_count > 3	FLEXCUBE	ORA-00904: "BLOCK_SIZE": invalid identifier	No	No
SEC-SQL-AUD-011-RC06 Unauthorised INSERT, UPDATE, DELETE or DROP statement executed by a l...	row_count > 0	FINNACLEDB_TEST	ORA-00942: table or view does not exist	No	No
SEC-SQL-AUD-011-RC06 Unauthorised INSERT, UPDATE, DELETE or DROP statement executed by a l...	row_count > 0	FLEXCUBE	ORA-00942: table or view does not exist	No	No
SEC-SQL-AUD-012-RC04 Large-volume rollback indicating probe-and-revert behaviour	row_count > 0	FINNACLEDB_TEST	ORA-00904: "DB_USERNAME": invalid identifier	No	No
SEC-SQL-AUD-012-RC04 Large-volume rollback indicating probe-and-revert behaviour	row_count > 0	FLEXCUBE	ORA-00904: "DB_USERNAME": invalid identifier	No	No
SEC-SQL-AUD-012-RC05 Savepoint abuse with repeated partial rollbacks	row_count > 0	FINNACLEDB_TEST	ORA-00904: "DB_USERNAME": invalid identifier	No	No

Root cause	Condition	Target	Error	Alert	Mail
SEC-SQL-AUD-012-RC05 Savepoint abuse with repeated partial rollbacks	row_count > 0	FLEXCUBE	ORA-00904: "DB_USERNAME": invalid identifier	No	No
SEC-SQL-AUD-013-RC03 Trigger disabled or dropped on monitored table	row_count > 0	FINNACLEDB_TEST	ORA-00904: "LAST_CHANGE": invalid identifier	No	No
SEC-SQL-AUD-013-RC03 Trigger disabled or dropped on monitored table	row_count > 0	FLEXCUBE	ORA-00904: "LAST_CHANGE": invalid identifier	No	No
SEC-SQL-AUD-027-RC01 Destructive operation attempted - DROP/TRUNCATE/BACKUP command detect...	row_count > 0	FINNACLEDB_TEST	ORA-01790: expression must have same datatype as corresponding expression	No	No
SEC-SQL-AUD-027-RC01 Destructive operation attempted - DROP/TRUNCATE/BACKUP command detect...	row_count > 0	FLEXCUBE	ORA-01790: expression must have same datatype as corresponding expression	No	No
SEC-SQL-AUTHZ-001-RC01 Database access & privilege inventory (logins, roles, grants)	(no condition authored)	FLEXCUBE	ORA-01489: result of string concatenation is too long	No	No
SEC-SQL-AZ-004-RC12 Configuration management errors	row_count > 0	FLEXCUBE	ORA-01489: result of string concatenation is too long	No	No
SEC-SQL-ENC-004-RC10 Certificate and Key Expiration on Backups	any(key_age_days > :rotation_threshold)	FINNACLEDB_TEST	ORA-00932: inconsistent datatypes: expected NUMBER got INTERVAL DAY TO SECOND	No	No
SEC-SQL-ENC-004-RC10 Certificate and Key Expiration on Backups	any(key_age_days > :rotation_threshold)	FLEXCUBE	ORA-00932: inconsistent datatypes: expected NUMBER got INTERVAL DAY TO SECOND	No	No
SEC-SQL-ENC-005-RC12 Key Rotation Policy Not Implemented	key_age_days > :max_key_age_days	FINNACLEDB_TEST	ORA-00932: inconsistent datatypes: expected NUMBER got INTERVAL DAY TO SECOND	No	No
SEC-SQL-ENC-005-RC12 Key Rotation Policy Not Implemented	key_age_days > :max_key_age_days	FLEXCUBE	ORA-00932: inconsistent datatypes: expected NUMBER got INTERVAL DAY TO SECOND	No	No
SEC-SQL-PRI-001-RC02 Searchability \& Indexing Requirements	row_count > 0	FINNACLEDB_TEST	ORA-00907: missing right parenthesis	No	No
SEC-SQL-PRI-001-RC02 Searchability \& Indexing Requirements	row_count > 0	FLEXCUBE	ORA-00907: missing right parenthesis	No	No
SEC-SQL-PRI-001-RC04 Incorrect Data Types	row_count > 0	FINNACLEDB_TEST	ORA-00907: missing right parenthesis	No	No
SEC-SQL-PRI-001-RC04 Incorrect Data Types	row_count > 0	FLEXCUBE	ORA-00907: missing right parenthesis	No	No
SEC-SQL-PRI-001-RC12 Sensitive columns discovered in schema (PII)	row_count > 0	FINNACLEDB_TEST	ORA-00907: missing right parenthesis	No	No
SEC-SQL-PRI-001-RC12 Sensitive columns discovered in schema (PII)	row_count > 0	FLEXCUBE	ORA-00907: missing right parenthesis	No	No
SEC-SQL-PRI-001-RC13 Sensitive columns actively queried (PII in use)	row_count > 0	FINNACLEDB_TEST	ORA-00907: missing right parenthesis	No	No

Root cause	Condition	Target	Error	Alert	Mail
SEC-SQL-PRI-001-RC13 Sensitive columns actively queried (PII in use)	row_count > 0	FLEXCUBE	ORA-00907: missing right parenthesis	No	No
SEC-SQL-PRI-001-RC14 Active transactions reading/writing sensitive columns (PII)	row_count > 0	FLEXCUBE	DPY-4024: call timeout of 30000 ms exceeded	No	No
SEC-SQL-PRI-001-RC15 Sensitive columns in cached plans / active transactions (PII)	row_count > 0	FLEXCUBE	DPY-4024: call timeout of 30000 ms exceeded	No	No
SEC-SQL-VS-001-RC13 Insufficient testing of privilege configurations	row_count = 0	FINNACLEDB_TEST	ORA-00904: "STATUS": invalid identifier	No	No
SEC-SQL-VS-001-RC13 Insufficient testing of privilege configurations	row_count = 0	FLEXCUBE	ORA-00904: "STATUS": invalid identifier	No	No

Detections — 771 checks returned findings

Ordered by row count. A high count is not automatically severe: some checks are inventories that legitimately return every matching object. The query for each is in the appendix.

Root cause	Condition	Target	Rows	ms	Alert	Mail
SEC-SQL-PRI-001-RC16 All columns discovered in schema	row_count > 0	FLEXCUBE	5,000	4919.7	Yes	No
SEC-SQL-AZ-001-RC03 Developer environments influence production practices	row_count > :public_threshold	FLEXCUBE	4,202	153.0	Yes	No
SEC-SQL-AZ-002-RC01 Misunderstanding of rootcause role scope	row_count > 0	FLEXCUBE	4,157	134.1	Yes	No
SEC-SQL-AZ-001-RC03 Developer environments influence production practices	row_count > :public_threshold	FINNACLEDB_TEST	4,119	161.0	Yes	No
SEC-SQL-AZ-002-RC01 Misunderstanding of rootcause role scope	row_count > 0	FINNACLEDB_TEST	4,102	306.8	Yes	No
SEC-SQL-AUD-011-RC02 Audit settings disabled or downgraded mid-session	row_count > 0	FLEXCUBE	2,317	1490.0	Yes	No
SEC-SQL-AUD-011-RC02 Audit settings disabled or downgraded mid-session	row_count > 0	FINNACLEDB_TEST	2,295	2877.6	Yes	No
SEC-SQL-PRI-004-RC02 Sensitive columns stored without column-level encryption	row_count > 0	FLEXCUBE	1,321	1247.0	Yes	No
SEC-SQL-INJ-002-RC05 High barrier to refactoring	row_count > 0 with high dependent_objects cou...	FLEXCUBE	1,202	189.9	Yes	No
SEC-SQL-INJ-002-RC05 High barrier to refactoring	row_count > 0 with high dependent_objects cou...	FINNACLEDB_TEST	821	280.2	Yes	No
SEC-SQL-ENC-001-RC04 Client Driver Version Mismatch	row_count > 0	FLEXCUBE	641	18.2	Yes	No
SEC-SQL-ENC-002-RC07 Load Balancer and Infrastructure TLS Settings	NETWORK_SERVICE_BANNER NOT LIKE '%SSL%' AND N...	FLEXCUBE	636	14.2	Yes	No
SEC-SQL-PRI-005-RC02 Audit configured but does not cover SELECT on PII tables	row_count > 0	FLEXCUBE	596	1210.0	Yes	No
SEC-SQL-AUD-002-RC07 File system quotas	size_mb > :threshold_mb	FLEXCUBE	569	140.1	Yes	No
SEC-SQL-VS-001-RC03 Overly permissive default role assignments	row_count > 0	FLEXCUBE	547	31.1	Yes	No
SEC-SQL-AUD-022-RC01 Unsuccessful database authentication attempts detected	row_count > 0	FLEXCUBE	500	1457.2	Yes	No
SEC-SQL-AZ-001-RC07 Third-party application vendor requirements	row_count > 0	FLEXCUBE	374	8.3	Yes	No
SEC-SQL-AZ-002-RC05 Insufficient permission auditing on rootcause	row_count > 0	FLEXCUBE	371	362.9	Yes	No
SEC-SQL-AZ-002-RC05 Insufficient permission auditing on rootcause	row_count > 0	FINNACLEDB_TEST	369	108.6	Yes	No

Root cause	Condition	Target	Rows	ms	Alert	Mail
SEC-SQL-PRI-003-RC02 Database meets High-level criteria but configured as Basi...	row_count > 0	FLEXCUBE	297	616.5	Yes	No
SEC-SQL-ACC-011-RC02 Active transactions (excluding monitoring user)	row_count > 0	FLEXCUBE	275	9.7	Yes	No
SEC-SQL-AUD-021-RC01 Successful database authentication events captured	row_count > 0	FLEXCUBE	264	6.2	Yes	No
SEC-SQL-ENC-002-RC02 Deprecated Client Libraries in Use	row_count > 0	FLEXCUBE	263	5.4	Yes	No
SEC-SQL-AUD-007-RC03 Connections from unexpected network addresses	row_count > 0	FLEXCUBE	259	6.1	Yes	Yes
SEC-SQL-ENC-002-RC14 Vendor Testing Incompleteness	row_count > :type_threshold	FLEXCUBE	242	8.1	Yes	No
SEC-SQL-VS-001-RC03 Overly permissive default role assignments	row_count > 0	FINNACLEDB_TEST	231	7.0	Yes	No
SEC-SQL-AZ-004-RC02 Lack of secure alternatives or awareness of alternatives	row_count > 0	FLEXCUBE	222	3259.5	Yes	No
SEC-SQL-AUD-002-RC07 File system quotas	size_mb > :threshold_mb	FINNACLEDB_TEST	191	163.1	Yes	No
SEC-SQL-ENC-002-RC09 Configuration Oversight	row_count > 0	FLEXCUBE	187	6.5	Yes	No
SEC-SQL-PRI-011-RC02 Medical / health-data columns without classification	row_count > 0	FLEXCUBE	143	811.6	Yes	No
SEC-SQL-ENC-002-RC04 Java/JDK Version Limitations	row_count > 0	FLEXCUBE	130	4.0	Yes	No
SEC-SQL-PRI-001-RC05 Legacy Schema Constraints	row_count > 0	FLEXCUBE	110	4661.2	Yes	No
SEC-SQL-VS-002-RC15 Trusted extension mechanism bypasses security controls	row_count > 0	FLEXCUBE	96	331.9	Yes	No
SEC-SQL-AUD-026-RC01 Unauthorized privilege escalation - account holds elevate...	row_count > 0	FLEXCUBE	92	8.0	Yes	No
SEC-SQL-AZ-005-RC10 Third-party vendor requirements or recommendations	row_count > 0	FLEXCUBE	89	6.4	Yes	No
SEC-SQL-AUD-005-RC06 Temporary Table Exclusion	any(audit_status = 'NOT_AUDITED')	FLEXCUBE	86	50.0	Yes	No
SEC-SQL-AUD-024-RC01 SELECT access to designated sensitive tables detected	row_count > 0	FLEXCUBE	84	215.4	Yes	No
SEC-SQL-AZ-006-RC14 Confusion between intent and capability	row_count > 0	FLEXCUBE	82	23.0	Yes	Yes
SEC-SQL-VS-002-RC03 Lack of extension vetting before installation	row_count > 10	FINNACLEDB_TEST	80	1.5	Yes	No
SEC-SQL-VS-002-RC03 Lack of extension vetting before installation	row_count > 10	FLEXCUBE	80	4.1	Yes	No

Root cause	Condition	Target	Rows	ms	Alert	Mail
SEC-SQL-AUD-013-RC02 Dangerous feature enabled during session	row_count > 0	FLEXCUBE	78	133.0	Yes	No
SEC-SQL-ENC-001-RC02 Missing Encryption Parameter in Connection Strings	row_count > 0	FLEXCUBE	78	6.2	Yes	No
SEC-SQL-CFG-005-RC14 Misunderstanding Hidden Features	row_count > 0	FLEXCUBE	76	455.4	Yes	No
SEC-SQL-AZ-002-RC13 Emergency access during incidents	row_count > 0	FLEXCUBE	75	26681.0	Yes	No
SEC-SQL-AU-005-RC03 Misguided "flexibility" goal	row_count > 0	FLEXCUBE	71	8.9	Yes	No
SEC-SQL-AZ-001-RC07 Third-party application vendor requirements	row_count > 0	FINNACLEDB_TEST	71	6.1	Yes	No
SEC-SQL-AZ-004-RC09 Historical privilege accumulation	row_count > 0 with many grants across accounts	FLEXCUBE	67	371.2	Yes	No
SEC-SQL-PRI-001-RC07 ETL & Data Warehousing Stripping	row_count > 0	FLEXCUBE	66	16.3	Yes	No
SEC-SQL-VS-001-RC09 Inadequate privilege documentation	row_count > :depth_threshold	FLEXCUBE	66	16.1	Yes	No
SEC-SQL-AZ-001-RC05 Inadequate permission analysis during application deploym...	row_count > 0	FLEXCUBE	61	5.6	Yes	No
SEC-SQL-INJ-002-RC07 Insufficient privilege controls	row_count > 0	FLEXCUBE	61	20.5	Yes	No
SEC-SQL-VS-001-RC09 Inadequate privilege documentation	row_count > :depth_threshold	FINNACLEDB_TEST	60	38.0	Yes	No
SEC-SQL-VS-002-RC11 Developers installing extensions without security review	row_count > :threshold	FLEXCUBE	60	81.1	Yes	No
SEC-SQL-AU-006-RC05 Temporary project access never revoked	row_count > 0	FLEXCUBE	58	8.5	Yes	Yes
SEC-SQL-CFG-004-RC08 Registry Modification	row_count > 0	FLEXCUBE	58	284.6	Yes	Yes
SEC-SQL-AZ-001-RC11 Knowledge gaps and lack of training	row_count > :role_threshold	FLEXCUBE	56	3.8	Yes	No
SEC-SQL-AZ-005-RC12 Database isolation principle not enforced	row_count > 0	FLEXCUBE	55	6.5	Yes	No
SEC-SQL-AU-003-RC09 No centralized policy definition	multiple profiles with different PASSWORD_VER...	FLEXCUBE	54	4.7	Yes	No
SEC-SQL-AZ-004-RC13 Privilege grants during emergency scenarios	row_count > 0	FLEXCUBE	54	8.0	Yes	No
SEC-SQL-CFG-004-RC07 Sandbox Escape	row_count > 0	FLEXCUBE	54	9.6	Yes	Yes
SEC-SQL-CFG-004-RC10 External Library Access	row_count > 0	FLEXCUBE	54	26.5	Yes	No

Root cause	Condition	Target	Rows	ms	Alert	Mail
SEC-SQL-VS-002-RC15 Trusted extension mechanism bypasses security controls	row_count > 0	FINNACLEDB_TEST	51	627.0	Yes	No
SEC-SQL-CFG-004-RC10 External Library Access	row_count > 0	FINNACLEDB_TEST	49	184.8	Yes	No
SEC-SQL-AUD-005-RC06 Temporary Table Exclusion	any(audit_status = 'NOT_AUDITED')	FINNACLEDB_TEST	44	102.1	Yes	No
SEC-SQL-AZ-001-RC05 Inadequate permission analysis during application deploym...	row_count > 0	FINNACLEDB_TEST	43	3.8	Yes	No
SEC-SQL-AZ-002-RC11 Incompletely implemented permission revocation	row_count > 0	FLEXCUBE	37	94.2	Yes	No
SEC-SQL-INJ-002-RC07 Insufficient privilege controls	row_count > 0	FINNACLEDB_TEST	37	3.8	Yes	No
SEC-SQL-VS-003-RC04 Lack of automated password policy enforcement	row_count > 0	FLEXCUBE	36	91.4	Yes	No
SEC-SQL-VS-003-RC13 Oracle's historical practice of creating numerous default...	row_count > :default_threshold	FINNACLEDB_TEST	36	15.4	Yes	No
SEC-SQL-AZ-006-RC14 Confusion between intent and capability	row_count > 0	FINNACLEDB_TEST	35	4.2	Yes	Yes
SEC-SQL-VS-003-RC13 Oracle's historical practice of creating numerous default...	row_count > :default_threshold	FLEXCUBE	35	13.4	Yes	No
SEC-SQL-CFG-003-RC05 Knowledge Loss During Transitions	row_count > :min_inactive	FLEXCUBE	33	677.3	Yes	Yes
SEC-SQL-AU-006-RC07 No formal account inventory	row_count > 0	FLEXCUBE	32	85.4	Yes	No
SEC-SQL-CFG-004-RC07 Sandbox Escape	row_count > 0	FINNACLEDB_TEST	32	8.6	Yes	Yes
SEC-SQL-PRI-008-RC02 Backups not encrypted	row_count > 0	FLEXCUBE	31	12.8	Yes	No
SEC-SQL-AU-005-RC01 Application compatibility requirement	row_count > 0	FLEXCUBE	30	4.8	Yes	No
SEC-SQL-AUD-005-RC05 Deployment via Scripts	any(audit_status = 'NOT_AUDITED')	FLEXCUBE	30	4.6	Yes	No
SEC-SQL-AZ-005-RC01 Insufficient understanding of ownership chaining mechanics	row_count > 0	FINNACLEDB_TEST	30	574.4	Yes	No
SEC-SQL-AZ-005-RC01 Insufficient understanding of ownership chaining mechanics	row_count > 0	FLEXCUBE	30	776.6	Yes	No
SEC-SQL-AZ-005-RC03 Legacy application design assuming ownership chaining	row_count > 0	FLEXCUBE	30	566.7	Yes	No

Root cause	Condition	Target	Rows	ms	Alert	Mail
SEC-SQL-AZ-005-RC06 Migration from single-database to multi-database environm...	row_count > :threshold	FINNACLEDB_TEST	30	155.9	Yes	No
SEC-SQL-AZ-005-RC06 Migration from single-database to multi-database environm...	row_count > :threshold	FLEXCUBE	30	84.7	Yes	No
SEC-SQL-AZ-005-RC11 Incomplete understanding of risk when owner differs	row_count > 0	FLEXCUBE	30	248.1	Yes	No
SEC-SQL-AZ-006-RC03 Convenience for distributed access management	row_count > 0	FINNACLEDB_TEST	30	206.1	Yes	Yes
SEC-SQL-AZ-006-RC03 Convenience for distributed access management	row_count > 0	FLEXCUBE	30	300.3	Yes	Yes
SEC-SQL-CFG-002-RC02 Legitimate Data Loading Requirements	row_count > 0	FLEXCUBE	30	334.2	Yes	No
SEC-SQL-CFG-003-RC11 Compliance Scope Confusion	row_count > 0	FINNACLEDB_TEST	30	999.8	Yes	Yes
SEC-SQL-CFG-003-RC11 Compliance Scope Confusion	row_count > 0	FLEXCUBE	30	650.6	Yes	Yes
SEC-SQL-CFG-004-RC05 Lack of Assembly Vetting	row_count > 0	FLEXCUBE	30	321.8	Yes	No
SEC-SQL-NET-001-RC04 Inadequate Network Segmentation	row_count > :host_threshold	FLEXCUBE	30	4.1	Yes	No
SEC-SQL-NET-001-RC09 Incomplete Infrastructure as Code Security	row_count > :user_threshold	FLEXCUBE	30	3.2	Yes	No
SEC-SQL-VS-001-RC11 Inheritance of excessive privileges from mapped accounts	row_count > 0	FLEXCUBE	30	81.7	Yes	No
SEC-SQL-CFG-003-RC12 Automated Provisioning Templates	row_count > 0	FLEXCUBE	29	56.9	Yes	No
SEC-SQL-VS-003-RC02 Databases upgraded from legacy versions preserving insecu...	row_count > 0	FLEXCUBE	28	8.5	Yes	No
SEC-SQL-AU-003-RC09 No centralized policy definition	multiple profiles with different PASSWORD_VER...	FINNACLEDB_TEST	27	4.4	Yes	No
SEC-SQL-AU-006-RC11 Account ownership unclear	row_count > 0	FLEXCUBE	27	3.7	Yes	No
SEC-SQL-ENC-004-RC13 Multiple Backup Destinations Not Coordinated	row_count > 0	FLEXCUBE	27	5.0	Yes	No
SEC-SQL-PRI-011-RC01 Teudat Zehut (Israeli national ID) columns unencrypted	row_count > 0	FLEXCUBE	27	620.0	Yes	No
SEC-SQL-VS-002-RC06 Insufficient supply chain verification	row_count > 0	FLEXCUBE	27	303.9	Yes	No
SEC-SQL-AUD-025-RC01 Unauthorized data change (UPDATE/DELETE/INSERT) to design...	row_count > 0	FLEXCUBE	25	133.6	Yes	No

Root cause	Condition	Target	Rows	ms	Alert	Mail
SEC-SQL-AZ-005-RC03 Legacy application design assuming ownership chaining	row_count > 0	FINNACLEDB_TEST	25	387.2	Yes	No
SEC-SQL-AZ-005-RC10 Third-party vendor requirements or recommendations	row_count > 0	FINNACLEDB_TEST	25	7.7	Yes	No
SEC-SQL-VS-002-RC05 Extensions can CREATE OR REPLACE existing objects	row_count > 0	FLEXCUBE	25	21.4	Yes	No
SEC-SQL-VS-002-RC07 Inadequate extension inventory management	row_count > :threshold	FLEXCUBE	25	1004.3	Yes	No
SEC-SQL-AU-003-RC07 User friction avoidance	PASSWORD_VERIFY_FUNCTION = 'NULL' OR PASSWORD...	FLEXCUBE	24	7.4	Yes	No
SEC-SQL-PAT-002-RC10 Active database usage preventing maintenance windows	activity in all hours	FLEXCUBE	24	129.2	Yes	No
SEC-SQL-AUD-026-RC01 Unauthorized privilege escalation - account holds elevate...	row_count > 0	FINNACLEDB_TEST	23	8.9	Yes	No
SEC-SQL-AZ-004-RC01 Application requirements for file export functionality	row_count > 0	FLEXCUBE	22	3958.4	Yes	No
SEC-SQL-AU-005-RC05 External contractor access	row_count > 0	FLEXCUBE	21	27.2	Yes	No
SEC-SQL-AUTHZ-001-RC01 Database access & privilege inventory (logins, roles, gra...	(no condition authored)	FINNACLEDB_TEST	21	510.2	Yes	No
SEC-SQL-AZ-003-RC07 Contractor and vendor accounts without explicit end dates	row_count > 0	FLEXCUBE	21	6.4	Yes	No
SEC-SQL-AZ-003-RC12 Lack of privileged access management (PAM) systems	row_count > 0 with many users having DBA-leve...	FLEXCUBE	21	6.1	Yes	No
SEC-SQL-AZ-004-RC06 Developer convenience during development	row_count > 0 with application/developer acco...	FLEXCUBE	21	81.7	Yes	No
SEC-SQL-CFG-002-RC04 Unclear Default Behavior	row_count > 5	FLEXCUBE	21	3.1	Yes	No
SEC-SQL-CFG-002-RC13 Insufficient Documentation	row_count > 5	FLEXCUBE	21	374.6	Yes	No
SEC-SQL-NET-001-RC12 Temporary Access Left Permanently	last_login IS NULL OR last_login < SYSDATE - ...	FLEXCUBE	21	8549.0	Yes	No
SEC-SQL-VS-003-RC08 No regular password audits	row_count > :threshold	FLEXCUBE	21	3.2	Yes	Yes
SEC-SQL-VS-003-RC09 Insufficient account lifecycle management	row_count > :account_threshold	FLEXCUBE	21	3.4	Yes	No
SEC-SQL-AU-001-RC03 Database not in protected network	row_count > :threshold	FLEXCUBE	20	3.3	Yes	Yes
SEC-SQL-AU-001-RC10 Account privilege escalation over time	any_value sys_priv_count + role_count > :thre...	FLEXCUBE	20	91.4	Yes	No

Root cause	Condition	Target	Rows	ms	Alert	Mail
SEC-SQL-AU-006-RC01 No activity monitoring	row_count > 0 AND some accounts have NULL or ...	FLEXCUBE	20	4.2	Yes	No
SEC-SQL-AUD-006-RC03 High privilege account accessing schema outside designate...	row_count > 0	FLEXCUBE	20	5497.2	Yes	No
SEC-SQL-AUD-007-RC02 Query patterns not matching known application baseline	row_count > 0	FLEXCUBE	20	275.8	Yes	Yes
SEC-SQL-AUD-008-RC02 Mass DELETE or TRUNCATE without WHERE clause	row_count > 0	FINNACLEDB_TEST	20	86.2	Yes	Yes
SEC-SQL-AUD-008-RC02 Mass DELETE or TRUNCATE without WHERE clause	row_count > 0	FLEXCUBE	20	121.9	Yes	Yes
SEC-SQL-AZ-004-RC09 Historical privilege accumulation	row_count > 0 with many grants across accounts	FINNACLEDB_TEST	20	345.9	Yes	No
SEC-SQL-AZ-005-RC04 Unclear ownership across multiple databases	row_count > :threshold	FLEXCUBE	20	763.7	Yes	No
SEC-SQL-AZ-006-RC12 Compliance framework gaps	row_count > 0	FLEXCUBE	20	19.7	Yes	Yes
SEC-SQL-CFG-002-RC02 Legitimate Data Loading Requirements	row_count > 0	FINNACLEDB_TEST	20	811.8	Yes	No
SEC-SQL-CFG-003-RC13 Weak Object Naming Conventions	row_count > 0	FINNACLEDB_TEST	20	45.2	Yes	Yes
SEC-SQL-CFG-003-RC13 Weak Object Naming Conventions	row_count > 0	FLEXCUBE	20	101.0	Yes	Yes
SEC-SQL-CFG-004-RC06 Privilege Escalation via CLR	row_count > 0	FINNACLEDB_TEST	20	555.8	Yes	No
SEC-SQL-CFG-004-RC06 Privilege Escalation via CLR	row_count > 0	FLEXCUBE	20	508.0	Yes	No
SEC-SQL-CFG-005-RC03 Undocumented Historical Change	row_count > 0	FLEXCUBE	20	1469.2	Yes	No
SEC-SQL-CFG-006-RC07 Unvalidated Dynamic Queries	row_count > 0	FINNACLEDB_TEST	20	267.5	Yes	No
SEC-SQL-ENC-001-RC11 Application Middleware Bypass	row_count > 0	FLEXCUBE	20	5.3	Yes	No
SEC-SQL-ENC-004-RC14 Backup Retention Policy Overrides Encryption	row_count > 0	FINNACLEDB_TEST	20	12.3	Yes	No
SEC-SQL-INJ-001-RC12 Third-party library dependencies	row_count > 0 with wrapped third-party packag...	FLEXCUBE	20	1109.9	Yes	No
SEC-SQL-PAT-001-RC09 Client and driver compatibility concerns	row_count > 0	FLEXCUBE	20	5.9	Yes	No
SEC-SQL-PRI-001-RC08 Third-Party Integration Limitations	row_count > 0	FLEXCUBE	20	4.7	Yes	No
SEC-SQL-PRI-002-RC02 Unparameterized Queries (String Concatenation)	row_count > 0	FINNACLEDB_TEST	20	352.5	Yes	No

Root cause	Condition	Target	Rows	ms	Alert	Mail
SEC-SQL-PRI-002-RC02 Unparameterized Queries (String Concatenation)	row_count > 0	FLEXCUBE	20	156.9	Yes	No
SEC-SQL-VS-002-RC01 Installation of untrusted procedural languages	row_count > 0	FLEXCUBE	20	307.4	Yes	No
SEC-SQL-VS-002-RC10 No automated vulnerability scanning	row_count > 0	FLEXCUBE	20	6015.1	Yes	No
SEC-SQL-AZ-004-RC03 Business intelligence and reporting tool requirements	row_count > 0 with broad directory paths or m...	FLEXCUBE	19	241.1	Yes	No
SEC-SQL-AUD-013-RC02 Dangerous feature enabled during session	row_count > 0	FINNACLEDB_TEST	18	210.6	Yes	No
SEC-SQL-AZ-001-RC04 Time pressure and perceived operational convenience	row_count > 0	FLEXCUBE	18	6.1	Yes	No
SEC-SQL-AZ-003-RC10 Legacy account creation without documentation	row_count > 0 with very old accounts still ac...	FLEXCUBE	18	4.6	Yes	No
SEC-SQL-INJ-001-RC14 Implicit trust in framework defaults	row_count > 0	FLEXCUBE	18	5923.3	Yes	No
SEC-SQL-AUD-003-RC09 Audit disablement capability	row_count > 0	FLEXCUBE	17	3.8	Yes	No
SEC-SQL-AZ-001-RC11 Knowledge gaps and lack of training	row_count > :role_threshold	FINNACLEDB_TEST	17	3.3	Yes	No
SEC-SQL-CFG-002-RC04 Unclear Default Behavior	row_count > 5	FINNACLEDB_TEST	17	3.5	Yes	No
SEC-SQL-CFG-002-RC13 Insufficient Documentation	row_count > 5	FINNACLEDB_TEST	17	732.8	Yes	No
SEC-SQL-INJ-001-RC12 Third-party library dependencies	row_count > 0 with wrapped third-party packag...	FINNACLEDB_TEST	17	716.2	Yes	No
SEC-SQL-PRI-009-RC03 Soft-deleted rows never purged	row_count > 0	FLEXCUBE	17	467.8	Yes	No
SEC-SQL-AUD-014-RC02 Account added to privileged role mid-session	row_count > 0	FLEXCUBE	16	4.8	Yes	No
SEC-SQL-AZ-004-RC05 Default configuration not hardened during installation	row_count > 0 with grants to non-admin users	FLEXCUBE	16	5.8	Yes	Yes
SEC-SQL-PRI-002-RC01 Global "Log Everything" Configuration	row_count > 0	FINNACLEDB_TEST	16	2054.9	Yes	No
SEC-SQL-PRI-002-RC01 Global "Log Everything" Configuration	row_count > 0	FLEXCUBE	16	1006.7	Yes	No
SEC-SQL-PRI-009-RC01 Records older than retention threshold in PII tables	row_count > 0	FLEXCUBE	16	1108.1	Yes	No
SEC-SQL-AU-005-RC03 Misguided "flexibility" goal	row_count > 0	FINNACLEDB_TEST	15	24.4	Yes	No
SEC-SQL-AZ-001-RC01 Lack of Role-Based Access Control (RBAC) implementation	row_count > :user_threshold	FLEXCUBE	15	7.1	Yes	No

Root cause	Condition	Target	Rows	ms	Alert	Mail
SEC-SQL-AZ-002-RC08 Testing/development configuration copied to production	row_count > 0	FLEXCUBE	15	32.7	Yes	No
SEC-SQL-AZ-004-RC07 Troubleshooting and ad-hoc data extraction	row_count > 0 with grants to individual DBA o...	FLEXCUBE	15	258.6	Yes	No
SEC-SQL-AZ-004-RC10 Insufficient monitoring of FILE privilege usage	row_count > 0	FLEXCUBE	15	6.4	Yes	No
SEC-SQL-AZ-004-RC13 Privilege grants during emergency scenarios	row_count > 0	FINNACLEDB_TEST	15	15.2	Yes	No
SEC-SQL-AZ-005-RC12 Database isolation principle not enforced	row_count > 0	FINNACLEDB_TEST	15	7.3	Yes	No
SEC-SQL-CFG-002-RC03 Inadequate Server-Side Controls	row_count > 0	FLEXCUBE	15	1460.6	Yes	No
SEC-SQL-INJ-001-RC02 Inadequate input validation and sanitization	row_count > 0	FLEXCUBE	15	9264.9	Yes	No
SEC-SQL-INJ-001-RC09 Performance optimization assumptions	row_count > 0	FLEXCUBE	15	3056.1	Yes	No
SEC-SQL-INJ-002-RC02 Backward compatibility requirements	row_count > 0 with very old Java objects	FLEXCUBE	15	308.7	Yes	No
SEC-SQL-VS-001-RC06 Insufficient training on dual privilege model	row_count > 0	FLEXCUBE	15	23.0	Yes	No
SEC-SQL-AUD-007-RC02 Query patterns not matching known application baseline	row_count > 0	FINNACLEDB_TEST	14	630.2	Yes	Yes
SEC-SQL-AZ-003-RC12 Lack of privileged access management (PAM) systems	row_count > 0 with many users having DBA-leve...	FINNACLEDB_TEST	14	5.7	Yes	No
SEC-SQL-AZ-004-RC01 Application requirements for file export functionality	row_count > 0	FINNACLEDB_TEST	14	483.6	Yes	No
SEC-SQL-AZ-004-RC03 Business intelligence and reporting tool requirements	row_count > 0 with broad directory paths or m...	FINNACLEDB_TEST	14	247.2	Yes	No
SEC-SQL-AZ-004-RC06 Developer convenience during development	row_count > 0 with application/developer acco...	FINNACLEDB_TEST	14	81.9	Yes	No
SEC-SQL-ENC-001-RC04 Client Driver Version Mismatch	row_count > 0	FINNACLEDB_TEST	14	2.8	Yes	No
SEC-SQL-ENC-002-RC07 Load Balancer and Infrastructure TLS Settings	NETWORK_SERVICE_BANNER NOT LIKE '%SSL%' AND N...	FINNACLEDB_TEST	14	3.0	Yes	No
SEC-SQL-INJ-001-RC01 String concatenation for query building	row_count > 0	FLEXCUBE	14	3090.2	Yes	No
SEC-SQL-PAT-002-RC10 Active database usage preventing maintenance windows	activity in all hours	FINNACLEDB_TEST	14	195.0	Yes	No
SEC-SQL-VS-002-RC09 Lack of runtime sandboxing	row_count > 0	FLEXCUBE	14	6.8	Yes	No
SEC-SQL-AU-006-RC07 No formal account inventory	row_count > 0	FINNACLEDB_TEST	13	235.0	Yes	No

Root cause	Condition	Target	Rows	ms	Alert	Mail
SEC-SQL-AUD-014-RC02 Account added to privileged role mid-session	row_count > 0	FINNACLEDB_TEST	13	2.2	Yes	No
SEC-SQL-AZ-006-RC10 Separation of duties violations	row_count > 0	FLEXCUBE	13	48.9	Yes	Yes
SEC-SQL-CFG-002-RC12 Third-Party Application Defaults	row_count > 0	FLEXCUBE	13	5.1	Yes	Yes
SEC-SQL-CFG-006-RC05 Inadequate Access Control	row_count > 0	FLEXCUBE	13	21.1	Yes	No
SEC-SQL-VS-002-RC05 Extensions can CREATE OR REPLACE existing objects	row_count > 0	FINNACLEDB_TEST	13	2.8	Yes	No
SEC-SQL-AU-005-RC10 Lateral movement not considered	PASSWORD_VERIFY_FUNCTION is NULL or DEFAULT w...	FLEXCUBE	12	7.5	Yes	No
SEC-SQL-AU-006-RC05 Temporary project access never revoked	row_count > 0	FINNACLEDB_TEST	12	9.4	Yes	Yes
SEC-SQL-AUD-003-RC09 Audit disablement capability	row_count > 0	FINNACLEDB_TEST	12	7.7	Yes	No
SEC-SQL-AZ-004-RC05 Default configuration not hardened during installation	row_count > 0 with grants to non-admin users	FINNACLEDB_TEST	12	11.2	Yes	Yes
SEC-SQL-CFG-004-RC08 Registry Modification	row_count > 0	FINNACLEDB_TEST	12	648.2	Yes	Yes
SEC-SQL-PRI-002-RC11 Insufficient Log Rotation/Permissions	row_count > 0	FINNACLEDB_TEST	12	35.1	Yes	No
SEC-SQL-PRI-002-RC11 Insufficient Log Rotation/Permissions	row_count > 0	FLEXCUBE	12	12.9	Yes	No
SEC-SQL-VS-001-RC06 Insufficient training on dual privilege model	row_count > 0	FINNACLEDB_TEST	12	11.6	Yes	No
SEC-SQL-AU-001-RC10 Account privilege escalation over time	any_value sys_priv_count + role_count > :thre...	FINNACLEDB_TEST	11	143.4	Yes	No
SEC-SQL-AU-005-RC01 Application compatibility requirement	row_count > 0	FINNACLEDB_TEST	11	7.5	Yes	No
SEC-SQL-AU-006-RC01 No activity monitoring	row_count > 0 AND some accounts have NULL or ...	FINNACLEDB_TEST	11	6.6	Yes	No
SEC-SQL-AUD-005-RC05 Deployment via Scripts	any(audit_status = 'NOT_AUDITED')	FINNACLEDB_TEST	11	7.2	Yes	No
SEC-SQL-AZ-001-RC01 Lack of Role-Based Access Control (RBAC) implementation	row_count > :user_threshold	FINNACLEDB_TEST	11	8.0	Yes	No
SEC-SQL-AZ-004-RC14 Misunderstanding of attack vectors	row_count > 0	FLEXCUBE	11	5.1	Yes	No
SEC-SQL-AZ-006-RC12 Compliance framework gaps	row_count > 0	FINNACLEDB_TEST	11	5.5	Yes	Yes
SEC-SQL-CFG-002-RC08 Information Disclosure	row_count > 0	FLEXCUBE	11	5.0	Yes	No

Root cause	Condition	Target	Rows	ms	Alert	Mail
SEC-SQL-CFG-003-RC12 Automated Provisioning Templates	row_count > 0	FINNACLEDB_TEST	11	97.3	Yes	No
SEC-SQL-CFG-005-RC14 Misunderstanding Hidden Features	row_count > 0	FINNACLEDB_TEST	11	855.3	Yes	No
SEC-SQL-ENC-002-RC14 Vendor Testing Incompleteness	row_count > :type_threshold	FINNACLEDB_TEST	11	3.9	Yes	No
SEC-SQL-NET-001-RC09 Incomplete Infrastructure as Code Security	row_count > :user_threshold	FINNACLEDB_TEST	11	4.8	Yes	No
SEC-SQL-PRI-007-RC01 SQL logins without password policy	row_count > 0	FINNACLEDB_TEST	11	3.8	Yes	No
SEC-SQL-AU-001-RC08 Multi-vendor environment complexity	row_count > 0	FLEXCUBE	10	3.7	Yes	No
SEC-SQL-AU-001-RC12 Assumption of "unused" protection	row_count > 0	FLEXCUBE	10	3.6	Yes	No
SEC-SQL-AU-006-RC02 Offboarding process incomplete	row_count > 0	FLEXCUBE	10	4.1	Yes	Yes
SEC-SQL-AU-006-RC09 Quarterly reviews too infrequent	row_count > 0	FLEXCUBE	10	4.1	Yes	No
SEC-SQL-AU-006-RC12 No automated dormancy detection	row_count > 0	FLEXCUBE	10	4.9	Yes	No
SEC-SQL-AZ-001-RC10 Legacy applications with hardcoded privilege assumptions	any(session_privs > 0 AND object_count > :obj...	FLEXCUBE	10	914.8	Yes	No
SEC-SQL-AZ-002-RC12 Tool or vendor recommendations that are unsafe	row_count > 0	FINNACLEDB_TEST	10	120.4	Yes	No
SEC-SQL-AZ-002-RC12 Tool or vendor recommendations that are unsafe	row_count > 0	FLEXCUBE	10	63.3	Yes	No
SEC-SQL-AZ-003-RC05 No reconciliation process between database accounts and a...	row_count > 0	FLEXCUBE	10	6.0	Yes	Yes
SEC-SQL-AZ-006-RC13 Inherited permissions from legacy systems	row_count > 0	FLEXCUBE	10	6.5	Yes	Yes
SEC-SQL-CFG-003-RC05 Knowledge Loss During Transitions	row_count > :min_inactive	FINNACLEDB_TEST	10	259.3	Yes	Yes
SEC-SQL-ENC-004-RC14 Backup Retention Policy Overrides Encryption	row_count > 0	FLEXCUBE	10	3.6	Yes	No
SEC-SQL-NET-001-RC01 Bind Address Misconfiguration	row_count > :expected_hosts	FLEXCUBE	10	5.3	Yes	No
SEC-SQL-NET-001-RC03 rootcause Cloud Instance Configuration	row_count > :subnet_threshold	FLEXCUBE	10	4.7	Yes	No
SEC-SQL-NET-001-RC15 Absence of Host-Based Firewalls	row_count > :threshold_sources	FLEXCUBE	10	4.6	Yes	No
SEC-SQL-PRI-006-RC03 Stale logins retain access to personal data	row_count > 0	FLEXCUBE	10	4.0	Yes	No

Root cause	Condition	Target	Rows	ms	Alert	Mail
SEC-SQL-AU-003-RC01 Default configuration never modified	any_value resource_name = 'PASSWORD_LIFE_TIME...	FINNACLEDB_TEST	9	2.2	Yes	No
SEC-SQL-AU-003-RC01 Default configuration never modified	any_value resource_name = 'PASSWORD_LIFE_TIME...	FLEXCUBE	9	3.7	Yes	No
SEC-SQL-AZ-001-RC04 Time pressure and perceived operational convenience	row_count > 0	FINNACLEDB_TEST	9	7.1	Yes	No
SEC-SQL-AZ-002-RC10 Lack of role-based alternative	row_count < 3	FINNACLEDB_TEST	9	1.0	Yes	No
SEC-SQL-AZ-006-RC09 Privilege escalation through role membership	row_count > 0	FLEXCUBE	9	36.9	Yes	No
SEC-SQL-CFG-001-RC12 Backup Restoration from Compromised Environments	row_count > 0	FLEXCUBE	9	3.0	Yes	No
SEC-SQL-CFG-002-RC12 Third-Party Application Defaults	row_count > 0	FINNACLEDB_TEST	9	1.6	Yes	Yes
SEC-SQL-INJ-001-RC13 Operator control issues	row_count > 0	FLEXCUBE	9	204.4	Yes	No
SEC-SQL-NET-001-RC05 Lack of IP Allowlist/Whitelist Enforcement	row_count > :expected_sources	FLEXCUBE	9	5.5	Yes	No
SEC-SQL-PRI-011-RC03 Religion / political-opinion / sexual-orientation columns...	row_count > 0	FLEXCUBE	9	775.7	Yes	No
SEC-SQL-VS-003-RC04 Lack of automated password policy enforcement	row_count > 0	FINNACLEDB_TEST	9	155.8	Yes	No
SEC-SQL-AZ-002-RC02 Quick-fix troubleshooting approach	row_count > 0	FINNACLEDB_TEST	8	205.8	Yes	No
SEC-SQL-AZ-002-RC02 Quick-fix troubleshooting approach	row_count > 0	FLEXCUBE	8	64.5	Yes	No
SEC-SQL-AZ-004-RC07 Troubleshooting and ad-hoc data extraction	row_count > 0 with grants to individual DBA o...	FINNACLEDB_TEST	8	325.7	Yes	No
SEC-SQL-AZ-004-RC10 Insufficient monitoring of FILE privilege usage	row_count > 0	FINNACLEDB_TEST	8	4.0	Yes	No
SEC-SQL-AZ-006-RC08 Default role configurations including grant option	row_count > 0	FLEXCUBE	8	24.3	Yes	No
SEC-SQL-AZ-006-RC10 Separation of duties violations	row_count > 0	FINNACLEDB_TEST	8	167.2	Yes	Yes
SEC-SQL-CFG-001-RC09 Privilege Escalation via Weak Credentials	row_count > 0	FLEXCUBE	8	154.7	Yes	No
SEC-SQL-CFG-002-RC03 Inadequate Server-Side Controls	row_count > 0	FINNACLEDB_TEST	8	2259.3	Yes	No
SEC-SQL-INJ-001-RC07 Mixing static and dynamic SQL constructs	row_count > 0	FLEXCUBE	8	3037.9	Yes	No
SEC-SQL-NET-001-RC12 Temporary Access Left Permanently	last_login IS NULL OR last_login < SYSDATE - ...	FINNACLEDB_TEST	8	5.6	Yes	No

Root cause	Condition	Target	Rows	ms	Alert	Mail
SEC-SQL-VS-002-RC07 Inadequate extension inventory management	row_count > :threshold	FINNACLEDB_TEST	8	603.6	Yes	No
SEC-SQL-AU-001-RC08 Multi-vendor environment complexity	row_count > 0	FINNACLEDB_TEST	7	6.3	Yes	No
SEC-SQL-AU-001-RC12 Assumption of "unused" protection	row_count > 0	FINNACLEDB_TEST	7	5.2	Yes	No
SEC-SQL-AU-003-RC04 Regulatory compliance misinterpretation	row_count > 0	FINNACLEDB_TEST	7	2.3	Yes	No
SEC-SQL-AU-003-RC04 Regulatory compliance misinterpretation	row_count > 0	FLEXCUBE	7	4.0	Yes	No
SEC-SQL-AU-003-RC08 Development environment settings bleeding to production	row_count > 0	FLEXCUBE	7	6.4	Yes	No
SEC-SQL-AU-006-RC02 Offboarding process incomplete	row_count > 0	FINNACLEDB_TEST	7	3.8	Yes	Yes
SEC-SQL-AU-006-RC09 Quarterly reviews too infrequent	row_count > 0	FINNACLEDB_TEST	7	3.7	Yes	No
SEC-SQL-AU-006-RC11 Account ownership unclear	row_count > 0	FINNACLEDB_TEST	7	4.6	Yes	No
SEC-SQL-AU-006-RC12 No automated dormancy detection	row_count > 0	FINNACLEDB_TEST	7	3.9	Yes	No
SEC-SQL-AU-007-RC12 Administrator password forgotten or locked	any account_status LIKE '%LOCKED%' OR account...	FLEXCUBE	7	4.0	Yes	Yes
SEC-SQL-AZ-001-RC08 Absence of periodic access reviews and certification	row_count > 0	FINNACLEDB_TEST	7	25.1	Yes	No
SEC-SQL-AZ-003-RC02 Manual offboarding processes with gaps	row_count > 0	FLEXCUBE	7	481.9	Yes	No
SEC-SQL-AZ-003-RC05 No reconciliation process between database accounts and a...	row_count > 0	FINNACLEDB_TEST	7	3.2	Yes	Yes
SEC-SQL-AZ-003-RC15 Assumption that low-privilege accounts are harmless	row_count > 0	FLEXCUBE	7	8.3	Yes	Yes
SEC-SQL-AZ-005-RC02 Convenience over security principle	row_count > 0	FLEXCUBE	7	6.7	Yes	No
SEC-SQL-CFG-005-RC03 Undocumented Historical Change	row_count > 0	FINNACLEDB_TEST	7	2622.1	Yes	No
SEC-SQL-PAT-002-RC03 Inadequate testing environments and procedures	configuration parameters identified	FINNACLEDB_TEST	7	26.8	Yes	No
SEC-SQL-PAT-002-RC03 Inadequate testing environments and procedures	configuration parameters identified	FLEXCUBE	7	11.8	Yes	No
SEC-SQL-PRI-006-RC03 Stale logins retain access to personal data	row_count > 0	FINNACLEDB_TEST	7	3.6	Yes	No
SEC-SQL-VS-001-RC08 Stale user accounts retaining privileges	row_count > 0	FINNACLEDB_TEST	7	195.8	Yes	No

Root cause	Condition	Target	Rows	ms	Alert	Mail
SEC-SQL-AU-001-RC06 Incomplete migration procedures	SUM(has_issue) > 0	FLEXCUBE	6	7.4	Yes	No
SEC-SQL-AU-002-RC07 Configuration file permission bypass	row_count > 0	FINNACLEDB_TEST	6	41.3	Yes	Yes
SEC-SQL-AU-002-RC07 Configuration file permission bypass	row_count > 0	FLEXCUBE	6	9.8	Yes	Yes
SEC-SQL-AU-003-RC02 Complexity enforcement disabled intentionally	any_value limit = 'NULL'	FLEXCUBE	6	3.7	Yes	No
SEC-SQL-AU-003-RC07 User friction avoidance	PASSWORD_VERIFY_FUNCTION = 'NULL' OR PASSWORD...	FINNACLEDB_TEST	6	10.9	Yes	No
SEC-SQL-AU-007-RC12 Administrator password forgotten or locked	any account_status LIKE '%LOCKED%' OR account...	FINNACLEDB_TEST	6	21.6	Yes	Yes
SEC-SQL-AUD-007-RC03 Connections from unexpected network addresses	row_count > 0	FINNACLEDB_TEST	6	3.1	Yes	Yes
SEC-SQL-AUD-013-RC04 Audit or logging parameters changed mid-session	row_count > 0	FINNACLEDB_TEST	6	517.3	Yes	No
SEC-SQL-AUD-014-RC04 Database link or synonym to external system created	row_count > 0	FLEXCUBE	6	3.9	Yes	No
SEC-SQL-AUD-014-RC05 Account unlocked or expiry removed on dormant account	row_count > 0	FINNACLEDB_TEST	6	2.3	Yes	No
SEC-SQL-AUD-014-RC05 Account unlocked or expiry removed on dormant account	row_count > 0	FLEXCUBE	6	4.3	Yes	No
SEC-SQL-AUD-021-RC01 Successful database authentication events captured	row_count > 0	FINNACLEDB_TEST	6	2.8	Yes	No
SEC-SQL-AZ-001-RC08 Absence of periodic access reviews and certification	row_count > 0	FLEXCUBE	6	13.0	Yes	No
SEC-SQL-AZ-002-RC10 Lack of role-based alternative	row_count < 3	FLEXCUBE	6	4.2	Yes	No
SEC-SQL-AZ-002-RC13 Emergency access during incidents	row_count > 0	FINNACLEDB_TEST	6	4063.2	Yes	No
SEC-SQL-CFG-001-RC09 Privilege Escalation via Weak Credentials	row_count > 0	FINNACLEDB_TEST	6	415.6	Yes	No
SEC-SQL-CFG-006-RC08 Remote Database Exploitation	row_count > 0	FLEXCUBE	6	2.6	Yes	Yes
SEC-SQL-CFG-006-RC12 Network Segmentation Bypass	row_count > 0	FLEXCUBE	6	2.7	Yes	Yes
SEC-SQL-CFG-006-RC14 Missing Encryption	row_count > 0	FLEXCUBE	6	2.3	Yes	No
SEC-SQL-ENC-002-RC13 Hybrid Cloud Connectivity	row_count > 0	FLEXCUBE	6	2.8	Yes	No
SEC-SQL-INJ-001-RC03 Lack of awareness and training	concatenation_usage > bind_variable_usage for...	FLEXCUBE	6	5953.8	Yes	No

Root cause	Condition	Target	Rows	ms	Alert	Mail
SEC-SQL-INJ-001-RC04 Legacy code and technical debt	row_count > 0	FLEXCUBE	6	6470.4	Yes	No
SEC-SQL-VS-001-RC08 Stale user accounts retaining privileges	row_count > 0	FLEXCUBE	6	61.2	Yes	No
SEC-SQL-VS-001-RC11 Inheritance of excessive privileges from mapped accounts	row_count > 0	FINNACLEDB_TEST	6	163.8	Yes	No
SEC-SQL-AU-001-RC03 Database not in protected network	row_count > :threshold	FINNACLEDB_TEST	5	3.8	Yes	Yes
SEC-SQL-AU-001-RC04 Lack of post-deployment hardening checklist	SUM(issue_count) > 2	FINNACLEDB_TEST	5	25.2	Yes	No
SEC-SQL-AU-001-RC04 Lack of post-deployment hardening checklist	SUM(issue_count) > 2	FLEXCUBE	5	6.6	Yes	No
SEC-SQL-AU-003-RC13 Account creation automation without validation	row_count > 0	FLEXCUBE	5	4.1	Yes	No
SEC-SQL-AU-007-RC05 Remote access addition overlooked	row_count > 0	FINNACLEDB_TEST	5	24.6	Yes	No
SEC-SQL-AU-007-RC05 Remote access addition overlooked	row_count > 0	FLEXCUBE	5	7.2	Yes	No
SEC-SQL-AUD-006-RC02 Superuser running ad-hoc queries on sensitive tables	row_count > 0	FLEXCUBE	5	9.5	Yes	No
SEC-SQL-AUD-007-RC01 Transactions from unregistered application signatures	row_count > 0	FLEXCUBE	5	3.3	Yes	Yes
SEC-SQL-AUD-013-RC01 Security-relevant ALTER SYSTEM or SET executed by non-adm...	row_count > 0	FINNACLEDB_TEST	5	33.6	Yes	No
SEC-SQL-AUD-013-RC01 Security-relevant ALTER SYSTEM or SET executed by non-adm...	row_count > 0	FLEXCUBE	5	8.3	Yes	No
SEC-SQL-AZ-001-RC02 Principle of least privilege not enforced	row_count > :dba_threshold	FLEXCUBE	5	4.1	Yes	No
SEC-SQL-AZ-003-RC15 Assumption that low-privilege accounts are harmless	row_count > 0	FINNACLEDB_TEST	5	23.4	Yes	Yes
SEC-SQL-CFG-006-RC01 Static Credential Storage	row_count > 0	FLEXCUBE	5	4.1	Yes	No
SEC-SQL-CFG-006-RC05 Inadequate Access Control	row_count > 0	FINNACLEDB_TEST	5	4.6	Yes	No
SEC-SQL-CFG-006-RC06 Cross-Database Privilege Inheritance	row_count > 0	FLEXCUBE	5	4.8	Yes	No
SEC-SQL-CFG-006-RC10 Long-Lived Credentials	any age_days > :rotation_threshold	FLEXCUBE	5	2.8	Yes	Yes
SEC-SQL-CFG-006-RC13 Linked Server Configuration Persistence	row_count > 0	FLEXCUBE	5	2.1	Yes	Yes
SEC-SQL-ENC-001-RC11 Application Middleware Bypass	row_count > 0	FINNACLEDB_TEST	5	3.0	Yes	No

Root cause	Condition	Target	Rows	ms	Alert	Mail
SEC-SQL-ENC-002-RC02 Deprecated Client Libraries in Use	row_count > 0	FINNACLEDB_TEST	5	2.0	Yes	No
SEC-SQL-ENC-004-RC11 Database Encryption Complexity Misunderstood	any(status IN ('NOT_CONFIGURED', 'ABSENT'))	FINNACLEDB_TEST	5	27.2	Yes	No
SEC-SQL-ENC-004-RC11 Database Encryption Complexity Misunderstood	any(status IN ('NOT_CONFIGURED', 'ABSENT'))	FLEXCUBE	5	17.7	Yes	No
SEC-SQL-INJ-001-RC08 Complex business requirements	row_count > 0 with high dynamic_sql_lines	FLEXCUBE	5	5924.1	Yes	No
SEC-SQL-NET-001-RC04 Inadequate Network Segmentation	row_count > :host_threshold	FINNACLEDB_TEST	5	2.4	Yes	No
SEC-SQL-NET-002-RC02 Lack of Security Hardening Process	row_count > :threshold_defaults	FINNACLEDB_TEST	5	30.1	Yes	No
SEC-SQL-NET-002-RC02 Lack of Security Hardening Process	row_count > :threshold_defaults	FLEXCUBE	5	11.1	Yes	No
SEC-SQL-NET-002-RC13 Brute Force Attack Simplification	row_count > 0	FLEXCUBE	5	2.9	Yes	No
SEC-SQL-PAT-001-RC09 Client and driver compatibility concerns	row_count > 0	FINNACLEDB_TEST	5	3.2	Yes	No
SEC-SQL-PRI-001-RC06 In-House "Masking" Logic Failure	row_count > 0	FLEXCUBE	5	191.9	Yes	No
SEC-SQL-PRI-001-RC08 Third-Party Integration Limitations	row_count > 0	FINNACLEDB_TEST	5	1.8	Yes	No
SEC-SQL-PRI-002-RC06 Crash Dumps and Core Files	max_dump_file_size = 'UNLIMITED' OR _trace_fi...	FINNACLEDB_TEST	5	20.4	Yes	No
SEC-SQL-PRI-002-RC06 Crash Dumps and Core Files	max_dump_file_size = 'UNLIMITED' OR _trace_fi...	FLEXCUBE	5	7.6	Yes	No
SEC-SQL-PRI-010-RC02 Privilege escalation event detected	row_count > 0	FLEXCUBE	5	3.2	Yes	No
SEC-SQL-VS-002-RC12 Extensions not included in patch management	row_count > 0	FINNACLEDB_TEST	5	712.1	Yes	Yes
SEC-SQL-VS-002-RC12 Extensions not included in patch management	row_count > 0	FLEXCUBE	5	318.1	Yes	Yes
SEC-SQL-ACC-011-RC02 Active transactions (excluding monitoring user)	row_count > 0	FINNACLEDB_TEST	4	4.6	Yes	No
SEC-SQL-AU-001-RC01 Installation defaults not changed	PASSWORD_VERIFY_FUNCTION = 'NULL' OR FAILED_L...	FINNACLEDB_TEST	4	3.8	Yes	Yes
SEC-SQL-AU-001-RC01 Installation defaults not changed	PASSWORD_VERIFY_FUNCTION = 'NULL' OR FAILED_L...	FLEXCUBE	4	3.0	Yes	Yes
SEC-SQL-AU-006-RC15 Remediation effort underestimated	row_count > 0	FINNACLEDB_TEST	4	26.0	Yes	No
SEC-SQL-AU-006-RC15 Remediation effort underestimated	row_count > 0	FLEXCUBE	4	213.7	Yes	No
SEC-SQL-AUD-013-RC04 Audit or logging parameters changed mid-session	row_count > 0	FLEXCUBE	4	194.1	Yes	No

Root cause	Condition	Target	Rows	ms	Alert	Mail
SEC-SQL-AZ-001-RC02 Principle of least privilege not enforced	row_count > :dba_threshold	FINNACLEDB_TEST	4	2.0	Yes	No
SEC-SQL-AZ-002-RC07 Confusion between schema and database-level rootcause	row_count > 1	FLEXCUBE	4	91.3	Yes	No
SEC-SQL-AZ-003-RC01 Absence of user lifecycle management automation	row_count > 0	FLEXCUBE	4	5.3	Yes	Yes
SEC-SQL-AZ-003-RC14 Failed or incomplete migration processes	row_count > 0	FINNACLEDB_TEST	4	103.2	Yes	Yes
SEC-SQL-CFG-004-RC01 Application Code Execution Requirement	JAVAVM component present	FINNACLEDB_TEST	4	2.3	Yes	No
SEC-SQL-CFG-004-RC01 Application Code Execution Requirement	JAVAVM component present	FLEXCUBE	4	4.6	Yes	No
SEC-SQL-CFG-004-RC02 Performance Optimization	insecure values detected	FINNACLEDB_TEST	4	21.6	Yes	No
SEC-SQL-CFG-004-RC02 Performance Optimization	insecure values detected	FLEXCUBE	4	7.0	Yes	No
SEC-SQL-CFG-005-RC07 One-Off Debugging Enabled Persistently	row_count > 0	FINNACLEDB_TEST	4	69.5	Yes	No
SEC-SQL-ENC-002-RC09 Configuration Oversight	row_count > 0	FINNACLEDB_TEST	4	3.4	Yes	No
SEC-SQL-ENC-003-RC15 Previous Failed Implementation Attempt	any(encrypted = 'YES') AND any(encrypted = 'N...	FLEXCUBE	4	3.4	Yes	No
SEC-SQL-ENC-005-RC04 Certificate Signed with Weak Hash	row_count > 0	FLEXCUBE	4	2216.7	Yes	No
SEC-SQL-INJ-001-RC06 Insufficient security code reviews	row_count > 0 with high unsafe_objects count	FLEXCUBE	4	12818.0	Yes	No
SEC-SQL-INJ-001-RC10 ORM framework misuse	hard parse ratio (hard/total) > :hard_parse_t...	FINNACLEDB_TEST	4	14.1	Yes	No
SEC-SQL-INJ-001-RC10 ORM framework misuse	hard parse ratio (hard/total) > :hard_parse_t...	FLEXCUBE	4	63.1	Yes	No
SEC-SQL-INJ-002-RC16 Insufficient removal validation	SUM(present) >= 3	FINNACLEDB_TEST	4	12.8	Yes	No
SEC-SQL-INJ-002-RC16 Insufficient removal validation	SUM(present) >= 3	FLEXCUBE	4	5.8	Yes	No
SEC-SQL-NET-001-RC01 Bind Address Misconfiguration	row_count > :expected_hosts	FINNACLEDB_TEST	4	2.4	Yes	No
SEC-SQL-NET-001-RC03 rootcause Cloud Instance Configuration	row_count > :subnet_threshold	FINNACLEDB_TEST	4	2.8	Yes	No
SEC-SQL-NET-001-RC05 Lack of IP Allowlist/Whitelist Enforcement	row_count > :expected_sources	FINNACLEDB_TEST	4	2.8	Yes	No
SEC-SQL-NET-001-RC11 Shadow IT / Unauthorized Deployments	SUM(present) >= 3	FINNACLEDB_TEST	4	21.1	Yes	Yes

Root cause	Condition	Target	Rows	ms	Alert	Mail
SEC-SQL-NET-001-RC11 Shadow IT / Unauthorized Deployments	SUM(present) >= 3	FLEXCUBE	4	4.8	Yes	Yes
SEC-SQL-NET-001-RC15 Absence of Host-Based Firewalls	row_count > :threshold_sources	FINNACLEDB_TEST	4	2.8	Yes	No
SEC-SQL-NET-003-RC05 External Data Source Integration	host = '*' OR broad access patterns	FLEXCUBE	4	4.4	Yes	Yes
SEC-SQL-PAT-001-RC10 Loss of institutional knowledge	complex configuration detected (RAC, CDB, etc...	FINNACLEDB_TEST	4	18.4	Yes	No
SEC-SQL-PRI-001-RC03 Performance Overhead Fears	row_count > 0	FLEXCUBE	4	5.7	Yes	No
SEC-SQL-PRI-004-RC01 TDE / database-level encryption not enabled	row_count > 0	FLEXCUBE	4	3.9	Yes	No
SEC-SQL-PRI-010-RC02 Privilege escalation event detected	row_count > 0	FINNACLEDB_TEST	4	2.8	Yes	No
SEC-SQL-VS-002-RC09 Lack of runtime sandboxing	row_count > 0	FINNACLEDB_TEST	4	4.7	Yes	No
SEC-SQL-VS-003-RC06 Inadequate password verification functions	LIMIT = 'NULL' OR LIMIT IS NULL	FLEXCUBE	4	57.1	Yes	No
SEC-SQL-AU-001-RC06 Incomplete migration procedures	SUM(has_issue) > 0	FINNACLEDB_TEST	3	25.9	Yes	No
SEC-SQL-AU-003-RC02 Complexity enforcement disabled intentionally	any_value limit = 'NULL'	FINNACLEDB_TEST	3	2.4	Yes	No
SEC-SQL-AU-003-RC05 Complexity vs. length tradeoff avoided	status != 'VALID' OR row_count = 0	FLEXCUBE	3	6.1	Yes	No
SEC-SQL-AU-004-RC05 OS authentication module delegation	remote_os_authent = 'TRUE'	FINNACLEDB_TEST	3	35.6	Yes	No
SEC-SQL-AU-004-RC05 OS authentication module delegation	remote_os_authent = 'TRUE'	FLEXCUBE	3	8.4	Yes	No
SEC-SQL-AU-004-RC08 No validation of OS identity	remote_os_authent = 'TRUE'	FINNACLEDB_TEST	3	38.2	Yes	No
SEC-SQL-AU-004-RC08 No validation of OS identity	remote_os_authent = 'TRUE'	FLEXCUBE	3	9.1	Yes	No
SEC-SQL-AU-005-RC10 Lateral movement not considered	PASSWORD_VERIFY_FUNCTION is NULL or DEFAULT w...	FINNACLEDB_TEST	3	11.0	Yes	No
SEC-SQL-AU-006-RC08 HR/Identity integration absent	ldap_directory_access = 'NONE' OR row_count = ...	FINNACLEDB_TEST	3	15.9	Yes	No
SEC-SQL-AU-006-RC08 HR/Identity integration absent	ldap_directory_access = 'NONE' OR row_count = ...	FLEXCUBE	3	7.0	Yes	No
SEC-SQL-AU-006-RC14 Audit trails insufficient	audit_trail = 'NONE' OR audit_sys_operations ...	FINNACLEDB_TEST	3	20.4	Yes	No
SEC-SQL-AU-006-RC14 Audit trails insufficient	audit_trail = 'NONE' OR audit_sys_operations ...	FLEXCUBE	3	7.8	Yes	No
SEC-SQL-AUD-001-RC05 Service account permission failures	row_count > 0	FINNACLEDB_TEST	3	23.3	Yes	No

Root cause	Condition	Target	Rows	ms	Alert	Mail
SEC-SQL-AUD-001-RC05 Service account permission failures	row_count > 0	FLEXCUBE	3	8.7	Yes	No
SEC-SQL-AUD-001-RC07 Misconfigured log destination	row_count > 0	FINNACLEDB_TEST	3	25.2	Yes	No
SEC-SQL-AUD-001-RC07 Misconfigured log destination	row_count > 0	FLEXCUBE	3	6.4	Yes	No
SEC-SQL-AUD-003-RC01 Insecure file permissions	audit_trail LIKE '%OS%' OR audit_trail LIKE '...	FINNACLEDB_TEST	3	34.5	Yes	No
SEC-SQL-AUD-003-RC01 Insecure file permissions	audit_trail LIKE '%OS%' OR audit_trail LIKE '...	FLEXCUBE	3	7.4	Yes	No
SEC-SQL-AUD-003-RC03 Lack of File Integrity Monitoring (FIM)	audit_trail IN ('OS', 'XML', 'XML,EXTENDED') ...	FINNACLEDB_TEST	3	35.9	Yes	No
SEC-SQL-AUD-003-RC03 Lack of File Integrity Monitoring (FIM)	audit_trail IN ('OS', 'XML', 'XML,EXTENDED') ...	FLEXCUBE	3	6.5	Yes	No
SEC-SQL-AZ-001-RC10 Legacy applications with hardcoded privilege assumptions	any(session_privs > 0 AND object_count > :obj...	FINNACLEDB_TEST	3	303.4	Yes	No
SEC-SQL-AZ-003-RC10 Legacy account creation without documentation	row_count > 0 with very old accounts still ac...	FINNACLEDB_TEST	3	2.6	Yes	No
SEC-SQL-AZ-003-RC14 Failed or incomplete migration processes	row_count > 0	FLEXCUBE	3	413.6	Yes	Yes
SEC-SQL-AZ-005-RC05 Multi-database applications without explicit permission a...	row_count > 0	FLEXCUBE	3	193.4	Yes	No
SEC-SQL-AZ-005-RC09 Testing configurations carried forward to production	SUM(count) > :threshold	FINNACLEDB_TEST	3	100.6	Yes	Yes
SEC-SQL-AZ-005-RC09 Testing configurations carried forward to production	SUM(count) > :threshold	FLEXCUBE	3	38.2	Yes	Yes
SEC-SQL-AZ-006-RC01 Overly broad delegation of grant rights	row_count > :threshold	FLEXCUBE	3	8.7	Yes	No
SEC-SQL-CFG-004-RC12 Insufficient Configuration Segregation	debug or trace parameters enabled	FINNACLEDB_TEST	3	32.5	Yes	No
SEC-SQL-CFG-004-RC12 Insufficient Configuration Segregation	debug or trace parameters enabled	FLEXCUBE	3	8.4	Yes	No
SEC-SQL-CFG-005-RC07 One-Off Debugging Enabled Persistently	row_count > 0	FLEXCUBE	3	16.2	Yes	No
SEC-SQL-CFG-006-RC04 Privilege Escalation via Link	row_count > 0	FLEXCUBE	3	4.9	Yes	No
SEC-SQL-ENC-001-RC14 Competing Priorities in Development Cycles	row_count > 0	FINNACLEDB_TEST	3	37.1	Yes	No
SEC-SQL-ENC-001-RC14 Competing Priorities in Development Cycles	row_count > 0	FLEXCUBE	3	20.0	Yes	No
SEC-SQL-ENC-003-RC05 Performance Impact Concerns	row_count > 0	FINNACLEDB_TEST	3	3.8	Yes	No

Root cause	Condition	Target	Rows	ms	Alert	Mail
SEC-SQL-ENC-003-RC05 Performance Impact Concerns	row_count > 0	FLEXCUBE	3	4.2	Yes	No
SEC-SQL-ENC-004-RC02 Backup Software Doesn't Support Encryption	row_count > 0	FLEXCUBE	3	3.4	Yes	No
SEC-SQL-INJ-001-RC05 Time pressure and deadline constraints	row_count > 0	FLEXCUBE	3	3398.0	Yes	No
SEC-SQL-INJ-002-RC10 Security compliance misalignment	FAIL_count > 0	FINNACLEDB_TEST	3	14.4	Yes	No
SEC-SQL-INJ-002-RC10 Security compliance misalignment	FAIL_count > 0	FLEXCUBE	3	5.7	Yes	No

Truncated at 400 of 771 detections to keep the report usable; the full set is in the source JSON.

Appendix — penetration queries

The exact SQL executed against the target, one entry per root cause (identical across both targets). This is the "query to penetrate" for each check.

SEC-SQL-ACC-010-RC07 After-hours transaction activity
Condition: row_count > 0 — Active transactions started outside business hours (before 07:00 or after 19:00) by a non-applicative login may indicate unauthorized or suspicious a...
Anomalous Transaction Activity FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT s.sid, s.serial#, s.username, s.osuser, s.machine, s.program, t.start_time, ROUND((SYSDATE - TO_DATE(t.start_time, 'MM/DD/YY HH24:MI:SS')) * 86400) AS duration_seconds, TO_NUMBER(TO_CHAR(TO_DATE(t.start_time, 'MM/DD/YY HH24:MI:SS'), 'HH24')) AS start_hour FROM v\$session s JOIN v\$transaction t ON t.ses_addr = s.saddr WHERE s.type = 'USER' AND (TO_NUMBER(TO_CHAR(TO_DATE(t.start_time, 'MM/DD/YY HH24:MI:SS'), 'HH24')) < 7 OR TO_NUMBER(TO_CHAR(TO_DATE(t.start_time, 'MM/DD/YY HH24:MI:SS'), 'HH24')) >= 19) AND UPPER(s.username) NOT IN ('SYS', 'SYSTEM', 'DBSNMP', 'SYSMAN', 'XDB', 'GSMADMIN_INTERNAL', 'APP_USER', 'SVC_APP', 'ETL_SERVICE') AND NOT (UPPER(COALESCE(s.program, '')) LIKE '%JDBC%' OR UPPER(COALESCE(s.program, '')) LIKE 'W3WP%' OR UPPER(COALESCE(s.program, '')) LIKE '%ODBC%' OR UPPER(COALESCE(s.program, '')) LIKE 'PYTHON%' OR UPPER(COALESCE(s.program, '')) LIKE 'NODE%' OR (COALESCE(s.program, '') <> '' AND UPPER(s.username) LIKE '%' UPPER(s.program) '%')) ORDER BY t.start_time
SEC-SQL-ACC-010-RC08 Privilege escalation attempts in transactions
Condition: row_count > 0 — Mutual wait pairs indicate circular lock dependency from inconsistent acquisition order
Anomalous Transaction Activity FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT s1.sid AS sid_a, s1.username AS user_a, s2.sid AS sid_b, s2.username AS user_b, s1.event AS wait_event, sq1.sql_text AS query_a, sq2.sql_text AS query_b FROM v\$session s1 JOIN v\$session s2 ON s1.blocking_session = s2.sid AND s2.blocking_session = s1.sid LEFT JOIN v\$sqlarea sq1 ON s1.sql_id = sq1.sql_id LEFT JOIN v\$sqlarea sq2 ON s2.sql_id = sq2.sql_id WHERE s1.blocking_session IS NOT NULL AND s2.blocking_session IS NOT NULL
SEC-SQL-ACC-010-RC09 Data exfiltration patterns in transactions
Condition: isolation_level IN ('SERIALIZABLE', 'REPEATABLE-READ') — SERIALIZABLE adds shared locks on every read; REPEATABLE-READ uses gap locks that increase deadlock risk
Anomalous Transaction Activity FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT value AS isolation_level FROM v\$parameter WHERE name = 'transaction_isolation'
SEC-SQL-ACC-010-RC10 Same login active from multiple hosts
Condition: row_count > 0 — Same login active from multiple hosts may indicate credential sharing or compromise; the queries column lists the recent SQL each session is running
Anomalous Transaction Activity FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT username AS login_name, COUNT(DISTINCT machine) AS host_count, LISTAGG(DISTINCT machine, ', ') WITHIN GROUP (ORDER BY machine) AS hosts, COUNT(*) AS session_count, LISTAGG(DISTINCT SUBSTR(sql_text, 1, 500), ' ') WITHIN GROUP (ORDER BY SUBSTR(sql_text, 1, 500)) AS queries FROM (SELECT s.username, s.machine, s.sid, q.sql_text FROM gv\$session s LEFT JOIN gv\$sql q ON q.sql_id = s.sql_id AND q.inst_id = s.inst_id WHERE s.username IS NOT NULL AND s.type = 'USER' AND s.machine IS NOT NULL) GROUP BY username HAVING COUNT(DISTINCT machine) > 1 ORDER BY host_count DESC
SEC-SQL-ACC-010-RC11 Schema reconnaissance activity
Condition: lock_wait_timeout_sec > 50 — Default is 50 seconds; values much higher delay deadlock detection and resolution
Anomalous Transaction Activity FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT value AS isolation_level FROM v\$parameter WHERE name = 'transaction_isolation'
SEC-SQL-ACC-010-RC12 Dormant account suddenly active
Condition: wait_seconds > 50 — Transactions waiting beyond the timeout threshold suggest deadlock detection is not triggering promptly
Anomalous Transaction Activity FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT s.sid AS trx_id, s.status AS trx_state, s.logon_time AS trx_started, s.seconds_in_wait AS wait_seconds, sq.sql_text AS trx_query FROM v\$session s LEFT JOIN v\$sqlarea sq ON s.sql_id = sq.sql_id WHERE s.blocking_session IS NOT NULL AND s.status = 'ACTIVE' ORDER BY s.seconds_in_wait DESC
SEC-SQL-ACC-010-RC13 Mass data modification in transactions

Condition: row_count > 0 — Tables in lock waits that have foreign key constraints may be propagating locks to parent/child tables
Anomalous Transaction Activity FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT o.object_name AS lock_table, c.table_name AS fk_table, c.constraint_name, r.table_name AS referenced_table FROM v\$lock l JOIN dba_objects o ON o.object_id = l.id1 JOIN all_constraints c ON c.table_name = o.object_name AND c.constraint_type = 'R' JOIN all_constraints r ON r.constraint_name = c.r_constraint_name AND r.owner = c.r_owner WHERE l.block > 0
SEC-SQL-ACC-011-RC02 Active transactions (excluding monitoring user)
Condition: row_count > 0 — Active transactions/sessions within the last 120 seconds (oracle)
Active Transactions Present FINNACLEDB_TEST: DETECTED 4 rows FLEXCUBE: DETECTED 275 rows alert: Yes mail: No
SELECT s.sid AS session_id, s.username AS login_name, s.machine AS host_name, s.program AS program_name, SYS_CONTEXT('USERENV','DB_NAME') AS database_name, t.xidusn '.' t.xidslot '.' t.xidsqn AS transaction_id, TO_CHAR(TO_DATE(t.start_time,'MM/DD/YY HH24:MI:SS'),'YYYY-MM-DD HH24:MI:SS') AS transaction_begin_time, s.last_call_et AS duration_seconds, s.status AS transaction_state, SUBSTR(sq.sql_text, 1, 4000) AS query_text FROM v\$session s LEFT JOIN v\$transaction t ON t.ses_addr = s.saddr LEFT JOIN v\$sql sq ON sq.sql_id = s.sql_id WHERE s.type = 'USER' AND s.username IS NOT NULL AND s.username NOT IN ('SYS','SYSTEM','DBSNMP','SYSMAN','XDB') AND (s.status = 'ACTIVE' OR t.ses_addr IS NOT NULL OR s.last_call_et <= 120)
SEC-SQL-AU-001-RC01 Installation defaults not changed
Condition: PASSWORD_VERIFY_FUNCTION = 'NULL' OR FAILED_LOGIN_ATTEMPTS = 'UNLIMITED' — Default profile has no password verification function or unlimited login attempts — not hardened
Default Admin Accounts Enabled FINNACLEDB_TEST: DETECTED 4 rows FLEXCUBE: DETECTED 4 rows alert: Yes mail: Yes (destructive)
SELECT profile, resource_name, limit FROM dba_profiles WHERE profile = 'DEFAULT' AND resource_name IN ('FAILED_LOGIN_ATTEMPTS', 'PASSWORD_LIFE_TIME', 'PASSWORD_LOCK_TIME', 'PASSWORD_VERIFY_FUNCTION') ORDER BY resource_name
SEC-SQL-AU-001-RC02 Installation mode misalignment
Condition: row_count > 0 — Oracle edition and version identified for mode analysis
Default Admin Accounts Enabled FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: Yes (destructive)
SELECT banner FROM v\$version WHERE ROWNUM = 1
SEC-SQL-AU-001-RC03 Database not in protected network
Condition: row_count > :threshold — Connections from many different hosts — database accessible from wide network range
Default Admin Accounts Enabled FINNACLEDB_TEST: DETECTED 5 rows FLEXCUBE: DETECTED 20 rows alert: Yes mail: Yes (destructive)
SELECT DISTINCT machine, program, COUNT(*) AS connection_count FROM v\$session WHERE type = 'USER' AND machine != (SELECT host_name FROM v\$instance) GROUP BY machine, program ORDER BY connection_count DESC FETCH FIRST 20 ROWS ONLY
SEC-SQL-AU-001-RC04 Lack of post-deployment hardening checklist
Condition: SUM(issue_count) > 2 — Multiple hardening gaps found — no systematic post-deployment security checklist was applied
Default Admin Accounts Enabled FINNACLEDB_TEST: DETECTED 5 rows FLEXCUBE: DETECTED 5 rows alert: Yes mail: No
SELECT 'default_passwords' AS check_item, COUNT(*) AS issue_count FROM dba_users_with_defpwd UNION ALL SELECT 'unlimited_login_attempts', COUNT(*) FROM dba_profiles WHERE resource_name = 'FAILED_LOGIN_ATTEMPTS' AND limit = 'UNLIMITED' UNION ALL SELECT 'no_password_verify', COUNT(*) FROM dba_profiles WHERE resource_name = 'PASSWORD_VERIFY_FUNCTION' AND (limit = 'NULL' OR limit IS NULL) UNION ALL SELECT 'audit_trail_none', CASE WHEN (SELECT value FROM v\$parameter WHERE name = 'audit_trail') = 'NONE' THEN 1 ELSE 0 END FROM dual UNION ALL SELECT 'sample_schemas_open', COUNT(*) FROM dba_users WHERE username IN ('SCOTT', 'HR', 'OE', 'SH', 'PM', 'IX', 'BI') AND account_status = 'OPEN'
SEC-SQL-AU-001-RC05 Knowledge gap on account purposes
Condition: row_count > :threshold — Multiple Oracle-maintained internal accounts left unlocked — likely not understanding which accounts are needed
Default Admin Accounts Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT username, account_status, profile, TO_CHAR(created, 'YYYY-MM-DD') AS created, default_tablespace, oracle_maintained FROM dba_users WHERE oracle_maintained = 'Y' AND account_status = 'OPEN' AND username NOT IN ('SYS', 'SYSTEM') ORDER BY username
SEC-SQL-AU-001-RC06 Incomplete migration procedures

Condition: SUM(has_issue) > 0 — Mixed audit modes, non-standard profiles, or orphaned objects suggest incomplete migration
Default Admin Accounts Enabled FINNACLEDB_TEST: DETECTED 3 rows FLEXCUBE: DETECTED 6 rows alert: Yes mail: No
SELECT 'mixed_audit_modes' AS check_item, CASE WHEN (SELECT value FROM v\$parameter WHERE name = 'audit_trail') != 'NONE' AND (SELECT value FROM v\$option WHERE parameter = 'Unified Auditing') = 'TRUE' THEN 1 ELSE 0 END AS has_issue FROM dual UNION ALL SELECT 'non_standard_profiles', COUNT(*) FROM dba_profiles WHERE profile NOT IN ('DEFAULT', 'ORA_STIG_PROFILE') AND resource_type = 'PASSWORD' GROUP BY profile HAVING COUNT(*) > 0 UNION ALL SELECT 'orphaned_db_links', COUNT(*) FROM dba_db_links WHERE owner NOT IN (SELECT username FROM dba_users WHERE account_status = 'OPEN')
SEC-SQL-AU-001-RC07 Recovery procedure reliance
Condition: row_count >= 0 — Review restore points that could be used to access historical data states
Default Admin Accounts Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT name, guarantee_flashback_database, storage_size, TO_CHAR(time, 'YYYY-MM-DD HH24:MI:SS') AS created FROM v\$restore_point ORDER BY time DESC
SEC-SQL-AU-001-RC08 Multi-vendor environment complexity
Condition: row_count > 0 — Active accounts with no recent login in multi-vendor environment indicate management gaps
Default Admin Accounts Enabled FINNACLEDB_TEST: DETECTED 7 rows FLEXCUBE: DETECTED 10 rows alert: Yes mail: No
SELECT username, account_status, authentication_type, TO_CHAR(last_login, 'YYYY-MM-DD HH24:MI:SS') AS last_login, ROUND(SYSDATE - NVL(CAST(last_login AS DATE), created)) AS days_inactive FROM dba_users WHERE oracle_maintained = 'N' AND account_status = 'OPEN' AND (last_login IS NULL OR last_login < SYSDATE - 90) ORDER BY days_inactive DESC
SEC-SQL-AU-001-RC09 No integration with identity provisioning
Condition: any_value authentication_type = 'PASSWORD' AND account_count > :threshold_accounts — Many password-authenticated accounts indicate lack of centralized identity provisioning
Default Admin Accounts Enabled FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT authentication_type, COUNT(*) AS account_count FROM dba_users WHERE oracle_maintained = 'N' AND account_status = 'OPEN' GROUP BY authentication_type ORDER BY account_count DESC
SEC-SQL-AU-001-RC10 Account privilege escalation over time
Condition: any_value sys_priv_count + role_count > :threshold_total — Users with high total privilege counts indicate privilege accumulation over time
Default Admin Accounts Enabled FINNACLEDB_TEST: DETECTED 11 rows FLEXCUBE: DETECTED 20 rows alert: Yes mail: No
SELECT u.username, u.created, u.last_login, (SELECT COUNT(*) FROM dba_sys_privs sp WHERE sp.grantee = u.username) AS sys_priv_count, (SELECT COUNT(*) FROM dba_role_privs rp WHERE rp.grantee = u.username) AS role_count, (SELECT COUNT(*) FROM dba_tab_privs tp WHERE tp.grantee = u.username) AS obj_priv_count FROM dba_users u WHERE u.oracle_maintained = 'N' AND u.account_status = 'OPEN' ORDER BY (SELECT COUNT(*) FROM dba_sys_privs sp WHERE sp.grantee = u.username) + (SELECT COUNT(*) FROM dba_role_privs rp WHERE rp.grantee = u.username) DESC FETCH FIRST 20 ROWS ONLY
SEC-SQL-AU-001-RC11 Inadequate change control
Condition: row_count = 0 — No audit policies for account changes indicates inadequate change control
Default Admin Accounts Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT policy_name, audit_option FROM audit_unified_policies WHERE audit_option IN ('CREATE USER', 'ALTER USER', 'DROP USER', 'GRANT', 'REVOKE') ORDER BY policy_name
SEC-SQL-AU-001-RC12 Assumption of "unused" protection
Condition: row_count > 0 — Active accounts with extended inactivity that are still open and usable
Default Admin Accounts Enabled FINNACLEDB_TEST: DETECTED 7 rows FLEXCUBE: DETECTED 10 rows alert: Yes mail: No
SELECT username, account_status, authentication_type, TO_CHAR(created, 'YYYY-MM-DD') AS created, TO_CHAR(last_login, 'YYYY-MM-DD HH24:MI:SS') AS last_login, ROUND(SYSDATE - NVL(CAST(last_login AS DATE), created)) AS days_since_activity FROM dba_users WHERE oracle_maintained = 'N' AND account_status = 'OPEN' AND (last_login IS NULL OR last_login < SYSDATE - 90) ORDER BY days_since_activity DESC
SEC-SQL-AU-002-RC01 Initialization flag usage
Condition: row_count > 0 — Default Oracle sample/utility accounts still open indicate initialization defaults not cleaned up

Empty/Blank Passwords Allowed FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT username, account_status, authentication_type, created FROM dba_users WHERE username IN ('SCOTT', 'HR', 'OE', 'SH', 'PM', 'IX', 'BI', 'DEMO', 'TEST', 'SAMPLE', 'DBSNMP', 'MGMT_VIEW', 'SYSMAN', 'APEX_PUBLIC_USER') AND account_status = 'OPEN' ORDER BY username
SEC-SQL-AU-002-RC03 Installation interruption
Condition: row_count > 0 — Accounts with default passwords confirm post-installation security hardening was not completed
Empty/Blank Passwords Allowed FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT username FROM dba_users_with_defpwd WHERE username IN (SELECT username FROM dba_users WHERE account_status = 'OPEN') ORDER BY username
SEC-SQL-AU-002-RC04 Test account forgotten in production
Condition: row_count > 0 — Accounts with test/temp/dev naming patterns in production indicate forgotten test accounts
Empty/Blank Passwords Allowed FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: Yes (destructive)
SELECT username, account_status, authentication_type, created, last_login FROM dba_users WHERE oracle_maintained = 'N' AND (UPPER(username) LIKE '%TEST%' OR UPPER(username) LIKE '%TEMP%' OR UPPER(username) LIKE '%DEV%' OR UPPER(username) LIKE '%QA%' OR UPPER(username) LIKE '%DEMO%' OR UPPER(username) LIKE '%SAMPLE%' OR UPPER(username) LIKE '%TMP%' OR UPPER(username) LIKE '%DEBUG%') AND account_status = 'OPEN' ORDER BY username
SEC-SQL-AU-002-RC05 Password never set in configuration
Condition: row_count > 0 — Accounts with no password or external authentication may be exploitable
Empty/Blank Passwords Allowed FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT username, account_status, authentication_type, password_versions, external_name, created FROM dba_users WHERE oracle_maintained = 'N' AND account_status = 'OPEN' AND (authentication_type = 'NONE' OR authentication_type = 'EXTERNAL' OR password_versions IS NULL) ORDER BY username
SEC-SQL-AU-002-RC06 Identity provisioning tool failure
Condition: row_count > 0 — Inactive globally/externally authenticated accounts suggest provisioning sync failure
Empty/Blank Passwords Allowed FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT username, account_status, authentication_type, external_name, TO_CHAR(created, 'YYYY-MM-DD') AS created, TO_CHAR(last_login, 'YYYY-MM-DD HH24:MI:SS') AS last_login, ROUND(SYSDATE - NVL(CAST(last_login AS DATE), created)) AS days_inactive FROM dba_users WHERE authentication_type IN ('GLOBAL', 'EXTERNAL') AND account_status = 'OPEN' AND (last_login IS NULL OR last_login < SYSDATE - 90) ORDER BY days_inactive DESC
SEC-SQL-AU-002-RC07 Configuration file permission bypass
Condition: row_count > 0 — Oracle configuration file paths identified for permission review
Empty/Blank Passwords Allowed FINNACLEDB_TEST: DETECTED 6 rows FLEXCUBE: DETECTED 6 rows alert: Yes mail: Yes (destructive)
SELECT name, value FROM v\$parameter WHERE name IN ('spfile', 'control_files', 'diagnostic_dest', 'audit_file_dest', 'background_dump_dest', 'user_dump_dest') AND value IS NOT NULL ORDER BY name
SEC-SQL-AU-002-RC09 Application provisioning defaults
Condition: row_count > 0 — Application accounts using default passwords confirm provisioning defaults not changed
Empty/Blank Passwords Allowed FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT username FROM dba_users_with_defpwd WHERE username IN (SELECT username FROM dba_users WHERE account_status = 'OPEN') ORDER BY username
SEC-SQL-AU-002-RC10 LDAP/directory integration incomplete
Condition: row_count > 0 — LDAP parameters configured but mixed auth types confirms incomplete integration
Empty/Blank Passwords Allowed FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No

SELECT name, value FROM v\$parameter WHERE name IN ('ldap_directory_access', 'ldap_directory_sysauth') ORDER BY name
SEC-SQL-AU-002-RC11 Batch user creation scripts
Condition: row_count > 0 — Multiple accounts created in the same minute indicate batch creation scripts
Empty/Blank Passwords Allowed FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: Yes (destructive)
SELECT TO_CHAR(created, 'YYYY-MM-DD HH24:MI') AS creation_window, COUNT(*) AS accounts_created, LISTAGG(username, ', ') WITHIN GROUP (ORDER BY username) AS usernames FROM dba_users WHERE oracle_maintained = 'N' GROUP BY TO_CHAR(created, 'YYYY-MM-DD HH24:MI') HAVING COUNT(*) > 3 ORDER BY creation_window DESC
SEC-SQL-AU-002-RC12 No pre-deployment security scanning
Condition: row_count = 0 — No security scanning tool accounts found in the database
Empty/Blank Passwords Allowed FINNACLEDB_TEST: DETECTED 0 rows FLEXCUBE: DETECTED 0 rows alert: Yes mail: Yes (destructive)
SELECT username, account_status FROM dba_users WHERE username IN ('DBSAT_ADMIN', 'DBSAT', 'SECURITY_SCAN', 'AUDIT_ADMIN') AND account_status != 'EXPIRED & LOCKED'
SEC-SQL-AU-003-RC01 Default configuration never modified
Condition: any_value resource_name = 'PASSWORD_LIFE_TIME' AND limit = 'UNLIMITED' — Default profile with unlimited password lifetime indicates never-modified password policy
Weak Password Policy FINNACLEDB_TEST: DETECTED 9 rows FLEXCUBE: DETECTED 9 rows alert: Yes mail: No
SELECT resource_name, limit FROM dba_profiles WHERE profile = 'DEFAULT' AND resource_type = 'PASSWORD' ORDER BY resource_name
SEC-SQL-AU-003-RC02 Complexity enforcement disabled intentionally
Condition: any_value limit = 'NULL' — Profiles without password verify function have no complexity enforcement
Weak Password Policy FINNACLEDB_TEST: DETECTED 3 rows FLEXCUBE: DETECTED 6 rows alert: Yes mail: No
SELECT profile, resource_name, limit FROM dba_profiles WHERE resource_name = 'PASSWORD_VERIFY_FUNCTION' ORDER BY profile
SEC-SQL-AU-003-RC03 Legacy system requirements
Condition: row_count > 0 — Case-insensitive logon or legacy protocol versions indicate legacy compatibility weakening security
Weak Password Policy FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('sec_case_sensitive_logon', 'sqlnet.allowed_logon_version_server', 'sqlnet.allowed_logon_version_client') AND (value = 'FALSE' OR value LIKE '%8%' OR value LIKE '%10%') ORDER BY name
SEC-SQL-AU-003-RC04 Regulatory compliance misinterpretation
Condition: row_count > 0 — Review password policy settings for compliance completeness
Weak Password Policy FINNACLEDB_TEST: DETECTED 7 rows FLEXCUBE: DETECTED 7 rows alert: Yes mail: No
SELECT profile, resource_name, limit FROM dba_profiles WHERE resource_name IN ('PASSWORD_LIFE_TIME', 'PASSWORD_REUSE_TIME', 'PASSWORD_REUSE_MAX', 'PASSWORD_LOCK_TIME', 'FAILED_LOGIN_ATTEMPTS', 'PASSWORD_GRACE_TIME', 'PASSWORD_VERIFY_FUNCTION') AND profile = 'DEFAULT' ORDER BY resource_name
SEC-SQL-AU-003-RC05 Complexity vs. length tradeoff avoided
Condition: status != 'VALID' OR row_count = 0 — Invalid or missing verification function means password policy is not actually enforced
Weak Password Policy FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 3 rows alert: Yes mail: No
SELECT object_name, object_type, status FROM dba_objects WHERE object_name IN (SELECT limit FROM dba_profiles WHERE resource_name = 'PASSWORD_VERIFY_FUNCTION' AND limit NOT IN ('NULL', 'DEFAULT')) AND object_type = 'FUNCTION' ORDER BY object_name
SEC-SQL-AU-003-RC07 User friction avoidance

Condition: PASSWORD_VERIFY_FUNCTION = 'NULL' OR PASSWORD_LIFE_TIME = 'UNLIMITED' OR FAILED_LOGIN_ATTEMPTS = 'UNLIMITED' — Relaxed password settings indicate policy was weakened to reduce user friction
Weak Password Policy FINNACLEDB_TEST: DETECTED 6 rows FLEXCUBE: DETECTED 24 rows alert: Yes mail: No
SELECT profile, resource_name, limit FROM dba_profiles WHERE resource_name IN ('PASSWORD_VERIFY_FUNCTION', 'PASSWORD_LIFE_TIME', 'PASSWORD_REUSE_TIME', 'PASSWORD_REUSE_MAX', 'PASSWORD_LOCK_TIME', 'FAILED_LOGIN_ATTEMPTS') AND profile IN (SELECT DISTINCT profile FROM dba_users WHERE account_status = 'OPEN') ORDER BY profile, resource_name
SEC-SQL-AU-003-RC08 Development environment settings bleeding to production
Condition: row_count > 0 — Permissive password settings (NULL verify, UNLIMITED lifetime) are typical of dev environments
Weak Password Policy FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 7 rows alert: Yes mail: No
SELECT profile, resource_name, limit FROM dba_profiles WHERE resource_name IN ('PASSWORD_VERIFY_FUNCTION', 'PASSWORD_LIFE_TIME', 'FAILED_LOGIN_ATTEMPTS') AND limit IN ('NULL', 'DEFAULT', 'UNLIMITED') AND profile IN (SELECT DISTINCT profile FROM dba_users WHERE account_status = 'OPEN') ORDER BY profile, resource_name
SEC-SQL-AU-003-RC09 No centralized policy definition
Condition: multiple profiles with different PASSWORD_VERIFY_FUNCTION or PASSWORD_LIFE_TIME — Inconsistent password settings across profiles indicates no centralized policy enforcement
Weak Password Policy FINNACLEDB_TEST: DETECTED 27 rows FLEXCUBE: DETECTED 54 rows alert: Yes mail: No
SELECT profile, resource_name, limit FROM dba_profiles WHERE resource_type = 'PASSWORD' ORDER BY profile, resource_name
SEC-SQL-AU-003-RC11 Lack of monitoring tools
Condition: AUDIT_TRAIL = 'NONE' OR no audit policy for login — No audit trail or no login audit policy means failed logins are not monitored
Weak Password Policy FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT parameter_name, parameter_value FROM dba_audit_mgmt_config_params WHERE parameter_name = 'AUDIT_TRAIL' UNION ALL SELECT 'UNIFIED_AUDIT' AS parameter_name, DECODE((SELECT value FROM v\$option WHERE parameter = 'Unified Auditing'), 'TRUE', 'ENABLED', 'DISABLED') AS parameter_value FROM dual
SEC-SQL-AU-003-RC13 Account creation automation without validation
Condition: row_count > 0 — Recently created accounts to check if they have proper password profiles assigned
Weak Password Policy FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 5 rows alert: Yes mail: No
SELECT username, profile, account_status, created, authentication_type FROM dba_users WHERE created > SYSDATE - 30 AND username NOT IN ('SYS', 'SYSTEM', 'DBSNMP', 'OUTLN') ORDER BY created DESC
SEC-SQL-AU-004-RC02 Oracle REMOTE_OS_ROLES enabled
Condition: value = 'TRUE' — REMOTE_OS_ROLES=TRUE allows remote OS users to receive database roles based on OS group membership
Remote OS Authentication Enabled FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name = 'remote_os_roles'
SEC-SQL-AU-004-RC03 OS service account access
Condition: row_count > 0 — Externally authenticated users bypass Oracle password authentication via OS identity
Remote OS Authentication Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT username, authentication_type, account_status, external_name FROM dba_users WHERE authentication_type = 'EXTERNAL' AND account_status = 'OPEN' ORDER BY username
SEC-SQL-AU-004-RC04 OS-integrated authentication trust configuration
Condition: row_count > 0 — Externally or globally authenticated users can bypass password checks from remote hosts
Remote OS Authentication Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No

SELECT u.username, u.authentication_type, u.external_name, u.account_status FROM dba_users u WHERE u.authentication_type IN ('EXTERNAL', 'GLOBAL') AND u.account_status = 'OPEN' ORDER BY u.username
SEC-SQL-AU-004-RC05 OS authentication module delegation
Condition: remote_os_authent = 'TRUE' — OS authentication delegation enabled for remote connections
Remote OS Authentication Enabled FINNACLEDB_TEST: DETECTED 3 rows FLEXCUBE: DETECTED 3 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('remote_os_authent', 'os_authent_prefix', 'remote_os_roles') ORDER BY name
SEC-SQL-AU-004-RC06 Kerberos/SSPI authentication without encryption
Condition: authentication includes kerberos but encryption is not REQUIRED — Kerberos authentication without mandatory encryption exposes authentication tokens
Remote OS Authentication Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('sqlnet.authentication_services', 'sqlnet.crypto_checksum_server', 'sqlnet.encryption_server', 'sqlnet.crypto_checksum_types_server', 'sqlnet.encryption_types_server') ORDER BY name
SEC-SQL-AU-004-RC07 Cross-system domain trusts
Condition: row_count > 0 — Globally authenticated users rely on directory service for authentication
Remote OS Authentication Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT username, authentication_type, external_name, account_status FROM dba_users WHERE authentication_type = 'GLOBAL' AND account_status = 'OPEN' ORDER BY username
SEC-SQL-AU-004-RC08 No validation of OS identity
Condition: remote_os_authent = 'TRUE' — REMOTE_OS_AUTHENT=TRUE accepts OS identity claims without cryptographic verification
Remote OS Authentication Enabled FINNACLEDB_TEST: DETECTED 3 rows FLEXCUBE: DETECTED 3 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('remote_os_authent', 'remote_os_roles', 'os_authent_prefix') ORDER BY name
SEC-SQL-AU-004-RC10 Assume-inside-firewall security model
Condition: ext_auth_users > 0 — Active externally authenticated users combined with remote OS auth indicate perimeter-trust model
Remote OS Authentication Enabled FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT COUNT(*) AS ext_auth_users FROM dba_users WHERE authentication_type = 'EXTERNAL' AND account_status = 'OPEN'
SEC-SQL-AU-004-RC11 LDAP with weak connection security
Condition: row_count > 0 — Active globally authenticated users confirm LDAP directory integration is in use
Remote OS Authentication Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT username, authentication_type, account_status FROM dba_users WHERE authentication_type = 'GLOBAL' AND account_status = 'OPEN'
SEC-SQL-AU-004-RC12 No OS authentication logging
Condition: no LOGON audit policy enabled OR audit_trail = 'NONE' — No audit of logon events means OS-authenticated sessions are not tracked
Remote OS Authentication Enabled FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT policy_name, audit_option, audit_condition FROM audit_unified_policies WHERE audit_option = 'LOGON' UNION ALL SELECT 'TRADITIONAL' AS policy_name, 'AUDIT_TRAIL' AS audit_option, value AS audit_condition FROM v\$parameter WHERE name = 'audit_trail'
SEC-SQL-AU-005-RC01 Application compatibility requirement
Condition: row_count > 0 — Password-authenticated accounts may exist for legacy application compatibility instead of OS/Kerberos auth

Mixed Mode Authentication FINNACLEDB_TEST: DETECTED 11 rows FLEXCUBE: DETECTED 30 rows alert: Yes mail: No
SELECT username, authentication_type, account_status, profile, created, last_login, common FROM dba_users WHERE authentication_type = 'PASSWORD' AND account_status = 'OPEN' AND username NOT IN ('SYS','SYSTEM','DBSNMP','OUTLN','XDB') AND oracle_maintained = 'N' ORDER BY username
SEC-SQL-AU-005-RC02 Multi-tenancy needs
Condition: row_count > 0 — Single accounts connecting from many distinct machines confirms shared credential usage
Mixed Mode Authentication FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT username, COUNT(DISTINCT machine) AS distinct_machines, COUNT(DISTINCT program) AS distinct_programs, COUNT(*) AS session_count FROM v\$session WHERE type = 'USER' AND username IS NOT NULL AND username NOT IN ('SYS','SYSTEM') GROUP BY username HAVING COUNT(DISTINCT machine) > 5 ORDER BY distinct_machines DESC
SEC-SQL-AU-005-RC03 Misguided "flexibility" goal
Condition: row_count > 0 — ANY-style privileges bypass schema isolation — a misguided flexibility approach
Mixed Mode Authentication FINNACLEDB_TEST: DETECTED 15 rows FLEXCUBE: DETECTED 71 rows alert: Yes mail: No
SELECT grantee, privilege FROM dba_sys_privs WHERE grantee NOT IN ('SYS','SYSTEM','DBA','IMP_FULL_DATABASE','EXP_FULL_DATABASE') AND privilege IN ('SELECT ANY TABLE', 'INSERT ANY TABLE', 'UPDATE ANY TABLE', 'DELETE ANY TABLE', 'CREATE ANY TABLE', 'DROP ANY TABLE', 'ALTER ANY TABLE', 'GRANT ANY PRIVILEGE', 'CREATE ANY PROCEDURE', 'EXECUTE ANY PROCEDURE') AND grantee IN (SELECT username FROM dba_users WHERE oracle_maintained = 'N') ORDER BY grantee, privilege
SEC-SQL-AU-005-RC04 Incomplete migration
Condition: multiple authentication_type values exist — Mixed authentication types suggest incomplete migration from password to external auth
Mixed Mode Authentication FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT authentication_type, COUNT(*) AS account_count FROM dba_users WHERE oracle_maintained = 'N' AND account_status = 'OPEN' GROUP BY authentication_type ORDER BY account_count DESC
SEC-SQL-AU-005-RC05 External contractor access
Condition: row_count > 0 — Accounts with unlimited password lifetime lack mandatory rotation and may persist after contractor departure
Mixed Mode Authentication FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 21 rows alert: Yes mail: No
SELECT u.username, u.account_status, u.created, u.expiry_date, p.resource_name, p.limit AS password_life_time FROM dba_users u JOIN dba_profiles p ON u.profile = p.profile WHERE p.resource_name = 'PASSWORD_LIFE_TIME' AND (p.limit = 'UNLIMITED' OR p.limit = 'DEFAULT') AND u.oracle_maintained = 'N' AND u.account_status = 'OPEN' ORDER BY u.created
SEC-SQL-AU-005-RC06 Development environment consistency
Condition: row_count > 0 — Development/test-named accounts in production suggest environment bleed
Mixed Mode Authentication FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT username, account_status, profile, created, last_login FROM dba_users WHERE oracle_maintained = 'N' AND (username LIKE '%DEV%' OR username LIKE '%TEST%' OR username LIKE '%QA%' OR username LIKE '%DEMO%' OR username LIKE '%SANDBOX%') AND account_status = 'OPEN' ORDER BY username
SEC-SQL-AU-005-RC07 Post-installation guidance misunderstood
Condition: row_count > 0 — Oracle-maintained accounts left OPEN (besides SYS/SYSTEM) indicate post-installation hardening not completed
Mixed Mode Authentication FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT username, account_status, profile, authentication_type FROM dba_users WHERE oracle_maintained = 'Y' AND account_status = 'OPEN' AND username NOT IN ('SYS','SYSTEM') ORDER BY username
SEC-SQL-AU-005-RC08 Service account password storage
Condition: row_count > 0 — Service accounts using password authentication may have credentials in connection strings or config files
Mixed Mode Authentication FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No

SELECT username, account_status, profile, authentication_type, created, last_login FROM dba_users WHERE oracle_maintained = 'N' AND authentication_type = 'PASSWORD' AND account_status = 'OPEN' AND (username LIKE '%SVC%' OR username LIKE '%SERVICE%' OR username LIKE '%APP%' OR username LIKE '%BATCH%' OR username LIKE '%ETL%' OR username LIKE '%API%') ORDER BY username
SEC-SQL-AU-005-RC10 Lateral movement not considered
Condition: PASSWORD_VERIFY_FUNCTION is NULL or DEFAULT with weak verification — Weak or absent password verification function allows simple passwords that may be reused across systems
Mixed Mode Authentication FINNACLEDB_TEST: DETECTED 3 rows FLEXCUBE: DETECTED 12 rows alert: Yes mail: No
SELECT p.profile, r.resource_name, r.limit FROM dba_profiles r JOIN (SELECT DISTINCT profile FROM dba_users WHERE account_status = 'OPEN' AND oracle_maintained = 'N') p ON r.profile = p.profile WHERE r.resource_name IN ('PASSWORD_REUSE_TIME', 'PASSWORD_REUSE_MAX', 'PASSWORD_VERIFY_FUNCTION') ORDER BY p.profile, r.resource_name
SEC-SQL-AU-005-RC11 Kerberos delegation configuration complexity
Condition: row_count > 0 OR row_count = 0 with kerberos params set — External authentication status shows Kerberos deployment state
Mixed Mode Authentication FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT username, authentication_type, external_name, account_status FROM dba_users WHERE authentication_type = 'EXTERNAL' AND account_status = 'OPEN' ORDER BY username
SEC-SQL-AU-005-RC12 Application connection string defaults
Condition: row_count > 0 — Application connections using dedicated database accounts likely use password-based connection strings
Mixed Mode Authentication FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT s.username, s.program, s.machine, s.osuser, COUNT(*) AS connection_count FROM v\$session s WHERE s.type = 'USER' AND s.username IS NOT NULL AND s.username NOT IN ('SYS','SYSTEM','DBSNMP') GROUP BY s.username, s.program, s.machine, s.osuser HAVING COUNT(*) > 5 ORDER BY connection_count DESC
SEC-SQL-AU-005-RC13 No automated compliance scanning
Condition: row_count > 0 AND comprehensive policies enabled — Enabled audit policies confirm automated compliance monitoring is in place
Mixed Mode Authentication FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT policy_name, enabled_option, entity_name, entity_type, success, failure FROM audit_unified_enabled_policies ORDER BY policy_name
SEC-SQL-AU-006-RC01 No activity monitoring
Condition: row_count > 0 AND some accounts have NULL or very old last_login — Accounts with no recent activity may be dormant and should be reviewed or locked
Dormant/Unused Accounts FINNACLEDB_TEST: DETECTED 11 rows FLEXCUBE: DETECTED 20 rows alert: Yes mail: No
SELECT username, account_status, TO_CHAR(last_login, 'YYYY-MM-DD HH24:MI:SS') AS last_login, TO_CHAR(created, 'YYYY-MM-DD') AS created, ROUND(SYSDATE - NVL(CAST(last_login AS DATE), created)) AS days_since_activity FROM dba_users WHERE oracle_maintained = 'N' AND account_status = 'OPEN' ORDER BY days_since_activity DESC NULLS FIRST FETCH FIRST 20 ROWS ONLY
SEC-SQL-AU-006-RC02 Offboarding process incomplete
Condition: row_count > 0 — Accounts with no login activity beyond threshold suggest incomplete offboarding
Dormant/Unused Accounts FINNACLEDB_TEST: DETECTED 7 rows FLEXCUBE: DETECTED 10 rows alert: Yes mail: Yes (destructive)
SELECT username, account_status, TO_CHAR(created, 'YYYY-MM-DD') AS created, TO_CHAR(last_login, 'YYYY-MM-DD HH24:MI:SS') AS last_login, profile, authentication_type, ROUND(SYSDATE - NVL(CAST(last_login AS DATE), created)) AS days_inactive FROM dba_users WHERE oracle_maintained = 'N' AND account_status = 'OPEN' AND (last_login IS NULL OR last_login < SYSDATE - 90) ORDER BY days_inactive DESC NULLS FIRST
SEC-SQL-AU-006-RC03 Service account lifecycle unknown
Condition: row_count > 0 — Service accounts with unlimited or very long password lifetime lack proper credential rotation
Dormant/Unused Accounts FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT u.username, u.profile, u.created, u.last_login, p.resource_name, p.limit FROM dba_users u JOIN dba_profiles p ON u.profile = p.profile WHERE u.oracle_maintained = 'N' AND u.account_status = 'OPEN' AND p.resource_name = 'PASSWORD_LIFE_TIME' AND (p.limit = 'UNLIMITED' OR CAST(p.limit AS INT) > 365) AND (u.username LIKE '%SVC%' OR u.username LIKE '%SERVICE%' OR u.username LIKE '%APP%' OR u.username LIKE '%BATCH%') ORDER BY u.username

SEC-SQL-AU-006-RC04 Contractor account cleanup forgotten
Condition: row_count > 0 — Inactive contractor-named accounts confirm cleanup was forgotten
Dormant/Unused Accounts FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT username, account_status, TO_CHAR(created, 'YYYY-MM-DD') AS created, TO_CHAR(last_login, 'YYYY-MM-DD HH24:MI:SS') AS last_login, profile, ROUND(SYSDATE - NVL(CAST(last_login AS DATE), created)) AS days_inactive FROM dba_users WHERE oracle_maintained = 'N' AND account_status = 'OPEN' AND (username LIKE '%CONTRACT%' OR username LIKE '%VENDOR%' OR username LIKE '%EXTERN%' OR username LIKE '%TEMP_%' OR username LIKE '%CONSULT%') AND NVL(CAST(last_login AS DATE), created) < SYSDATE - 60 ORDER BY days_inactive DESC
SEC-SQL-AU-006-RC05 Temporary project access never revoked
Condition: row_count > 0 — Powerful system privileges granted directly to users suggest temporary grants not revoked
Dormant/Unused Accounts FINNACLEDB_TEST: DETECTED 12 rows FLEXCUBE: DETECTED 58 rows alert: Yes mail: Yes (destructive)
SELECT grantee, privilege, admin_option FROM dba_sys_privs WHERE grantee IN (SELECT username FROM dba_users WHERE oracle_maintained = 'N' AND account_status = 'OPEN') AND privilege IN ('SELECT ANY TABLE', 'INSERT ANY TABLE', 'DELETE ANY TABLE', 'CREATE ANY TABLE', 'DROP ANY TABLE', 'ALTER ANY TABLE', 'EXECUTE ANY PROCEDURE', 'CREATE SESSION', 'DBA') AND grantee NOT IN ('SYS', 'SYSTEM') ORDER BY grantee, privilege
SEC-SQL-AU-006-RC06 Test/dev accounts migrated to production
Condition: row_count > 0 — Test/dev-named accounts in production indicate environment migration without cleanup
Dormant/Unused Accounts FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: Yes (destructive)
SELECT username, account_status, profile, created, last_login, authentication_type FROM dba_users WHERE oracle_maintained = 'N' AND account_status = 'OPEN' AND (username LIKE 'TEST%' OR username LIKE '%_TEST' OR username LIKE 'DEV%' OR username LIKE '%_DEV' OR username LIKE 'QA%' OR username LIKE 'UAT%' OR username LIKE 'DEMO%' OR username LIKE 'SAMPLE%' OR username LIKE 'SCOTT' OR username LIKE 'HR') ORDER BY username
SEC-SQL-AU-006-RC07 No formal account inventory
Condition: row_count > 0 — Full account inventory with privilege counts provides baseline for account management review
Dormant/Unused Accounts FINNACLEDB_TEST: DETECTED 13 rows FLEXCUBE: DETECTED 32 rows alert: Yes mail: No
SELECT u.username, u.account_status, u.authentication_type, u.profile, u.created, u.last_login, u.expiry_date, u.oracle_maintained, COUNT(DISTINCT sp.privilege) AS sys_privs, COUNT(DISTINCT rp.granted_role) AS roles FROM dba_users u LEFT JOIN dba_sys_privs sp ON u.username = sp.grantee LEFT JOIN dba_role_privs rp ON u.username = rp.grantee WHERE u.account_status = 'OPEN' GROUP BY u.username, u.account_status, u.authentication_type, u.profile, u.created, u.last_login, u.expiry_date, u.oracle_maintained ORDER BY u.oracle_maintained, u.username
SEC-SQL-AU-006-RC08 HR/Identity integration absent
Condition: ldap_directory_access = 'NONE' OR row_count = 0 — No LDAP directory access configured confirms missing HR/Identity integration
Dormant/Unused Accounts FINNACLEDB_TEST: DETECTED 3 rows FLEXCUBE: DETECTED 3 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('ldap_directory_access', 'os_authent_prefix', 'remote_os_authent') ORDER BY name
SEC-SQL-AU-006-RC09 Quarterly reviews too infrequent
Condition: row_count > 0 — Active accounts with no recent login indicate dormant accounts needing quarterly review
Dormant/Unused Accounts FINNACLEDB_TEST: DETECTED 7 rows FLEXCUBE: DETECTED 10 rows alert: Yes mail: No
SELECT u.username, TO_CHAR(u.created, 'YYYY-MM-DD') AS created, u.account_status, TO_CHAR(u.last_login, 'YYYY-MM-DD HH24:MI:SS') AS last_login, u.profile, ROUND(SYSDATE - NVL(CAST(u.last_login AS DATE), u.created)) AS days_since_activity FROM dba_users u WHERE u.account_status = 'OPEN' AND u.username NOT IN ('SYS', 'SYSTEM', 'DBSNMP', 'OUTLN', 'MDSYS', 'XDB') AND (u.last_login IS NULL OR u.last_login < SYSDATE - 90) ORDER BY days_since_activity DESC
SEC-SQL-AU-006-RC10 Shared service accounts misclassified
Condition: row_count > 0 — Accounts connecting from many machines or programs indicate shared service accounts
Dormant/Unused Accounts FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT username, COUNT(DISTINCT machine) AS unique_machines, COUNT(DISTINCT program) AS unique_programs, COUNT(*) AS total_sessions FROM v\$session WHERE username IS NOT NULL AND type = 'USER' GROUP BY username HAVING COUNT(DISTINCT machine) > 5 OR COUNT(DISTINCT program) > 5 ORDER BY unique_machines DESC

SEC-SQL-AU-006-RC11 Account ownership unclear
Condition: row_count > 0 — Application accounts without standardized naming or metadata indicate unclear ownership
Dormant/Unused Accounts FINNACLEDB_TEST: DETECTED 7 rows FLEXCUBE: DETECTED 27 rows alert: Yes mail: No
SELECT u.username, u.created, u.account_status, u.profile, u.default_tablespace, u.authentication_type FROM dba_users u WHERE u.account_status = 'OPEN' AND u.username NOT IN (SELECT DISTINCT grantee FROM dba_role_privs WHERE granted_role = 'DBA') AND u.username NOT LIKE '%SYS%' AND u.username NOT LIKE 'APEX%' AND u.username NOT IN ('SYS', 'SYSTEM', 'DBSNMP', 'OUTLN', 'MDSYS', 'XDB', 'WMSYS', 'APQOSSYS', 'ANONYMOUS') ORDER BY u.created
SEC-SQL-AU-006-RC12 No automated dormancy detection
Condition: row_count > 0 — Active accounts with 90+ days of inactivity confirm dormant accounts are not being automatically detected
Dormant/Unused Accounts FINNACLEDB_TEST: DETECTED 7 rows FLEXCUBE: DETECTED 10 rows alert: Yes mail: No
SELECT username, TO_CHAR(last_login, 'YYYY-MM-DD HH24:MI:SS') AS last_login, account_status, ROUND(SYSDATE - NVL(CAST(last_login AS DATE), created)) AS days_inactive FROM dba_users WHERE account_status = 'OPEN' AND (last_login IS NULL OR last_login < SYSDATE - 90) AND username NOT IN ('SYS', 'SYSTEM', 'DBSNMP', 'OUTLN') ORDER BY days_inactive DESC FETCH FIRST 20 ROWS ONLY
SEC-SQL-AU-006-RC13 Privilege accumulation over time
Condition: row_count > 0 — Users with many system privileges indicate privilege accumulation
Dormant/Unused Accounts FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT grantee, COUNT(DISTINCT privilege) AS sys_priv_count FROM dba_sys_privs WHERE grantee NOT IN ('SYS', 'SYSTEM', 'DBA', 'IMP_FULL_DATABASE', 'EXP_FULL_DATABASE') AND grantee IN (SELECT username FROM dba_users WHERE account_status = 'OPEN') GROUP BY grantee HAVING COUNT(DISTINCT privilege) > 10 ORDER BY sys_priv_count DESC
SEC-SQL-AU-006-RC14 Audit trails insufficient
Condition: audit_trail = 'NONE' OR audit_sys_operations = 'FALSE' — Disabled auditing indicates insufficient audit trails for access review
Dormant/Unused Accounts FINNACLEDB_TEST: DETECTED 3 rows FLEXCUBE: DETECTED 3 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('audit_trail', 'audit_sys_operations', 'unified_audit_sga_queue_size') ORDER BY name
SEC-SQL-AU-006-RC15 Remediation effort underestimated
Condition: row_count > 0 — Scope assessment showing large number of direct grants indicates significant remediation effort
Dormant/Unused Accounts FINNACLEDB_TEST: DETECTED 4 rows FLEXCUBE: DETECTED 4 rows alert: Yes mail: No
SELECT 'Total users' AS metric, COUNT(*) AS value FROM dba_users WHERE account_status = 'OPEN' AND username NOT LIKE '%SYS%' UNION ALL SELECT 'Users with DBA role', COUNT(*) FROM dba_role_privs WHERE granted_role = 'DBA' AND grantee IN (SELECT username FROM dba_users WHERE account_status = 'OPEN') UNION ALL SELECT 'Direct object grants', COUNT(*) FROM dba_tab_privs WHERE grantee IN (SELECT username FROM dba_users WHERE account_status = 'OPEN') AND grantee NOT IN ('SYS', 'SYSTEM', 'PUBLIC') UNION ALL SELECT 'System privilege grants', COUNT(*) FROM dba_sys_privs WHERE grantee IN (SELECT username FROM dba_users WHERE account_status = 'OPEN') AND grantee NOT IN ('SYS', 'SYSTEM')
SEC-SQL-AU-007-RC01 Default trust-based local authentication
Condition: row_count > 0 — Externally authenticated users combined with remote_os_authent=TRUE pose a security risk
Trust Authentication Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT username, account_status, authentication_type, external_name FROM dba_users WHERE authentication_type = 'EXTERNAL' ORDER BY username
SEC-SQL-AU-007-RC02 Administrator convenience during setup
Condition: row_count > 0 — Accounts with default passwords indicate setup convenience settings not hardened
Trust Authentication Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT username FROM dba_users_with_defpwd WHERE username IN (SELECT username FROM dba_users WHERE account_status = 'OPEN')
SEC-SQL-AU-007-RC03 Peer authentication misunderstood as secure

Condition: row_count > 0 — Multiple externally authenticated users indicate reliance on OS authentication
Trust Authentication Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT username, authentication_type, external_name, account_status FROM dba_users WHERE authentication_type = 'EXTERNAL' AND account_status = 'OPEN' ORDER BY username
SEC-SQL-AU-007-RC04 Development environment settings
Condition: row_count > 0 — Insecure parameters indicate development settings not hardened for production
Trust Authentication Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE (name = 'remote_os_authent' AND value = 'TRUE') OR (name = 'sec_case_sensitive_logon' AND value = 'FALSE') OR (name = 'remote_login_passwordfile' AND value = 'NONE') OR (name = '07_DICTIONARY_ACCESSIBILITY' AND value = 'TRUE') OR (name = 'remote_os_roles' AND value = 'TRUE') ORDER BY name
SEC-SQL-AU-007-RC05 Remote access addition overlooked
Condition: row_count > 0 — Listener configuration review for remote access security controls
Trust Authentication Enabled FINNACLEDB_TEST: DETECTED 5 rows FLEXCUBE: DETECTED 5 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('local_listener', 'remote_listener', 'dispatchers', 'sec_protocol_error_trace_action', 'sec_protocol_error_further_action') ORDER BY name
SEC-SQL-AU-007-RC06 Incomplete hardening documentation
Condition: row_count > 0 — Multiple unhardened parameters suggest incomplete hardening documentation was followed
Trust Authentication Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE (name = 'audit_trail' AND value = 'NONE') OR (name = 'sec_return_server_release_banner' AND value = 'TRUE') OR (name = 'sql92_security' AND value = 'FALSE') OR (name = 'remote_os_authent' AND value = 'TRUE') OR (name = '07_DICTIONARY_ACCESSIBILITY' AND value = 'TRUE') ORDER BY name
SEC-SQL-AU-007-RC07 Configuration file management tools
Condition: row_count = 0 OR any encryption = 'REJECTED' — Missing or rejected encryption settings indicate configuration files not managed for security
Trust Authentication Enabled FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: Yes (destructive)
SELECT name, value FROM v\$parameter WHERE name IN ('sqlnet.encryption_server', 'sqlnet.crypto_checksum_server', 'sqlnet.encryption_types_server') OR name LIKE '%encryption%' OR name LIKE '%crypto%' ORDER BY name
SEC-SQL-AU-007-RC09 Recovery procedure reliance
Condition: remote_login_passwordfile = 'NONE' — No password file means recovery relies on OS authentication only, which may be insecure
Trust Authentication Enabled FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: Yes (destructive)
SELECT name, value FROM v\$parameter WHERE name IN ('remote_login_passwordfile', 'sec_case_sensitive_logon') ORDER BY name
SEC-SQL-AU-007-RC10 Container/Kubernetes secrets practice
Condition: row_count > 0 — Connections from container-like hostnames indicate Kubernetes/Docker deployment
Trust Authentication Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT DISTINCT s.machine, s.program, s.username FROM v\$session s WHERE s.type = 'USER' AND (s.machine LIKE '%pod%' OR s.machine LIKE '%container%' OR s.machine LIKE '%docker%' OR s.machine LIKE '%k8s%' OR LENGTH(s.machine) = 12) ORDER BY s.machine
SEC-SQL-AU-007-RC11 Single-node assumption
Condition: unique_clients > :client_threshold — Many unique client machines indicate multi-node access pattern
Trust Authentication Enabled FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT COUNT(DISTINCT machine) AS unique_clients, COUNT(DISTINCT username) AS unique_users FROM v\$session WHERE type = 'USER' AND username IS NOT NULL

SEC-SQL-AU-007-RC12 Administrator password forgotten or locked
Condition: any account_status LIKE '%LOCKED%' OR account_status LIKE '%EXPIRED%' — Locked or expired DBA accounts indicate administrator access issues
Trust Authentication Enabled FINNACLEDB_TEST: DETECTED 6 rows FLEXCUBE: DETECTED 7 rows alert: Yes mail: Yes (destructive)
SELECT username, account_status, lock_date, expiry_date FROM dba_users WHERE username IN ('SYS', 'SYSTEM') OR username IN (SELECT grantee FROM dba_role_privs WHERE granted_role = 'DBA') ORDER BY username
SEC-SQL-AU-007-RC13 Documentation templates outdated
Condition: row_count > 0 — Active audit policies — verify these match documented security templates
Trust Authentication Enabled FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT policy_name, enabled_option, entity_name, entity_type, success, failure FROM audit_unified_enabled_policies ORDER BY policy_name
SEC-SQL-AU-007-RC14 No configuration drift detection
Condition: row_count > 0 — Modified security parameters without documentation — indicates configuration drift
Trust Authentication Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT name, value, ismodified FROM v\$parameter WHERE ismodified != 'FALSE' AND name IN ('audit_trail', 'audit_sys_operations', 'remote_login_passwordfile', 'sec_case_sensitive_logon', 'remote_os_authent', 'o7_dictionary_accessibility', 'utl_file_dir', 'sec_protocol_error_trace_action') ORDER BY name
SEC-SQL-AUD-001-RC01 Global logging switch is disabled
Condition: value = 'NONE' — AUDIT_TRAIL set to NONE — all auditing is completely disabled
Audit Logging Disabled FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('audit_trail', 'audit_sys_operations') ORDER BY name
SEC-SQL-AUD-001-RC02 Missing audit plugin/extension
Condition: row_count = 0 — No specialized audit components (Audit Vault, Database Vault) installed
Audit Logging Disabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT comp_id, comp_name, version, status FROM dba_registry WHERE comp_id IN ('AUDIT_VAULT', 'DV') OR comp_name LIKE '%Audit%' ORDER BY comp_name
SEC-SQL-AUD-001-RC03 Audit policy set to "NONE"
Condition: row_count = 0 — No unified audit policies enabled — audit capture is effectively set to NONE
Audit Logging Disabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT policy_name, enabled_option, entity_name, entity_type, success, failure FROM audit_unified_enabled_policies ORDER BY policy_name
SEC-SQL-AUD-001-RC04 License restrictions
Condition: row_count > 0 — Oracle edition and options — some audit features require Enterprise Edition or additional licenses
Audit Logging Disabled FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT banner FROM v\$version WHERE ROWNUM = 1 UNION ALL SELECT parameter ': ' value FROM v\$option WHERE parameter IN ('Unified Auditing', 'Database Vault', 'Label Security') ORDER BY 1
SEC-SQL-AUD-001-RC05 Service account permission failures
Condition: row_count > 0 — Audit file destination — check OS permissions on this directory
Audit Logging Disabled FINNACLEDB_TEST: DETECTED 3 rows FLEXCUBE: DETECTED 3 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('audit_file_dest', 'audit_trail', 'audit_syslog_level') ORDER BY name

SEC-SQL-AUD-001-RC06 Performance-based disablement
Condition: enabled_policies < :min_policies — Very few audit policies enabled — may indicate performance-based disablement
Audit Logging Disabled FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT COUNT(*) AS enabled_policies FROM audit_unified_enabled_policies
SEC-SQL-AUD-001-RC07 Misconfigured log destination
Condition: row_count > 0 — Audit destination settings — verify path exists and is writable
Audit Logging Disabled FINNACLEDB_TEST: DETECTED 3 rows FLEXCUBE: DETECTED 3 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('audit_file_dest', 'audit_trail', 'audit_syslog_level') ORDER BY name
SEC-SQL-AUD-001-RC08 Default "Secure by Default" settings
Condition: row_count = 0 — No custom audit policies — only Oracle predefined policies exist
Audit Logging Disabled FINNACLEDB_TEST: DETECTED 0 rows FLEXCUBE: DETECTED 0 rows alert: Yes mail: No
SELECT policy_name, audit_option FROM audit_unified_policies WHERE policy_name NOT LIKE 'ORA_%' ORDER BY policy_name
SEC-SQL-AUD-001-RC09 Conflicting configuration files
Condition: row_count > 0 — SPFILE and runtime audit parameters differ — conflicting configuration sources
Audit Logging Disabled FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT sp.name, sp.value AS spfile_value, p.value AS runtime_value FROM v\$spparameter sp JOIN v\$parameter p ON sp.name = p.name WHERE sp.name IN ('audit_trail', 'audit_sys_operations', 'audit_file_dest') AND sp.value IS NOT NULL AND sp.value != p.value ORDER BY sp.name
SEC-SQL-AUD-001-RC10 Plugin initialization failure
Condition: account_status != 'OPEN' — AUDSYS schema locked or expired — unified audit infrastructure cannot function
Audit Logging Disabled FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT username, account_status, default_tablespace FROM dba_users WHERE username = 'AUDSYS'
SEC-SQL-AUD-001-RC11 Container/Ephemeral storage misconfiguration
Condition: row_count > 1 — Multi-tenant (CDB/PDB) environment — audit policies must be configured per PDB
Audit Logging Disabled FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT con_id, name, open_mode FROM v\$containers ORDER BY con_id
SEC-SQL-AUD-001-RC12 Audit process crash
Condition: row_count > 0 — Audit-related errors in alert log indicate audit process failures
Audit Logging Disabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT originating_timestamp, message_text FROM v\$diag_alert_ext WHERE (message_text LIKE '%audit%' OR message_text LIKE '%AUDSYS%') AND (message_text LIKE '%error%' OR message_text LIKE '%fail%' OR message_text LIKE '%crash%' OR message_text LIKE '%ORA-%') AND originating_timestamp > SYSDATE - 7 ORDER BY originating_timestamp DESC FETCH FIRST 20 ROWS ONLY
SEC-SQL-AUD-002-RC01 Aggressive rotation configuration
Condition: retention_days < :min_retention_days — Audit trail retention shorter than compliance requirement — records purged too aggressively
Inadequate Log Retention FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No

SEC-SQL-AUD-003-RC02 Internal audit tables are mutable
Condition: row_count > 0 — Non-SYS users have DML privileges on audit tables — audit data can be tampered
Audit Trail Modification Risk FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT GRANTEE, PRIVILEGE, TABLE_NAME, GRANTOR FROM ALL_TAB_PRIVS WHERE TABLE_NAME IN ('AUD\$', 'FGA_LOG\$') AND PRIVILEGE IN ('DELETE', 'UPDATE', 'INSERT', 'ALTER') AND GRANTEE NOT IN ('SYS', 'SYSTEM')
SEC-SQL-AUD-003-RC03 Lack of File Integrity Monitoring (FIM)
Condition: audit_trail IN ('OS', 'XML', 'XML,EXTENDED') OR audit_sys_operations = 'TRUE' — OS-level audit files are being generated — FIM should protect them
Audit Trail Modification Risk FINNACLEDB_TEST: DETECTED 3 rows FLEXCUBE: DETECTED 3 rows alert: Yes mail: No
SELECT NAME, VALUE FROM V\$PARAMETER WHERE NAME IN ('audit_trail', 'audit_file_dest', 'audit_sys_operations')
SEC-SQL-AUD-003-RC04 Service account ownership
Condition: audit_trail IN ('OS', 'XML', 'XML,EXTENDED') — File-based auditing means oracle OS user can modify audit files
Audit Trail Modification Risk FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT NAME, VALUE FROM V\$PARAMETER WHERE NAME IN ('audit_trail', 'audit_file_dest')
SEC-SQL-AUD-003-RC05 Missing "Append-Only" attributes
Condition: audit_mode IN ('OS', 'XML', 'XML,EXTENDED') — File-based audit without OS-level immutability attributes
Audit Trail Modification Risk FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT VALUE AS audit_mode FROM V\$PARAMETER WHERE NAME = 'audit_trail'
SEC-SQL-AUD-003-RC06 Audit logs not shipped remotely
Condition: audit_syslog_level IS NULL OR audit_syslog_level = '' — Audit syslog shipping not configured — audit data only on local server
Audit Trail Modification Risk FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT NAME, VALUE FROM V\$PARAMETER WHERE NAME IN ('audit_trail', 'audit_syslog_level')
SEC-SQL-AUD-003-RC07 Non-repudiation features disabled
Condition: row_count > 0 — Fine-grained audit policies exist
Audit Trail Modification Risk FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT OBJECT_SCHEMA, OBJECT_NAME, POLICY_NAME, POLICY_TEXT, ENABLED, POLICY_COLUMN FROM DBA_AUDIT_POLICIES WHERE ENABLED = 'YES'
SEC-SQL-AUD-003-RC08 Admin accounts shared/untracked
Condition: row_count > 0 — Multiple concurrent sessions from shared admin accounts
Audit Trail Modification Risk FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT s.USERNAME, s.OSUSER, s.MACHINE, s.PROGRAM, COUNT(*) AS session_count FROM V\$SESSION s WHERE s.USERNAME IN ('SYS', 'SYSTEM') OR s.SID IN (SELECT SID FROM V\$SESSION WHERE SCHEMANAME = 'SYS') GROUP BY s.USERNAME, s.OSUSER, s.MACHINE, s.PROGRAM HAVING COUNT(*) > 1 ORDER BY session_count DESC
SEC-SQL-AUD-003-RC09 Audit disablement capability
Condition: row_count > 0 — Non-standard users have audit administration privileges
Audit Trail Modification Risk FINNACLEDB_TEST: DETECTED 12 rows FLEXCUBE: DETECTED 17 rows alert: Yes mail: No
SELECT GRANTEE, PRIVILEGE FROM DBA_SYS_PRIVS WHERE PRIVILEGE IN ('AUDIT ANY', 'AUDIT SYSTEM', 'ADMINISTER DATABASE TRIGGER') AND GRANTEE NOT IN ('SYS', 'SYSTEM', 'DBA') ORDER BY GRANTEE

SEC-SQL-AUD-004-RC01 Logging level set to "Errors Only"
Condition: row_count < :min_policies — Very few audit policies enabled — insufficient coverage
Missing Login Failure Logs FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT POLICY_NAME, ENABLED_OPTION, SUCCESS, FAILURE FROM AUDIT_UNIFIED_ENABLED_POLICIES ORDER BY POLICY_NAME
SEC-SQL-AUD-004-RC02 "Successful Logins Only" filter
Condition: FAILURE = 'NO' OR row_count = 0 — Failed logon auditing not enabled — missing brute force detection
Missing Login Failure Logs FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT POLICY_NAME, ENABLED_OPTION, SUCCESS, FAILURE FROM AUDIT_UNIFIED_ENABLED_POLICIES WHERE POLICY_NAME LIKE '%LOGON%' ORDER BY POLICY_NAME
SEC-SQL-AUD-004-RC03 Connection string errors
Condition: sessions_audited = 0 — No logon audit records — cannot track connection string or authentication issues
Missing Login Failure Logs FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT COUNT(*) AS sessions_audited, COUNT(DISTINCT AUTHENTICATION_TYPE) AS auth_types, COUNT(DISTINCT DBPROXY_USERNAME) AS proxy_users FROM UNIFIED_AUDIT_TRAIL WHERE EVENT_TIMESTAMP > SYSTIMESTAMP - INTERVAL '1' DAY AND ACTION_NAME = 'LOGON'
SEC-SQL-AUD-004-RC04 Generic "Audit All" disabled
Condition: row_count < :min_policies — Key Oracle predefined audit policies not enabled
Missing Login Failure Logs FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT POLICY_NAME FROM AUDIT_UNIFIED_ENABLED_POLICIES WHERE POLICY_NAME IN ('ORA_SECURECONFIG', 'ORA_LOGON_FAILURES', 'ORA_ACCOUNT_MGMT', 'ORA_DATABASE_PARAMETER', 'ORA_DV_AUDPOL', 'ORA_DV_AUDPOL2', 'ORA_RAS_POLICY_MGMT', 'ORA_RAS_SESSION_MGMT', 'ORA_CIS_RECOMMENDATIONS')
SEC-SQL-AUD-004-RC05 Log sampling/Throttling
Condition: row_count > 0 — Audit policies with conditional evaluation — events may be sampled or filtered
Missing Login Failure Logs FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT POLICY_NAME, AUDIT_CONDITION, CONDITION_EVAL_OPT FROM AUDIT_UNIFIED_POLICIES WHERE AUDIT_CONDITION IS NOT NULL AND AUDIT_CONDITION != 'NONE'
SEC-SQL-AUD-004-RC06 Misinterpreted Error Codes
Condition: row_count > 0 — Non-zero return codes in audit trail — potential security events that may be misinterpreted
Missing Login Failure Logs FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT RETURN_CODE, ACTION_NAME, COUNT(*) AS occurrences FROM UNIFIED_AUDIT_TRAIL WHERE RETURN_CODE != 0 AND EVENT_TIMESTAMP > SYSTIMESTAMP - INTERVAL '7' DAY GROUP BY RETURN_CODE, ACTION_NAME ORDER BY occurrences DESC FETCH FIRST 20 ROWS ONLY
SEC-SQL-AUD-004-RC07 Internal vs External Authentication
Condition: row_count = 0 — No logon audit records capturing authentication type — gap in tracking
Missing Login Failure Logs FINNACLEDB_TEST: DETECTED 0 rows FLEXCUBE: clean alert: Yes mail: No
SELECT AUTHENTICATION_TYPE, COUNT(*) AS logon_count FROM UNIFIED_AUDIT_TRAIL WHERE ACTION_NAME = 'LOGON' AND EVENT_TIMESTAMP > SYSTIMESTAMP - INTERVAL '7' DAY GROUP BY AUTHENTICATION_TYPE ORDER BY logon_count DESC
SEC-SQL-AUD-004-RC08 Application-side suppression
Condition: row_count > 0 — Users excluded from audit policies may allow application-side audit suppression
Missing Login Failure Logs FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No

SELECT entity_name, policy_name, enabled_option FROM audit_unified_enabled_policies WHERE entity_type = 'USER' AND enabled_option LIKE 'EXCEPT%' ORDER BY entity_name
SEC-SQL-AUD-005-RC01 DML-focused Audit Policy
Condition: row_count > 0 — DML audit actions are configured — need to verify DDL is also covered
DDL Logging Disabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT policy_name, audit_option, audit_option_type, object_schema, object_name FROM audit_unified_policies WHERE audit_option_type = 'STANDARD ACTION' AND audit_option IN ('SELECT', 'INSERT', 'UPDATE', 'DELETE') ORDER BY policy_name
SEC-SQL-AUD-005-RC02 Granularity settings
Condition: value IN ('NONE', 'OS') — Audit trail is set to NONE or OS-only, which may lack the granularity needed for compliance (no SQL text capture)
DDL Logging Disabled FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name = 'audit_trail' ORDER BY name
SEC-SQL-AUD-005-RC03 Privileged User Exclusion
Condition: row_count > 0 — Privileged users explicitly excluded from unified audit policies, creating an audit gap for the most powerful accounts
DDL Logging Disabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT policy_name, entity_name, entity_type, enabled_option FROM audit_unified_enabled_policies WHERE entity_type = 'USER' AND entity_name IN ('SYS', 'SYSTEM', 'SYSMAN', 'DBSNMP') AND enabled_option = 'EXCEPT' ORDER BY policy_name
SEC-SQL-AUD-005-RC04 Logging "Write" vs "DDL"
Condition: row_count = 0 — No unified audit policies covering DDL operations — schema changes are not being logged
DDL Logging Disabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT policy_name, audit_option, object_schema, object_name FROM audit_unified_policies WHERE audit_option IN ('CREATE TABLE', 'ALTER TABLE', 'DROP TABLE', 'CREATE INDEX', 'DROP INDEX', 'CREATE VIEW', 'DROP VIEW', 'CREATE PROCEDURE', 'DROP PROCEDURE', 'CREATE TRIGGER', 'DROP TRIGGER') ORDER BY policy_name
SEC-SQL-AUD-005-RC05 Deployment via Scripts
Condition: any(audit_status = 'NOT_AUDITED') — Application or deployment users not covered by any user-specific audit policy may run DDL scripts without being audited
DDL Logging Disabled FINNACLEDB_TEST: DETECTED 11 rows FLEXCUBE: DETECTED 30 rows alert: Yes mail: No
SELECT u.username, u.account_status, u.created, CASE WHEN EXISTS (SELECT 1 FROM audit_unified_enabled_policies p WHERE p.entity_name = u.username AND p.entity_type = 'USER') THEN 'AUDITED' ELSE 'NOT_AUDITED' END AS audit_status FROM dba_users u WHERE u.username NOT IN ('SYS', 'SYSTEM', 'DBSNMP', 'OUTLN', 'XDB') AND u.account_status = 'OPEN' AND u.oracle_maintained = 'N' ORDER BY u.username
SEC-SQL-AUD-005-RC06 Temporary Table Exclusion
Condition: any(audit_status = 'NOT_AUDITED') — Global temporary tables not covered by object-specific audit policies may contain sensitive data that goes unaudited
DDL Logging Disabled FINNACLEDB_TEST: DETECTED 44 rows FLEXCUBE: DETECTED 86 rows alert: Yes mail: No
SELECT t.owner, t.table_name, t.temporal, CASE WHEN EXISTS (SELECT 1 FROM audit_unified_policies p WHERE p.object_schema = t.owner AND p.object_name = t.table_name) THEN 'AUDITED' ELSE 'NOT_AUDITED' END AS audit_status FROM dba_tables t WHERE t.temporal = 'Y' AND t.owner NOT IN ('SYS', 'SYSTEM') ORDER BY t.owner, t.table_name
SEC-SQL-AUD-005-RC07 Extension/Plugin Limitations
Condition: row_count = 0 — No audit policies covering EXECUTE on PL/SQL packages — sensitive DBMS packages (DBMS_SQL, UTL_FILE, UTL_HTTP) go unaudited
DDL Logging Disabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT policy_name, audit_option FROM audit_unified_policies WHERE audit_option_type = 'STANDARD ACTION' AND (audit_option LIKE 'EXECUTE%' OR audit_option LIKE '%PROCEDURE%' OR audit_option LIKE '%PACKAGE%') ORDER BY policy_name

SEC-SQL-AUD-005-RC08 Statement Truncation
Condition: any(name='audit_trail' AND value IN ('DB', 'OS', 'NONE')) — Traditional audit trail without EXTENDED option does not capture SQL text — statements are not recorded
DDL Logging Disabled FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('audit_trail', 'unified_audit_sga_queue_size') ORDER BY name
SEC-SQL-AUD-006-RC01 Privileged account executing DML outside maintenance window
Condition: row_count > 0 — DBA-role account has active DML session
High Privilege User Suspicious Transactions FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT s.SID, s.SERIAL#, s.USERNAME, s.OSUSER, s.MACHINE, s.STATUS, q.SQL_TEXT, s.LOGON_TIME FROM V\$SESSION s JOIN V\$SQL q ON s.SQL_ID = q.SQL_ID WHERE s.STATUS = 'ACTIVE' AND q.COMMAND_TYPE IN (2,6,7,9) AND s.USERNAME IN (SELECT GRANTEE FROM DBA_ROLE_PRIVS WHERE GRANTED_ROLE = 'DBA')
SEC-SQL-AUD-006-RC02 Superuser running ad-hoc queries on sensitive tables
Condition: row_count > 0 — DBA account querying sensitive object names
High Privilege User Suspicious Transactions FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 5 rows alert: Yes mail: No
SELECT s.USERNAME, q.SQL_TEXT, s.MACHINE, s.LOGON_TIME FROM V\$SESSION s JOIN V\$SQL q ON s.SQL_ID = q.SQL_ID WHERE s.USERNAME IN (SELECT GRANTEE FROM DBA_ROLE_PRIVS WHERE GRANTED_ROLE = 'DBA') AND (UPPER(q.SQL_TEXT) LIKE '%PASSWORD%' OR UPPER(q.SQL_TEXT) LIKE '%CREDENTIAL%' OR UPPER(q.SQL_TEXT) LIKE '%SALARY%' OR UPPER(q.SQL_TEXT) LIKE '%API_KEY%')
SEC-SQL-AUD-006-RC03 High privilege account accessing schema outside designated scope
Condition: row_count > 0 — DBA account accessing objects in schemas it does not own
High Privilege User Suspicious Transactions FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 20 rows alert: Yes mail: No
SELECT s.USERNAME, o.OWNER AS accessed_schema, q.SQL_TEXT FROM V\$SESSION s JOIN V\$SQL q ON s.SQL_ID = q.SQL_ID JOIN ALL_OBJECTS o ON UPPER(q.SQL_TEXT) LIKE '%' o.OBJECT_NAME '%' WHERE s.USERNAME IN (SELECT GRANTEE FROM DBA_ROLE_PRIVS WHERE GRANTED_ROLE = 'DBA') AND o.OWNER != s.USERNAME AND o.OWNER NOT IN ('SYS','SYSTEM','OUTLN','DBSNMP') FETCH FIRST 20 ROWS ONLY
SEC-SQL-AUD-006-RC04 Service account performing interactive or manual transactions
Condition: row_count > 0 — Service account connected via interactive Oracle client tool
High Privilege User Suspicious Transactions FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT USERNAME, PROGRAM, MACHINE, TERMINAL, LOGON_TIME, STATUS FROM V\$SESSION WHERE (UPPER(PROGRAM) LIKE '%SQLPLUS%' OR UPPER(PROGRAM) LIKE '%SQLDEV%' OR UPPER(PROGRAM) LIKE '%TOAD%' OR UPPER(PROGRAM) LIKE '%DBEAVER%') AND (UPPER(USERNAME) LIKE '%SVC%' OR UPPER(USERNAME) LIKE '%SERVICE%' OR UPPER(USERNAME) LIKE '%APP%' OR UPPER(USERNAME) LIKE '%API%' OR UPPER(USERNAME) LIKE '%ETL%')
SEC-SQL-AUD-006-RC05 Privilege escalation during active session
Condition: row_count > 0 — Non-DBA account granted or activated elevated role in last 24 hours
High Privilege User Suspicious Transactions FINNACLEDB_TEST: FAILED FLEXCUBE: FAILED alert: No mail: No
SELECT DB_USER, ACTION_NAME, OBJ_NAME, TIMESTAMP, RETURN_CODE FROM DBA_AUDIT_TRAIL WHERE ACTION_NAME IN ('GRANT ROLE','SET ROLE','GRANT PRIVILEGE') AND DB_USER NOT IN (SELECT GRANTEE FROM DBA_ROLE_PRIVS WHERE GRANTED_ROLE = 'DBA') AND TIMESTAMP > SYSDATE - 1
SEC-SQL-AUD-007-RC01 Transactions from unregistered application signatures
Condition: row_count > 0 — Sessions with no CLIENT_INFO registration suggest rogue or unregistered tool
Unknown Transaction Patterns FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 5 rows alert: Yes mail: Yes (destructive)
SELECT USERNAME, CLIENT_INFO, MACHINE, COUNT(*) AS sessions FROM V\$SESSION WHERE TYPE = 'USER' AND CLIENT_INFO IS NULL AND STATUS = 'ACTIVE' GROUP BY USERNAME, CLIENT_INFO, MACHINE
SEC-SQL-AUD-007-RC02 Query patterns not matching known application baseline
Condition: row_count > 0 — Single-execution query with unusually high buffer gets -- not in baseline

Unknown Transaction Patterns FINNACLEDB_TEST: DETECTED 14 rows FLEXCUBE: DETECTED 20 rows alert: Yes mail: Yes (destructive)
SELECT SQL_ID, SQL_TEXT, EXECUTIONS, BUFFER_GETS, DISK_READS, FIRST_LOAD_TIME, LAST_ACTIVE_TIME FROM V\$SQL WHERE EXECUTIONS = 1 AND BUFFER_GETS > 100000 AND PARSING_USER_ID NOT IN (SELECT USER_ID FROM DBA_USERS WHERE USERNAME IN ('SYS','SYSTEM')) ORDER BY BUFFER_GETS DESC FETCH FIRST 20 ROWS ONLY
SEC-SQL-AUD-007-RC03 Connections from unexpected network addresses
Condition: row_count > 0 — Session from machine not matching internal domain pattern
Unknown Transaction Patterns FINNACLEDB_TEST: DETECTED 6 rows FLEXCUBE: DETECTED 259 rows alert: Yes mail: Yes (destructive)
SELECT USERNAME, MACHINE, TERMINAL, PROGRAM, LOGON_TIME, STATUS FROM V\$SESSION WHERE TYPE = 'USER' AND MACHINE NOT LIKE '%.internal' AND MACHINE NOT LIKE '%.corp' AND MACHINE NOT IN (SELECT HOST_NAME FROM V\$INSTANCE)
SEC-SQL-AUD-007-RC04 Off-hours transaction activity from non-automated accounts
Condition: row_count > 0 — Human Oracle account active outside 06:00-20:00 UTC
Unknown Transaction Patterns FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT USERNAME, PROGRAM, MACHINE, LOGON_TIME, STATUS, TO_NUMBER(TO_CHAR(SYS_EXTRACT_UTC(SYSTIMESTAMP), 'HH24')) AS utc_hour FROM V\$SESSION WHERE TYPE = 'USER' AND STATUS = 'ACTIVE' AND USERNAME NOT IN (SELECT USERNAME FROM DBA_USERS WHERE REGEXP_LIKE(USERNAME, '(SVC SERVICE APP ETL BATCH MONITOR BACKUP)')) AND TO_NUMBER(TO_CHAR(SYS_EXTRACT_UTC(SYSTIMESTAMP), 'HH24')) NOT BETWEEN 6 AND 20
SEC-SQL-AUD-007-RC05 Execution of previously unseen stored procedures or functions
Condition: row_count > 0 — Procedure/function/trigger created within last 7 days by non-SYS account
Unknown Transaction Patterns FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT OWNER, OBJECT_NAME, OBJECT_TYPE, CREATED, LAST_DDL_TIME, STATUS FROM DBA_OBJECTS WHERE OBJECT_TYPE IN ('PROCEDURE', 'FUNCTION', 'PACKAGE', 'PACKAGE BODY', 'TRIGGER') AND CREATED > SYSDATE - 7 AND OWNER NOT IN ('SYS', 'SYSTEM', 'OUTLN', 'DBSNMP', 'APEX_PUBLIC_USER')
SEC-SQL-AUD-008-RC01 Bulk data export or mass SELECT on sensitive tables
Condition: row_count > 0 — Data Pump export or UTL_FILE write session active
Risky Transaction Execution FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: Yes (destructive)
SELECT s.USERNAME, s.PROGRAM, s.MODULE, q.SQL_TEXT, s.LAST_CALL_ET, s.STATUS FROM V\$SESSION s JOIN V\$SQL q ON s.SQL_ID = q.SQL_ID WHERE (UPPER(s.PROGRAM) LIKE '%EXPDP%' OR UPPER(s.PROGRAM) LIKE '%DATA_PUMP%' OR UPPER(q.SQL_TEXT) LIKE '%UTL_FILE%') AND s.TYPE = 'USER'
SEC-SQL-AUD-008-RC02 Mass DELETE or TRUNCATE without WHERE clause
Condition: row_count > 0 — Large table segments exist -- truncation risk is high
Risky Transaction Execution FINNACLEDB_TEST: DETECTED 20 rows FLEXCUBE: DETECTED 20 rows alert: Yes mail: Yes (destructive)
SELECT OWNER, SEGMENT_NAME, BLOCKS * 8192 / 1024 / 1024 AS size_mb FROM DBA_SEGMENTS WHERE SEGMENT_TYPE = 'TABLE' AND BLOCKS > 1000 AND OWNER NOT IN ('SYS','SYSTEM','OUTLN') ORDER BY BLOCKS DESC FETCH FIRST 20 ROWS ONLY
SEC-SQL-AUD-008-RC03 DDL statements executed in production by non-DBA accounts
Condition: row_count > 0 — DDL executed by non-DBA account in the last hour per audit trail
Risky Transaction Execution FINNACLEDB_TEST: FAILED FLEXCUBE: FAILED alert: No mail: No
SELECT DB_USER, ACTION_NAME, OBJ_OWNER, OBJ_NAME, TIMESTAMP FROM DBA_AUDIT_TRAIL WHERE ACTION_NAME IN ('CREATE TABLE', 'ALTER TABLE', 'DROP TABLE', 'CREATE INDEX', 'DROP INDEX', 'CREATE VIEW', 'CREATE PROCEDURE', 'ALTER PROCEDURE', 'DROP PROCEDURE') AND DB_USER NOT IN (SELECT GRANTEE FROM DBA_ROLE_PRIVS WHERE GRANTED_ROLE = 'DBA') AND TIMESTAMP > SYSDATE - 1/24
SEC-SQL-AUD-008-RC04 Transactions accessing credential or encryption key tables
Condition: row_count > 0 — TDE keys exist in wallet -- unauthorized access by non-DBA session is critical risk
Risky Transaction Execution FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: Yes (destructive)

SELECT * FROM V\$ENCRYPTION_WALLET
SEC-SQL-AUD-008-RC05 Cross-schema transactions outside application scope
Condition: row_count > 0 — Query references tables from more than 2 non-SYS schemas simultaneously
Risky Transaction Execution FINNACLEDB_TEST: FAILED FLEXCUBE: FAILED alert: No mail: No
SELECT s.USERNAME, q.SQL_TEXT, q.EXECUTIONS, q.LAST_ACTIVE_TIME FROM V\$SESSION s JOIN V\$SQL q ON s.SQL_ID = q.SQL_ID WHERE s.TYPE = 'USER' AND s.STATUS = 'ACTIVE' AND (SELECT COUNT(DISTINCT OWNER) FROM DBA_OBJECTS WHERE OBJECT_NAME = ANY(SELECT REGEXP_SUBSTR(UPPER(q.SQL_TEXT), '[A-Z][A-Z0-9_]+', 1, LEVEL) FROM DUAL CONNECT BY LEVEL <= REGEXP_COUNT(UPPER(q.SQL_TEXT), '[A-Z][A-Z0-9_]+')) AND OBJECT_TYPE = 'TABLE' AND OWNER NOT IN ('SYS', 'SYSTEM')) > 2
SEC-SQL-AUD-009-RC01 Tautology patterns detected in active query text
Condition: row_count > 0 — Tautology SQL confirmed in Oracle audit trail
SQL Injection Runtime Indicators FINNACLEDB_TEST: FAILED FLEXCUBE: FAILED alert: No mail: No
SELECT DB_USER, SQL_TEXT, TIMESTAMP, RETURNCODE FROM DBA_AUDIT_TRAIL WHERE (UPPER(SQL_TEXT) LIKE '%OR 1=1%' OR UPPER(SQL_TEXT) LIKE '%OR TRUE%') AND TIMESTAMP > SYSDATE - 1
SEC-SQL-AUD-009-RC02 Stacked or chained query execution detected
Condition: row_count > 0 — Stacked query confirmed in Oracle audit trail
SQL Injection Runtime Indicators FINNACLEDB_TEST: FAILED FLEXCUBE: FAILED alert: No mail: No
SELECT DB_USER, SQL_TEXT, TIMESTAMP FROM DBA_AUDIT_TRAIL WHERE REGEXP_LIKE(UPPER(SQL_TEXT), ';' s*(SELECT DROP INSERT)) AND TIMESTAMP > SYSDATE - 1
SEC-SQL-AUD-009-RC03 UNION-based probing across unknown tables
Condition: row_count > 0 — UNION injection targeting Oracle catalog confirmed in audit trail
SQL Injection Runtime Indicators FINNACLEDB_TEST: FAILED FLEXCUBE: FAILED alert: No mail: No
SELECT DB_USER, SQL_TEXT, TIMESTAMP FROM DBA_AUDIT_TRAIL WHERE REGEXP_LIKE(UPPER(SQL_TEXT), 'UNION.+SELECT') AND (UPPER(SQL_TEXT) LIKE '%ALL_TABLES%' OR UPPER(SQL_TEXT) LIKE '%DBA_USERS%') AND TIMESTAMP > SYSDATE - 1
SEC-SQL-AUD-009-RC04 Time-delay injection functions detected in active queries
Condition: row_count > 0 — Active session stalled on Oracle delay package — time-based injection in progress
SQL Injection Runtime Indicators FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT USERNAME, STATUS, LAST_CALL_ET, PROGRAM, SQL_ID FROM V\$SESSION WHERE STATUS = 'ACTIVE' AND LAST_CALL_ET > 3 AND SQL_ID IN (SELECT sql_id FROM V\$SQL WHERE UPPER(sql_text) LIKE '%DBMS_LOCK.SLEEP%' OR UPPER(sql_text) LIKE '%DBMS_PIPE%')
SEC-SQL-AUD-009-RC05 Comment-based obfuscation or encoding in query text
Condition: row_count > 0 — Obfuscated SQL confirmed in Oracle audit trail
SQL Injection Runtime Indicators FINNACLEDB_TEST: FAILED FLEXCUBE: FAILED alert: No mail: No
SELECT DB_USER, SQL_TEXT, TIMESTAMP FROM DBA_AUDIT_TRAIL WHERE (REGEXP_LIKE(SQL_TEXT, 'CHR\s*([0-9]+\)\s*\ \'', 'i') OR SQL_TEXT LIKE '%/*%/%') AND TIMESTAMP > SYSDATE - 1
SEC-SQL-AUD-010-RC01 Excessive information_schema or catalog enumeration
Condition: row_count > 0 — Unified audit confirms high-volume dictionary access -- schema enumeration
Database Reconnaissance Activity FINNACLEDB_TEST: FAILED FLEXCUBE: FAILED alert: No mail: No
SELECT db_username, count(*) AS dict_accesses, max(event_timestamp) AS last_seen FROM unified_audit_trail WHERE (object_name LIKE 'ALL_TABLES' OR object_name LIKE 'DBA_COLUMNS' OR object_name LIKE 'ALL_OBJECTS') AND event_timestamp > systimestamp - interval '1' hour GROUP BY db_username HAVING count(*) > 50
SEC-SQL-AUD-010-RC02 System principal and user enumeration by non-admin account

Condition: row_count > 0 — User view access confirmed in unified audit trail
Database Reconnaissance Activity FINNACLEDB_TEST: FAILED FLEXCUBE: FAILED alert: No mail: No
SELECT db_username, count(*) AS user_view_accesses, max(event_timestamp) AS last_seen FROM unified_audit_trail WHERE object_name IN ('DBA_USERS','ALL_USERS') AND event_timestamp > systimestamp - interval '1' hour GROUP BY db_username ORDER BY user_view_accesses DESC
SEC-SQL-AUD-010-RC03 Mass table and column listing within a single session
Condition: row_count > 0 — Repeated ALL_COLUMNS access in audit trail -- mass enumeration confirmed
Database Reconnaissance Activity FINNACLEDB_TEST: FAILED FLEXCUBE: FAILED alert: No mail: No
SELECT db_username, count(*) AS accesses, max(event_timestamp) AS last_seen FROM unified_audit_trail WHERE object_name IN ('ALL_COLUMNS','DBA_TAB_COLUMNS') AND event_timestamp > systimestamp - interval '30' minute GROUP BY db_username HAVING count(*) > 10
SEC-SQL-AUD-010-RC05 Permission and privilege discovery queries
Condition: row_count > 0 — Burst of privilege view accesses in audit trail -- systematic privilege mapping confirmed
Database Reconnaissance Activity FINNACLEDB_TEST: FAILED FLEXCUBE: FAILED alert: No mail: No
SELECT db_username, count(*) AS priv_view_accesses, max(event_timestamp) AS last_seen FROM unified_audit_trail WHERE object_name IN ('DBA_SYS_PRIVS','DBA_TAB_PRIVS','DBA_ROLE_PRIVS') AND event_timestamp > systimestamp - interval '30' minute GROUP BY db_username HAVING count(*) > 10
SEC-SQL-AUD-011-RC02 Audit settings disabled or downgraded mid-session
Condition: row_count > 0 — Unified audit policies found disabled or inactive -- confirm tampering
Audit and Log Tampering FINNACLEDB_TEST: DETECTED 2295 rows FLEXCUBE: DETECTED 2317 rows alert: Yes mail: No
SELECT policy_name, enabled_option, entity_name FROM audit_unified_enabled_policies WHERE enabled_option = 'NONE' UNION ALL SELECT policy_name, NULL, NULL FROM audit_unified_policies WHERE policy_name NOT IN (SELECT policy_name FROM audit_unified_enabled_policies)
SEC-SQL-AUD-011-RC03 Audit or log table truncated or dropped
Condition: value_check — Verify AUD\$ row count is not unexpectedly zero -- zero rows may indicate recent truncation
Audit and Log Tampering FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT count(*) AS aud_row_count, max(ntimestamp#) AS latest_entry FROM sys.aud\$
SEC-SQL-AUD-011-RC04 Trigger disabled immediately before DML operation
Condition: row_count > 0 — Disabled triggers on non-system tables found -- verify authorization and correlate with DML
Audit and Log Tampering FINNACLEDB_TEST: FAILED FLEXCUBE: FAILED alert: No mail: No
SELECT owner, trigger_name, table_name, status, last_change FROM dba_triggers WHERE status = 'DISABLED' AND owner NOT IN ('SYS','SYSTEM') ORDER BY last_change DESC
SEC-SQL-AUD-011-RC05 Error log or event buffer cleared during session
Condition: row_count > 3 — More than 3 log switches in last hour -- unusually high frequency may indicate deliberate log cycling
Audit and Log Tampering FINNACLEDB_TEST: FAILED FLEXCUBE: FAILED alert: No mail: No
SELECT sequence#, first_time, next_time, blocks, block_size FROM v\$log_history WHERE first_time > SYSDATE - 1/24 ORDER BY first_time DESC
SEC-SQL-AUD-011-RC06 Unauthorised INSERT, UPDATE, DELETE or DROP statement executed by a login
Condition: row_count > 0 — A login executed a statement it is not authorised to run
Audit and Log Tampering FINNACLEDB_TEST: FAILED FLEXCUBE: FAILED alert: No mail: No
SELECT * FROM monitoring.v_sec_sql_aud_011_rc06 ORDER BY entry_date DESC

SEC-SQL-AUD-012-RC01 Transaction open beyond threshold with no activity
Condition: row_count > 0 — Blocked Oracle sessions exist -- long-open transaction is causing active lock contention
Long-Running Uncommitted Transactions FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: Yes (destructive)
SELECT count(*) AS blocked_sessions FROM v\$session WHERE blocking_session IS NOT NULL
SEC-SQL-AUD-012-RC02 Exclusive locks held on large tables for extended period
Condition: row_count > 0 — Blocked Oracle sessions confirm exclusive lock is causing active downstream contention
Long-Running Uncommitted Transactions FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: Yes (destructive)
SELECT count(*) AS blocked_sessions FROM v\$session WHERE blocking_session IS NOT NULL
SEC-SQL-AUD-012-RC03 Idle-in-transaction session blocking downstream operations
Condition: row_count > 0 — Inactive Oracle session holding locks with downstream waiters -- idle-in-transaction blocking pattern
Long-Running Uncommitted Transactions FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT s.sid, s.serial#, s.username, s.status, s.last_call_et, count(w.sid) AS blocked_sessions FROM v\$session s JOIN v\$lock l ON l.sid = s.sid LEFT JOIN v\$session w ON w.blocking_session = s.sid WHERE s.status = 'INACTIVE' AND l.block = 1 AND s.last_call_et > 300 GROUP BY s.sid, s.serial#, s.username, s.status, s.last_call_et HAVING count(w.sid) > 0
SEC-SQL-AUD-012-RC04 Large-volume rollback indicating probe-and-revert behaviour
Condition: row_count > 0 — DML and ROLLBACK counts in audit trail -- compare to identify disproportionate rollback ratio
Long-Running Uncommitted Transactions FINNACLEDB_TEST: FAILED FLEXCUBE: FAILED alert: No mail: No
SELECT db_username, action_name, count(*) AS occurrences FROM unified_audit_trail WHERE action_name IN ('INSERT','UPDATE','DELETE','ROLLBACK') AND event_timestamp > systimestamp - interval '30' minute GROUP BY db_username, action_name ORDER BY db_username, action_name
SEC-SQL-AUD-012-RC05 Savepoint abuse with repeated partial rollbacks
Condition: row_count > 0 — More than 20 SAVEPOINT or ROLLBACK TO SAVEPOINT events in 10 minutes -- savepoint abuse pattern
Long-Running Uncommitted Transactions FINNACLEDB_TEST: FAILED FLEXCUBE: FAILED alert: No mail: No
SELECT db_username, action_name, count(*) AS event_count, max(event_timestamp) AS last_seen FROM unified_audit_trail WHERE action_name IN ('SAVEPOINT','ROLLBACK TO SAVEPOINT') AND event_timestamp > systimestamp - interval '10' minute GROUP BY db_username, action_name HAVING count(*) > 20 ORDER BY db_username
SEC-SQL-AUD-013-RC01 Security-relevant ALTER SYSTEM or SET executed by non-admin
Condition: row_count > 0 — Review security-critical parameter values -- verify none changed from approved baseline
Security Configuration Change During Active Session FINNACLEDB_TEST: DETECTED 5 rows FLEXCUBE: DETECTED 5 rows alert: Yes mail: No
SELECT name, value, description FROM v\$parameter WHERE name IN ('audit_trail','os_authent_prefix','remote_os_authent', 'utl_file_dir','o7_dictionary_accessibility','remote_login_passwordfile','sec_case_sensitive_logon')
SEC-SQL-AUD-013-RC02 Dangerous feature enabled during session
Condition: row_count > 0 — Non-DBA accounts hold EXECUTE on dangerous packages -- verify each grant is authorized
Security Configuration Change During Active Session FINNACLEDB_TEST: DETECTED 18 rows FLEXCUBE: DETECTED 78 rows alert: Yes mail: No
SELECT grantee, table_name, privilege, grantable FROM dba_tab_privs WHERE table_name IN ('UTL_FILE','UTL_HTTP','UTL_SMTP','UTL_TCP', 'DBMS_SCHEDULER','DBMS_ADVISOR','DBMS_JAVA', 'DBMS_XMLQUERY') AND privilege = 'EXECUTE' AND grantee NOT IN ('SYS','SYSTEM','DBA')
SEC-SQL-AUD-013-RC03 Trigger disabled or dropped on monitored table
Condition: row_count > 0 — Disabled triggers on application-owned tables -- verify authorization and check for unaudited DML

Security Configuration Change During Active Session FINNACLEDB_TEST: FAILED FLEXCUBE: FAILED alert: No mail: No
SELECT owner, trigger_name, table_name, status, trigger_type, last_change FROM dba_triggers WHERE status = 'DISABLED' AND owner NOT IN ('SYS','SYSTEM','AUDSYS') ORDER BY last_change DESC
SEC-SQL-AUD-013-RC04 Audit or logging parameters changed mid-session
Condition: row_count > 0 — Audit policy disabling command detected in V\$SQL within last hour
Security Configuration Change During Active Session FINNACLEDB_TEST: DETECTED 6 rows FLEXCUBE: DETECTED 4 rows alert: Yes mail: No
SELECT s.username, s.sid, q.sql_text, q.last_active_time FROM v\$sql q JOIN v\$session s ON q.parsing_user_id = s.user# WHERE (UPPER(q.sql_text) LIKE '%NOAUDIT%' OR UPPER(q.sql_text) LIKE '%ALTER AUDIT POLICY%DISABLE%' OR UPPER(q.sql_text) LIKE '%ALTER SYSTEM SET AUDIT_TRAIL%NONE%') AND q.last_active_time > SYSDATE - 1/24
SEC-SQL-AUD-013-RC05 Network or authentication configuration modified at runtime
Condition: row_count > 0 — Database links created within last hour -- verify each is authorized and required
Security Configuration Change During Active Session FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT owner, db_link, username, host, created FROM dba_db_links WHERE created > SYSDATE - 1/24 ORDER BY created DESC
SEC-SQL-AUD-014-RC01 New user or login created during application session
Condition: row_count > 0 — Oracle accounts created in last hour -- verify each was created through an authorized process
User and Role Manipulation During Session FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT username, account_status, created, profile FROM dba_users WHERE created > SYSDATE - 1/24 ORDER BY created DESC
SEC-SQL-AUD-014-RC02 Account added to privileged role mid-session
Condition: row_count > 0 — Current DBA/SYSDBA role grantees -- verify against approved role grant baseline
User and Role Manipulation During Session FINNACLEDB_TEST: DETECTED 13 rows FLEXCUBE: DETECTED 16 rows alert: Yes mail: No
SELECT grantee, granted_role, admin_option, default_role FROM dba_role_privs WHERE granted_role IN ('DBA','SYSDBA','SYSOPER','IMP_FULL_DATABASE','EXP_FULL_DATABASE') ORDER BY granted_role, grantee
SEC-SQL-AUD-014-RC03 Password changed on another account during active session
Condition: row_count > 0 — Dormant accounts with future expiry dates -- may indicate password expiry removal to reactivate dormant credentials
User and Role Manipulation During Session FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT username, account_status, expiry_date, last_login, profile FROM dba_users WHERE expiry_date > SYSDATE AND last_login < SYSDATE - 90 ORDER BY last_login DESC
SEC-SQL-AUD-014-RC04 Database link or synonym to external system created
Condition: row_count > 0 — Database links present -- verify each is in the approved inventory and was not created without authorization
User and Role Manipulation During Session FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 6 rows alert: Yes mail: No
SELECT owner, db_link, username, host, created FROM dba_db_links ORDER BY created DESC
SEC-SQL-AUD-014-RC05 Account unlocked or expiry removed on dormant account
Condition: row_count > 0 — Open accounts that have not logged in for 90 days -- potential recently unlocked dormant accounts
User and Role Manipulation During Session FINNACLEDB_TEST: DETECTED 6 rows FLEXCUBE: DETECTED 6 rows alert: Yes mail: No
SELECT username, account_status, expiry_date, lock_date, last_login FROM dba_users WHERE account_status = 'OPEN' AND last_login < SYSDATE - 90 ORDER BY last_login DESC FETCH FIRST 20 ROWS ONLY
SEC-SQL-AUD-015-RC01 Recent schema changes or DML detected via SQL Server Audit
Condition: row_count > 0 — Audit setup row present (setup_ok=1 means DBDOME_DML_POL and DBDOME_DDL_POL policies enabled; setup_ok=0 means AUDIT_ADMIN privilege missing or anoth...

Schema and DML Activity Tracking FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: Yes (destructive)
SELECT CASE WHEN COUNT(*) >= 2 THEN 1 ELSE 0 END AS setup_ok, COUNT(*) AS enabled_server_audits, COUNT(*) AS enabled_server_specs, 'DBDOME_DML_POL,DBDOME_DDL_POL' AS audit_name, 'unified_audit_trail' AS audit_directory, CASE WHEN COUNT(*) >= 2 THEN NULL ELSE 'Manual setup required (AUDIT_ADMIN privilege): create and enable DBDOME_DML_POL and DBDOME_DDL_POL audit policies. See seed script header for the exact statements.' END AS error_message FROM audit_unified_enabled_policies WHERE policy_name IN ('DBDOME_DML_POL','DBDOME_DDL_POL')
SEC-SQL-AUD-020-RC01 Database or schema dropped (audit trail)
Condition: row_count > 0 — A database/schema drop is present in the audit trail (last 24h).
Destructive DDL/DML detected via audit trail FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT event_timestamp AS event_time, dbusername AS login_name, object_schema AS database_name, object_name, sql_text AS statement FROM unified_audit_trail WHERE event_timestamp > SYSTIMESTAMP - INTERVAL '1' DAY AND (action_name IN ('DROP PLUGGABLE DATABASE','DROP TABLESPACE') OR UPPER(sql_text) LIKE '%DROP DATABASE%') ORDER BY event_timestamp DESC FETCH FIRST 500 ROWS ONLY
SEC-SQL-AUD-020-RC02 Table dropped (audit trail)
Condition: row_count > 0 — A DROP TABLE is present in the audit trail (last 24h).
Destructive DDL/DML detected via audit trail FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT event_timestamp AS event_time, dbusername AS login_name, object_schema AS database_name, object_name, sql_text AS statement FROM unified_audit_trail WHERE event_timestamp > SYSTIMESTAMP - INTERVAL '1' DAY AND (action_name = 'DROP TABLE') ORDER BY event_timestamp DESC FETCH FIRST 500 ROWS ONLY
SEC-SQL-AUD-020-RC03 User, login or role dropped (audit trail)
Condition: row_count > 0 — A user/login/role drop is present in the audit trail (last 24h).
Destructive DDL/DML detected via audit trail FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT event_timestamp AS event_time, dbusername AS login_name, object_schema AS database_name, object_name, sql_text AS statement FROM unified_audit_trail WHERE event_timestamp > SYSTIMESTAMP - INTERVAL '1' DAY AND (action_name IN ('DROP USER','DROP ROLE')) ORDER BY event_timestamp DESC FETCH FIRST 500 ROWS ONLY
SEC-SQL-AUD-020-RC04 Data deleted from a table (audit trail)
Condition: row_count > 0 — A DELETE/TRUNCATE is present in the audit trail (last 24h).
Destructive DDL/DML detected via audit trail FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT event_timestamp AS event_time, dbusername AS login_name, object_schema AS database_name, object_name, sql_text AS statement FROM unified_audit_trail WHERE event_timestamp > SYSTIMESTAMP - INTERVAL '1' DAY AND (action_name IN ('DELETE','TRUNCATE TABLE')) ORDER BY event_timestamp DESC FETCH FIRST 500 ROWS ONLY
SEC-SQL-AUD-021-RC01 Successful database authentication events captured
Condition: row_count > 0 — Successful database authentication events are present
Successful database authentication monitoring FINNACLEDB_TEST: DETECTED 6 rows FLEXCUBE: DETECTED 264 rows alert: Yes mail: No
SELECT TO_CHAR(s.logon_time,'YYYY-MM-DD HH24:MI:SS') AS event_time, s.username AS login_name, s.machine AS host_name, s.osuser AS os_user, s.status AS auth_info, s.program AS program FROM v\$session s WHERE s.type='USER' AND s.username IS NOT NULL AND s.username NOT IN ('SYS','SYSTEM','DBSNMP','SYSMAN','XDB') ORDER BY s.logon_time DESC FETCH FIRST 500 ROWS ONLY
SEC-SQL-AUD-022-RC01 Unsuccessful database authentication attempts detected
Condition: row_count > 0 — Failed database authentication attempts are present
Unsuccessful database authentication attempts FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 500 rows alert: Yes mail: No
SELECT TO_CHAR(event_timestamp,'YYYY-MM-DD HH24:MI:SS') AS event_time, dbusername AS login_name, userhost AS host_name, os_username AS os_user, 'ORA-' LPAD(TO_CHAR(return_code),5,'0') AS error_info, TO_CHAR(return_code) AS attempts FROM unified_audit_trail WHERE action_name='LOGON' AND return_code <> 0 AND event_timestamp > SYSTIMESTAMP - INTERVAL '1' DAY ORDER BY event_timestamp DESC FETCH FIRST 500 ROWS ONLY
SEC-SQL-AUD-023-RC01 User account created (CREATE USER/LOGIN/ROLE)
Condition: row_count > 0 — User account created (CREATE USER/LOGIN/ROLE) detected

DBA and administrator activity monitoring FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT TO_CHAR(event_timestamp,'YYYY-MM-DD HH24:MI:SS') AS event_time, dbusername AS login_name, userhost AS host_name, os_username AS os_user, action_name AS auth_info, sql_text AS statement FROM unified_audit_trail WHERE (action_name IN ('CREATE USER','CREATE ROLE')) AND event_timestamp > SYSTIMESTAMP - INTERVAL '1' DAY ORDER BY event_timestamp DESC FETCH FIRST 500 ROWS ONLY
SEC-SQL-AUD-023-RC02 Privileges granted (possible excessive grant)
Condition: row_count > 0 — Privileges granted (possible excessive grant) detected
DBA and administrator activity monitoring FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT TO_CHAR(event_timestamp,'YYYY-MM-DD HH24:MI:SS') AS event_time, dbusername AS login_name, userhost AS host_name, os_username AS os_user, action_name AS auth_info, sql_text AS statement FROM unified_audit_trail WHERE (action_name LIKE '%GRANT%') AND event_timestamp > SYSTIMESTAMP - INTERVAL '1' DAY ORDER BY event_timestamp DESC FETCH FIRST 500 ROWS ONLY
SEC-SQL-AUD-023-RC03 User account dropped (DROP USER/LOGIN/ROLE)
Condition: row_count > 0 — User account dropped (DROP USER/LOGIN/ROLE) detected
DBA and administrator activity monitoring FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT TO_CHAR(event_timestamp,'YYYY-MM-DD HH24:MI:SS') AS event_time, dbusername AS login_name, userhost AS host_name, os_username AS os_user, action_name AS auth_info, sql_text AS statement FROM unified_audit_trail WHERE (action_name IN ('DROP USER','DROP ROLE')) AND event_timestamp > SYSTIMESTAMP - INTERVAL '1' DAY ORDER BY event_timestamp DESC FETCH FIRST 500 ROWS ONLY
SEC-SQL-AUD-024-RC01 SELECT access to designated sensitive tables detected
Condition: row_count > 0 — SELECT access to a designated sensitive table detected
Sensitive business-data access monitoring FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 84 rows alert: Yes mail: No
SELECT TO_CHAR(last_active_time,'YYYY-MM-DD HH24:MI:SS') AS event_time, parsing_schema_name AS login_name, SYS_CONTEXT('USERENV','DB_NAME') AS host_name, CAST(NULL AS VARCHAR2(30)) AS os_user, 'plan cache' AS auth_info, SUBSTR(sql_text,1,4000) AS statement FROM v\$sql WHERE command_type = 3 -- SELECT AND (UPPER(sql_text) LIKE '%FSDH_CUSTOMER_BVN%' OR UPPER(sql_text) LIKE '%STTM_CUSTOMER%' OR UPPER(sql_text) LIKE '%STTM_CUST_ACCOUNT%' OR UPPER(sql_text) LIKE '%STTM_PERSONAL%' OR UPPER(sql_text) LIKE '%STTM_CORPORATE%') ORDER BY last_active_time DESC FETCH FIRST 500 ROWS ONLY
SEC-SQL-AUD-025-RC01 Unauthorized data change (UPDATE/DELETE/INSERT) to designated sensitive tables detected
Condition: row_count > 0 — Data change (UPDATE/DELETE/INSERT) to a designated sensitive table detected
Unauthorized sensitive-data change monitoring FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 25 rows alert: Yes mail: No
SELECT TO_CHAR(last_active_time,'YYYY-MM-DD HH24:MI:SS') AS event_time, parsing_schema_name AS login_name, SYS_CONTEXT('USERENV','DB_NAME') AS host_name, CAST(NULL AS VARCHAR2(30)) AS os_user, CASE command_type WHEN 6 THEN 'UPDATE' WHEN 7 THEN 'DELETE' WHEN 2 THEN 'INSERT' END AS auth_info, SUBSTR(sql_text,1,4000) AS statement FROM v\$sql WHERE command_type IN (6,7,2) -- UPDATE / DELETE / INSERT (data change) AND (UPPER(sql_text) LIKE '%FSDH_CUSTOMER_BVN%' OR UPPER(sql_text) LIKE '%STTM_CUSTOMER%' OR UPPER(sql_text) LIKE '%STTM_CUST_ACCOUNT%' OR UPPER(sql_text) LIKE '%STTM_PERSONAL%' OR UPPER(sql_text) LIKE '%STTM_CORPORATE%') ORDER BY last_active_time DESC FETCH FIRST 500 ROWS ONLY
SEC-SQL-AUD-026-RC01 Unauthorized privilege escalation - account holds elevated/administrative privileges
Condition: row_count > 0 — A non-baseline account holds elevated/administrative privileges (possible unauthorized privilege assignment)
Unauthorized privilege escalation monitoring FINNACLEDB_TEST: DETECTED 23 rows FLEXCUBE: DETECTED 92 rows alert: Yes mail: No
SELECT * FROM (SELECT TO_CHAR(SYSTIMESTAMP,'YYYY-MM-DD HH24:MI:SS') AS event_time, grantee AS login_name, SYS_CONTEXT('USERENV','DB_NAME') AS host_name, CAST(NULL AS VARCHAR2(30)) AS os_user, 'ROLE ' granted_role AS auth_info, 'Account ' grantee ' holds elevated role ' granted_role CASE WHEN admin_option='YES' THEN ' (WITH ADMIN OPTION)' ELSE '' END AS statement FROM dba_role_privs WHERE granted_role IN ('DBA','PDB_DBA','IMP_FULL_DATABASE','EXP_FULL_DATABASE','DATAPUMP_IMP_FULL_DATABASE','DATAPUMP_EXP_FULL_DATABASE','AUDIT_ADMIN') UNION ALL SELECT TO_CHAR(SYSTIMESTAMP,'YYYY-MM-DD HH24:MI:SS'), grantee, SYS_CONTEXT('USERENV','DB_NAME'), CAST(NULL AS VARCHAR2(30)), 'PRIV ' privilege, 'Account ' grantee ' holds elevated privilege ' privilege CASE WHEN admin_option='YES' THEN ' (WITH ADMIN OPTION)' ELSE '' END FROM dba_sys_privs WHERE privilege IN ('ALTER SYSTEM','ALTER DATABASE','BECOME USER','GRANT ANY PRIVILEGE','GRANT ANY ROLE','GRANT ANY OBJECT PRIVILEGE','CREATE ANY TABLE','DROP ANY TABLE', 'SELECT ANY TABLE','INSERT ANY TABLE','UPDATE ANY TABLE','DELETE ANY TABLE','CREATE USER', 'ALTER USER','DROP USER','CREATE ANY PROCEDURE','ALTER ANY TABLE','AUDIT SYSTEM','ALTER PROFILE')) t WHERE login_name IN (SELECT username FROM dba_users WHERE oracle_maintained='N') ORDER BY login_name, auth_info FETCH FIRST 500 ROWS ONLY
SEC-SQL-AUD-027-RC01 Destructive operation attempted - DROP/TRUNCATE/BACKUP command detected (including failed attempts)
Condition: row_count > 0 — A destructive command (DROP/TRUNCATE/PURGE, incl. failed attempts) was executed or attempted in the last 24 hours
Destructive operation monitoring FINNACLEDB_TEST: FAILED FLEXCUBE: FAILED alert: No mail: No

SELECT * FROM (SELECT TO_CHAR(event_timestamp, 'YYYY-MM-DD HH24:MI:SS') AS event_time, dbusername AS login_name, SYS_CONTEXT('USERENV','DB_NAME') AS host_name, os_username AS os_user, action_name CASE WHEN return_code <> 0 THEN ' (FAILED ORA-' return_code ')' ELSE '' END AS auth_info, NVL(object_schema '.' object_name, '-') ' :: ' NVL(SUBSTR(sql_text, 1, 200), '-') AS statement FROM unified_audit_trail WHERE event_timestamp >= SYSTIMESTAMP - INTERVAL '1' DAY AND (action_name LIKE 'DROP%' OR action_name LIKE 'TRUNCATE%' OR action_name LIKE 'PURGE%') UNION ALL SELECT SUBSTR(droptime, 1, 10) ' ' SUBSTR(droptime, 12, 8), owner, SYS_CONTEXT('USERENV','DB_NAME'), CAST(NULL AS VARCHAR2(30)), 'DROP ' type ' (recyclebin)', 'Dropped object ' owner '.' original_name FROM dba_recyclebin WHERE TO_DATE(droptime, 'YYYY-MM-DD:HH24:MI:SS') >= SYSDATE - 1) t ORDER BY event_time DESC FETCH FIRST 500 ROWS ONLY
SEC-SQL-AUTHZ-001-RC01 Database access & privilege inventory (logins, roles, grants)
Condition: (no condition authored)
Database Access & Privilege Inventory FINNACLEDB_TEST: DETECTED 21 rows FLEXCUBE: FAILED alert: Yes mail: No
SELECT sys_context('USERENV','SERVER_HOST') AS server, sys_context('USERENV','DB_NAME') AS database_name, u.username AS login_name, u.username AS database_user, 'USER' AS principal_type, (SELECT LISTAGG(rp.granted_role, ', ') WITHIN GROUP (ORDER BY rp.granted_role) FROM dba_role_privs rp WHERE rp.grantee = u.username) AS roles, (SELECT LISTAGG(sp.privilege, ', ') WITHIN GROUP (ORDER BY sp.privilege) FROM dba_sys_privs sp WHERE sp.grantee = u.username) AS permissions, u.account_status AS login_status FROM dba_users u WHERE u.oracle_maintained = 'N' ORDER BY u.username
SEC-SQL-AZ-001-RC01 Lack of Role-Based Access Control (RBAC) implementation
Condition: row_count > :user_threshold — Many users with direct system privilege grants instead of role-based assignments indicates lack of RBAC
Excessive Administrative Privileges FINNACLEDB_TEST: DETECTED 11 rows FLEXCUBE: DETECTED 15 rows alert: Yes mail: No
SELECT grantee, COUNT(*) AS direct_privs FROM dba_sys_privs WHERE grantee NOT IN (SELECT role FROM dba_roles) AND grantee NOT IN ('SYS', 'SYSTEM', 'PUBLIC') GROUP BY grantee HAVING COUNT(*) > 5 ORDER BY direct_privs DESC
SEC-SQL-AZ-001-RC02 Principle of least privilege not enforced
Condition: row_count > :dba_threshold — Multiple non-system users have the DBA role, which grants all system privileges
Excessive Administrative Privileges FINNACLEDB_TEST: DETECTED 4 rows FLEXCUBE: DETECTED 5 rows alert: Yes mail: No
SELECT grantee, granted_role, admin_option FROM dba_role_privs WHERE granted_role = 'DBA' AND grantee NOT IN ('SYS', 'SYSTEM') ORDER BY grantee
SEC-SQL-AZ-001-RC03 Developer environments influence production practices
Condition: row_count > :public_threshold — Object privileges granted to PUBLIC indicate overly permissive access patterns typical of development environments
Excessive Administrative Privileges FINNACLEDB_TEST: DETECTED 4119 rows FLEXCUBE: DETECTED 4202 rows alert: Yes mail: No
SELECT privilege, table_name, grantor FROM dba_tab_privs WHERE grantee = 'PUBLIC' AND grantor NOT IN ('SYS', 'SYSTEM') ORDER BY grantor, table_name
SEC-SQL-AZ-001-RC04 Time pressure and perceived operational convenience
Condition: row_count > 0 — Users with multiple ANY privileges suggest convenience-driven grants that bypass proper schema-specific access control
Excessive Administrative Privileges FINNACLEDB_TEST: DETECTED 9 rows FLEXCUBE: DETECTED 18 rows alert: Yes mail: No
SELECT grantee, COUNT(*) AS any_priv_count FROM dba_sys_privs WHERE privilege LIKE '%ANY%' AND grantee NOT IN ('SYS', 'SYSTEM', 'DBA') AND grantee NOT IN (SELECT role FROM dba_roles WHERE role LIKE '%DBA%' OR role LIKE '%ADMIN%') GROUP BY grantee HAVING COUNT(*) > 3 ORDER BY any_priv_count DESC
SEC-SQL-AZ-001-RC05 Inadequate permission analysis during application deployment
Condition: row_count > 0 — Users with broad schema management ANY privileges suggest deployment accounts with excessive permissions
Excessive Administrative Privileges FINNACLEDB_TEST: DETECTED 43 rows FLEXCUBE: DETECTED 61 rows alert: Yes mail: No
SELECT grantee, privilege, admin_option FROM dba_sys_privs WHERE grantee NOT IN ('SYS', 'SYSTEM') AND privilege IN ('CREATE ANY TABLE', 'DROP ANY TABLE', 'ALTER ANY TABLE', 'CREATE ANY INDEX', 'DROP ANY INDEX', 'CREATE ANY VIEW', 'DROP ANY VIEW', 'CREATE ANY PROCEDURE', 'DROP ANY PROCEDURE') ORDER BY grantee, privilege
SEC-SQL-AZ-001-RC06 Multi-schema and cross-database application requirements
Condition: row_count > 0 — Users with object privileges across many schemas indicate complex cross-schema access requirements
Excessive Administrative Privileges FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 1 rows alert: Yes mail: No

SELECT grantee, COUNT(DISTINCT owner) AS schema_count, COUNT(*) AS total_grants FROM dba_tab_privs WHERE grantee NOT IN ('SYS', 'SYSTEM', 'PUBLIC') AND grantee NOT IN (SELECT role FROM dba_roles) GROUP BY grantee HAVING COUNT(DISTINCT owner) > 3 ORDER BY schema_count DESC
SEC-SQL-AZ-001-RC07 Third-party application vendor requirements
Condition: row_count > 0 — Users with broad system privileges or deprecated roles (RESOURCE, CONNECT) suggest vendor-dictated privilege grants
Excessive Administrative Privileges FINNACLEDB_TEST: DETECTED 71 rows FLEXCUBE: DETECTED 374 rows alert: Yes mail: No
SELECT grantee, privilege, admin_option FROM dba_sys_privs WHERE grantee NOT IN ('SYS', 'SYSTEM') AND (privilege LIKE '%ANY%' OR privilege IN ('DBA', 'RESOURCE', 'CONNECT')) AND grantee NOT IN (SELECT role FROM dba_roles) ORDER BY grantee, privilege
SEC-SQL-AZ-001-RC08 Absence of periodic access reviews and certification
Condition: row_count > 0 — Active accounts not logged in for extended periods still have privileges, indicating no periodic access review
Excessive Administrative Privileges FINNACLEDB_TEST: DETECTED 7 rows FLEXCUBE: DETECTED 6 rows alert: Yes mail: No
SELECT u.username, TO_CHAR(u.created, 'YYYY-MM-DD') AS created, u.account_status, TO_CHAR(u.last_login, 'YYYY-MM-DD HH24:MI:SS') AS last_login, ROUND(SYSDATE - NVL(CAST(u.last_login AS DATE), u.created)) AS days_since_login, (SELECT COUNT(*) FROM dba_sys_privs sp WHERE sp.grantee = u.username) AS sys_privs, (SELECT COUNT(*) FROM dba_role_privs rp WHERE rp.grantee = u.username) AS roles FROM dba_users u WHERE u.oracle_maintained = 'N' AND u.account_status = 'OPEN' AND NVL(CAST(u.last_login AS DATE), u.created) < SYSDATE - 90 ORDER BY days_since_login DESC
SEC-SQL-AZ-001-RC09 Employee role transitions without privilege adjustment
Condition: row_count > 0 — Users with many roles may have accumulated privileges from multiple job functions without cleanup
Excessive Administrative Privileges FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT grantee, COUNT(*) AS role_count, LISTAGG(granted_role, ', ') WITHIN GROUP (ORDER BY granted_role) AS roles FROM dba_role_privs WHERE grantee NOT IN (SELECT role FROM dba_roles) AND grantee NOT IN ('SYS', 'SYSTEM') GROUP BY grantee HAVING COUNT(*) > 5 ORDER BY role_count DESC
SEC-SQL-AZ-001-RC10 Legacy applications with hardcoded privilege assumptions
Condition: any(session_privs > 0 AND object_count > :obj_threshold) — Schema owner accounts with CREATE SESSION indicate applications connecting directly as the schema owner — a legacy pattern
Excessive Administrative Privileges FINNACLEDB_TEST: DETECTED 3 rows FLEXCUBE: DETECTED 10 rows alert: Yes mail: No
SELECT u.username, u.account_status, u.authentication_type, (SELECT COUNT(*) FROM dba_sys_privs WHERE grantee = u.username AND privilege IN ('CREATE SESSION', 'UNLIMITED TABLESPACE')) AS session_privs, (SELECT COUNT(*) FROM dba_objects WHERE owner = u.username) AS object_count FROM dba_users u WHERE u.oracle_maintained = 'N' AND u.account_status = 'OPEN' AND EXISTS (SELECT 1 FROM dba_objects WHERE owner = u.username AND object_type IN ('TABLE', 'VIEW', 'PACKAGE')) ORDER BY object_count DESC FETCH FIRST 10 ROWS ONLY
SEC-SQL-AZ-001-RC11 Knowledge gaps and lack of training
Condition: row_count > :role_threshold — Use of deprecated Oracle roles (CONNECT, RESOURCE) suggests knowledge gaps — these roles grant more privileges than expected
Excessive Administrative Privileges FINNACLEDB_TEST: DETECTED 17 rows FLEXCUBE: DETECTED 56 rows alert: Yes mail: No
SELECT grantee, granted_role FROM dba_role_privs WHERE granted_role IN ('CONNECT', 'RESOURCE', 'DBA') AND grantee NOT IN ('SYS', 'SYSTEM') AND grantee NOT IN (SELECT role FROM dba_roles) ORDER BY granted_role, grantee
SEC-SQL-AZ-001-RC12 Vendor default configurations
Condition: row_count > 0 — Oracle-maintained default accounts are not locked — these should be expired and locked after installation
Excessive Administrative Privileges FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: Yes (destructive)
SELECT username, account_status, profile, created, authentication_type FROM dba_users WHERE oracle_maintained = 'Y' AND account_status NOT IN ('EXPIRED & LOCKED', 'LOCKED') ORDER BY username
SEC-SQL-AZ-001-RC13 No centralized identity and access governance
Condition: row_count > 0 — Enterprise users (GLOBAL/EXTERNAL authentication) exist — indicates some centralized identity integration
Excessive Administrative Privileges FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No

SELECT username, authentication_type, external_name FROM dba_users WHERE authentication_type IN ('GLOBAL', 'EXTERNAL') ORDER BY username
SEC-SQL-AZ-001-RC14 Shared service and application accounts
Condition: any(sys_priv_count > 5 OR role_count > 3) — Shared accounts have elevated privileges — individual attribution is impossible
Excessive Administrative Privileges FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT u.username, COUNT(DISTINCT p.privilege) AS sys_priv_count, COUNT(DISTINCT r.granted_role) AS role_count FROM dba_users u LEFT JOIN dba_sys_privs p ON u.username = p.grantee LEFT JOIN dba_role_privs r ON u.username = r.grantee WHERE u.oracle_maintained = 'N' AND u.username IN (SELECT username FROM v\$session WHERE type = 'USER' GROUP BY username HAVING COUNT(DISTINCT machine) > 3 OR COUNT(DISTINCT program) > 3) GROUP BY u.username ORDER BY sys_priv_count DESC
SEC-SQL-AZ-002-RC01 Misunderstanding of rootcause role scope
Condition: row_count > 0 — PUBLIC has object privileges on non-system schemas — all users can access these objects
rootcause Role Excessive Access FINNACLEDB_TEST: DETECTED 4102 rows FLEXCUBE: DETECTED 4157 rows alert: Yes mail: No
SELECT owner, table_name, privilege, grantor FROM dba_tab_privs WHERE grantee = 'PUBLIC' AND owner NOT IN ('SYS', 'PUBLIC', 'SYSTEM') ORDER BY owner, table_name
SEC-SQL-AZ-002-RC02 Quick-fix troubleshooting approach
Condition: row_count > 0 — PUBLIC has EXECUTE on sensitive packages — these are common quick-fix grants that should be role-based
rootcause Role Excessive Access FINNACLEDB_TEST: DETECTED 8 rows FLEXCUBE: DETECTED 8 rows alert: Yes mail: No
SELECT owner, table_name, privilege FROM dba_tab_privs WHERE grantee = 'PUBLIC' AND privilege = 'EXECUTE' AND table_name IN ('UTL_FILE', 'UTL_HTTP', 'UTL_TCP', 'UTL_SMTP', 'DBMS_SQL', 'DBMS_SYS_SQL', 'DBMS_LOB', 'DBMS_BACKUP_RESTORE', 'DBMS_SCHEDULER', 'DBMS_JOB') ORDER BY table_name
SEC-SQL-AZ-002-RC03 Temporary workarounds becoming permanent
Condition: custom_role_count < 3 — Very few custom roles exist — indicates PUBLIC is used as a shortcut instead of proper RBAC
rootcause Role Excessive Access FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT COUNT(*) AS custom_role_count FROM dba_roles WHERE oracle_maintained = 'N' AND role NOT IN ('DBA', 'CONNECT', 'RESOURCE', 'EXP_FULL_DATABASE', 'IMP_FULL_DATABASE')
SEC-SQL-AZ-002-RC04 Legacy database practices
Condition: total_public_grants > 50 — PUBLIC has a large number of grants — likely accumulated over multiple upgrades without cleanup
rootcause Role Excessive Access FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: Yes (destructive)
SELECT COUNT(*) AS total_public_grants FROM (SELECT privilege FROM dba_sys_privs WHERE grantee = 'PUBLIC' UNION ALL SELECT privilege FROM dba_tab_privs WHERE grantee = 'PUBLIC')
SEC-SQL-AZ-002-RC05 Insufficient permission auditing on rootcause
Condition: row_count > 0 — GRANT/REVOKE operations are being audited — provides visibility into PUBLIC permission changes
rootcause Role Excessive Access FINNACLEDB_TEST: DETECTED 369 rows FLEXCUBE: DETECTED 371 rows alert: Yes mail: No
SELECT policy_name, audit_option FROM audit_unified_policies WHERE audit_option IN ('GRANT', 'REVOKE') UNION ALL SELECT policy_name, audit_option FROM audit_unified_policies WHERE audit_option_type = 'SYSTEM PRIVILEGE' AND audit_option = 'GRANT ANY PRIVILEGE' ORDER BY 1
SEC-SQL-AZ-002-RC06 Default view permissions in system objects
Condition: row_count > 0 — PUBLIC has SELECT on DBA_ dictionary views — all users can view sensitive metadata
rootcause Role Excessive Access FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT owner, table_name, privilege FROM dba_tab_privs WHERE grantee = 'PUBLIC' AND privilege = 'SELECT' AND owner IN ('SYS', 'SYSTEM') AND table_name NOT LIKE 'V_\$_' AND table_name NOT LIKE 'GV_\$_' AND table_name LIKE 'DBA_%' ORDER BY owner, table_name
SEC-SQL-AZ-002-RC07 Confusion between schema and database-level rootcause

Condition: row_count > 1 — PUBLIC has grants across multiple user schemas — indicates confusion about PUBLIC scope across the entire database
rootcause Role Excessive Access FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 4 rows alert: Yes mail: No
SELECT owner, COUNT(*) AS grant_count FROM dba_tab_privs WHERE grantee = 'PUBLIC' AND owner NOT IN (SELECT username FROM dba_users WHERE oracle_maintained = 'Y') GROUP BY owner HAVING COUNT(*) > 5 ORDER BY grant_count DESC
SEC-SQL-AZ-002-RC08 Testing/development configuration copied to production
Condition: row_count > 0 — PUBLIC has DML privileges (INSERT/UPDATE/DELETE) on user tables — typical of dev/test configurations, not production
rootcause Role Excessive Access FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 15 rows alert: Yes mail: No
SELECT owner, table_name, privilege FROM dba_tab_privs WHERE grantee = 'PUBLIC' AND privilege IN ('INSERT', 'UPDATE', 'DELETE') AND owner NOT IN (SELECT username FROM dba_users WHERE oracle_maintained = 'Y') ORDER BY owner, table_name
SEC-SQL-AZ-002-RC09 Bulk permission scripts without proper restriction
Condition: row_count > 0 — All tables in a schema are granted to PUBLIC — confirms a bulk script pattern
rootcause Role Excessive Access FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT t.owner, COUNT(t.table_name) AS total_tables, COUNT(p.table_name) AS granted_tables FROM dba_tables t LEFT JOIN dba_tab_privs p ON t.owner = p.owner AND t.table_name = p.table_name AND p.grantee = 'PUBLIC' WHERE t.owner NOT IN (SELECT username FROM dba_users WHERE oracle_maintained = 'Y') GROUP BY t.owner HAVING COUNT(p.table_name) > 0 AND COUNT(p.table_name) = COUNT(t.table_name) ORDER BY total_tables DESC
SEC-SQL-AZ-002-RC10 Lack of role-based alternative
Condition: row_count < 3 — Very few custom roles defined — indicates PUBLIC is being used instead of proper RBAC
rootcause Role Excessive Access FINNACLEDB_TEST: DETECTED 9 rows FLEXCUBE: DETECTED 6 rows alert: Yes mail: No
SELECT role, authentication_type FROM dba_roles WHERE oracle_maintained = 'N' ORDER BY role
SEC-SQL-AZ-002-RC11 Incompletely implemented permission revocation
Condition: row_count > 0 — PUBLIC still has grants on user objects — revocation may be incomplete
rootcause Role Excessive Access FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 37 rows alert: Yes mail: No
SELECT owner, privilege, COUNT(*) AS remaining_grants FROM dba_tab_privs WHERE grantee = 'PUBLIC' AND owner NOT IN (SELECT username FROM dba_users WHERE oracle_maintained = 'Y') GROUP BY owner, privilege ORDER BY owner, privilege
SEC-SQL-AZ-002-RC12 Tool or vendor recommendations that are unsafe
Condition: row_count > 0 — PUBLIC has EXECUTE on commonly misrecommended packages — third-party tools often suggest this unsafe configuration
rootcause Role Excessive Access FINNACLEDB_TEST: DETECTED 10 rows FLEXCUBE: DETECTED 10 rows alert: Yes mail: No
SELECT owner, table_name, privilege FROM dba_tab_privs WHERE grantee = 'PUBLIC' AND privilege = 'EXECUTE' AND table_name IN ('UTL_FILE', 'UTL_HTTP', 'UTL_TCP', 'UTL_SMTTP', 'DBMS_JAVA', 'DBMS_JAVA_TEST', 'DBMS_SQL', 'DBMS_LOB', 'DBMS_XMLGEN', 'DBMS_RANDOM', 'DBMS_SCHEDULER', 'DBMS_PIPE', 'DBMS_ALERT') ORDER BY table_name
SEC-SQL-AZ-002-RC13 Emergency access during incidents
Condition: row_count > 0 — GRANT to PUBLIC operations have no subsequent REVOKE — emergency grants were never cleaned up
rootcause Role Excessive Access FINNACLEDB_TEST: DETECTED 6 rows FLEXCUBE: DETECTED 75 rows alert: Yes mail: No
SELECT g.event_timestamp AS grant_time, g.dbusername AS granted_by, g.sql_text AS grant_sql FROM unified_audit_trail g WHERE g.action_name = 'GRANT' AND g.sql_text LIKE '%PUBLIC%' AND g.event_timestamp > SYSDATE - 90 AND NOT EXISTS (SELECT 1 FROM unified_audit_trail r WHERE r.action_name = 'REVOKE' AND r.sql_text LIKE '%PUBLIC%' AND r.event_timestamp > g.event_timestamp) ORDER BY g.event_timestamp DESC
SEC-SQL-AZ-003-RC01 Absence of user lifecycle management automation
Condition: row_count > 0 — Profiles with UNLIMITED idle time — no automatic session termination for inactive users

Orphaned User Accounts FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 4 rows alert: Yes mail: Yes (destructive)
SELECT profile, limit AS idle_time_limit FROM dba_profiles WHERE resource_name = 'IDLE_TIME' AND limit IN ('UNLIMITED', 'DEFAULT') ORDER BY profile
SEC-SQL-AZ-003-RC02 Manual offboarding processes with gaps
Condition: row_count > 0 — Expired accounts still own database objects — offboarding did not include object reassignment or cleanup
Orphaned User Accounts FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 7 rows alert: Yes mail: No
SELECT u.username, u.account_status, u.created, COUNT(o.object_name) AS owned_objects FROM dba_users u LEFT JOIN dba_objects o ON u.username = o.owner WHERE u.oracle_maintained = 'N' AND u.account_status LIKE '%EXPIRED%' GROUP BY u.username, u.account_status, u.created HAVING COUNT(o.object_name) > 0 ORDER BY owned_objects DESC
SEC-SQL-AZ-003-RC03 Delayed or forgotten deprovisioning
Condition: row_count > 0 — Accounts inactive for 180+ days still open — indicates severely delayed deprovisioning
Orphaned User Accounts FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT username, TO_CHAR(last_login, 'YYYY-MM-DD HH24:MI:SS') AS last_login, TO_CHAR(created, 'YYYY-MM-DD') AS created, ROUND(SYSDATE - NVL(CAST(last_login AS DATE), created)) AS days_inactive FROM dba_users WHERE oracle_maintained = 'N' AND account_status = 'OPEN' AND authentication_type = 'DATABASE' AND NVL(CAST(last_login AS DATE), created) < SYSDATE - 180 ORDER BY days_inactive DESC
SEC-SQL-AZ-003-RC04 Lack of integration between HR systems and database access
Condition: all(authentication_type = 'DATABASE') — All users use DATABASE authentication — no integration with external identity systems (LDAP, AD, Kerberos)
Orphaned User Accounts FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT authentication_type, COUNT(*) AS user_count FROM dba_users WHERE oracle_maintained = 'N' AND account_status = 'OPEN' GROUP BY authentication_type ORDER BY user_count DESC
SEC-SQL-AZ-003-RC05 No reconciliation process between database accounts and authorized users
Condition: row_count > 0 — Active accounts dormant for extended periods indicate missing reconciliation process
Orphaned User Accounts FINNACLEDB_TEST: DETECTED 7 rows FLEXCUBE: DETECTED 10 rows alert: Yes mail: Yes (destructive)
SELECT username, account_status, created, last_login, ROUND(EXTRACT(DAY FROM (SYSTIMESTAMP - CAST(NVL(last_login, created) AS TIMESTAMP)))) AS days_dormant FROM dba_users WHERE oracle_maintained = 'N' AND account_status = 'OPEN' AND (last_login IS NULL OR last_login < SYSDATE - 90) ORDER BY days_dormant DESC
SEC-SQL-AZ-003-RC06 Service and application accounts without ownership tracking
Condition: row_count > 0 with service-type account names — Service accounts without clear naming conventions or recent activity lack ownership tracking
Orphaned User Accounts FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT username, account_status, created, last_login, profile, authentication_type FROM dba_users WHERE oracle_maintained = 'N' AND account_status = 'OPEN' AND (username LIKE '%SVC%' OR username LIKE '%APP%' OR username LIKE '%SERVICE%' OR username LIKE '%API%' OR username LIKE '%BATCH%' OR username LIKE '%ETL%' OR authentication_type = 'EXTERNAL') ORDER BY last_login NULLS FIRST
SEC-SQL-AZ-003-RC07 Contractor and vendor accounts without explicit end dates
Condition: row_count > 0 — Accounts without password expiration or explicit expiry dates may persist indefinitely
Orphaned User Accounts FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 21 rows alert: Yes mail: No
SELECT u.username, u.account_status, u.created, u.expiry_date, u.profile, p.limit AS password_life_time FROM dba_users u JOIN dba_profiles p ON u.profile = p.profile WHERE u.oracle_maintained = 'N' AND u.account_status = 'OPEN' AND p.resource_name = 'PASSWORD_LIFE_TIME' AND (p.limit = 'UNLIMITED' OR p.limit = 'DEFAULT') AND u.expiry_date IS NULL ORDER BY u.created ASC
SEC-SQL-AZ-003-RC08 Shared database accounts and credentials
Condition: row_count > 0 — Accounts logged in from many distinct machines simultaneously suggest shared credentials
Orphaned User Accounts FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 1 rows alert: Yes mail: Yes (destructive)

SELECT username, COUNT(DISTINCT machine) AS distinct_machines, COUNT(DISTINCT program) AS distinct_programs, COUNT(*) AS current_sessions FROM v\$session WHERE type = 'USER' AND username IS NOT NULL GROUP BY username HAVING COUNT(DISTINCT machine) > 5 ORDER BY distinct_machines DESC
SEC-SQL-AZ-003-RC09 Cross-database and application account complexity
Condition: large number of total_accounts relative to expected user base — Many more database accounts than expected active users indicates cross-database account sprawl
Orphaned User Accounts FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: Yes (destructive)
SELECT COUNT(*) AS total_accounts, SUM(CASE WHEN account_status = 'OPEN' THEN 1 ELSE 0 END) AS active_accounts, SUM(CASE WHEN account_status LIKE '%LOCKED%' THEN 1 ELSE 0 END) AS locked_accounts, SUM(CASE WHEN account_status LIKE '%EXPIRED%' THEN 1 ELSE 0 END) AS expired_accounts, SUM(CASE WHEN created > SYSDATE - 90 THEN 1 ELSE 0 END) AS recently_created FROM dba_users WHERE oracle_maintained = 'N'
SEC-SQL-AZ-003-RC10 Legacy account creation without documentation
Condition: row_count > 0 with very old accounts still active — Accounts created years ago and still active without clear purpose indicate legacy undocumented accounts
Orphaned User Accounts FINNACLEDB_TEST: DETECTED 3 rows FLEXCUBE: DETECTED 18 rows alert: Yes mail: No
SELECT username, account_status, created, last_login, profile, authentication_type, ROUND(SYSDATE - created) AS age_days FROM dba_users WHERE oracle_maintained = 'N' AND created < SYSDATE - 730 AND account_status = 'OPEN' ORDER BY created ASC
SEC-SQL-AZ-003-RC11 Database environment fragmentation
Condition: row_count > 0 — Dev/test/QA named accounts in production indicate environment fragmentation
Orphaned User Accounts FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: Yes (destructive)
SELECT username, account_status, created, last_login, profile FROM dba_users WHERE oracle_maintained = 'N' AND (username LIKE '%DEV%' OR username LIKE '%TEST%' OR username LIKE '%QA%' OR username LIKE '%STAGE%' OR username LIKE '%DEMO%' OR username LIKE '%TEMP%' OR username LIKE '%SANDBOX%') AND account_status = 'OPEN' ORDER BY username
SEC-SQL-AZ-003-RC12 Lack of privileged access management (PAM) systems
Condition: row_count > 0 with many users having DBA-level roles — Multiple users with direct DBA role grants without PAM indicates unmanaged privileged access
Orphaned User Accounts FINNACLEDB_TEST: DETECTED 14 rows FLEXCUBE: DETECTED 21 rows alert: Yes mail: No
SELECT grantee, granted_role, admin_option, delegate_option, default_role FROM dba_role_privs WHERE granted_role IN ('DBA', 'SYSDBA', 'SYSOPER', 'DATAPUMP_EXP_FULL_DATABASE', 'DATAPUMP_IMP_FULL_DATABASE', 'EXP_FULL_DATABASE', 'IMP_FULL_DATABASE') AND grantee NOT IN ('SYS', 'SYSTEM') ORDER BY grantee, granted_role
SEC-SQL-AZ-003-RC13 Application default accounts never updated
Condition: row_count > 0 — Default Oracle sample/feature accounts still open indicate they were never secured after installation
Orphaned User Accounts FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT username, account_status, created, last_login, profile FROM dba_users WHERE username IN ('SCOTT', 'HR', 'OE', 'SH', 'PM', 'IX', 'BI', 'APEX_PUBLIC_USER', 'FLOWS_FILES', 'APEX_050000', 'GSMUSER', 'XS\$NULL', 'MDDATA', 'SPATIAL_CSW_ADMIN_USR', 'SPATIAL_WFS_ADMIN_USR', 'LBACSYS', 'ORDPLUGINS', 'SI_INFORMTN_SCHEMA', 'OLAPSYS', 'OWBSYS', 'ORDDATA', 'CTXSYS', 'MDSYS') AND account_status = 'OPEN' ORDER BY username
SEC-SQL-AZ-003-RC14 Failed or incomplete migration processes
Condition: row_count > 0 — Active accounts with no meaningful privileges or objects suggest migration artifacts
Orphaned User Accounts FINNACLEDB_TEST: DETECTED 4 rows FLEXCUBE: DETECTED 3 rows alert: Yes mail: Yes (destructive)
SELECT u.username, u.account_status, u.created, u.last_login FROM dba_users u WHERE u.oracle_maintained = 'N' AND u.account_status = 'OPEN' AND NOT EXISTS (SELECT 1 FROM dba_tab_privs WHERE grantee = u.username) AND NOT EXISTS (SELECT 1 FROM dba_role_privs WHERE grantee = u.username AND granted_role NOT IN ('CONNECT', 'RESOURCE')) AND NOT EXISTS (SELECT 1 FROM dba_objects WHERE owner = u.username) ORDER BY u.created
SEC-SQL-AZ-003-RC15 Assumption that low-privilege accounts are harmless
Condition: row_count > 0 — Dormant low-privilege accounts still expand attack surface and should be locked
Orphaned User Accounts FINNACLEDB_TEST: DETECTED 5 rows FLEXCUBE: DETECTED 7 rows alert: Yes mail: Yes (destructive)

SELECT u.username, u.account_status, u.created, u.last_login, LISTAGG(rp.granted_role, ', ') WITHIN GROUP (ORDER BY rp.granted_role) AS roles FROM dba_users u LEFT JOIN dba_role_privs rp ON u.username = rp.grantee WHERE u.oracle_maintained = 'N' AND u.account_status = 'OPEN' AND (u.last_login IS NULL OR u.last_login < SYSDATE - 180) AND NOT EXISTS (SELECT 1 FROM dba_role_privs WHERE grantee = u.username AND granted_role IN ('DBA', 'RESOURCE')) GROUP BY u.username, u.account_status, u.created, u.last_login ORDER BY u.last_login NULLS FIRST
SEC-SQL-AZ-004-RC01 Application requirements for file export functionality
Condition: row_count > 0 — Accounts with directory object or UTL_FILE access can read/write files on the server filesystem
File System Access Privilege FINNACLEDB_TEST: DETECTED 14 rows FLEXCUBE: DETECTED 22 rows alert: Yes mail: No
SELECT grantee, privilege, table_name AS directory_name FROM dba_tab_privs WHERE (table_name IN (SELECT directory_name FROM dba_directories) OR table_name = 'UTL_FILE') AND privilege IN ('READ', 'WRITE', 'EXECUTE') AND grantee NOT IN ('SYS', 'SYSTEM', 'PUBLIC') ORDER BY grantee, table_name
SEC-SQL-AZ-004-RC02 Lack of secure alternatives or awareness of alternatives
Condition: row_count > 0 — Application code using UTL_FILE may lack awareness of more secure alternatives
File System Access Privilege FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 222 rows alert: Yes mail: No
SELECT owner, name, type, line FROM ALL_SOURCE WHERE UPPER(text) LIKE '%UTL_FILE%' AND owner NOT IN ('SYS', 'SYSTEM', 'XDB') ORDER BY owner, name, line
SEC-SQL-AZ-004-RC03 Business intelligence and reporting tool requirements
Condition: row_count > 0 with broad directory paths or many grantees — Directory objects pointing to broad filesystem paths with multiple grantees indicate BI tool filesystem access
File System Access Privilege FINNACLEDB_TEST: DETECTED 14 rows FLEXCUBE: DETECTED 19 rows alert: Yes mail: No
SELECT d.directory_name, d.directory_path, p.grantee, p.privilege FROM dba_directories d LEFT JOIN dba_tab_privs p ON d.directory_name = p.table_name WHERE d.directory_path NOT LIKE '%/oracle%admin%' AND d.directory_path NOT LIKE '%/oracle%diag%' ORDER BY d.directory_name, p.grantee
SEC-SQL-AZ-004-RC04 Inadequate file operation path restrictions
Condition: row_count > 0 — Directory objects pointing to root, /tmp, or Oracle home paths provide excessive filesystem access
File System Access Privilege FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT directory_name, directory_path FROM dba_directories WHERE directory_path IN ('/', '/tmp', '/etc', '/var', '/home') OR directory_path LIKE '/oracle%/dbs%' OR directory_path LIKE '%ORACLE_HOME%' OR LENGTH(directory_path) < 5 ORDER BY directory_path
SEC-SQL-AZ-004-RC05 Default configuration not hardened during installation
Condition: row_count > 0 with grants to non-admin users — Default directory objects with grants to non-admin users indicate post-install hardening was not done
File System Access Privilege FINNACLEDB_TEST: DETECTED 12 rows FLEXCUBE: DETECTED 16 rows alert: Yes mail: Yes (destructive)
SELECT d.directory_name, d.directory_path, p.grantee, p.privilege FROM dba_directories d LEFT JOIN dba_tab_privs p ON d.directory_name = p.table_name WHERE d.directory_name IN ('DATA_PUMP_DIR', 'ORACLE_OCM_CONFIG_DIR', 'ORACLE_OCM_CONFIG_DIR2', 'XMLDIR', 'XSDDIR', 'ORACLE_BASE') OR d.directory_path LIKE '%/admin/%' ORDER BY d.directory_name
SEC-SQL-AZ-004-RC06 Developer convenience during development
Condition: row_count > 0 with application/developer accounts having file access — Non-DBA accounts with directory or UTL_FILE privileges suggest development grants propagated to production
File System Access Privilege FINNACLEDB_TEST: DETECTED 14 rows FLEXCUBE: DETECTED 21 rows alert: Yes mail: No
SELECT grantee, privilege, table_name FROM dba_tab_privs WHERE table_name IN (SELECT directory_name FROM dba_directories) AND grantee NOT IN ('SYS', 'SYSTEM', 'DBA') AND privilege IN ('READ', 'WRITE') UNION ALL SELECT grantee, privilege, table_name FROM dba_tab_privs WHERE table_name = 'UTL_FILE' AND privilege = 'EXECUTE' AND grantee NOT IN ('SYS', 'SYSTEM', 'PUBLIC') ORDER BY grantee
SEC-SQL-AZ-004-RC07 Troubleshooting and ad-hoc data extraction
Condition: row_count > 0 with grants to individual DBA or developer accounts — Individual accounts with directory grants suggest troubleshooting grants not revoked
File System Access Privilege FINNACLEDB_TEST: DETECTED 8 rows FLEXCUBE: DETECTED 15 rows alert: Yes mail: No

SELECT grantee, owner, table_name, privilege, grantor FROM dba_tab_privs WHERE table_name IN (SELECT directory_name FROM dba_directories) AND grantee NOT IN ('SYS', 'SYSTEM') ORDER BY grantee, table_name
SEC-SQL-AZ-004-RC08 Third-party tool or vendor requirements
Condition: row_count > 0 — ETL or backup tool accounts with directory privileges indicate vendor-required file access
File System Access Privilege FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT p.grantee, p.privilege, p.table_name, d.directory_path FROM dba_tab_privs p JOIN dba_directories d ON p.table_name = d.directory_name WHERE p.grantee NOT IN ('SYS', 'SYSTEM') AND (p.grantee LIKE '%INFORMATICA%' OR p.grantee LIKE '%DATASTAGE%' OR p.grantee LIKE '%GOLDENGATE%' OR p.grantee LIKE '%OGG%' OR p.grantee LIKE '%SSIS%' OR p.grantee LIKE '%TALEND%' OR p.grantee LIKE '%ODI%') ORDER BY p.grantee, d.directory_path
SEC-SQL-AZ-004-RC09 Historical privilege accumulation
Condition: row_count > 0 with many grants across accounts — Accumulated filesystem and network access grants across many accounts indicate privilege sprawl
File System Access Privilege FINNACLEDB_TEST: DETECTED 20 rows FLEXCUBE: DETECTED 67 rows alert: Yes mail: No
SELECT grantee, privilege, table_name AS object_name, 'DIRECTORY' AS grant_type FROM dba_tab_privs WHERE table_name IN (SELECT directory_name FROM dba_directories) AND grantee NOT IN ('SYS', 'SYSTEM') UNION ALL SELECT grantee, privilege, table_name AS object_name, 'PACKAGE' AS grant_type FROM dba_tab_privs WHERE table_name IN ('UTL_FILE', 'DBMS_LOB', 'UTL_TCP', 'UTL_HTTP', 'UTL_SMTP', 'UTL_INADDR') AND privilege = 'EXECUTE' AND grantee NOT IN ('SYS', 'SYSTEM', 'PUBLIC') ORDER BY grantee, object_name
SEC-SQL-AZ-004-RC10 Insufficient monitoring of FILE privilege usage
Condition: row_count > 0 — Non-DBA users have READ/WRITE access to Oracle directories — file system access granted
File System Access Privilege FINNACLEDB_TEST: DETECTED 8 rows FLEXCUBE: DETECTED 15 rows alert: Yes mail: No
SELECT grantee, table_name AS directory_name, privilege FROM dba_tab_privs WHERE owner = 'SYS' AND table_name IN (SELECT directory_name FROM dba_directories) AND privilege IN ('READ', 'WRITE', 'EXECUTE') AND grantee NOT IN ('SYS', 'SYSTEM', 'DBA') ORDER BY grantee, table_name
SEC-SQL-AZ-004-RC11 Shared service accounts with FILE privilege
Condition: row_count > 0 — Shared or service accounts have directory object access — reduces file access accountability
File System Access Privilege FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT grantee, table_name AS directory_name, privilege FROM dba_tab_privs WHERE owner = 'SYS' AND table_name IN (SELECT directory_name FROM dba_directories) AND grantee IN (SELECT username FROM dba_users WHERE username LIKE '%SERVICE%' OR username LIKE '%APP%' OR username LIKE '%SHARED%' OR username LIKE '%GENERIC%' OR username LIKE '%BATCH%') ORDER BY grantee
SEC-SQL-AZ-004-RC12 Configuration management errors
Condition: row_count > 0 — Users with many individually granted system privileges — likely manual grants without role-based approach
File System Access Privilege FINNACLEDB_TEST: clean FLEXCUBE: FAILED alert: Yes mail: No
SELECT grantee, COUNT(DISTINCT privilege) AS priv_count, LISTAGG(privilege, ', ') WITHIN GROUP (ORDER BY privilege) AS privileges FROM dba_sys_privs WHERE grantee IN (SELECT username FROM dba_users WHERE oracle_maintained = 'N' AND account_status = 'OPEN') AND grantee NOT IN ('SYS', 'SYSTEM') GROUP BY grantee HAVING COUNT(DISTINCT privilege) > 15 ORDER BY priv_count DESC FETCH FIRST 20 ROWS ONLY
SEC-SQL-AZ-004-RC13 Privilege grants during emergency scenarios
Condition: row_count > 0 — Powerful system privileges granted to non-admin accounts — possibly emergency grants not revoked
File System Access Privilege FINNACLEDB_TEST: DETECTED 15 rows FLEXCUBE: DETECTED 54 rows alert: Yes mail: No
SELECT grantee, privilege, admin_option FROM dba_sys_privs WHERE privilege IN ('ALTER SYSTEM', 'ALTER DATABASE', 'SELECT ANY TABLE', 'CREATE ANY TABLE', 'DROP ANY TABLE', 'GRANT ANY PRIVILEGE', 'EXEMPT ACCESS POLICY', 'BECOME USER') AND grantee NOT IN ('SYS', 'SYSTEM', 'DBA') AND grantee IN (SELECT username FROM dba_users WHERE oracle_maintained = 'N') ORDER BY grantee
SEC-SQL-AZ-004-RC14 Misunderstanding of attack vectors
Condition: row_count > 0 — Directory objects point to sensitive OS paths — file access attack surface not understood
File System Access Privilege FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 11 rows alert: Yes mail: No

SELECT directory_name, directory_path, (SELECT COUNT(*) FROM dba_tab_privs tp WHERE tp.table_name = d.directory_name AND tp.owner = 'SYS') AS grantee_count FROM dba_directories d WHERE directory_path LIKE '/etc%' OR directory_path LIKE '/tmp%' OR directory_path LIKE '/home%' OR directory_path LIKE '/var%' OR directory_path LIKE 'C:\%' ORDER BY directory_name
SEC-SQL-AZ-005-RC01 Insufficient understanding of ownership chaining mechanics
Condition: row_count > 0 — Objects reference other schemas — ownership chain allows access without direct grants on referenced objects
Cross-Database Ownership Chaining FINNACLEDB_TEST: DETECTED 30 rows FLEXCUBE: DETECTED 30 rows alert: Yes mail: No
SELECT owner, object_name, object_type, (SELECT COUNT(*) FROM dba_tab_privs tp WHERE tp.owner = o.owner AND tp.table_name = o.object_name) AS grantee_count FROM dba_objects o WHERE object_type IN ('VIEW', 'PROCEDURE', 'FUNCTION', 'PACKAGE') AND owner NOT IN ('SYS', 'SYSTEM', 'MDSYS', 'CTXSYS', 'XDB') AND EXISTS (SELECT 1 FROM dba_dependencies d WHERE d.owner = o.owner AND d.name = o.object_name AND d.referenced_owner != o.owner) ORDER BY owner, object_name FETCH FIRST 30 ROWS ONLY
SEC-SQL-AZ-005-RC02 Convenience over security principle
Condition: row_count > 0 — Non-admin users have multiple ANY privileges — convenience grants bypassing least privilege
Cross-Database Ownership Chaining FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 7 rows alert: Yes mail: No
SELECT grantee, COUNT(*) AS any_priv_count, LISTAGG(privilege, ', ') WITHIN GROUP (ORDER BY privilege) AS any_privileges FROM dba_sys_privs WHERE privilege LIKE '%ANY%' AND grantee NOT IN ('SYS', 'SYSTEM', 'DBA', 'IMP_FULL_DATABASE', 'DATAPUMP_IMP_FULL_DATABASE', 'EXP_FULL_DATABASE') AND grantee IN (SELECT username FROM dba_users WHERE oracle_maintained = 'N') GROUP BY grantee HAVING COUNT(*) > 3 ORDER BY any_priv_count DESC
SEC-SQL-AZ-005-RC03 Legacy application design assuming ownership chaining
Condition: row_count > 0 — Definer-rights PL/SQL objects access other schemas — relying on ownership chaining for cross-schema access
Cross-Database Ownership Chaining FINNACLEDB_TEST: DETECTED 25 rows FLEXCUBE: DETECTED 30 rows alert: Yes mail: No
SELECT o.owner, o.object_name, o.object_type, d.referenced_owner, d.referenced_name, d.referenced_type FROM ALL_OBJECTS o JOIN ALL_DEPENDENCIES d ON o.owner = d.owner AND o.object_name = d.name WHERE o.object_type IN ('PROCEDURE', 'FUNCTION', 'PACKAGE', 'PACKAGE BODY') AND o.owner NOT IN ('SYS', 'SYSTEM') AND d.referenced_owner != o.owner AND d.referenced_owner NOT IN ('SYS', 'SYSTEM', 'PUBLIC') ORDER BY o.owner, o.object_name FETCH FIRST 30 ROWS ONLY
SEC-SQL-AZ-005-RC04 Unclear ownership across multiple databases
Condition: row_count > :threshold — Complex cross-schema dependency web — ownership boundaries are unclear
Cross-Database Ownership Chaining FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 20 rows alert: Yes mail: No
SELECT owner AS dependent_schema, referenced_owner AS referenced_schema, COUNT(*) AS dependency_count FROM ALL_DEPENDENCIES WHERE owner NOT IN ('SYS', 'SYSTEM', 'PUBLIC', 'MDSYS', 'CTXSYS', 'XDB') AND referenced_owner NOT IN ('SYS', 'SYSTEM', 'PUBLIC', 'MDSYS', 'CTXSYS', 'XDB') AND owner != referenced_owner GROUP BY owner, referenced_owner ORDER BY dependency_count DESC FETCH FIRST 20 ROWS ONLY
SEC-SQL-AZ-005-RC05 Multi-database applications without explicit permission architecture
Condition: row_count > 0 — Heavy cross-schema object grants — multi-schema application without clear permission design
Cross-Database Ownership Chaining FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 3 rows alert: Yes mail: No
SELECT grantor, grantee, COUNT(*) AS grant_count FROM ALL_TAB_PRIVS WHERE grantor NOT IN ('SYS', 'SYSTEM') AND grantee NOT IN ('SYS', 'SYSTEM', 'PUBLIC') AND grantor != grantee GROUP BY grantor, grantee HAVING COUNT(*) > 20 ORDER BY grant_count DESC FETCH FIRST 20 ROWS ONLY
SEC-SQL-AZ-005-RC06 Migration from single-database to multi-database environments
Condition: row_count > :threshold — Many PUBLIC grants on application objects — convenience grants from schema migration without security review
Cross-Database Ownership Chaining FINNACLEDB_TEST: DETECTED 30 rows FLEXCUBE: DETECTED 30 rows alert: Yes mail: No
SELECT owner, table_name, privilege, grantor, grantable FROM dba_tab_privs WHERE grantee = 'PUBLIC' AND owner NOT IN ('SYS', 'SYSTEM', 'MDSYS', 'CTXSYS', 'XDB', 'WMSYS') ORDER BY owner, table_name FETCH FIRST 30 ROWS ONLY
SEC-SQL-AZ-005-RC08 Lack of audit or monitoring of cross-database access
Condition: audited_objects = 0 — No object-level auditing configured — cross-schema access is completely invisible

Cross-Database Ownership Chaining FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT COUNT(*) AS audited_objects FROM dba_obj_audit_opts WHERE owner NOT IN ('SYS', 'SYSTEM')
SEC-SQL-AZ-005-RC09 Testing configurations carried forward to production
Condition: SUM(count) > :threshold — Multiple testing artifacts found — test configurations carried to production
Cross-Database Ownership Chaining FINNACLEDB_TEST: DETECTED 3 rows FLEXCUBE: DETECTED 3 rows alert: Yes mail: Yes (destructive)
SELECT 'test_schemas' AS check_item, COUNT(*) AS count FROM dba_users WHERE username LIKE '%TEST%' OR username LIKE '%DEV%' OR username LIKE '%QA%' AND account_status = 'OPEN' UNION ALL SELECT 'unlimited_profile_limits', COUNT(*) FROM dba_profiles WHERE profile = 'DEFAULT' AND limit = 'UNLIMITED' AND resource_name IN ('FAILED_LOGIN_ATTEMPTS', 'PASSWORD_LIFE_TIME') UNION ALL SELECT 'public_grants_on_sensitive', COUNT(*) FROM dba_tab_privs WHERE grantee = 'PUBLIC' AND privilege IN ('SELECT', 'INSERT', 'UPDATE', 'DELETE') AND owner NOT IN ('SYS', 'SYSTEM')
SEC-SQL-AZ-005-RC10 Third-party vendor requirements or recommendations
Condition: row_count > 0 — Application accounts with overly broad ANY privileges — likely vendor-requested convenience grants
Cross-Database Ownership Chaining FINNACLEDB_TEST: DETECTED 25 rows FLEXCUBE: DETECTED 89 rows alert: Yes mail: No
SELECT grantee, privilege, admin_option FROM dba_sys_privs WHERE grantee IN (SELECT username FROM dba_users WHERE oracle_maintained = 'N' AND username NOT IN ('SYS', 'SYSTEM')) AND privilege IN ('SELECT ANY TABLE', 'CREATE ANY TABLE', 'ALTER ANY TABLE', 'DROP ANY TABLE', 'CREATE ANY INDEX', 'UNLIMITED TABLESPACE', 'SELECT ANY DICTIONARY') ORDER BY grantee, privilege
SEC-SQL-AZ-005-RC11 Incomplete understanding of risk when owner differs
Condition: row_count > 0 — Definer-rights procedures access data in other schemas — callers get access through procedure owner's privileges
Cross-Database Ownership Chaining FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 30 rows alert: Yes mail: No
SELECT o.owner AS procedure_owner, o.object_name, d.referenced_owner AS data_owner, d.referenced_name AS accessed_object, d.referenced_type FROM ALL_OBJECTS o JOIN ALL_DEPENDENCIES d ON o.owner = d.owner AND o.object_name = d.name WHERE o.object_type IN ('PROCEDURE', 'FUNCTION', 'PACKAGE BODY') AND o.owner != d.referenced_owner AND d.referenced_owner NOT IN ('SYS', 'SYSTEM', 'PUBLIC') AND o.owner NOT IN ('SYS', 'SYSTEM') AND d.referenced_type IN ('TABLE', 'VIEW') ORDER BY o.owner, o.object_name FETCH FIRST 30 ROWS ONLY
SEC-SQL-AZ-005-RC12 Database isolation principle not enforced
Condition: row_count > 0 — ANY table privileges break schema isolation — users can access any schema's data
Cross-Database Ownership Chaining FINNACLEDB_TEST: DETECTED 15 rows FLEXCUBE: DETECTED 55 rows alert: Yes mail: No
SELECT grantee, privilege FROM dba_sys_privs WHERE privilege IN ('SELECT ANY TABLE', 'INSERT ANY TABLE', 'UPDATE ANY TABLE', 'DELETE ANY TABLE', 'ALTER ANY TABLE', 'DROP ANY TABLE') AND grantee NOT IN ('SYS', 'SYSTEM', 'DBA', 'IMP_FULL_DATABASE', 'DATAPUMP_IMP_FULL_DATABASE') AND grantee IN (SELECT username FROM dba_users WHERE oracle_maintained = 'N') ORDER BY grantee, privilege
SEC-SQL-AZ-006-RC01 Overly broad delegation of grant rights
Condition: row_count > :threshold — Many object privileges with GRANT OPTION — users can re-grant access without DBA involvement
Grant Option Abuse FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 3 rows alert: Yes mail: No
SELECT grantee, owner, table_name, privilege, grantable FROM dba_tab_privs WHERE grantable = 'YES' AND grantee NOT IN ('SYS', 'SYSTEM', 'DBA') AND grantee IN (SELECT username FROM dba_users WHERE oracle_maintained = 'N') ORDER BY grantee, owner, table_name FETCH FIRST 30 ROWS ONLY
SEC-SQL-AZ-006-RC02 Inadequate understanding of WITH GRANT OPTION implications
Condition: row_count > 0 — Non-DBA users have granted privileges to others WITH GRANT OPTION — indirect access chains exist
Grant Option Abuse FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT tp.grantor, tp.grantee, tp.owner '.' tp.table_name AS object, tp.privilege, tp.grantable FROM dba_tab_privs tp WHERE tp.grantor NOT IN ('SYS', 'SYSTEM') AND tp.grantable = 'YES' AND tp.grantee NOT IN ('SYS', 'SYSTEM', 'PUBLIC') ORDER BY tp.grantor, tp.grantee FETCH FIRST 30 ROWS ONLY
SEC-SQL-AZ-006-RC03 Convenience for distributed access management
Condition: row_count > 0 — Users are re-granting privileges they received — convenience delegation instead of role-based access

Grant Option Abuse FINNACLEDB_TEST: DETECTED 30 rows FLEXCUBE: DETECTED 30 rows alert: Yes mail: Yes (destructive)
SELECT tp.grantor, tp.grantee, tp.owner '.' tp.table_name AS object, tp.privilege FROM dba_tab_privs tp WHERE tp.grantor NOT IN ('SYS', 'SYSTEM') AND tp.grantor IN (SELECT grantee FROM dba_tab_privs WHERE grantable = 'YES') AND tp.grantee NOT IN ('SYS', 'SYSTEM', 'PUBLIC') ORDER BY tp.grantor, tp.grantee FETCH FIRST 30 ROWS ONLY
SEC-SQL-AZ-006-RC04 No approval workflow for permission delegation
Condition: row_count > 0 — Non-DBA users performing GRANT operations — permission delegation without centralized approval
Grant Option Abuse FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT db_user AS granting_user, COUNT(*) AS grant_count, MIN(TO_CHAR(extended_timestamp, 'YYYY-MM-DD')) AS first_grant, MAX(TO_CHAR(extended_timestamp, 'YYYY-MM-DD')) AS last_grant FROM dba_common_audit_trail WHERE statement_type = 'GRANT' AND db_user NOT IN ('SYS', 'SYSTEM') AND extended_timestamp > SYSTIMESTAMP - INTERVAL '90' DAY GROUP BY db_user ORDER BY grant_count DESC
SEC-SQL-AZ-006-RC05 Inadequate monitoring of grant activity
Condition: row_count = 0 — No audit policies covering GRANT/REVOKE operations indicates monitoring gap
Grant Option Abuse FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT policy_name, audit_option FROM audit_unified_policies WHERE audit_option IN ('GRANT', 'REVOKE', 'ALTER USER') ORDER BY policy_name
SEC-SQL-AZ-006-RC06 Lack of restrictions on re-delegation
Condition: row_count > 0 — Object privileges with GRANT OPTION allow unrestricted re-delegation
Grant Option Abuse FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT grantee, owner, table_name, privilege, grantable FROM dba_tab_privs WHERE grantable = 'YES' AND grantee NOT IN ('SYS', 'SYSTEM', 'DBA', 'PUBLIC') AND owner NOT IN ('SYS', 'SYSTEM') ORDER BY grantee, owner, table_name
SEC-SQL-AZ-006-RC07 Grant option on sensitive privileges
Condition: row_count > 0 — ADMIN OPTION on sensitive privileges enables cascading privilege escalation
Grant Option Abuse FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: clean alert: Yes mail: No
SELECT grantee, privilege, admin_option FROM dba_sys_privs WHERE admin_option = 'YES' AND privilege IN ('CREATE USER', 'ALTER USER', 'DROP USER', 'GRANT ANY PRIVILEGE', 'GRANT ANY ROLE', 'ALTER SYSTEM', 'CREATE ANY TABLE', 'DROP ANY TABLE', 'SELECT ANY TABLE', 'CREATE ANY PROCEDURE', 'EXECUTE ANY PROCEDURE', 'ALTER DATABASE') AND grantee NOT IN ('SYS', 'SYSTEM', 'DBA') ORDER BY grantee, privilege
SEC-SQL-AZ-006-RC08 Default role configurations including grant option
Condition: row_count > 0 — Non-DBA roles with ADMIN OPTION privileges create hidden delegation paths
Grant Option Abuse FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 8 rows alert: Yes mail: No
SELECT role, privilege, admin_option FROM role_sys_privs WHERE admin_option = 'YES' AND role NOT IN ('DBA', 'IMP_FULL_DATABASE', 'EXP_FULL_DATABASE', 'DATAPUMP_EXP_FULL_DATABASE', 'DATAPUMP_IMP_FULL_DATABASE') ORDER BY role, privilege
SEC-SQL-AZ-006-RC09 Privilege escalation through role membership
Condition: row_count > 0 — Nested role grants containing powerful roles confirm escalation through role membership
Grant Option Abuse FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 9 rows alert: Yes mail: No
SELECT role AS parent_role, granted_role AS child_role, admin_option FROM role_role_privs WHERE granted_role IN ('DBA', 'RESOURCE', 'CONNECT') OR granted_role IN (SELECT role FROM role_sys_privs WHERE privilege LIKE '%ANY%') ORDER BY role, granted_role
SEC-SQL-AZ-006-RC10 Separation of duties violations
Condition: row_count > 0 — Users with conflicting privilege pairs violate separation of duties
Grant Option Abuse FINNACLEDB_TEST: DETECTED 8 rows FLEXCUBE: DETECTED 13 rows alert: Yes mail: Yes (destructive)

SELECT sp1.grantee, sp1.privilege AS privilege_1, sp2.privilege AS privilege_2 FROM dba_sys_privs sp1 JOIN dba_sys_privs sp2 ON sp1.grantee = sp2.grantee WHERE sp1.grantee NOT IN ('SYS', 'SYSTEM', 'DBA') AND ((sp1.privilege = 'CREATE USER' AND sp2.privilege = 'AUDIT SYSTEM') OR (sp1.privilege = 'ALTER SYSTEM' AND sp2.privilege = 'SELECT ANY TABLE') OR (sp1.privilege = 'CREATE ANY PROCEDURE' AND sp2.privilege = 'GRANT ANY PRIVILEGE') OR (sp1.privilege = 'ALTER DATABASE' AND sp2.privilege = 'CREATE USER') OR (sp1.privilege = 'DROP ANY TABLE' AND sp2.privilege = 'CREATE ANY TABLE')) ORDER BY sp1.grantee
SEC-SQL-AZ-006-RC11 Implicit grant options in template scripts
Condition: row_count > 0 — Multiple users with many ADMIN OPTION privileges suggest template-based provisioning with implicit GRANT OPTION
Grant Option Abuse FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT grantee, COUNT(*) AS admin_priv_count FROM dba_sys_privs WHERE admin_option = 'YES' AND grantee NOT IN ('SYS', 'SYSTEM', 'DBA') GROUP BY grantee HAVING COUNT(*) > 3 ORDER BY admin_priv_count DESC
SEC-SQL-AZ-006-RC12 Compliance framework gaps
Condition: row_count > 0 — Users with many ANY privileges violate least-privilege compliance requirements
Grant Option Abuse FINNACLEDB_TEST: DETECTED 11 rows FLEXCUBE: DETECTED 20 rows alert: Yes mail: Yes (destructive)
SELECT grantee, COUNT(*) AS any_priv_count, LISTAGG(privilege, ', ') WITHIN GROUP (ORDER BY privilege) AS any_privileges FROM dba_sys_privs WHERE privilege LIKE '%ANY%' AND grantee NOT IN ('SYS', 'SYSTEM', 'DBA') GROUP BY grantee HAVING COUNT(*) > 3 ORDER BY any_priv_count DESC
SEC-SQL-AZ-006-RC13 Inherited permissions from legacy systems
Condition: row_count > 0 — Legacy role contents may include unnecessary privileges from older Oracle versions
Grant Option Abuse FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 10 rows alert: Yes mail: Yes (destructive)
SELECT role, privilege FROM role_sys_privs WHERE role IN ('CONNECT', 'RESOURCE') ORDER BY role, privilege
SEC-SQL-AZ-006-RC14 Confusion between intent and capability
Condition: row_count > 0 — ANY TABLE privileges grant far more access than typically intended
Grant Option Abuse FINNACLEDB_TEST: DETECTED 35 rows FLEXCUBE: DETECTED 82 rows alert: Yes mail: Yes (destructive)
SELECT sp.grantee, sp.privilege FROM dba_sys_privs sp WHERE sp.privilege IN ('SELECT ANY TABLE', 'INSERT ANY TABLE', 'UPDATE ANY TABLE', 'DELETE ANY TABLE') AND sp.grantee NOT IN ('SYS', 'SYSTEM', 'DBA') ORDER BY sp.grantee, sp.privilege
SEC-SQL-CFG-001-RC02 Third-Party Vendor Requirements
Condition: row_count > 0 — Insecure parameter settings that may be required by third-party vendors
OS Command Execution Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT name, value, isdefault FROM v\$parameter WHERE name IN ('remote_os_authent', 'remote_os_roles', 'sec_case_sensitive_logon', '07_DICTIONARY_ACCESSIBILITY', 'sql92_security', 'audit_trail') AND ((name = 'remote_os_authent' AND UPPER(value) = 'TRUE') OR (name = 'remote_os_roles' AND UPPER(value) = 'TRUE') OR (name = 'sec_case_sensitive_logon' AND UPPER(value) = 'FALSE') OR (name = '07_DICTIONARY_ACCESSIBILITY' AND UPPER(value) = 'TRUE') OR (name = 'sql92_security' AND UPPER(value) = 'FALSE') OR (name = 'audit_trail' AND UPPER(value) = 'NONE')) ORDER BY name
SEC-SQL-CFG-001-RC03 Administrative Troubleshooting Convenience
Condition: row_count > 0 — Non-default troubleshooting parameters may indicate security weakened for convenience
OS Command Execution Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT name, value, isdefault FROM v\$parameter WHERE name IN ('sql_trace', 'timed_statistics', 'max_dump_file_size', 'remote_os_authent', 'utl_file_dir', '_trace_files_public') AND isdefault = 'FALSE' ORDER BY name
SEC-SQL-CFG-001-RC04 Inadequate Change Control
Condition: row_count > :threshold_modified — Many modified parameters without change control audit indicates inadequate configuration management
OS Command Execution Enabled FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: clean alert: Yes mail: No

SELECT name, value, isdefault, ismodified FROM v\$parameter WHERE ismodified != 'FALSE' AND name NOT LIKE '%_oracle%' ORDER BY name
SEC-SQL-CFG-001-RC05 SQL Injection Vulnerability
Condition: row_count > 0 — Public execute on procedures with dynamic SQL confirms SQL injection vulnerability
OS Command Execution Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT grantee, owner, table_name, privilege FROM dba_tab_privs WHERE privilege = 'EXECUTE' AND grantee = 'PUBLIC' AND owner NOT IN ('SYS', 'SYSTEM', 'XDB', 'MDSYS', 'CTXSYS', 'ORDSYS') AND table_name IN (SELECT DISTINCT name FROM dba_source WHERE owner NOT IN ('SYS', 'SYSTEM') AND type IN ('PROCEDURE', 'FUNCTION', 'PACKAGE') AND UPPER(text) LIKE '%EXECUTE IMMEDIATE%') ORDER BY owner, table_name
SEC-SQL-CFG-001-RC07 Lack of Monitoring and Auditing
Condition: row_count < 2 — Fewer than 2 enabled audit policies indicates severely insufficient monitoring
OS Command Execution Enabled FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT policy_name, enabled_option, entity_name FROM audit_unified_enabled_policies ORDER BY policy_name
SEC-SQL-CFG-001-RC08 Knowledge Gap on Alternatives
Condition: row_count > 0 — Insecure parameter settings that have modern secure alternatives
OS Command Execution Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT name, value FROM v\$parameter WHERE (name = 'remote_os_authent' AND UPPER(value) = 'TRUE') OR (name = 'utl_file_dir' AND value IS NOT NULL AND value != '') OR (name = '07_DICTIONARY_ACCESSIBILITY' AND UPPER(value) = 'TRUE') OR (name = 'sec_case_sensitive_logon' AND UPPER(value) = 'FALSE') ORDER BY name
SEC-SQL-CFG-001-RC09 Privilege Escalation via Weak Credentials
Condition: row_count > 0 — Privileged accounts without password complexity enforcement are vulnerable to credential compromise
OS Command Execution Enabled FINNACLEDB_TEST: DETECTED 6 rows FLEXCUBE: DETECTED 8 rows alert: Yes mail: No
SELECT u.username, u.profile, p.limit AS verify_function FROM dba_users u JOIN dba_profiles p ON p.profile = u.profile AND p.resource_name = 'PASSWORD_VERIFY_FUNCTION' WHERE u.oracle_maintained = 'N' AND u.account_status = 'OPEN' AND (p.limit = 'NULL' OR p.limit IS NULL) AND EXISTS (SELECT 1 FROM dba_sys_privs sp WHERE sp.grantee = u.username AND sp.privilege LIKE '%ANY%') ORDER BY u.username
SEC-SQL-CFG-001-RC10 Lateral Movement Capability
Condition: row_count > 0 — Database links provide lateral movement paths to other databases
OS Command Execution Enabled FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT owner, db_link, username, host FROM ALL_DB_LINKS WHERE owner NOT IN ('SYS', 'SYSTEM') ORDER BY owner, db_link
SEC-SQL-CFG-001-RC11 Persistence Mechanisms
Condition: row_count > 0 — Database-level triggers from non-system users can provide attacker persistence
OS Command Execution Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT owner, trigger_name, trigger_type, triggering_event, status FROM ALL_TRIGGERS WHERE base_object_type = 'DATABASE' AND status = 'ENABLED' AND owner NOT IN ('SYS', 'SYSTEM') ORDER BY owner, trigger_name
SEC-SQL-CFG-001-RC12 Backup Restoration from Compromised Environments
Condition: row_count > 0 — Accounts created near restore operations may have been introduced through compromised backup
OS Command Execution Enabled FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 9 rows alert: Yes mail: No
SELECT username, created, account_status, authentication_type FROM dba_users WHERE oracle_maintained = 'N' AND created > SYSDATE - 90 ORDER BY created DESC
SEC-SQL-CFG-002-RC02 Legitimate Data Loading Requirements

Condition: row_count > 0 — Directory objects with READ/WRITE privileges enable file-based data loading
Local File Loading Enabled FINNACLEDB_TEST: DETECTED 20 rows FLEXCUBE: DETECTED 30 rows alert: Yes mail: No
SELECT d.directory_name, d.directory_path, p.grantee, p.privilege FROM dba_directories d LEFT JOIN dba_tab_privs p ON p.table_name = d.directory_name AND p.type = 'DIRECTORY' WHERE d.directory_name NOT IN ('ORACLE_HOME', 'ORACLE_BASE') ORDER BY d.directory_name, p.grantee
SEC-SQL-CFG-002-RC03 Inadequate Server-Side Controls
Condition: row_count > 0 — Non-DBA users with directory READ/WRITE access may lack compensating controls
Local File Loading Enabled FINNACLEDB_TEST: DETECTED 8 rows FLEXCUBE: DETECTED 15 rows alert: Yes mail: No
SELECT grantee, privilege, table_name FROM dba_tab_privs WHERE table_name IN (SELECT directory_name FROM dba_directories) AND privilege IN ('READ', 'WRITE', 'EXECUTE') AND grantee NOT IN ('SYS', 'SYSTEM', 'DBA') ORDER BY grantee, table_name
SEC-SQL-CFG-002-RC04 Unclear Default Behavior
Condition: row_count > 5 — Many directory objects suggest broad file access configuration that may not be fully reviewed
Local File Loading Enabled FINNACLEDB_TEST: DETECTED 17 rows FLEXCUBE: DETECTED 21 rows alert: Yes mail: No
SELECT directory_name, directory_path FROM dba_directories ORDER BY directory_name
SEC-SQL-CFG-002-RC06 Unvalidated User Input
Condition: row_count > 0 — Non-DBA users with EXECUTE on file/network packages can access server resources
Local File Loading Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT grantee, privilege, table_name FROM ALL_TAB_PRIVS WHERE table_name IN ('UTL_FILE', 'UTL_HTTP', 'UTL_TCP', 'UTL_SMTP', 'DBMS_LOB', 'UTL_INADDR', 'DBMS_JAVA', 'DBMS_JAVA_TEST') AND privilege = 'EXECUTE' AND grantee NOT IN ('SYS', 'SYSTEM', 'DBA', 'PUBLIC') ORDER BY grantee, table_name
SEC-SQL-CFG-002-RC08 Information Disclosure
Condition: row_count > 0 — Directory objects pointing to sensitive system paths enable information disclosure
Local File Loading Enabled FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 11 rows alert: Yes mail: No
SELECT directory_name, directory_path FROM dba_directories WHERE directory_path LIKE '/etc%' OR directory_path LIKE '/home%' OR directory_path LIKE '%/conf%' OR directory_path LIKE '%password%' OR directory_path LIKE '%config%' ORDER BY directory_name
SEC-SQL-CFG-002-RC09 Implicit Trust Model
Condition: row_count > 0 — Directory objects granted to many users indicate implicit trust model for file access
Local File Loading Enabled FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT d.directory_name, d.directory_path, COUNT(DISTINCT p.grantee) AS grantee_count, LISTAGG(p.grantee, ', ') WITHIN GROUP (ORDER BY p.grantee) AS grantees FROM dba_directories d LEFT JOIN dba_tab_privs p ON p.table_name = d.directory_name AND p.type = 'DIRECTORY' GROUP BY d.directory_name, d.directory_path HAVING COUNT(DISTINCT p.grantee) > 3 ORDER BY grantee_count DESC
SEC-SQL-CFG-002-RC11 Backup Retention
Condition: row_count > 0 — Directory objects with dev/test paths suggest leftover configuration from environment promotion
Local File Loading Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT directory_name, directory_path FROM dba_directories WHERE directory_path LIKE '%dev%' OR directory_path LIKE '%test%' OR directory_path LIKE '%tmp%' OR directory_path LIKE '%temp%' ORDER BY directory_name
SEC-SQL-CFG-002-RC12 Third-Party Application Defaults
Condition: row_count > 0 — Non-standard directory objects may be from third-party application defaults
Local File Loading Enabled FINNACLEDB_TEST: DETECTED 9 rows FLEXCUBE: DETECTED 13 rows alert: Yes mail: Yes (destructive)

SELECT directory_name, directory_path FROM dba_directories WHERE directory_name NOT IN ('DATA_PUMP_DIR', 'ORACLE_HOME', 'ORACLE_BASE', 'ADMIN_DIR', 'OPATCH_INST_DIR', 'OPATCH_LOG_DIR', 'OPATCH_SCRIPT_DIR', 'XMLDIR', 'XSDDIR') ORDER BY directory_name
SEC-SQL-CFG-002-RC13 Insufficient Documentation
Condition: row_count > 5 — Many directory objects suggest accumulated configuration that may lack documentation
Local File Loading Enabled FINNACLEDB_TEST: DETECTED 17 rows FLEXCUBE: DETECTED 21 rows alert: Yes mail: No
SELECT d.directory_name, d.directory_path, COUNT(p.grantee) AS grant_count FROM dba_directories d LEFT JOIN dba_tab_privs p ON p.table_name = d.directory_name AND p.type = 'DIRECTORY' GROUP BY d.directory_name, d.directory_path ORDER BY d.directory_name
SEC-SQL-CFG-003-RC01 Default Installation Artifacts
Condition: row_count > 0 — Oracle sample schemas present in the database from default installation
Sample/Test Databases Present FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT username, account_status, default_tablespace, created FROM dba_users WHERE username IN ('HR', 'OE', 'PM', 'IX', 'SH', 'BI', 'SCOTT', 'ADAMS', 'BLAKE', 'CLARK', 'JONES') ORDER BY username
SEC-SQL-CFG-003-RC02 Development Environment Cloning
Condition: row_count > 0 — Users with dev/test/QA names in production suggest database was cloned from development
Sample/Test Databases Present FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: Yes (destructive)
SELECT username, account_status, default_tablespace, created FROM dba_users WHERE (UPPER(username) LIKE '%DEV%' OR UPPER(username) LIKE '%TEST%' OR UPPER(username) LIKE '%QA%' OR UPPER(username) LIKE '%STAGING%') ORDER BY username
SEC-SQL-CFG-003-RC03 Backup Restoration Error
Condition: row_count > 0 — Sample schema objects present after restore indicate backup included unnecessary schemas
Sample/Test Databases Present FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT owner, COUNT(*) AS object_count FROM ALL_OBJECTS WHERE owner IN ('HR', 'OE', 'PM', 'IX', 'SH', 'BI', 'SCOTT') GROUP BY owner ORDER BY owner
SEC-SQL-CFG-003-RC04 Migration Process Gaps
Condition: row_count > 0 — Migration-related user accounts suggest incomplete cleanup after database migration
Sample/Test Databases Present FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT username, account_status, default_tablespace, created FROM dba_users WHERE UPPER(username) LIKE '%MIGRATE%' OR UPPER(username) LIKE '%IMPORT%' OR UPPER(username) LIKE '%CONVERT%' OR UPPER(username) LIKE '%LEGACY%' ORDER BY username
SEC-SQL-CFG-003-RC05 Knowledge Loss During Transitions
Condition: row_count > :min_inactive — Many non-Oracle locked/expired accounts suggest accumulated schemas from past team members
Sample/Test Databases Present FINNACLEDB_TEST: DETECTED 10 rows FLEXCUBE: DETECTED 33 rows alert: Yes mail: Yes (destructive)
SELECT u.username, u.account_status, u.created, u.profile, (SELECT COUNT(*) FROM dba_objects o WHERE o.owner = u.username) AS object_count FROM dba_users u WHERE u.oracle_maintained = 'N' AND u.username NOT IN ('SYS', 'SYSTEM') AND u.account_status != 'OPEN' ORDER BY u.created
SEC-SQL-CFG-003-RC06 Inadequate Cleanup Procedures
Condition: row_count > 0 — Test/demo/sample schemas still present indicate inadequate cleanup procedures
Sample/Test Databases Present FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 1 rows alert: Yes mail: Yes (destructive)
SELECT username, account_status, authentication_type, created FROM dba_users WHERE username IN ('HR', 'OE', 'PM', 'IX', 'SH', 'BI', 'SCOTT') OR (UPPER(username) LIKE '%TEST%' OR UPPER(username) LIKE '%DEMO%' OR UPPER(username) LIKE '%SAMPLE%') ORDER BY username

SEC-SQL-CFG-003-RC07 Application Configuration Pointing to Wrong Database
Condition: row_count > 0 — Paired prod/test schemas in the same database suggest configuration confusion risk
Sample/Test Databases Present FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT username, account_status, default_tablespace, created FROM dba_users WHERE oracle_maintained = 'N' AND (UPPER(username) LIKE '%PROD%' OR UPPER(username) LIKE '%LIVE%') AND username IN (SELECT REPLACE(REPLACE(UPPER(username), 'PROD', 'TEST'), 'LIVE', 'TEST') FROM dba_users) ORDER BY username
SEC-SQL-CFG-003-RC08 Demonstration and Training Use
Condition: row_count > 0 — Unlocked sample schema accounts in production likely left from training or demos
Sample/Test Databases Present FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT username, account_status, default_tablespace, created, profile FROM dba_users WHERE username IN ('HR', 'OE', 'PM', 'IX', 'SH', 'BI', 'SCOTT') AND account_status = 'OPEN' ORDER BY username
SEC-SQL-CFG-003-RC09 Documentation Failures
Condition: custom_schemas > :expected_max — More custom schemas than expected suggests accumulated undocumented schemas
Sample/Test Databases Present FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT COUNT(*) AS total_schemas, COUNT(CASE WHEN account_status = 'OPEN' THEN 1 END) AS open_schemas, COUNT(CASE WHEN account_status != 'OPEN' THEN 1 END) AS non_open_schemas, COUNT(CASE WHEN oracle_maintained = 'N' THEN 1 END) AS custom_schemas FROM dba_users
SEC-SQL-CFG-003-RC10 Insufficient RBAC Controls
Condition: row_count > 0 — Grants on sample schema objects allow unnecessary access to test data and structures
Sample/Test Databases Present FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT grantee, owner, table_name, privilege FROM dba_tab_privs WHERE owner IN ('HR', 'OE', 'PM', 'IX', 'SH', 'BI', 'SCOTT') AND grantee NOT IN ('SYS', 'SYSTEM', 'DBA') ORDER BY owner, grantee
SEC-SQL-CFG-003-RC11 Compliance Scope Confusion
Condition: row_count > 0 — Cross-schema object grants may cross compliance boundaries if schemas are not properly isolated
Sample/Test Databases Present FINNACLEDB_TEST: DETECTED 30 rows FLEXCUBE: DETECTED 30 rows alert: Yes mail: Yes (destructive)
SELECT grantee, owner, table_name, privilege, grantable FROM dba_tab_privs WHERE grantee NOT IN ('SYS', 'SYSTEM', 'PUBLIC') AND owner NOT IN ('SYS', 'SYSTEM', 'DBSNMP', 'OUTLN', 'XDB') AND grantee != owner ORDER BY grantee, owner FETCH FIRST 30 ROWS ONLY
SEC-SQL-CFG-003-RC12 Automated Provisioning Templates
Condition: row_count > 0 — Powerful default roles (DBA, RESOURCE) granted widely suggest templated provisioning without least privilege
Sample/Test Databases Present FINNACLEDB_TEST: DETECTED 11 rows FLEXCUBE: DETECTED 29 rows alert: Yes mail: No
SELECT grantee, granted_role, default_role, admin_option FROM dba_role_privs WHERE grantee IN (SELECT username FROM dba_users WHERE oracle_maintained = 'N' AND account_status = 'OPEN') AND granted_role IN ('DBA', 'RESOURCE', 'CONNECT', 'SELECT_CATALOG_ROLE', 'EXECUTE_CATALOG_ROLE', 'DELETE_CATALOG_ROLE') ORDER BY grantee, granted_role
SEC-SQL-CFG-003-RC13 Weak Object Naming Conventions
Condition: row_count > 0 — User objects with system-style naming (DBA_ V\$, DBMS_) could cause confusion or be used for impersonation
Sample/Test Databases Present FINNACLEDB_TEST: DETECTED 20 rows FLEXCUBE: DETECTED 20 rows alert: Yes mail: Yes (destructive)
SELECT owner, object_name, object_type FROM ALL_OBJECTS WHERE owner NOT IN ('SYS', 'SYSTEM', 'DBSNMP', 'OUTLN', 'XDB', 'CTXSYS', 'MDSYS') AND (object_name LIKE 'DBA_%' OR object_name LIKE 'V\$%' OR object_name LIKE 'GV\$%' OR object_name LIKE 'ALL_%' OR object_name LIKE 'USER_%' OR object_name LIKE 'DBMS_%') ORDER BY owner, object_name FETCH FIRST 20 ROWS ONLY
SEC-SQL-CFG-003-RC14 Multi-Tenant Instance Management
Condition: CDB configuration identified — Determines if this is a CDB (multitenant) architecture requiring PDB isolation checks

Sample/Test Databases Present FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: Yes (destructive)
SELECT name, value FROM v\$parameter WHERE name = 'enable_pluggable_database'
SEC-SQL-CFG-004-RC01 Application Code Execution Requirement
Condition: JAVAVM component present — Oracle Java VM is installed, enabling Java stored procedures and classes within the database
In-Database Code Execution Enabled FINNACLEDB_TEST: DETECTED 4 rows FLEXCUBE: DETECTED 4 rows alert: Yes mail: No
SELECT comp_id, comp_name, version, status FROM dba_registry WHERE comp_id IN ('JAVAVM', 'XML', 'CATJAVA', 'ORDIM') ORDER BY comp_id
SEC-SQL-CFG-004-RC02 Performance Optimization
Condition: insecure values detected — Parameters like utl_file_dir, remote_os_authent, or o7_dictionary_accessibility weaken security for performance/convenience
In-Database Code Execution Enabled FINNACLEDB_TEST: DETECTED 4 rows FLEXCUBE: DETECTED 4 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('utl_file_dir', 'plsql_native_library_dir', 'remote_os_authent', 'remote_login_passwordfile', 'sec_case_sensitive_logon', 'remote_os_roles', 'o7_dictionary_accessibility') ORDER BY name
SEC-SQL-CFG-004-RC03 Third-Party Tool Dependencies
Condition: row_count > 0 AND sys_privs is high — Third-party tool accounts with many system privileges create unnecessary security exposure
In-Database Code Execution Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT s.username, s.program, COUNT(DISTINCT sp.privilege) AS sys_privs, COUNT(DISTINCT rp.granted_role) AS roles FROM v\$session s LEFT JOIN dba_sys_privs sp ON s.username = sp.grantee LEFT JOIN dba_role_privs rp ON s.username = rp.grantee WHERE s.type = 'USER' AND s.username IS NOT NULL AND s.username NOT IN ('SYS','SYSTEM') AND (s.program LIKE '%OEM%' OR s.program LIKE '%monitor%' OR s.program LIKE '%backup%' OR s.program LIKE '%nagios%' OR s.program LIKE '%zabbix%' OR s.program LIKE '%toad%' OR s.program LIKE '%SQL Developer%') GROUP BY s.username, s.program ORDER BY sys_privs DESC
SEC-SQL-CFG-004-RC05 Lack of Assembly Vetting
Condition: row_count > 0 — User-owned Java objects in the database should be reviewed for security implications
In-Database Code Execution Enabled FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 30 rows alert: Yes mail: No
SELECT owner, object_name, object_type, status FROM ALL_OBJECTS WHERE object_type IN ('JAVA CLASS', 'JAVA SOURCE', 'JAVA RESOURCE') AND owner NOT IN ('SYS', 'SYSTEM', 'DBSNMP', 'OUTLN', 'XDB', 'CTXSYS', 'MDSYS', 'ORDSYS', 'ORDDATA', 'WMSYS') ORDER BY owner, object_type, object_name FETCH FIRST 30 ROWS ONLY
SEC-SQL-CFG-004-RC06 Privilege Escalation via CLR
Condition: row_count > 0 — DEFINER-rights procedures execute with the owner's privileges, creating potential escalation paths
In-Database Code Execution Enabled FINNACLEDB_TEST: DETECTED 20 rows FLEXCUBE: DETECTED 20 rows alert: Yes mail: No
SELECT owner, object_name, object_type, authid FROM ALL_PROCEDURES WHERE authid = 'DEFINER' AND owner NOT IN ('SYS', 'SYSTEM', 'DBSNMP', 'OUTLN', 'XDB', 'CTXSYS', 'MDSYS', 'WMSYS') AND object_type IN ('PROCEDURE', 'FUNCTION', 'PACKAGE') ORDER BY owner, object_name FETCH FIRST 20 ROWS ONLY
SEC-SQL-CFG-004-RC07 Sandbox Escape
Condition: row_count > 0 — Dangerous Java permissions (Runtime, File, Socket) granted to non-system users enable sandbox escape
In-Database Code Execution Enabled FINNACLEDB_TEST: DETECTED 32 rows FLEXCUBE: DETECTED 54 rows alert: Yes mail: Yes (destructive)
SELECT x.grantee, x.type_name, x.name, x.action, x.enabled FROM XMLTABLE('/ROWSET/ROW' PASSING XMLTYPE(NVL(DBMS_XMLGEN.getxml(CASE WHEN (SELECT COUNT(*) FROM all_objects WHERE object_name = 'DBA_JAVA_POLICY' AND object_type IN ('VIEW','SYNONYM')) > 0 THEN q'[SELECT grantee, type_name, name, action, enabled FROM dba_java_policy WHERE type_name IN ('java.lang.RuntimePermission', 'java.io.FilePermission', 'java.net.SocketPermission') AND grantee NOT IN ('SYS','SYSTEM','JAVASYSPRIV') AND enabled = 'ENABLED' ORDER BY grantee, type_name]' ELSE q'[SELECT NULL AS grantee, NULL AS type_name, NULL AS name, NULL AS action, NULL AS enabled FROM dual WHERE 1=0]' END), '<ROWSET/>')) COLUMNS grantee VARCHAR2(128) PATH 'GRANTEE', type_name VARCHAR2(256) PATH 'TYPE_NAME', name VARCHAR2(4000) PATH 'NAME', action VARCHAR2(256) PATH 'ACTION', enabled VARCHAR2(40) PATH 'ENABLED') x
SEC-SQL-CFG-004-RC08 Registry Modification
Condition: row_count > 0 — EXECUTE grants on OS-interacting packages enable file system, network, and OS command access from the database

In-Database Code Execution Enabled FINNACLEDB_TEST: DETECTED 12 rows FLEXCUBE: DETECTED 58 rows alert: Yes mail: Yes (destructive)
SELECT grantee, privilege, table_name AS object_name FROM dba_tab_privs WHERE table_name IN ('UTL_FILE', 'DBMS_SCHEDULER', 'UTL_TCP', 'UTL_HTTP', 'UTL_SMTP', 'UTL_INADDR', 'DBMS_JAVA', 'DBMS_JAVA_TEST') AND privilege = 'EXECUTE' AND grantee NOT IN ('SYS', 'SYSTEM', 'PUBLIC', 'DBA') AND owner = 'SYS' ORDER BY grantee, table_name
SEC-SQL-CFG-004-RC09 Network-Based Attacks
Condition: row_count > 0 — Database links with stored credentials provide lateral movement paths between databases
In-Database Code Execution Enabled FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT owner, db_link, username, host FROM ALL_DB_LINKS WHERE owner NOT IN ('SYS', 'SYSTEM') ORDER BY owner, db_link
SEC-SQL-CFG-004-RC10 External Library Access
Condition: row_count > 0 — User-owned external libraries allow loading shared objects (.so/.dll) into the database process
In-Database Code Execution Enabled FINNACLEDB_TEST: DETECTED 49 rows FLEXCUBE: DETECTED 54 rows alert: Yes mail: No
SELECT owner, library_name, file_spec, dynamic, status FROM dba_libraries WHERE owner NOT IN ('SYS', 'SYSTEM', 'DBSNMP', 'OUTLN', 'XDB') ORDER BY owner, library_name
SEC-SQL-CFG-004-RC11 Post-Exploitation Persistence
Condition: row_count > 0 — Database-level triggers (LOGON, STARTUP, SERVERERROR) can be used for persistent backdoor execution
In-Database Code Execution Enabled FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT owner, trigger_name, trigger_type, triggering_event, table_name, status FROM dba_triggers WHERE triggering_event LIKE '%LOGON%' OR triggering_event LIKE '%STARTUP%' OR triggering_event LIKE '%SERVERERROR%' ORDER BY owner, trigger_name
SEC-SQL-CFG-004-RC12 Insufficient Configuration Segregation
Condition: debug or trace parameters enabled — Debug/trace parameters enabled in production suggest development configuration bleed
In-Database Code Execution Enabled FINNACLEDB_TEST: DETECTED 3 rows FLEXCUBE: DETECTED 3 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('plsql_debug', 'sql_trace', 'timed_statistics', 'plsql_optimize_level', 'plsql_warnings', 'tracefile_identifier') AND (value IS NOT NULL AND value != 'FALSE' AND value != '0') ORDER BY name
SEC-SQL-CFG-004-RC13 Monitoring Gaps
Condition: Unified Auditing is TRUE — Unified Auditing must be enabled for comprehensive security event monitoring
In-Database Code Execution Enabled FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT parameter, value FROM v\$option WHERE parameter = 'Unified Auditing'
SEC-SQL-CFG-005-RC01 Troubleshooting Discovery
Condition: row_count > 0 — Security parameters in insecure state suggest troubleshooting changes not reverted
Debug/Trace Flags Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT name, value, ismodified FROM v\$parameter WHERE name IN ('remote_os_authent', 'o7_dictionary_accessibility', 'sec_case_sensitive_logon', 'audit_trail', 'remote_os_roles', 'sql92_security') AND (value = 'TRUE' AND name IN ('remote_os_authent', 'o7_dictionary_accessibility', 'remote_os_roles') OR value = 'FALSE' AND name IN ('sec_case_sensitive_logon', 'sql92_security')) ORDER BY name
SEC-SQL-CFG-005-RC02 Vendor/Support Recommendation
Condition: row_count > 0 — Non-default security parameters may have been changed per support recommendations without full security review
Debug/Trace Flags Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No

SELECT name, value, isdefault, ismodified, description FROM v\$parameter WHERE isdefault = 'FALSE' AND name IN ('sec_case_sensitive_logon', 'sec_max_failed_login_attempts', 'sec_protocol_error_trace_action', 'sec_return_server_release_banner', 'audit_sys_operations', 'unified_audit_sga_queue_size', 'remote_os_authent', 'remote_os_roles', 'o7_dictionary_accessibility') ORDER BY name
SEC-SQL-CFG-005-RC03 Undocumented Historical Change
Condition: row_count > 0 — ALTER SYSTEM SET commands in alert log show recent parameter changes that should be documented
Debug/Trace Flags Enabled FINNACLEDB_TEST: DETECTED 7 rows FLEXCUBE: DETECTED 20 rows alert: Yes mail: No
SELECT originating_timestamp, message_text FROM v\$diag_alert_ext WHERE message_text LIKE '%ALTER SYSTEM SET%' AND originating_timestamp > SYSDATE - 90 ORDER BY originating_timestamp DESC FETCH FIRST 20 ROWS ONLY
SEC-SQL-CFG-005-RC04 Perceived Performance Optimization
Condition: row_count > 0 — Security parameters disabled for performance reasons indicate misguided optimization
Debug/Trace Flags Enabled FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE (name = 'audit_trail' AND value = 'NONE') OR (name = 'db_block_checking' AND value = 'FALSE') OR (name = 'db_block_checksum' AND value = 'OFF') OR (name = 'sec_case_sensitive_logon' AND value = 'FALSE') OR (name = 'sql92_security' AND value = 'FALSE') ORDER BY name
SEC-SQL-CFG-005-RC05 Dangerous Flag Combinations
Condition: row_count >= 2 — Multiple insecure parameters together create compound security vulnerabilities
Debug/Trace Flags Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE (name = 'remote_os_authent' AND value = 'TRUE') OR (name = 'remote_os_roles' AND value = 'TRUE') OR (name = 'O7_DICTIONARY_ACCESSIBILITY' AND value = 'TRUE') OR (name = 'utl_file_dir' AND value != '') OR (name = '_trace_files_public' AND value = 'TRUE') ORDER BY name
SEC-SQL-CFG-005-RC06 Experimentation in Production
Condition: row_count > 0 — Non-default hidden (underscore) parameters indicate experimental settings in production
Debug/Trace Flags Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT name, value, description FROM v\$parameter WHERE name LIKE '_%%' ESCAPE '\' AND isdefault = 'FALSE' ORDER BY name
SEC-SQL-CFG-005-RC07 One-Off Debugging Enabled Persistently
Condition: row_count > 0 — Debug or trace parameters left enabled in production indicate one-off debugging not cleaned up
Debug/Trace Flags Enabled FINNACLEDB_TEST: DETECTED 4 rows FLEXCUBE: DETECTED 3 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE (name LIKE '%debug%' AND value != '0' AND value != 'FALSE') OR (name LIKE '%trace%' AND value NOT IN ('0', 'FALSE', 'NONE', 'OFF')) AND name NOT IN ('diagnostic_dest', 'tracefile_identifier')) OR (name = 'sql_trace' AND value = 'TRUE') OR (name = 'timed_statistics' AND value = 'TRUE') OR (name = 'statistics_level' AND value = 'ALL') ORDER BY name
SEC-SQL-CFG-005-RC08 Configuration Drift
Condition: row_count > :drift_threshold — Many modified parameters indicate significant configuration drift from baseline
Debug/Trace Flags Enabled FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: clean alert: Yes mail: No
SELECT name, value, ismodified, isdefault, description FROM v\$parameter WHERE ismodified != 'FALSE' ORDER BY name
SEC-SQL-CFG-005-RC09 Undocumented Server Requirements
Condition: row_count > 0 — Parameters set for backward compatibility suggest undocumented legacy application requirements
Debug/Trace Flags Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE (name = 'sec_case_sensitive_logon' AND value = 'FALSE') OR (name = 'compatible' AND value LIKE '11%') OR (name = 'cursor_sharing' AND value = 'FORCE') OR (name = 'optimizer_features_enable' AND value < '19') ORDER BY name

SEC-SQL-CFG-005-RC10 Verbose Logging for "Security"
Condition: row_count > 0 — Diagnostic destination contains trace files that may contain sensitive data
Debug/Trace Flags Enabled FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name = 'diagnostic_dest'
SEC-SQL-CFG-005-RC12 Lazy Optimization
Condition: row_count > 0 — Quick-fix optimizer parameters indicate lazy optimization that may have unintended security or stability effects
Debug/Trace Flags Enabled FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE (name = 'cursor_sharing' AND value = 'FORCE') OR (name = 'optimizer_index_caching' AND value != '0') OR (name = 'optimizer_index_cost_adj' AND TO_NUMBER(value) < 50) OR (name = 'skip_unusable_indexes' AND value = 'TRUE') ORDER BY name
SEC-SQL-CFG-005-RC13 Lack of Audit Trail
Condition: row_count = 0 — No auditing on ALTER SYSTEM means configuration changes are not tracked
Debug/Trace Flags Enabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT policy_name, audit_option, audit_option_type FROM audit_unified_policies WHERE audit_option LIKE '%ALTER SYSTEM%' OR audit_option = 'ALTER SYSTEM' UNION ALL SELECT 'Traditional' AS policy_name, audit_option, 'STANDARD' FROM dba_stmt_audit_opts WHERE audit_option = 'ALTER SYSTEM'
SEC-SQL-CFG-005-RC14 Misunderstanding Hidden Features
Condition: row_count > 0 — Grants on powerful internal packages to non-system users indicate potential security risks from misunderstood features
Debug/Trace Flags Enabled FINNACLEDB_TEST: DETECTED 11 rows FLEXCUBE: DETECTED 76 rows alert: Yes mail: No
SELECT grantee, privilege, table_name FROM dba_tab_privs WHERE table_name IN ('DBMS_SYS_SQL', 'DBMS_BACKUP_RESTORE', 'DBMS_FILE_TRANSFER', 'UTL_FILE', 'UTL_TCP', 'UTL_HTTP', 'UTL_SMTP', 'DBMS_ADVISOR', 'UTL_INADDR', 'DBMS_JAVA') AND grantee NOT IN ('SYS', 'SYSTEM', 'DBA', 'PUBLIC') ORDER BY table_name, grantee
SEC-SQL-CFG-006-RC01 Static Credential Storage
Condition: row_count > 0 — Database links with stored usernames have static credentials in the data dictionary
Database Links Unsecured FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 5 rows alert: Yes mail: No
SELECT owner, db_link, username, host, created FROM dba_db_links WHERE username IS NOT NULL ORDER BY owner, db_link
SEC-SQL-CFG-006-RC02 Overprivileged Link Credentials
Condition: row_count > 0 — Database links using SYS, SYSTEM, or admin accounts have overprivileged credentials
Database Links Unsecured FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT owner, db_link, username, host FROM dba_db_links WHERE username IN ('SYS', 'SYSTEM', 'DBA') OR username LIKE '%ADMIN%' ORDER BY owner, db_link
SEC-SQL-CFG-006-RC03 rootcause Database Link Exposure
Condition: row_count > 0 — PUBLIC database links give every database user access to the remote database
Database Links Unsecured FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT owner, db_link, username, host, created FROM dba_db_links WHERE owner = 'PUBLIC' ORDER BY db_link
SEC-SQL-CFG-006-RC04 Privilege Escalation via Link
Condition: row_count > 0 — Database links where remote user has more privileges than link owner enable privilege escalation
Database Links Unsecured FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 3 rows alert: Yes mail: No

SELECT dl.owner AS link_owner, dl.db_link, dl.username AS remote_user, dl.host, (SELECT COUNT(*) FROM dba_sys_privs WHERE grantee = dl.owner) AS owner_priv_count FROM dba_db_links dl WHERE dl.username IS NOT NULL AND dl.owner != 'PUBLIC' ORDER BY dl.owner, dl.db_link
SEC-SQL-CFG-006-RC05 Inadequate Access Control
Condition: row_count > 0 — Non-DBA users with CREATE DATABASE LINK privilege can create uncontrolled remote connections
Database Links Unsecured FINNACLEDB_TEST: DETECTED 5 rows FLEXCUBE: DETECTED 13 rows alert: Yes mail: No
SELECT grantee, privilege FROM dba_sys_privs WHERE privilege IN ('CREATE DATABASE LINK', 'CREATE PUBLIC DATABASE LINK') AND grantee NOT IN ('SYS', 'SYSTEM', 'DBA') ORDER BY grantee
SEC-SQL-CFG-006-RC06 Cross-Database Privilege Inheritance
Condition: row_count > 0 — Fixed-user database links create privilege trust chains between databases
Database Links Unsecured FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 5 rows alert: Yes mail: No
SELECT dl1.owner, dl1.db_link AS link1, dl1.host AS target1, dl1.username AS remote_user FROM dba_db_links dl1 WHERE dl1.username IS NOT NULL ORDER BY dl1.owner
SEC-SQL-CFG-006-RC07 Unvalidated Dynamic Queries
Condition: row_count > 0 — Dynamic SQL referencing database links creates SQL injection risk on remote databases
Database Links Unsecured FINNACLEDB_TEST: DETECTED 20 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT sql_id, SUBSTR(sql_text, 1, 300) AS sql_preview FROM v\$sql WHERE sql_text LIKE '%@%' AND (sql_text LIKE '%EXECUTE IMMEDIATE%' OR sql_text LIKE '%DBMS_SQL%') AND parsing_schema_name NOT IN ('SYS', 'SYSTEM') ORDER BY last_active_time DESC FETCH FIRST 20 ROWS ONLY
SEC-SQL-CFG-006-RC08 Remote Database Exploitation
Condition: row_count > 0 — All database links represent potential attack vectors to remote databases
Database Links Unsecured FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 6 rows alert: Yes mail: Yes (destructive)
SELECT owner, db_link, username, host, created FROM dba_db_links ORDER BY owner, db_link
SEC-SQL-CFG-006-RC09 Credential Exposure via Monitoring
Condition: row_count > 0 — CREATE/ALTER DATABASE LINK with passwords visible in shared pool exposes credentials to anyone who can query V\$SQL
Database Links Unsecured FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT sql_id, SUBSTR(sql_text, 1, 200) AS sql_preview FROM v\$sql WHERE sql_text LIKE '%CREATE%DATABASE%LINK%IDENTIFIED%BY%' OR sql_text LIKE '%ALTER%DATABASE%LINK%IDENTIFIED%BY%' ORDER BY last_active_time DESC FETCH FIRST 10 ROWS ONLY
SEC-SQL-CFG-006-RC10 Long-Lived Credentials
Condition: any age_days > :rotation_threshold — Database links older than rotation threshold have potentially stale credentials
Database Links Unsecured FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 5 rows alert: Yes mail: Yes (destructive)
SELECT owner, db_link, username, host, created, ROUND(SYSDATE - created) AS age_days FROM dba_db_links WHERE username IS NOT NULL ORDER BY age_days DESC
SEC-SQL-CFG-006-RC11 Scope Confusion
Condition: row_count > 0 — Public database links with embedded credentials — any user can use these stored credentials
Database Links Unsecured FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT owner, db_link, username, host FROM dba_db_links WHERE username IS NOT NULL AND owner = 'PUBLIC' ORDER BY db_link
SEC-SQL-CFG-006-RC12 Network Segmentation Bypass
Condition: row_count > 0 — Database links found — each link represents a network path that bypasses application-level segmentation

Database Links Unsecured FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 6 rows alert: Yes mail: Yes (destructive)
SELECT owner, db_link, username, host FROM dba_db_links ORDER BY owner, db_link
SEC-SQL-CFG-006-RC13 Linked Server Configuration Persistence
Condition: row_count > 0 — Database links with stored credentials — passwords persist in data dictionary
Database Links Unsecured FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 5 rows alert: Yes mail: Yes (destructive)
SELECT owner, db_link, username, host, created FROM dba_db_links WHERE username IS NOT NULL ORDER BY owner, db_link
SEC-SQL-CFG-006-RC14 Missing Encryption
Condition: row_count > 0 — Database links without TCPS protocol or SECURITY clause — data transmitted unencrypted
Database Links Unsecured FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 6 rows alert: Yes mail: No
SELECT db_link, owner, host FROM dba_db_links WHERE host NOT LIKE '%TCPS%' AND host NOT LIKE '%(SECURITY%)' ORDER BY owner, db_link
SEC-SQL-CFG-006-RC15 Mutual Authentication Gaps
Condition: row_count = 0 — No wallet or authentication settings configured for mutual TLS
Database Links Unsecured FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name LIKE '%authentication%' OR name LIKE '%wallet%' ORDER BY name
SEC-SQL-CFG-006-RC16 Catalog-Level Exposure
Condition: row_count > 0 — Database links using privileged accounts expose full remote data dictionary
Database Links Unsecured FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT owner, db_link, username, host FROM dba_db_links WHERE username IN ('SYS', 'SYSTEM', 'SYSDBA') OR username IS NULL ORDER BY owner, db_link
SEC-SQL-ENC-001-RC01 Default Configuration Not Changed
Condition: row_count = 0 — No encryption banner in session connection info — connections are unencrypted
Data-in-Transit Unencrypted FINNACLEDB_TEST: DETECTED 0 rows FLEXCUBE: DETECTED 0 rows alert: Yes mail: No
SELECT network_service_banner FROM v\$session_connect_info WHERE sid = SYS_CONTEXT('USERENV', 'SID') AND network_service_banner LIKE '%Encryption%'
SEC-SQL-ENC-001-RC02 Missing Encryption Parameter in Connection Strings
Condition: row_count > 0 — Active sessions without encryption banners — connecting without encryption parameters
Data-in-Transit Unencrypted FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 78 rows alert: Yes mail: No
SELECT s.sid, s.serial#, s.username, s.program, s.machine, sci.network_service_banner FROM v\$session s LEFT JOIN v\$session_connect_info sci ON s.sid = sci.sid AND sci.network_service_banner LIKE '%Encryption%' WHERE s.type = 'USER' AND s.status = 'ACTIVE' AND sci.sid IS NULL ORDER BY s.username
SEC-SQL-ENC-001-RC03 Backward Compatibility Requirements
Condition: value = 'ACCEPTED' OR row_count = 0 — Encryption set to ACCEPTED (or not set) — allows both encrypted and unencrypted connections for backward compatibility
Data-in-Transit Unencrypted FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name LIKE '%encryption%server%' OR name LIKE '%crypto%server%' ORDER BY name
SEC-SQL-ENC-001-RC04 Client Driver Version Mismatch
Condition: row_count > 0 — Client driver information — older drivers may not support TLS 1.2/1.3

Data-in-Transit Unencrypted FINNACLEDB_TEST: DETECTED 14 rows FLEXCUBE: DETECTED 641 rows alert: Yes mail: No
SELECT s.sid, s.username, s.program, s.machine, sci.client_driver FROM v\$session s LEFT JOIN v\$session_connect_info sci ON s.sid = sci.sid WHERE s.type = 'USER' ORDER BY sci.client_driver
SEC-SQL-ENC-001-RC05 Cost and Performance Concerns
Condition: row_count > 0 — Encryption explicitly REJECTED — deliberately disabled, possibly for performance reasons
Data-in-Transit Unencrypted FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE (name LIKE '%encryption%' AND value = 'REJECTED') OR (name LIKE '%crypto%' AND value = 'REJECTED') ORDER BY name
SEC-SQL-ENC-001-RC06 Lack of DBA Encryption Knowledge
Condition: row_count = 0 OR isdefault = 'TRUE' — All encryption parameters at default values — no encryption configuration has been applied
Data-in-Transit Unencrypted FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT name, value, isdefault FROM v\$parameter WHERE name IN ('sqlnet.encryption_server', 'sqlnet.encryption_client', 'sqlnet.encryption_types_server', 'wallet_root', 'tde_configuration') ORDER BY name
SEC-SQL-ENC-001-RC07 Certificate Management Complexity
Condition: status != 'OPEN' OR row_count = 0 — Oracle wallet not open or not configured — TLS certificate management incomplete
Data-in-Transit Unencrypted FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT wrl_type, wrl_parameter, status, wallet_type, wallet_order FROM v\$encryption_wallet
SEC-SQL-ENC-001-RC08 Self-Signed Certificate Issues
Condition: row_count > 0 — Listener configuration — check for TCPS protocol usage with certificate validation
Data-in-Transit Unencrypted FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('local_listener', 'remote_listener') ORDER BY name
SEC-SQL-ENC-001-RC09 Network Architecture Assumptions
Condition: encryption_params = 0 — No encryption parameters configured — relying entirely on network security for protection
Data-in-Transit Unencrypted FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT COUNT(*) AS encryption_params FROM v\$parameter WHERE (name LIKE '%encryption%' OR name LIKE '%crypto%' OR name LIKE '%ssl%' OR name LIKE '%wallet%') AND value IS NOT NULL AND value != 'FALSE'
SEC-SQL-ENC-001-RC10 Lack of Security Requirements
Condition: row_count > 2 — Multiple security parameters at default/disabled values — security requirements may not be defined
Data-in-Transit Unencrypted FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('audit_trail', 'audit_sys_operations', 'sec_case_sensitive_logon', 'sec_max_failed_login_attempts', 'sec_protocol_error_further_action') AND (value IS NULL OR value = 'FALSE' OR value = 'NONE') ORDER BY name
SEC-SQL-ENC-001-RC11 Application Middleware Bypass
Condition: row_count > 0 — Middleware connections — check if encryption is active for application server connections
Data-in-Transit Unencrypted FINNACLEDB_TEST: DETECTED 5 rows FLEXCUBE: DETECTED 20 rows alert: Yes mail: No
SELECT s.program, s.machine, COUNT(*) AS session_count, MIN(sci.network_service_banner) AS encryption_info FROM v\$session s LEFT JOIN v\$session_connect_info sci ON s.sid = sci.sid AND sci.network_service_banner LIKE '%Encryption%' WHERE s.type = 'USER' GROUP BY s.program, s.machine ORDER BY session_count DESC FETCH FIRST 20 ROWS ONLY
SEC-SQL-ENC-001-RC12 Testing and Development Environment Carryover

Condition: row_count > 0 — Development-style security settings found — configuration may have been carried from test/dev environment
Data-in-Transit Unencrypted FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE (name = 'remote_os_authent' AND value = 'TRUE') OR (name = 'o7_dictionary_accessibility' AND value = 'TRUE') OR (name = 'sec_case_sensitive_logon' AND value = 'FALSE') OR (name = 'audit_trail' AND value = 'NONE') ORDER BY name
SEC-SQL-ENC-001-RC13 Inadequate Monitoring and Visibility
Condition: row_count = 0 — No audit policies monitoring encryption status — unencrypted connections go undetected
Data-in-Transit Unencrypted FINNACLEDB_TEST: DETECTED 0 rows FLEXCUBE: DETECTED 0 rows alert: Yes mail: No
SELECT policy_name, audit_condition FROM audit_unified_policies WHERE policy_name LIKE '%ENCRYPT%' OR audit_condition LIKE '%encryption%' ORDER BY policy_name
SEC-SQL-ENC-001-RC14 Competing Priorities in Development Cycles
Condition: row_count > 0 — Encryption and audit status summary — multiple NOT_CONFIGURED items indicate security was deprioritized
Data-in-Transit Unencrypted FINNACLEDB_TEST: DETECTED 3 rows FLEXCUBE: DETECTED 3 rows alert: Yes mail: No
SELECT 'network_encryption' AS check_type, CASE WHEN EXISTS (SELECT 1 FROM v\$parameter WHERE name LIKE '%encryption%server%' AND value = 'REQUIRED') THEN 'CONFIGURED' ELSE 'NOT_CONFIGURED' END AS status FROM dual UNION ALL SELECT 'tde_encryption' AS check_type, CASE WHEN EXISTS (SELECT 1 FROM v\$encryption_wallet WHERE status = 'OPEN') THEN 'CONFIGURED' ELSE 'NOT_CONFIGURED' END AS status FROM dual UNION ALL SELECT 'audit_configured' AS check_type, CASE WHEN EXISTS (SELECT 1 FROM audit_unified_enabled_policies) THEN 'CONFIGURED' ELSE 'NOT_CONFIGURED' END AS status FROM dual
SEC-SQL-ENC-002-RC01 Legacy Application Dependency on TLS 1.0/1.1
Condition: SSL_VERSION includes 1.0 or 1.1 — Deprecatd TLS versions allowed in sqlnet.ora configuration
Obsolete TLS Versions FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT 'Check sqlnet.ora for SSL_VERSION parameter' AS action, 'File: \$ORACLE_HOME/network/admin/sqlnet.ora' AS location, 'SSL_VERSION should be 1.2 or higher' AS requirement FROM DUAL
SEC-SQL-ENC-002-RC02 Deprecatd Client Libraries in Use
Condition: row_count > 0 — Active client sessions captured — check DRIVER_INFO for older client versions
Obsolete TLS Versions FINNACLEDB_TEST: DETECTED 5 rows FLEXCUBE: DETECTED 263 rows alert: Yes mail: No
SELECT s.SID, s.PROGRAM, s.MODULE, s.CLIENT_INFO, s.MACHINE FROM V\$SESSION s WHERE s.TYPE = 'USER' AND s.USERNAME IS NOT NULL ORDER BY s.PROGRAM
SEC-SQL-ENC-002-RC04 Java/JDK Version Limitations
Condition: row_count > 0 — JDBC connections detected — check driver version for TLS capability
Obsolete TLS Versions FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 130 rows alert: Yes mail: No
SELECT s.SID, s.PROGRAM, s.MODULE, s.CLIENT_INFO, s.MACHINE FROM V\$SESSION s WHERE s.TYPE = 'USER' AND (s.PROGRAM LIKE '%JDBC%' OR s.PROGRAM LIKE '%java%' OR s.MODULE LIKE '%JDBC%') ORDER BY s.PROGRAM
SEC-SQL-ENC-002-RC06 Operating System Default Restrictions
Condition: row_count > 0 — Oracle Native Network Encryption settings captured
Obsolete TLS Versions FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT NAME, VALUE FROM V\$PARAMETER WHERE NAME IN ('sqlnet.encryption_server', 'sqlnet.encryption_client', 'sqlnet.crypto_checksum_server', 'sqlnet.crypto_checksum_client', 'sqlnet.encryption_types_server', 'sqlnet.crypto_checksum_types_server')
SEC-SQL-ENC-002-RC07 Load Balancer and Infrastructure TLS Settings
Condition: NETWORK_SERVICE_BANNER NOT LIKE '%SSL%' AND NETWORK_SERVICE_BANNER NOT LIKE '%TLS%' — Connections without TLS — may be arriving unencrypted from load balancer
Obsolete TLS Versions FINNACLEDB_TEST: DETECTED 14 rows FLEXCUBE: DETECTED 636 rows alert: Yes mail: No

SELECT s.SID, s.PROGRAM, s.MACHINE, ns.NETWORK_SERVICE_BANNER FROM V\$SESSION s JOIN V\$SESSION_CONNECT_INFO ns ON s.SID = ns.SID WHERE s.TYPE = 'USER' AND s.USERNAME IS NOT NULL ORDER BY s.SID
SEC-SQL-ENC-002-RC08 Lack of TLS 1.2 Configuration Knowledge
Condition: ssl_sessions = 0 — No SSL/TLS sessions — TLS not configured for Oracle Net
Obsolete TLS Versions FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT COUNT(*) AS ssl_sessions FROM V\$SESSION_CONNECT_INFO WHERE NETWORK_SERVICE_BANNER LIKE '%SSL%' OR NETWORK_SERVICE_BANNER LIKE '%TLS%'
SEC-SQL-ENC-002-RC09 Configuration Oversight
Condition: row_count > 0 — SSL/crypto connections detected — check cipher suites in sqlnet.ora
Obsolete TLS Versions FINNACLEDB_TEST: DETECTED 4 rows FLEXCUBE: DETECTED 187 rows alert: Yes mail: No
SELECT s.SID, s.PROGRAM, ci.NETWORK_SERVICE_BANNER FROM V\$SESSION s JOIN V\$SESSION_CONNECT_INFO ci ON s.SID = ci.SID WHERE ci.NETWORK_SERVICE_BANNER LIKE '%SSL%' OR ci.NETWORK_SERVICE_BANNER LIKE '%Crypto%' ORDER BY s.SID
SEC-SQL-ENC-002-RC10 Intentional Compatibility Decisions
Condition: SSL_VERSION includes multiple versions — Explicit multi-version TLS configuration — intentional compatibility choice
Obsolete TLS Versions FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT 'Check sqlnet.ora for SSL_VERSION=(1.0,1.1,1.2)' AS check_item, 'Multi-version TLS is an intentional compatibility decision' AS interpretation FROM DUAL
SEC-SQL-ENC-002-RC11 No Automated Configuration Audit
Condition: row_count = 0 — No audit policy enabled for parameter changes — TLS config changes unaudited
Obsolete TLS Versions FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT POLICY_NAME FROM AUDIT_UNIFIED_ENABLED_POLICIES WHERE POLICY_NAME IN ('ORA_SECURECONFIG', 'ORA_DATABASE_PARAMETER')
SEC-SQL-ENC-002-RC12 Database Version Limitations
Condition: VERSION < '12.1' — Oracle version older than 12.1 has limited TLS 1.2 support
Obsolete TLS Versions FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT BANNER_FULL, BANNER_LEGACY, BANNER FROM V\$VERSION WHERE BANNER LIKE 'Oracle Database%'
SEC-SQL-ENC-002-RC13 Hybrid Cloud Connectivity
Condition: row_count > 0 — Database links exist — cross-environment connections may have different TLS requirements
Obsolete TLS Versions FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 6 rows alert: Yes mail: No
SELECT OWNER, DB_LINK, HOST, CREATED FROM DBA_DB_LINKS ORDER BY CREATED DESC
SEC-SQL-ENC-002-RC14 Vendor Testing Incompleteness
Condition: row_count > :type_threshold — Many distinct client types — TLS testing may not cover all
Obsolete TLS Versions FINNACLEDB_TEST: DETECTED 11 rows FLEXCUBE: DETECTED 242 rows alert: Yes mail: No
SELECT DISTINCT s.PROGRAM, s.MODULE, ci.NETWORK_SERVICE_BANNER, COUNT(*) AS connections FROM V\$SESSION s LEFT JOIN V\$SESSION_CONNECT_INFO ci ON s.SID = ci.SID WHERE s.TYPE = 'USER' AND s.USERNAME IS NOT NULL GROUP BY s.PROGRAM, s.MODULE, ci.NETWORK_SERVICE_BANNER ORDER BY connections DESC
SEC-SQL-ENC-003-RC01 Not Aware of TDE Feature
Condition: row_count = 0 — No column-level encryption either — TDE completely unused
Transparent Data Encryption Disabled FINNACLEDB_TEST: DETECTED 0 rows FLEXCUBE: DETECTED 0 rows alert: Yes mail: No

SELECT TABLE_NAME, COLUMN_NAME, ENCRYPTION_ALG FROM DBA_ENCRYPTED_COLUMNS
SEC-SQL-ENC-003-RC02 Licensing Restrictions
Condition: row_count = 0 — Advanced Security Option not installed — TDE requires licensing
Transparent Data Encryption Disabled FINNACLEDB_TEST: DETECTED 0 rows FLEXCUBE: DETECTED 0 rows alert: Yes mail: Yes (destructive)
SELECT COMP_NAME, STATUS, VERSION FROM DBA_REGISTRY WHERE COMP_NAME LIKE '%Advanced Security%'
SEC-SQL-ENC-003-RC03 Missing License Entitlements
Condition: CURRENTLY_USED = 'FALSE' OR row_count = 0 — TDE feature not currently used — licensing may be blocking deployment
Transparent Data Encryption Disabled FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT NAME, DETECTED_USAGES, CURRENTLY_USED, FIRST_USAGE_DATE, LAST_USAGE_DATE FROM DBA_FEATURE_USAGE_STATISTICS WHERE NAME LIKE '%Transparent Data Encryption%' OR NAME LIKE '%Tablespace Encryption%'
SEC-SQL-ENC-003-RC04 Key Management System Not Available
Condition: STATUS != 'OPEN' OR row_count = 0 — TDE keystore not open or not configured — encryption operations will fail
Transparent Data Encryption Disabled FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: Yes (destructive)
SELECT WRL_TYPE, WRL_PARAMETER, STATUS, WALLET_TYPE, KEYSTORE_MODE FROM V\$ENCRYPTION_WALLET
SEC-SQL-ENC-003-RC05 Performance Impact Concerns
Condition: row_count > 0 — CPU info captured — Intel AES-NI or equivalent needed for TDE performance
Transparent Data Encryption Disabled FINNACLEDB_TEST: DETECTED 3 rows FLEXCUBE: DETECTED 3 rows alert: Yes mail: No
SELECT STAT_NAME, VALUE FROM V\$OSSTAT WHERE STAT_NAME IN ('NUM_CPUS', 'NUM_CPU_CORES', 'NUM_CPU_SOCKETS')
SEC-SQL-ENC-003-RC06 Database Design Incompatibility
Condition: row_count > 0 — External tables exist — these bypass TDE (data files outside the database)
Transparent Data Encryption Disabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT OWNER, TABLE_NAME, TYPE_NAME FROM DBA_EXTERNAL_TABLES WHERE OWNER NOT IN ('SYS', 'SYSTEM')
SEC-SQL-ENC-003-RC07 Application Testing Not Completed
Condition: row_count > 1 — Mix of encrypted and unencrypted tablespaces — partial TDE deployment
Transparent Data Encryption Disabled FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT ENCRYPTED, COUNT(*) AS ts_count FROM DBA_TABLESPACES WHERE CONTENTS = 'PERMANENT' AND TABLESPACE_NAME NOT IN ('SYSTEM', 'SYSAUX') GROUP BY ENCRYPTED
SEC-SQL-ENC-003-RC08 Backup and Recovery Complexity
Condition: row_count > 0 — Encrypted tablespaces exist — RMAN backups require the wallet for restore, adding backup/recovery complexity
Transparent Data Encryption Disabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT tablespace_name, encrypted, status FROM dba_tablespaces WHERE encrypted = 'YES' ORDER BY tablespace_name
SEC-SQL-ENC-003-RC09 Administrative Overhead
Condition: row_count > 0 — Encryption keys exist requiring ongoing management — key rotation, backup, and lifecycle management
Transparent Data Encryption Disabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No

SELECT key_id, keystore_type, activation_time, creator, creator_dbname FROM v\$encryption_keys ORDER BY activation_time DESC
SEC-SQL-ENC-003-RC10 Temporary workspace encryption complications
Condition: row_count > 0 — Encrypted permanent tablespaces exist while temp is unencrypted — temp data spills create encryption gap
Transparent Data Encryption Disabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT tablespace_name, encrypted FROM dba_tablespaces WHERE encrypted = 'YES' AND contents = 'PERMANENT'
SEC-SQL-ENC-003-RC11 High Availability and Replication Issues
Condition: row_count > 0 — Database is part of a Data Guard configuration — TDE wallet must be synchronized with standby
Transparent Data Encryption Disabled FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: Yes (destructive)
SELECT database_role, protection_mode, dataguard_broker FROM v\$database WHERE database_role IN ('PRIMARY', 'PHYSICAL STANDBY')
SEC-SQL-ENC-003-RC12 Cloud Database Settings
Condition: any(status = 'NOT_AVAILABLE' OR status = 'CLOSED') — TDE wallet is not available or closed — in cloud environments, encryption may need explicit configuration through the cloud console
Transparent Data Encryption Disabled FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT wrl_type, status, wallet_type FROM v\$encryption_wallet
SEC-SQL-ENC-003-RC13 Development/Test Database Assumption
Condition: any(status IN ('NOT_AVAILABLE', 'CLOSED', 'UNDEFINED')) — No TDE encryption is configured — this database has no data-at-rest encryption
Transparent Data Encryption Disabled FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT status, wallet_type FROM v\$encryption_wallet
SEC-SQL-ENC-003-RC14 Lack of Compliance Driver
Condition: row_count = 0 — No column-level encryption is configured — no data elements are individually protected
Transparent Data Encryption Disabled FINNACLEDB_TEST: DETECTED 0 rows FLEXCUBE: DETECTED 0 rows alert: Yes mail: No
SELECT table_name, column_name, encryption_alg FROM ALL_ENCRYPTED_COLUMNS WHERE owner NOT IN ('SYS', 'SYSTEM') ORDER BY table_name
SEC-SQL-ENC-003-RC15 Previous Failed Implementation Attempt
Condition: any(encrypted = 'YES') AND any(encrypted = 'NO') — Mixed encryption state (some tablespaces encrypted, others not) indicates an incomplete TDE implementation
Transparent Data Encryption Disabled FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 4 rows alert: Yes mail: No
SELECT tablespace_name, encrypted FROM dba_tablespaces WHERE contents = 'PERMANENT' AND tablespace_name NOT IN ('SYSTEM', 'SYSAUX') ORDER BY tablespace_name
SEC-SQL-ENC-003-RC16 Wallets and Key Storage Issues
Condition: row_count > 0 — Encryption keys that have not been backed up — losing the wallet means permanent data loss for encrypted tablespaces
Transparent Data Encryption Disabled FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT key_id, activation_time, backed_up, creator_dbname FROM v\$encryption_keys WHERE backed_up = 'NO' ORDER BY activation_time DESC
SEC-SQL-ENC-004-RC01 Feature Not Enabled at Backup Time
Condition: any(encrypted = 'NO') — Recent RMAN backups were not encrypted, leaving backup media vulnerable to unauthorized access
Unencrypted Backups FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT input_type, status, start_time, output_bytes_display FROM v\$rman_backup_job_details WHERE start_time > SYSDATE - 30 ORDER BY start_time DESC FETCH FIRST 10 ROWS ONLY

SEC-SQL-ENC-004-RC02 Backup Software Doesn't Support Encryption
Condition: row_count > 0 — RMAN device/channel configuration retrieved — check if backup target supports encryption
Unencrypted Backups FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 3 rows alert: Yes mail: No
SELECT conf#, name, value FROM v\$rman_configuration WHERE name LIKE '%DEVICE%' OR name LIKE '%CHANNEL%' ORDER BY name
SEC-SQL-ENC-004-RC03 Key Management for Backups Not Established
Condition: any(backed_up = 'NO') — Encryption keys not backed up — if the wallet is lost, encrypted backups cannot be restored
Unencrypted Backups FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT key_id, activation_time, backed_up, creator_dbname FROM v\$encryption_keys ORDER BY activation_time DESC
SEC-SQL-ENC-004-RC05 Performance Impact on Backup Windows
Condition: any(encrypted = 'YES' AND avg_minutes > :duration_threshold) — Encrypted backups are taking significantly longer than expected, potentially exceeding the backup window
Unencrypted Backups FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT input_type, ROUND(AVG(elapsed_seconds)/60, 1) AS avg_minutes, ROUND(MAX(elapsed_seconds)/60, 1) AS max_minutes, COUNT(*) AS backup_count FROM v\$rman_backup_job_details WHERE start_time > SYSDATE - 30 AND status = 'COMPLETED' GROUP BY input_type ORDER BY input_type
SEC-SQL-ENC-004-RC06 Backup Storage Assumptions
Condition: row_count > 0 — Backup destination configuration retrieved — check if storage locations provide adequate physical security
Unencrypted Backups FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT conf#, name, value FROM v\$rman_configuration WHERE name LIKE '%DEST%' OR name LIKE '%FORMAT%' ORDER BY name
SEC-SQL-ENC-004-RC07 Regulatory Requirement Not Mapped to Backups
Condition: backup_enc = 'BACKUP_NOT_ENCRYPTED' — RMAN backup encryption is not configured, even though data-at-rest encryption may or may not be in place
Unencrypted Backups FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT CASE WHEN EXISTS (SELECT 1 FROM v\$rman_configuration WHERE name LIKE '%ENCRYPTION%') THEN 'BACKUP_ENCRYPTED' ELSE 'BACKUP_NOT_ENCRYPTED' END AS backup_enc, CASE WHEN EXISTS (SELECT 1 FROM dba_tablespaces WHERE encrypted = 'YES') THEN 'DATA_ENCRYPTED' ELSE 'DATA_NOT_ENCRYPTED' END AS data_enc FROM dual
SEC-SQL-ENC-004-RC08 Off-Site Backup Storage Not Recognized as High Risk
Condition: row_count > 0 — Backup destinations include remote/cloud/tape targets that may traverse untrusted networks or storage
Unencrypted Backups FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT conf#, name, value FROM v\$rman_configuration WHERE name LIKE '%DEST%' AND (value LIKE '%nfs%' OR value LIKE '%s3%' OR value LIKE '%oss%' OR value LIKE '%tape%' OR value LIKE '%sbt%' OR value LIKE '%cloud%') ORDER BY name
SEC-SQL-ENC-004-RC09 Backup Recovery Testing Not Performed
Condition: row_count = 0 — No RMAN restore or validate operations found in the check period — backup recoverability has not been tested
Unencrypted Backups FINNACLEDB_TEST: DETECTED 0 rows FLEXCUBE: clean alert: Yes mail: No
SELECT operation, status, start_time, end_time, object_type FROM v\$rman_status WHERE operation IN ('RESTORE', 'VALIDATE', 'RECOVER') AND start_time > SYSDATE - 90 ORDER BY start_time DESC
SEC-SQL-ENC-004-RC10 Certificate and Key Expiration on Backups
Condition: any(key_age_days > :rotation_threshold) — Encryption keys are older than the recommended rotation period, increasing risk of compromise
Unencrypted Backups FINNACLEDB_TEST: FAILED FLEXCUBE: FAILED alert: No mail: No

SELECT key_id, activation_time, ROUND(SYSDATE - activation_time) AS key_age_days, backed_up, creator_dbname FROM v\$encryption_keys ORDER BY activation_time
SEC-SQL-ENC-004-RC11 Database Encryption Complexity Misunderstood
Condition: any(status IN ('NOT_CONFIGURED', 'ABSENT')) — Encryption implementation is incomplete — some components are configured while others are missing, suggesting complexity barriers
Unencrypted Backups FINNACLEDB_TEST: DETECTED 5 rows FLEXCUBE: DETECTED 5 rows alert: Yes mail: No
SELECT 'TDE Wallet' AS component, CASE WHEN (SELECT status FROM v\$encryption_wallet WHERE ROWNUM = 1) = 'OPEN' THEN 'CONFIGURED' ELSE 'NOT_CONFIGURED' END AS status FROM dual UNION ALL SELECT 'Encrypted Tablespaces', CASE WHEN EXISTS (SELECT 1 FROM dba_tablespaces WHERE encrypted = 'YES') THEN 'PRESENT' ELSE 'ABSENT' END FROM dual UNION ALL SELECT 'Column Encryption', CASE WHEN EXISTS (SELECT 1 FROM dba_encrypted_columns) THEN 'PRESENT' ELSE 'ABSENT' END FROM dual UNION ALL SELECT 'RMAN Backup Encryption', CASE WHEN EXISTS (SELECT 1 FROM v\$rman_configuration WHERE name LIKE '%ENCRYPTION%') THEN 'CONFIGURED' ELSE 'NOT_CONFIGURED' END FROM dual UNION ALL SELECT 'Network Encryption', CASE WHEN (SELECT value FROM v\$parameter WHERE name = 'sec_protocol_version') IS NOT NULL THEN 'CONFIGURED' ELSE 'NOT_CONFIGURED' END FROM dual
SEC-SQL-ENC-004-RC12 No Centralized Key Management Service
Condition: row_count > 0 — Encrypted tablespaces found — TDE is in use and keys need centralized management
Unencrypted Backups FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT tablespace_name, encrypted, encrypted, status FROM dba_tablespaces WHERE encrypted = 'YES' ORDER BY tablespace_name
SEC-SQL-ENC-004-RC13 Multiple Backup Destinations Not Coordinated
Condition: row_count > 0 — Backup pieces found — check if all are encrypted consistently
Unencrypted Backups FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 27 rows alert: Yes mail: No
SELECT DISTINCT device_type, completion_time, encrypted, status FROM v\$backup_piece WHERE completion_time > SYSDATE - 30 ORDER BY completion_time DESC FETCH FIRST 30 ROWS ONLY
SEC-SQL-ENC-004-RC14 Backup Retention Policy Overrides Encryption
Condition: row_count > 0 — Old unencrypted backups still retained — encryption policy not enforced retroactively
Unencrypted Backups FINNACLEDB_TEST: DETECTED 20 rows FLEXCUBE: DETECTED 10 rows alert: Yes mail: No
SELECT recid, stamp, device_type, handle, tag, completion_time, encrypted, bytes / 1024 / 1024 AS size_mb FROM v\$backup_piece WHERE encrypted = 'NO' AND status = 'A' AND completion_time < SYSDATE - 90 ORDER BY completion_time FETCH FIRST 20 ROWS ONLY
SEC-SQL-ENC-004-RC15 Developer and QA Databases Using Production Backups
Condition: row_count > 0 — TDE-encrypted columns found — data becomes accessible if wallet is copied with backup
Unencrypted Backups FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT owner, table_name, column_name, encryption_alg, salt FROM ALL_ENCRYPTED_COLUMNS WHERE owner NOT IN ('SYS', 'SYSTEM') ORDER BY owner, table_name
SEC-SQL-ENC-004-RC16 Backup Scripts Written Before Encryption Features Existed
Condition: unencrypted > 0 — Recent unencrypted backups confirm encryption not applied to backup scripts
Unencrypted Backups FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT COUNT(*) AS total_backups, SUM(CASE WHEN encrypted = 'NO' THEN 1 ELSE 0 END) AS unencrypted, SUM(CASE WHEN encrypted = 'YES' THEN 1 ELSE 0 END) AS encrypted FROM v\$backup_piece WHERE status = 'A' AND completion_time > SYSDATE - 30
SEC-SQL-ENC-005-RC01 Default/Legacy Cipher Suites Not Disabled
Condition: row_count > 0 — Active session encryption algorithms retrieved — check for weak algorithms
Weak Encryption Keys/Ciphers FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT network_service_banner FROM v\$session_connect_info WHERE sid = SYS_CONTEXT('USERENV', 'SID') AND (network_service_banner LIKE '%Encryption%' OR network_service_banner LIKE '%Crypto%') ORDER BY network_service_banner

SEC-SQL-ENC-005-RC02 Short Key Lengths Selected for Performance
Condition: row_count > 0 — Column-level TDE using less than AES256 — shorter key for performance
Weak Encryption Keys/Ciphers FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT owner, table_name, column_name, encryption_alg FROM ALL_ENCRYPTED_COLUMNS WHERE encryption_alg NOT LIKE 'AES 256%' AND owner NOT IN ('SYS', 'SYSTEM') ORDER BY owner, table_name
SEC-SQL-ENC-005-RC03 Deprecated Algorithm Still in Use
Condition: row_count > 0 — 3DES encryption still used in TDE — deprecated algorithm
Weak Encryption Keys/Ciphers FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT tablespace_name, encrypted FROM dba_tablespaces WHERE encrypted = 'YES' AND encrypted LIKE '%3DES%' UNION ALL SELECT table_name, encryption_alg FROM dba_encrypted_columns WHERE encryption_alg LIKE '%3DES%' ORDER BY 1
SEC-SQL-ENC-005-RC04 Certificate Signed with Weak Hash
Condition: row_count > 0 — Certificate or SSL errors that may relate to weak hash algorithms
Weak Encryption Keys/Ciphers FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 4 rows alert: Yes mail: No
SELECT originating_timestamp, message_text FROM v\$diag_alert_ext WHERE (message_text LIKE '%certificate%' OR message_text LIKE '%SSL%' OR message_text LIKE '%ORA-29024%' OR message_text LIKE '%ORA-28860%' OR message_text LIKE '%weak%' OR message_text LIKE '%SHA1%') AND originating_timestamp > SYSDATE - 90 ORDER BY originating_timestamp DESC FETCH FIRST 20 ROWS ONLY
SEC-SQL-ENC-005-RC05 Self-Signed Certificates with Weak Parameters
Condition: row_count > 0 — TLS/SSL active — certificate parameters need verification
Weak Encryption Keys/Ciphers FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT sid, network_service_banner FROM v\$session_connect_info WHERE sid = SYS_CONTEXT('USERENV', 'SID') AND (network_service_banner LIKE '%SSL%' OR network_service_banner LIKE '%TLS%') ORDER BY network_service_banner
SEC-SQL-ENC-005-RC06 Key Generation with Insufficient Entropy
Condition: row_count > 0 — Wallet configuration retrieved — software keystore on VM may have lower entropy
Weak Encryption Keys/Ciphers FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT wrl_type, wrl_parameter, status, wallet_type, keystore_mode FROM v\$encryption_wallet
SEC-SQL-ENC-005-RC07 Cipher Suite Configuration Not Updated
Condition: row_count > 0 — Network encryption cipher configuration retrieved for review
Weak Encryption Keys/Ciphers FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('sqlnet.encryption_types_server', 'sqlnet.encryption_types_client', 'sqlnet.crypto_checksum_types_server', 'sqlnet.crypto_checksum_types_client', 'ssl_cipher_suites') ORDER BY name
SEC-SQL-ENC-005-RC08 No Cipher Suite Hardening Standard
Condition: any value IS NULL OR any value = 'ACCEPTED' — Default encryption settings — no hardening applied, any algorithm accepted
Weak Encryption Keys/Ciphers FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('sqlnet.encryption_server', 'sqlnet.encryption_client', 'sqlnet.encryption_types_server', 'sqlnet.encryption_types_client', 'sqlnet.crypto_checksum_server', 'sqlnet.crypto_checksum_client') ORDER BY name
SEC-SQL-ENC-005-RC09 RSA Key Exchange Still Enabled
Condition: value IS NULL OR value LIKE '%RSA%' — SSL cipher suites not restricted or include RSA key exchange

Weak Encryption Keys/Ciphers FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name = 'ssl_cipher_suites'
SEC-SQL-ENC-005-RC10 CBC Mode Still Allowed in Configuration
Condition: row_count > 0 — CBC mode ciphers allowed or no cipher restriction — vulnerable to padding attacks
Weak Encryption Keys/Ciphers FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name = 'ssl_cipher_suites' AND (value LIKE '%CBC%' OR value IS NULL) ORDER BY name
SEC-SQL-ENC-005-RC11 TLS 1.2 Allows Weak Ciphers
Condition: value NOT LIKE '%1.2%' OR value LIKE '%1.0%' OR value LIKE '%1.1%' — Older TLS versions still allowed — enables cipher downgrade
Weak Encryption Keys/Ciphers FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name = 'ssl_version'
SEC-SQL-ENC-005-RC12 Key Rotation Policy Not Implemented
Condition: key_age_days > :max_key_age_days — TDE master key is older than policy allows — rotation overdue
Weak Encryption Keys/Ciphers FINNACLEDB_TEST: FAILED FLEXCUBE: FAILED alert: No mail: No
SELECT key_id, activation_time, ROUND(SYSDATE - activation_time) AS key_age_days FROM v\$encryption_keys WHERE activation_time = (SELECT MIN(activation_time) FROM v\$encryption_keys)
SEC-SQL-ENC-005-RC13 Asymmetric Key Weak Parameters
Condition: row_count > 0 — Encryption key infrastructure retrieved — key parameters need verification
Weak Encryption Keys/Ciphers FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT wrl_type, status, wallet_type, keystore_mode FROM v\$encryption_wallet UNION ALL SELECT 'TDE_KEY' AS wrl_type, TO_CHAR(activation_time, 'YYYY-MM-DD') AS status, keystore_type AS wallet_type, creator AS keystore_mode FROM v\$encryption_keys ORDER BY 1
SEC-SQL-ENC-005-RC14 No Monitoring of Negotiated Ciphers
Condition: row_count = 0 — No logon audit policy — negotiated ciphers not being tracked
Weak Encryption Keys/Ciphers FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT policy_name, audit_option FROM audit_unified_policies WHERE audit_option LIKE '%LOGON%' OR policy_name LIKE '%LOGON%' ORDER BY policy_name
SEC-SQL-ENC-005-RC15 Compliance Requirements Misunderstood
Condition: unencrypted_ts > 0 — Not all tablespaces encrypted — partial TDE deployment may not meet compliance
Weak Encryption Keys/Ciphers FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT (SELECT COUNT(*) FROM dba_tablespaces WHERE encrypted = 'YES') AS encrypted_ts, (SELECT COUNT(*) FROM dba_tablespaces WHERE encrypted = 'NO' AND contents = 'PERMANENT' AND tablespace_name NOT IN ('SYSTEM', 'SYSAUX')) AS unencrypted_ts, (SELECT COUNT(*) FROM dba_encrypted_columns) AS encrypted_columns, (SELECT status FROM v\$encryption_wallet WHERE ROWNUM = 1) AS wallet_status FROM dual
SEC-SQL-ENC-005-RC16 Vendor Default Cipher Suites Used Without Review
Condition: encryption banner shows weak algorithms or no encryption — Active session using weak or no encryption confirms default cipher suites in use
Weak Encryption Keys/Ciphers FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT network_service_banner FROM v\$session_connect_info WHERE sid = SYS_CONTEXT('USERENV', 'SID') AND network_service_banner IS NOT NULL
SEC-SQL-INJ-001-RC01 String concatenation for query building

Condition: row_count > 0 — DBMS_SQL.PARSE with concatenated queries is a SQL injection vector
Dynamic SQL in Stored Procedures FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 14 rows alert: Yes mail: No
SELECT owner, name, type, line, text FROM ALL_SOURCE WHERE UPPER(text) LIKE '%DBMS_SQL%' AND (UPPER(text) LIKE '% %' OR UPPER(text) LIKE '%PARSE%') AND owner NOT IN ('SYS', 'SYSTEM', 'XDB', 'MDSYS', 'CTXSYS', 'WMSYS', 'ORDSYS') AND type IN ('PROCEDURE', 'FUNCTION', 'PACKAGE BODY') ORDER BY owner, name, line FETCH FIRST 20 ROWS ONLY
SEC-SQL-INJ-001-RC02 Inadequate input validation and sanitization
Condition: row_count > 0 — PL/SQL objects using dynamic SQL without DBMS_ASSERT indicate inadequate input validation
Dynamic SQL in Stored Procedures FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 15 rows alert: Yes mail: No
SELECT DISTINCT owner, name, type FROM ALL_SOURCE WHERE (UPPER(text) LIKE '%EXECUTE IMMEDIATE%' OR UPPER(text) LIKE '%DBMS_SQL.PARSE%') AND owner NOT IN ('SYS', 'SYSTEM', 'XDB', 'MDSYS', 'CTXSYS') AND type IN ('PROCEDURE', 'FUNCTION', 'PACKAGE BODY') AND name NOT IN (SELECT DISTINCT name FROM ALL_SOURCE WHERE UPPER(text) LIKE '%DBMS_ASSERT%' AND owner NOT IN ('SYS', 'SYSTEM')) ORDER BY owner, name
SEC-SQL-INJ-001-RC03 Lack of awareness and training
Condition: concatenation_usage > bind_variable_usage for any owner — More concatenation than bind variable usage across schemas suggests widespread lack of awareness
Dynamic SQL in Stored Procedures FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 6 rows alert: Yes mail: No
SELECT owner, COUNT(DISTINCT name) AS objects_with_dynamic_sql, SUM(CASE WHEN UPPER(text) LIKE '%USING%' THEN 1 ELSE 0 END) AS bind_variable_usage, SUM(CASE WHEN UPPER(text) LIKE '% %' AND (UPPER(text) LIKE '%EXECUTE IMMEDIATE%' OR UPPER(text) LIKE '%DBMS_SQL%') THEN 1 ELSE 0 END) AS concatenation_usage FROM ALL_SOURCE WHERE (UPPER(text) LIKE '%EXECUTE IMMEDIATE%' OR UPPER(text) LIKE '%DBMS_SQL%') AND owner NOT IN ('SYS', 'SYSTEM', 'XDB', 'MDSYS', 'CTXSYS', 'WMSYS') AND type IN ('PROCEDURE', 'FUNCTION', 'PACKAGE BODY') GROUP BY owner ORDER BY concatenation_usage DESC
SEC-SQL-INJ-001-RC04 Legacy code and technical debt
Condition: row_count > 0 — Old PL/SQL objects with concatenated dynamic SQL represent legacy code technical debt
Dynamic SQL in Stored Procedures FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 6 rows alert: Yes mail: No
SELECT o.owner, o.object_name, o.object_type, o.created, o.last_ddl_time, ROUND(SYSDATE - o.created) AS age_days FROM ALL_OBJECTS o WHERE o.object_type IN ('PROCEDURE', 'FUNCTION', 'PACKAGE BODY') AND o.owner NOT IN ('SYS', 'SYSTEM', 'XDB', 'MDSYS', 'CTXSYS') AND o.created < SYSDATE - 1095 AND EXISTS (SELECT 1 FROM ALL_SOURCE s WHERE s.owner = o.owner AND s.name = o.object_name AND (UPPER(s.text) LIKE '%EXECUTE IMMEDIATE% %' OR UPPER(s.text) LIKE '%DBMS_SQL% %')) ORDER BY o.created ASC FETCH FIRST 20 ROWS ONLY
SEC-SQL-INJ-001-RC05 Time pressure and deadline constraints
Condition: row_count > 0 — Recently modified PL/SQL with unsafe dynamic SQL patterns suggest shortcuts from time pressure
Dynamic SQL in Stored Procedures FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 3 rows alert: Yes mail: No
SELECT o.owner, o.object_name, o.object_type, o.last_ddl_time FROM ALL_OBJECTS o WHERE o.object_type IN ('PROCEDURE', 'FUNCTION', 'PACKAGE BODY', 'TRIGGER') AND o.owner NOT IN ('SYS', 'SYSTEM', 'XDB', 'MDSYS', 'CTXSYS') AND o.last_ddl_time > SYSDATE - 90 AND EXISTS (SELECT 1 FROM ALL_SOURCE s WHERE s.owner = o.owner AND s.name = o.object_name AND UPPER(s.text) LIKE '%EXECUTE IMMEDIATE% %' AND UPPER(s.text) NOT LIKE '%DBMS_ASSERT%' AND UPPER(s.text) NOT LIKE '%USING%') ORDER BY o.last_ddl_time DESC
SEC-SQL-INJ-001-RC06 Insufficient security code reviews
Condition: row_count > 0 with high unsafe_objects count — High concentration of unsafe dynamic SQL across schemas suggests code review process gaps
Dynamic SQL in Stored Procedures FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 4 rows alert: Yes mail: No
SELECT owner, COUNT(DISTINCT name) AS total_plsql_objects, SUM(CASE WHEN EXISTS (SELECT 1 FROM ALL_SOURCE s2 WHERE s2.owner = s1.owner AND s2.name = s1.name AND (UPPER(s2.text) LIKE '%EXECUTE IMMEDIATE% %' OR UPPER(s2.text) LIKE '%DBMS_SQL% %')) THEN 1 ELSE 0 END) AS unsafe_objects FROM (SELECT DISTINCT owner, name FROM ALL_SOURCE WHERE owner NOT IN ('SYS', 'SYSTEM', 'XDB', 'MDSYS', 'CTXSYS') AND type IN ('PROCEDURE', 'FUNCTION', 'PACKAGE BODY')) s1 GROUP BY owner HAVING SUM(CASE WHEN EXISTS (SELECT 1 FROM ALL_SOURCE s2 WHERE s2.owner = s1.owner AND s2.name = s1.name AND (UPPER(s2.text) LIKE '%EXECUTE IMMEDIATE% %' OR UPPER(s2.text) LIKE '%DBMS_SQL% %')) THEN 1 ELSE 0 END) > 0 ORDER BY unsafe_objects DESC
SEC-SQL-INJ-001-RC07 Mixing static and dynamic SQL constructs
Condition: row_count > 0 — Dynamic WHERE clause construction with concatenation creates SQL injection risk even within prepared statements
Dynamic SQL in Stored Procedures FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 8 rows alert: Yes mail: No
SELECT owner, name, type, line, text FROM ALL_SOURCE WHERE UPPER(text) LIKE '%WHERE% %' AND (UPPER(text) LIKE '%EXECUTE IMMEDIATE%' OR UPPER(text) LIKE '%OPEN%FOR%') AND owner NOT IN ('SYS', 'SYSTEM', 'XDB', 'MDSYS', 'CTXSYS') AND type IN ('PROCEDURE', 'FUNCTION', 'PACKAGE BODY') ORDER BY owner, name, line FETCH FIRST 20 ROWS ONLY

SEC-SQL-INJ-001-RC08 Complex business requirements
Condition: row_count > 0 with high dynamic_sql_lines — PL/SQL objects with many dynamic SQL lines indicate complex business requirements driving dynamic SQL usage
Dynamic SQL in Stored Procedures FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 5 rows alert: Yes mail: No
SELECT owner, name, type, COUNT(*) AS dynamic_sql_lines FROM ALL_SOURCE WHERE (UPPER(text) LIKE '%EXECUTE IMMEDIATE%' OR UPPER(text) LIKE '%DBMS_SQL%') AND owner NOT IN ('SYS', 'SYSTEM', 'XDB', 'MDSYS', 'CTXSYS') AND type IN ('PROCEDURE', 'FUNCTION', 'PACKAGE BODY') GROUP BY owner, name, type HAVING COUNT(*) > 5 ORDER BY dynamic_sql_lines DESC
SEC-SQL-INJ-001-RC09 Performance optimization assumptions
Condition: row_count > 0 — Static SQL executed via EXECUTE IMMEDIATE suggests misguided performance optimization
Dynamic SQL in Stored Procedures FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 15 rows alert: Yes mail: No
SELECT owner, name, type, line, text FROM ALL_SOURCE WHERE UPPER(text) LIKE '%EXECUTE IMMEDIATE%SELECT%' AND UPPER(text) NOT LIKE '% %' AND owner NOT IN ('SYS', 'SYSTEM', 'XDB', 'MDSYS', 'CTXSYS') AND type IN ('PROCEDURE', 'FUNCTION', 'PACKAGE BODY') ORDER BY owner, name, line FETCH FIRST 20 ROWS ONLY
SEC-SQL-INJ-001-RC10 ORM framework misuse
Condition: hard_parse_ratio (hard/total) > :hard_parse_threshold — High hard parse ratio suggests applications using literal SQL instead of parameterized queries
Dynamic SQL in Stored Procedures FINNACLEDB_TEST: DETECTED 4 rows FLEXCUBE: DETECTED 4 rows alert: Yes mail: No
SELECT name, value FROM v\$sysstat WHERE name IN ('parse count (total)', 'parse count (hard)', 'parse count (failures)', 'execute count') ORDER BY name
SEC-SQL-INJ-001-RC11 Absence of security testing tools
Condition: unsafe_objects > :threshold — Many PL/SQL objects with unsafe dynamic SQL and no DBMS_ASSERT suggests no SAST tooling in the pipeline
Dynamic SQL in Stored Procedures FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT COUNT(DISTINCT owner '.' name) AS unsafe_objects FROM ALL_SOURCE WHERE (UPPER(text) LIKE '%EXECUTE IMMEDIATE% %' OR UPPER(text) LIKE '%DBMS_SQL.PARSE%') AND UPPER(text) NOT LIKE '%DBMS_ASSERT%' AND owner NOT IN ('SYS', 'SYSTEM', 'XDB', 'MDSYS', 'CTXSYS', 'WMSYS', 'ORDSYS') AND type IN ('PROCEDURE', 'FUNCTION', 'PACKAGE BODY', 'TRIGGER')
SEC-SQL-INJ-001-RC12 Third-party library dependencies
Condition: row_count > 0 with wrapped third-party packages — Wrapped (obfuscated) PL/SQL packages cannot be inspected for SQL injection patterns
Dynamic SQL in Stored Procedures FINNACLEDB_TEST: DETECTED 17 rows FLEXCUBE: DETECTED 20 rows alert: Yes mail: No
SELECT owner, object_name, object_type FROM ALL_OBJECTS WHERE object_type IN ('PACKAGE BODY', 'PROCEDURE', 'FUNCTION') AND owner NOT IN ('SYS', 'SYSTEM', 'XDB', 'MDSYS', 'CTXSYS', 'WMSYS', 'ORDSYS') AND NOT EXISTS (SELECT 1 FROM ALL_SOURCE WHERE owner = ALL_OBJECTS.owner AND name = ALL_OBJECTS.object_name AND type = ALL_OBJECTS.object_type AND ROWNUM = 1 AND text NOT LIKE '%wrapped%') ORDER BY owner, object_name FETCH FIRST 20 ROWS ONLY
SEC-SQL-INJ-001-RC13 Operator control issues
Condition: row_count > 0 — OS command execution objects or EXECUTABLE scheduler jobs indicate shell access from database
Dynamic SQL in Stored Procedures FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 9 rows alert: Yes mail: No
SELECT owner, object_name, object_type FROM dba_objects WHERE (object_name LIKE '%OS_COMMAND%' OR object_name LIKE '%SHELL%' OR object_name LIKE '%HOST_COMMAND%') AND object_type IN ('PROCEDURE', 'FUNCTION', 'JAVA CLASS', 'JAVA SOURCE') AND owner NOT IN ('SYS', 'SYSTEM') UNION ALL SELECT owner, job_name, job_type FROM dba_scheduler_jobs WHERE job_type = 'EXECUTABLE' AND owner NOT IN ('SYS', 'SYSTEM') ORDER BY 1, 2
SEC-SQL-INJ-001-RC14 Implicit trust in framework defaults
Condition: row_count > 0 — Stored procedures with internal dynamic SQL are not inherently safe from SQL injection
Dynamic SQL in Stored Procedures FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 18 rows alert: Yes mail: No
SELECT owner, name, type, COUNT(*) AS dynamic_sql_count FROM ALL_SOURCE WHERE (UPPER(text) LIKE '%EXECUTE IMMEDIATE%' OR UPPER(text) LIKE '%DBMS_SQL%') AND type IN ('PROCEDURE', 'FUNCTION', 'PACKAGE BODY') AND owner NOT IN ('SYS', 'SYSTEM', 'XDB', 'MDSYS', 'CTXSYS', 'WMSYS') GROUP BY owner, name, type ORDER BY dynamic_sql_count DESC FETCH FIRST 20 ROWS ONLY
SEC-SQL-INJ-001-RC15 Incomplete parameter binding

Condition: row_count > 0 — EXECUTE IMMEDIATE with both concatenation (for identifiers) and USING (for values) indicates partial binding
Dynamic SQL in Stored Procedures FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT owner, name, type, line, text FROM ALL_SOURCE WHERE UPPER(text) LIKE '%EXECUTE IMMEDIATE% %USING%' AND owner NOT IN ('SYS', 'SYSTEM', 'XDB', 'MDSYS', 'CTXSYS') AND type IN ('PROCEDURE', 'FUNCTION', 'PACKAGE BODY') ORDER BY owner, name, line FETCH FIRST 20 ROWS ONLY
SEC-SQL-INJ-002-RC02 Backward compatibility requirements
Condition: row_count > 0 with very old Java objects — Old Java stored procedures maintained for backward compatibility likely include OS execution capabilities
Extended Stored Procedures in Use FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 15 rows alert: Yes mail: No
SELECT o.owner, o.object_name, o.object_type, o.created, o.last_ddl_time, ROUND(SYSDATE - o.created) AS age_days FROM ALL_OBJECTS o WHERE o.object_type IN ('JAVA CLASS', 'JAVA SOURCE') AND o.owner NOT IN ('SYS', 'SYSTEM', 'XDB', 'ORDSYS', 'MDSYS') AND o.created < SYSDATE - 1095 ORDER BY o.created ASC FETCH FIRST 15 ROWS ONLY
SEC-SQL-INJ-002-RC05 High barrier to refactoring
Condition: row_count > 0 with high dependent_objects count — Many objects depending on Java/external procedure execution indicate high refactoring cost
Extended Stored Procedures in Use FINNACLEDB_TEST: DETECTED 821 rows FLEXCUBE: DETECTED 1202 rows alert: Yes mail: No
SELECT d.referenced_owner, d.referenced_name, d.referenced_type, COUNT(*) AS dependent_objects FROM ALL_DEPENDENCIES d WHERE d.referenced_type IN ('JAVA CLASS', 'JAVA SOURCE') AND d.referenced_owner NOT IN ('SYS', 'SYSTEM') GROUP BY d.referenced_owner, d.referenced_name, d.referenced_type HAVING COUNT(*) > 3 ORDER BY dependent_objects DESC
SEC-SQL-INJ-002-RC06 Operational automation dependencies
Condition: row_count > 0 — EXECUTABLE scheduler jobs directly invoke OS commands for operational tasks
Extended Stored Procedures in Use FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT owner, job_name, job_type, job_action, enabled, state, last_start_date, next_run_date FROM dba_scheduler_jobs WHERE job_type = 'EXECUTABLE' AND owner NOT IN ('SYS', 'SYSTEM') ORDER BY owner, job_name
SEC-SQL-INJ-002-RC07 Insufficient privilege controls
Condition: row_count > 0 — Non-admin users with OS execution privileges indicate excessive privilege grants
Extended Stored Procedures in Use FINNACLEDB_TEST: DETECTED 37 rows FLEXCUBE: DETECTED 61 rows alert: Yes mail: No
SELECT grantee, privilege FROM dba_sys_privs WHERE privilege IN ('CREATE EXTERNAL JOB', 'CREATE ANY JOB', 'MANAGE SCHEDULER') AND grantee NOT IN ('SYS', 'SYSTEM', 'DBA', 'SCHEDULER_ADMIN') UNION ALL SELECT grantee, type_name ':' name ':' action AS privilege FROM dba_java_policy WHERE (type_name = 'java.lang.RuntimePermission' OR (type_name = 'java.io.FilePermission' AND action LIKE '%execute%')) AND grantee NOT IN ('SYS', 'SYSTEM', 'JAVA_ADMIN') ORDER BY grantee
SEC-SQL-INJ-002-RC08 Missing monitoring and enforcement
Condition: non_bind_sql_count > :threshold — SQL statements with injection-like patterns found — monitoring and enforcement are missing
Extended Stored Procedures in Use FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT COUNT(*) AS non_bind_sql_count FROM v\$sql WHERE executions = 1 AND parsing_schema_name NOT IN ('SYS', 'SYSTEM') AND (sql_text LIKE '%' OR %' OR sql_text LIKE '%' AND %' OR sql_text LIKE '%1=1%') AND ROWNUM <= 50
SEC-SQL-INJ-002-RC09 Assumption of network isolation
Condition: sql_firewall = 'FALSE' AND db_vault = 'FALSE' AND audit_trail = 'NONE' — No database-level SQL injection protections enabled — relying only on network isolation
Extended Stored Procedures in Use FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT 'sql_firewall' AS protection, value FROM v\$option WHERE parameter = 'SQL Firewall' UNION ALL SELECT 'db_vault', value FROM v\$option WHERE parameter = 'Oracle Database Vault' UNION ALL SELECT 'audit_trail', value FROM v\$parameter WHERE name = 'audit_trail'
SEC-SQL-INJ-002-RC10 Security compliance misalignment
Condition: FAIL_count > 0 — One or more compliance security checks failed — SQL injection risk not addressed per compliance standards

Extended Stored Procedures in Use FINNACLEDB_TEST: DETECTED 3 rows FLEXCUBE: DETECTED 3 rows alert: Yes mail: No
SELECT 'audit_enabled' AS check_item, CASE WHEN value != 'NONE' THEN 'PASS' ELSE 'FAIL' END AS status FROM v\$parameter WHERE name = 'audit_trail' UNION ALL SELECT 'password_complexity', CASE WHEN limit != 'NULL' AND limit IS NOT NULL THEN 'PASS' ELSE 'FAIL' END FROM dba_profiles WHERE profile = 'DEFAULT' AND resource_name = 'PASSWORD_VERIFY_FUNCTION' UNION ALL SELECT 'failed_login_limit', CASE WHEN limit != 'UNLIMITED' THEN 'PASS' ELSE 'FAIL' END FROM dba_profiles WHERE profile = 'DEFAULT' AND resource_name = 'FAILED_LOGIN_ATTEMPTS'
SEC-SQL-INJ-002-RC12 Third-party application requirements
Condition: row_count > 0 — Third-party applications generating unique SQL per execution — likely using string concatenation instead of bind variables
Extended Stored Procedures in Use FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT s.username, s.program, s.module, COUNT(*) AS one_exec_sql, SUM(q.sharable_mem) / 1024 / 1024 AS wasted_mb FROM v\$session s JOIN v\$sql q ON s.sql_id = q.sql_id WHERE q.executions = 1 AND s.type = 'USER' AND s.username NOT IN ('SYS', 'SYSTEM') GROUP BY s.username, s.program, s.module HAVING COUNT(*) > 100 ORDER BY one_exec_sql DESC FETCH FIRST 10 ROWS ONLY
SEC-SQL-INJ-002-RC13 Knowledge loss and staff turnover
Condition: SUM(count) > 1 — Multiple security features dormant or degraded — suggests knowledge loss from staff turnover
Extended Stored Procedures in Use FINNACLEDB_TEST: DETECTED 3 rows FLEXCUBE: DETECTED 3 rows alert: Yes mail: No
SELECT 'dormant_audit_policies' AS indicator, COUNT(*) AS count FROM audit_unified_enabled_policies WHERE policy_name NOT LIKE 'ORA_%' UNION ALL SELECT 'stale_password_profiles', COUNT(*) FROM dba_profiles WHERE resource_name = 'PASSWORD_VERIFY_FUNCTION' AND (limit = 'NULL' OR limit IS NULL) UNION ALL SELECT 'no_recent_security_patches', CASE WHEN (SELECT MAX(action_time) FROM dba_registry_history WHERE action = 'APPLY') < SYSDATE - 365 THEN 1 ELSE 0 END FROM dual
SEC-SQL-INJ-002-RC16 Insufficient removal validation
Condition: SUM(present) >= 3 — Multiple security control layers are absent — protections removed without alternative controls
Extended Stored Procedures in Use FINNACLEDB_TEST: DETECTED 4 rows FLEXCUBE: DETECTED 4 rows alert: Yes mail: No
SELECT 'no_sql_firewall' AS gap, CASE WHEN (SELECT value FROM v\$option WHERE parameter = 'SQL Firewall') = 'FALSE' THEN 1 ELSE 0 END AS present FROM dual UNION ALL SELECT 'no_db_vault', CASE WHEN (SELECT value FROM v\$option WHERE parameter = 'Oracle Database Vault') = 'FALSE' THEN 1 ELSE 0 END FROM dual UNION ALL SELECT 'no_audit', CASE WHEN (SELECT value FROM v\$parameter WHERE name = 'audit_trail') = 'NONE' THEN 1 ELSE 0 END FROM dual UNION ALL SELECT 'no_vpd', CASE WHEN (SELECT COUNT(*) FROM dba_policies WHERE enable = 'YES') = 0 THEN 1 ELSE 0 END FROM dual
SEC-SQL-NET-001-RC01 Bind Address Misconfiguration
Condition: row_count > :expected_hosts — Connections from many different hosts — listener is accepting connections from a wide network range
Database Exposed to rootcause Internet FINNACLEDB_TEST: DETECTED 4 rows FLEXCUBE: DETECTED 10 rows alert: Yes mail: No
SELECT DISTINCT s.machine, COUNT(*) AS connections FROM v\$session s WHERE s.type = 'USER' GROUP BY s.machine ORDER BY connections DESC
SEC-SQL-NET-001-RC02 Overly Permissive Security Group/Firewall Rules
Condition: row_count > 0 — Failed login attempts from multiple hosts — brute force attempts due to permissive network access
Database Exposed to rootcause Internet FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT userhost, COUNT(*) AS failed_attempts FROM dba_audit_trail WHERE action_name = 'LOGON' AND returncode != 0 AND timestamp > SYSDATE - 7 GROUP BY userhost HAVING COUNT(*) > 10 ORDER BY failed_attempts DESC FETCH FIRST 20 ROWS ONLY
SEC-SQL-NET-001-RC03 rootcause Cloud Instance Configuration
Condition: row_count > :subnet_threshold — Connections from many different network segments — cloud instance may have public IP
Database Exposed to rootcause Internet FINNACLEDB_TEST: DETECTED 4 rows FLEXCUBE: DETECTED 10 rows alert: Yes mail: No
SELECT DISTINCT SUBSTR(machine, 1, 30) AS host_pattern, COUNT(*) AS connection_count FROM v\$session WHERE type = 'USER' GROUP BY SUBSTR(machine, 1, 30) ORDER BY connection_count DESC
SEC-SQL-NET-001-RC04 Inadequate Network Segmentation
Condition: row_count > :host_threshold — Connections from many different hosts — database accessible from multiple network segments
Database Exposed to rootcause Internet FINNACLEDB_TEST: DETECTED 5 rows FLEXCUBE: DETECTED 30 rows alert: Yes mail: No

SELECT DISTINCT machine, program, username, COUNT(*) AS sessions FROM v\$session WHERE type = 'USER' AND machine IS NOT NULL GROUP BY machine, program, username ORDER BY sessions DESC FETCH FIRST 30 ROWS ONLY
SEC-SQL-NET-001-RC05 Lack of IP Allowlist/Whitelist Enforcement
Condition: row_count > :expected_sources — More connection sources than expected — no IP allowlist restricting database access
Database Exposed to rootcause Internet FINNACLEDB_TEST: DETECTED 4 rows FLEXCUBE: DETECTED 9 rows alert: Yes mail: No
SELECT DISTINCT machine, COUNT(*) AS session_count FROM v\$session WHERE type = 'USER' AND machine NOT IN (SELECT host_name FROM v\$instance) GROUP BY machine ORDER BY session_count DESC
SEC-SQL-NET-001-RC06 Disabled or Absent Firewall Rules
Condition: row_count > :threshold — Multiple failed logins from different hosts — listener port is accessible without firewall filtering
Database Exposed to rootcause Internet FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 3 rows alert: Yes mail: No
SELECT userhost, username, COUNT(*) AS failed_count FROM dba_audit_trail WHERE action_name = 'LOGON' AND returncode IN (1017, 28000, 28001) AND timestamp > SYSDATE - 7 GROUP BY userhost, username ORDER BY failed_count DESC FETCH FIRST 20 ROWS ONLY
SEC-SQL-NET-001-RC07 Missing Virtual Private Network (VPN) Requirements
Condition: unique_hosts > :vpn_threshold — Many unique connection sources — suggests direct access without VPN aggregation
Database Exposed to rootcause Internet FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT COUNT(DISTINCT machine) AS unique_hosts, COUNT(DISTINCT program) AS unique_programs, COUNT(*) AS total_sessions FROM v\$session WHERE type = 'USER'
SEC-SQL-NET-001-RC08 Cloud Provider Shared Responsibility Model Misunderstanding
Condition: NO_count > 1 — Multiple database-level security features disabled — relying solely on cloud infrastructure controls
Database Exposed to rootcause Internet FINNACLEDB_TEST: DETECTED 3 rows FLEXCUBE: DETECTED 3 rows alert: Yes mail: No
SELECT 'tde_enabled' AS feature, CASE WHEN (SELECT COUNT(*) FROM v\$encryption_wallet WHERE status = 'OPEN') > 0 THEN 'YES' ELSE 'NO' END AS status FROM dual UNION ALL SELECT 'network_encryption', CASE WHEN (SELECT value FROM v\$parameter WHERE name = 'sqlnet.encryption_server') IS NOT NULL THEN 'YES' ELSE 'NO' END FROM dual UNION ALL SELECT 'audit_enabled', CASE WHEN (SELECT value FROM v\$parameter WHERE name = 'audit_trail') != 'NONE' THEN 'YES' ELSE 'NO' END FROM dual
SEC-SQL-NET-001-RC09 Incomplete Infrastructure as Code Security
Condition: row_count > :user_threshold — Many user accounts with various creation dates — suggests manual user management not driven by IaC
Database Exposed to rootcause Internet FINNACLEDB_TEST: DETECTED 11 rows FLEXCUBE: DETECTED 30 rows alert: Yes mail: No
SELECT username, profile, default_tablespace, TO_CHAR(created, 'YYYY-MM-DD') AS created FROM dba_users WHERE oracle_maintained = 'N' AND username NOT IN ('SYS', 'SYSTEM') AND account_status = 'OPEN' ORDER BY created DESC
SEC-SQL-NET-001-RC10 No Network Access Control List (NACL) Restrictions
Condition: row_count = 0 OR host = '*' — No network ACLs or wildcard ACL — database can connect to any external host without restriction
Database Exposed to rootcause Internet FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 3 rows alert: Yes mail: No
SELECT host, lower_port, upper_port, acl FROM dba_network_acls ORDER BY host
SEC-SQL-NET-001-RC11 Shadow IT / Unauthorized Deployments
Condition: SUM(present) >= 3 — Multiple management gaps — database appears to be unmanaged, possibly shadow IT
Database Exposed to rootcause Internet FINNACLEDB_TEST: DETECTED 4 rows FLEXCUBE: DETECTED 4 rows alert: Yes mail: Yes (destructive)
SELECT 'no_patches' AS indicator, CASE WHEN (SELECT COUNT(*) FROM dba_registry_history WHERE action = 'APPLY') = 0 THEN 1 ELSE 0 END AS present FROM dual UNION ALL SELECT 'default_passwords', CASE WHEN (SELECT COUNT(*) FROM dba_users_with_defpwd) > 3 THEN 1 ELSE 0 END FROM dual UNION ALL SELECT 'no_backup', CASE WHEN (SELECT COUNT(*) FROM v\$rman_configuration) = 0 THEN 1 ELSE 0 END FROM dual UNION ALL SELECT 'no_auditing', CASE WHEN (SELECT value FROM v\$parameter WHERE name = 'audit_trail') = 'NONE' THEN 1 ELSE 0 END FROM dual

SEC-SQL-NET-001-RC12 Temporary Access Left Permanently
Condition: last_login IS NULL OR last_login < SYSDATE - 90 — Active accounts with no recent login activity — dormant accounts that should have been removed
Database Exposed to rootcause Internet FINNACLEDB_TEST: DETECTED 8 rows FLEXCUBE: DETECTED 21 rows alert: Yes mail: No
SELECT u.username, u.account_status, TO_CHAR(u.created, 'YYYY-MM-DD') AS created, (SELECT MAX(TO_CHAR(timestamp, 'YYYY-MM-DD')) FROM dba_audit_trail a WHERE a.username = u.username AND a.action_name = 'LOGON' AND a.returncode = 0) AS last_login FROM dba_users u WHERE u.oracle_maintained = 'N' AND u.account_status = 'OPEN' AND u.username NOT IN ('SYS', 'SYSTEM') AND u.created < SYSDATE - 180 ORDER BY u.created
SEC-SQL-NET-001-RC13 DNS Misconfiguration
Condition: row_count > 0 — Database links use DNS hostnames instead of IP addresses — vulnerable to DNS poisoning
Database Exposed to rootcause Internet FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT db_link, host FROM ALL_DB_LINKS WHERE host LIKE '%.%' AND host NOT LIKE '%.%.%.%' ORDER BY db_link
SEC-SQL-NET-001-RC14 No Default-Deny Network Policy
Condition: NOT_CONFIGURED_count >= 2 — Multiple network access controls not configured — default-allow policy in effect
Database Exposed to rootcause Internet FINNACLEDB_TEST: DETECTED 3 rows FLEXCUBE: DETECTED 3 rows alert: Yes mail: No
SELECT 'tcp_validnode_checking' AS control, CASE WHEN (SELECT COUNT(*) FROM v\$parameter WHERE name LIKE '%valid%node%' AND value IS NOT NULL) > 0 THEN 'CONFIGURED' ELSE 'NOT_CONFIGURED' END AS status FROM dual UNION ALL SELECT 'network_acls', CASE WHEN (SELECT COUNT(*) FROM dba_network_acls) > 0 THEN 'CONFIGURED' ELSE 'NOT_CONFIGURED' END FROM dual UNION ALL SELECT 'login_profile', CASE WHEN (SELECT limit FROM dba_profiles WHERE profile = 'DEFAULT' AND resource_name = 'FAILED_LOGIN_ATTEMPTS') != 'UNLIMITED' THEN 'CONFIGURED' ELSE 'NOT_CONFIGURED' END FROM dual
SEC-SQL-NET-001-RC15 Absence of Host-Based Firewalls
Condition: row_count > :threshold_sources — Connections from many different hosts suggest no host-based firewall restriction
Database Exposed to rootcause Internet FINNACLEDB_TEST: DETECTED 4 rows FLEXCUBE: DETECTED 10 rows alert: Yes mail: No
SELECT machine, COUNT(*) AS connection_count, COUNT(DISTINCT username) AS distinct_users FROM v\$session WHERE type = 'USER' AND machine IS NOT NULL GROUP BY machine ORDER BY connection_count DESC
SEC-SQL-NET-001-RC16 Infrastructure Drift from Code
Condition: row_count > :threshold_modified — Many runtime-modified parameters suggest drift from declarative configuration
Database Exposed to rootcause Internet FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: clean alert: Yes mail: No
SELECT name, value, isdefault, ismodified FROM v\$parameter WHERE ismodified != 'FALSE' AND name NOT LIKE '%_oracle%' AND name NOT LIKE '%diagnostic%' ORDER BY name
SEC-SQL-NET-002-RC01 Default Configuration Never Changed
Condition: row_count > 0 — Check listener configuration for default port usage
Default Ports in Use FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name = 'local_listener' ORDER BY name
SEC-SQL-NET-002-RC02 Lack of Security Hardening Process
Condition: row_count > :threshold_defaults — Many security parameters at defaults indicate no hardening process was applied
Default Ports in Use FINNACLEDB_TEST: DETECTED 5 rows FLEXCUBE: DETECTED 5 rows alert: Yes mail: No
SELECT name, value, isdefault FROM v\$parameter WHERE name IN ('audit_trail', 'audit_sys_operations', 'sec_case_sensitive_logon', 'remote_os_authent', '07_DICTIONARY_ACCESSIBILITY', 'remote_login_passwordfile', 'sql92_security', 'sec_max_failed_login_attempts') AND isdefault = 'TRUE' ORDER BY name
SEC-SQL-NET-002-RC03 Convenience and Standardization
Condition: row_count > 0 — Sample accounts still open confirm convenience-driven configuration approach

Default Ports in Use FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT username, account_status FROM dba_users WHERE username IN ('SCOTT', 'HR', 'OE', 'SH', 'DEMO') AND account_status = 'OPEN' ORDER BY username
SEC-SQL-NET-002-RC04 Documentation Focuses on Default Values
Condition: row_count > 0 — Tutorial/documentation sample accounts still active in production
Default Ports in Use FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT username, account_status FROM dba_users WHERE username IN ('SCOTT', 'HR', 'OE', 'SH', 'PM', 'IX', 'BI') AND account_status = 'OPEN' ORDER BY username
SEC-SQL-NET-002-RC05 Port Scanning Facilitation
Condition: row_count > 0 — Listener configuration exposed on known ports facilitates scanning
Default Ports in Use FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('local_listener', 'remote_listener', 'dispatchers') AND value IS NOT NULL ORDER BY name
SEC-SQL-NET-002-RC06 Application Connection String Hardcoding
Condition: distinct_programs > :threshold_programs — Many distinct programs connecting suggests widespread hardcoded connection configurations
Default Ports in Use FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT COUNT(DISTINCT program) AS distinct_programs, COUNT(DISTINCT machine) AS distinct_machines FROM v\$session WHERE type = 'USER'
SEC-SQL-NET-002-RC07 Monitoring and Observability Tied to Defaults
Condition: row_count > 0 — Monitoring tools connected that may have default port assumptions
Default Ports in Use FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT s.username, s.program, s.module, s.machine, COUNT(*) AS session_count FROM v\$session s WHERE s.program LIKE '%OMS%' OR s.program LIKE '%agent%' OR s.program LIKE '%emctl%' OR s.program LIKE '%oem%' OR s.program LIKE '%nagios%' OR s.program LIKE '%zabbix%' OR s.program LIKE '%datadog%' OR s.username IN ('DBSNMP', 'SYSMAN', 'OEM_MONITOR') GROUP BY s.username, s.program, s.module, s.machine ORDER BY session_count DESC
SEC-SQL-NET-002-RC08 No Port Randomization or Obscurity Strategy
Condition: row_count > 0 — Listener on standard Oracle ports without obscurity measures
Default Ports in Use FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name = 'local_listener' AND (value LIKE '%1521%' OR value LIKE '%1526%' OR value LIKE '%2483%' OR value LIKE '%2484%' OR value IS NULL) ORDER BY name
SEC-SQL-NET-002-RC09 Infrastructure as Code Using Defaults
Condition: row_count > 0 — Default listener configuration suggests IaC template defaults in use
Default Ports in Use FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: clean alert: Yes mail: No
SELECT name, value, isdefault FROM v\$parameter WHERE name IN ('local_listener', 'dispatchers') AND (isdefault = 'TRUE' OR value LIKE '%1521%') ORDER BY name
SEC-SQL-NET-002-RC10 Assumption Port Numbers Alone Provide Security
Condition: row_count > 0 — Listener configured - check for additional network security beyond port
Default Ports in Use FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: clean alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('local_listener') AND value IS NOT NULL ORDER BY name
SEC-SQL-NET-002-RC11 No Centralized Port Management
Condition: row_count > 1 — Multiple listener endpoints suggest uncoordinated port management

Default Ports in Use FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('local_listener', 'remote_listener', 'dispatchers') AND value IS NOT NULL ORDER BY name
SEC-SQL-NET-002-RC12 Version Disclosure via Banners
Condition: row_count > 0 — Detailed version information is available and likely exposed through listener
Default Ports in Use FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT banner, banner_full, banner_legacy FROM v\$version
SEC-SQL-NET-002-RC13 Brute Force Attack Simplification
Condition: row_count > 0 — Unlimited login attempts or default lockout makes brute force feasible
Default Ports in Use FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 5 rows alert: Yes mail: No
SELECT profile, resource_name, limit FROM dba_profiles WHERE resource_name IN ('FAILED_LOGIN_ATTEMPTS', 'PASSWORD_LOCK_TIME') AND (limit = 'UNLIMITED' OR limit = 'DEFAULT') ORDER BY profile, resource_name
SEC-SQL-NET-002-RC14 Lack of Port Rotation Practices
Condition: uptime_days > :threshold_days — Long uptime on same configuration indicates no port rotation practices
Default Ports in Use FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT instance_name, startup_time, ROUND(SYSDATE - startup_time) AS uptime_days, version FROM v\$instance
SEC-SQL-NET-002-RC15 Automated Scanning Vulnerability
Condition: row_count > 0 — Listener on well-known ports is easily discovered by automated scanners
Default Ports in Use FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name = 'local_listener' AND (value LIKE '%1521%' OR value LIKE '%1526%' OR value IS NULL) ORDER BY name
SEC-SQL-NET-003-RC01 Default Allow Firewall Posture
Condition: distinct_hosts > :threshold_hosts — Connections from many distinct hosts suggest default-allow network posture
Unrestricted Outbound Connections FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT COUNT(DISTINCT machine) AS distinct_hosts, COUNT(DISTINCT program) AS distinct_programs, COUNT(*) AS total_sessions FROM v\$session WHERE type = 'USER'
SEC-SQL-NET-003-RC02 Egress Filtering Not Implemented
Condition: row_count > 0 — Database links to external systems confirm outbound data paths without egress control
Unrestricted Outbound Connections FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT owner, db_link, host FROM ALL_DB_LINKS WHERE host IS NOT NULL ORDER BY owner, db_link
SEC-SQL-NET-003-RC03 Database Features Requiring Internet Access
Condition: row_count > 0 — Network ACLs targeting non-private IP addresses confirm external internet access configuration
Unrestricted Outbound Connections FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT host, lower_port, upper_port, acl FROM dba_network_acls WHERE host NOT LIKE '10.%' AND host NOT LIKE '172.%' AND host NOT LIKE '192.168.%' AND host != 'localhost' ORDER BY host
SEC-SQL-NET-003-RC04 External data access configuration
Condition: row_count > 0 — Database links enable outbound connections to external databases without network policy restrictions

Unrestricted Outbound Connections FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT owner, db_link, username, host, created FROM ALL_DB_LINKS ORDER BY owner, db_link
SEC-SQL-NET-003-RC05 External Data Source Integration
Condition: host = '*' OR broad access patterns — Broad network ACL entries allow outbound connections to any host
Unrestricted Outbound Connections FINNACLEDB_TEST: DETECTED 3 rows FLEXCUBE: DETECTED 4 rows alert: Yes mail: Yes (destructive)
SELECT na.host, na.lower_port, na.upper_port, na.acl, nap.principal, nap.privilege, nap.is_grant FROM dba_network_acls na JOIN dba_network_acl_privileges nap ON na.acl = nap.acl ORDER BY na.host
SEC-SQL-NET-003-RC10 DNS Unrestricted Outbound Access
Condition: row_count > 0 — DNS resolve privileges granted through network ACLs enable outbound DNS communication
Unrestricted Outbound Connections FINNACLEDB_TEST: DETECTED 3 rows FLEXCUBE: DETECTED 3 rows alert: Yes mail: No
SELECT acl, principal, privilege, is_grant FROM dba_network_acl_privileges WHERE privilege = 'resolve' ORDER BY principal
SEC-SQL-NET-003-RC12 Legacy System Requirements
Condition: row_count > 0 — Database links to non-local hosts indicate outbound connectivity requirements
Unrestricted Outbound Connections FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT owner, db_link, host FROM ALL_DB_LINKS WHERE host NOT LIKE '%localhost%' AND host NOT LIKE '%127.0.0.1%' AND host NOT LIKE '%(LOCAL)%' ORDER BY owner, db_link
SEC-SQL-NET-003-RC14 Database Replication to Cloud
Condition: row_count > 0 — Active standby destinations indicate outbound replication connections
Unrestricted Outbound Connections FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT dest_id, dest_name, status, target, destination FROM v\$archive_dest WHERE status != 'INACTIVE' AND target = 'STANDBY' ORDER BY dest_id
SEC-SQL-NET-003-RC15 Monitoring and Logging Exfiltration
Condition: row_count > 0 — Network ACLs for monitoring ports or wildcard hosts indicate unrestricted telemetry access
Unrestricted Outbound Connections FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT host, lower_port, upper_port, acl FROM dba_network_acls WHERE lower_port IN (443, 8443, 9090, 9100, 5044, 8086) OR host = '*' ORDER BY host
SEC-SQL-NET-003-RC16 No Default-Deny Egress Policy
Condition: host = '*' OR lower_port IS NULL — Wildcard host or unrestricted port in network ACLs indicates no default-deny egress policy
Unrestricted Outbound Connections FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT host, lower_port, upper_port, acl, (SELECT COUNT(DISTINCT principal) FROM dba_network_acl_privileges nap WHERE nap.acl = na.acl AND nap.is_grant = 'true') AS grant_count FROM dba_network_acls na WHERE host = '*' OR lower_port IS NULL ORDER BY host
SEC-SQL-PAT-001-RC01 Lack of proactive EOL tracking and monitoring
Condition: last patch older than :months_threshold months — No recent patches combined with older version confirms lack of proactive EOL tracking
End-of-Life Database Version FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT patch_id, action, status, action_time, description FROM dba_registry_sqlpatch WHERE action = 'APPLY' AND status = 'SUCCESS' ORDER BY action_time DESC FETCH FIRST 5 ROWS ONLY
SEC-SQL-PAT-001-RC02 Deferred major version upgrades due to technical complexity
Condition: created date suggests database has been running for years without major upgrade — Long-running database on older version confirms deferred major version upgrade

End-of-Life Database Version FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT dbid, name, created, log_mode, open_mode FROM v\$database
SEC-SQL-PAT-001-RC04 Extended downtime requirements for migration
Condition: version is significantly behind current LTS — HA database on old version confirms downtime fears are delaying migration
End-of-Life Database Version FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT version FROM v\$instance
SEC-SQL-PAT-001-RC05 Application dependency on deprecated database features
Condition: row_count > 0 — Active usage of deprecated features confirms application dependency blocking upgrades
End-of-Life Database Version FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: Yes (destructive)
SELECT name, detected_usages, currently_used, first_usage_date, last_usage_date FROM dba_feature_usage_statistics WHERE currently_used = 'TRUE' AND detected_usages > 0 AND (name LIKE '%Deprecated%' OR name LIKE '%obsolete%' OR name IN ('Oracle Streams', 'Advanced Replication', 'Rule Manager', 'Expression Filter')) ORDER BY last_usage_date DESC
SEC-SQL-PAT-001-RC06 Unfamiliar upgrade procedures and architectural changes
Condition: invalid_components > 0 OR custom_schemas > :complexity_threshold — Invalid components, many custom schemas, or lagging COMPATIBLE parameter indicate upgrade complexity
End-of-Life Database Version FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT (SELECT COUNT(*) FROM dba_registry WHERE status != 'VALID') AS invalid_components, (SELECT COUNT(*) FROM dba_objects WHERE status = 'INVALID' AND owner NOT IN ('SYS', 'SYSTEM')) AS invalid_objects, (SELECT COUNT(DISTINCT owner) FROM dba_objects WHERE owner NOT IN (SELECT username FROM dba_users WHERE oracle_maintained = 'Y')) AS custom_schemas, (SELECT value FROM v\$parameter WHERE name = 'compatible') AS compatible_param, (SELECT version FROM v\$instance) AS current_version FROM dual
SEC-SQL-PAT-001-RC07 Inadequate testing environments that mimic production
Condition: patch_history available — Current patch level establishes baseline for testing environment comparison
End-of-Life Database Version FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT patch_id, action, status, action_time, description FROM dba_registry_sqlpatch ORDER BY action_time DESC FETCH FIRST 10 ROWS ONLY
SEC-SQL-PAT-001-RC08 Resource constraints and competing priorities
Condition: last_patch is older than :max_patch_age_days — Last successful patch application date indicates patch currency
End-of-Life Database Version FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT patch_id, action, status, action_time, description FROM dba_registry_sqlpatch WHERE action = 'APPLY' AND status = 'SUCCESS' ORDER BY action_time DESC FETCH FIRST 1 ROW ONLY
SEC-SQL-PAT-001-RC09 Client and driver compatibility concerns
Condition: row_count > 0 — Diverse client programs connecting to the database may have compatibility requirements
End-of-Life Database Version FINNACLEDB_TEST: DETECTED 5 rows FLEXCUBE: DETECTED 20 rows alert: Yes mail: No
SELECT DISTINCT s.program, s.machine, COUNT(*) AS connections FROM v\$session s WHERE s.type = 'USER' AND s.username IS NOT NULL GROUP BY s.program, s.machine ORDER BY connections DESC FETCH FIRST 20 ROWS ONLY
SEC-SQL-PAT-001-RC10 Loss of institutional knowledge
Condition: complex configuration detected (RAC, CDB, etc.) — Complex configurations require specialized patching knowledge that may be lost with staff turnover
End-of-Life Database Version FINNACLEDB_TEST: DETECTED 4 rows FLEXCUBE: DETECTED 3 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('cluster_database', 'enable_pluggable_database', 'db_domain', 'compatible', 'nls_characterset') AND value IS NOT NULL ORDER BY name
SEC-SQL-PAT-001-RC11 Legacy application portfolio constraints

Condition: row_count > 0 — Legacy or deprecated features in use may create compatibility risks with newer patches
End-of-Life Database Version FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT name, detected_usages, currently_used, first_usage_date, last_usage_date, description FROM dba_feature_usage_statistics WHERE currently_used = 'TRUE' AND name IN ('Advanced Replication', 'Streams', 'Workspace Manager', 'XML DB Repository', 'Oracle Multimedia', 'Client Identifier', 'Fine-grained Auditing', 'Oracle Spatial') ORDER BY detected_usages DESC
SEC-SQL-PAT-001-RC12 Incremental neglect over multiple release cycles
Condition: days_since_patch > :max_days — Long gap since last patch confirms incremental neglect over multiple release cycles
End-of-Life Database Version FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT TO_CHAR(MAX(action_time), 'YYYY-MM-DD HH24:MI:SS') AS last_patch_date, ROUND(SYSDATE - CAST(MAX(action_time) AS DATE)) AS days_since_patch FROM dba_registry_sqlpatch WHERE action = 'APPLY' AND status = 'SUCCESS'
SEC-SQL-PAT-001-RC13 Lack of automatic failure during deployment processes
Condition: row_count > 0 — Failed patches in history that were not automatically detected and remediated
End-of-Life Database Version FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT patch_id, action, status, action_time, description FROM dba_registry_sqlpatch WHERE status != 'SUCCESS' ORDER BY action_time DESC
SEC-SQL-PAT-001-RC14 Vendor licensing and support service uncertainty
Condition: edition information returned — Oracle edition (Enterprise, Standard, XE) determines patch availability and support terms
End-of-Life Database Version FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT banner FROM v\$version WHERE banner LIKE 'Oracle%'
SEC-SQL-PAT-002-RC01 Incomplete asset discovery and inventory
Condition: database properties returned — Database identification properties for comprehensive asset tracking
Missing Critical Security Patches FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT name, db_unique_name, platform_name, created, dbid, cdb FROM v\$database
SEC-SQL-PAT-002-RC02 Extended CVE disclosure-to-patch timelines
Condition: days_since_last_cpu > :max_days — Gap since last CPU/RU application indicates window of vulnerability exposure
Missing Critical Security Patches FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT TO_CHAR(MAX(action_time), 'YYYY-MM-DD HH24:MI:SS') AS last_cpu_date, ROUND(SYSDATE - CAST(MAX(action_time) AS DATE)) AS days_since_last_cpu FROM dba_registry_sqlpatch WHERE action = 'APPLY' AND status = 'SUCCESS' AND (description LIKE '%Database%Update%' OR description LIKE '%CPU%' OR description LIKE '%PSU%' OR description LIKE '%RU%')
SEC-SQL-PAT-002-RC03 Inadequate testing environments and procedures
Condition: configuration parameters identified — Production configuration parameters that test environments must match for valid patch testing
Missing Critical Security Patches FINNACLEDB_TEST: DETECTED 7 rows FLEXCUBE: DETECTED 7 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('cluster_database', 'enable_pluggable_database', 'compatible', 'nls_characterset', 'optimizer_features_enable', 'cursor_sharing', 'parallel_max_servers', 'db_block_size') AND value IS NOT NULL ORDER BY name
SEC-SQL-PAT-002-RC04 Fear of patch-induced breaking changes
Condition: optimizer_features_enable or _fix_control is set — Explicit optimizer feature controls suggest previous patch-induced plan regressions
Missing Critical Security Patches FINNACLEDB_TEST: DETECTED 3 rows FLEXCUBE: DETECTED 3 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('optimizer_features_enable', 'optimizer_adaptive_plans', 'optimizer_adaptive_statistics', '_fix_control') ORDER BY name

SEC-SQL-PAT-002-RC05 Long approval workflows and change control processes
Condition: irregular patch intervals — Inconsistent gaps between patches suggest variable approval timelines
Missing Critical Security Patches FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT action_time, patch_id, description FROM dba_registry_sqlpatch WHERE action = 'APPLY' AND status = 'SUCCESS' ORDER BY action_time DESC FETCH FIRST 10 ROWS ONLY
SEC-SQL-PAT-002-RC06 Resource constraints and manual patching processes
Condition: configuration returned — Number of PDBs and RAC configuration determines patching complexity and resource requirements
Missing Critical Security Patches FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT (SELECT COUNT(*) FROM v\$pdb WHERE con_id > 2) AS pdb_count, (SELECT value FROM v\$parameter WHERE name = 'cluster_database') AS is_rac, (SELECT value FROM v\$parameter WHERE name = 'enable_pluggable_database') AS is_cdb FROM dual
SEC-SQL-PAT-002-RC07 Patch validation failures in staging
Condition: row_count > 0 — Patch failures or rollbacks in history indicate validation issues that may recur
Missing Critical Security Patches FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT patch_id, action, status, action_time, description FROM dba_registry_sqlpatch WHERE status IN ('WITH ERRORS', 'ROLLBACK', 'ROLLBACK SUCCESS') ORDER BY action_time DESC
SEC-SQL-PAT-002-RC08 Vendor patch delays for specific platforms or configurations
Condition: platform information returned — Platform identification determines patch availability timeline
Missing Critical Security Patches FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT name, platform_id, platform_name FROM v\$database
SEC-SQL-PAT-002-RC09 Incompatible dependency chains
Condition: patch inventory available — Current patch inventory determines compatibility with new patches
Missing Critical Security Patches FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT patch_id, action, status, action_time, description FROM dba_registry_sqlpatch ORDER BY action_time DESC FETCH FIRST 15 ROWS ONLY
SEC-SQL-PAT-002-RC10 Active database usage preventing maintenance windows
Condition: activity in all hours — Consistent activity across all hours indicates no natural maintenance window
Missing Critical Security Patches FINNACLEDB_TEST: DETECTED 14 rows FLEXCUBE: DETECTED 24 rows alert: Yes mail: No
SELECT TO_CHAR(sample_time, 'HH24') AS hour, COUNT(*) AS active_sessions FROM v\$active_session_history WHERE sample_time > SYSDATE - 7 GROUP BY TO_CHAR(sample_time, 'HH24') ORDER BY hour
SEC-SQL-PAT-002-RC11 Lack of patch rollback and recovery procedures
Condition: flashback_on = 'YES' AND log_mode = 'ARCHIVELOG' — Flashback Database enabled with ARCHIVELOG mode allows guaranteed restore point creation
Missing Critical Security Patches FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT flashback_on, log_mode, force_logging FROM v\$database
SEC-SQL-PAT-002-RC12 Critical patches for rarely-exploited vulnerabilities
Condition: patch history available for gap analysis — Patch application history reveals whether comprehensive CPUs or only selective patches are applied
Missing Critical Security Patches FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT patch_id, action_time, description FROM dba_registry_sqlpatch WHERE action = 'APPLY' AND status = 'SUCCESS' ORDER BY action_time DESC FETCH FIRST 10 ROWS ONLY

SEC-SQL-PAT-002-RC13 Incomplete patches that bypass exploits
Condition: any status != 'SUCCESS' — Failed or incomplete patch applications indicate potentially incomplete security fixes
Missing Critical Security Patches FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT patch_id, patch_uid, target_version AS version, status, action, action_time, description FROM dba_registry_sqlpatch ORDER BY action_time DESC FETCH FIRST 20 ROWS ONLY
SEC-SQL-PAT-002-RC14 Delayed CVE scoring and analysis
Condition: days_since_patch > :patch_threshold — Database not patched within threshold period may be exposed to known CVEs
Missing Critical Security Patches FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT TO_CHAR(MAX(action_time), 'YYYY-MM-DD HH24:MI:SS') AS last_patch_date, ROUND(SYSDATE - CAST(MAX(action_time) AS DATE)) AS days_since_patch FROM dba_registry_sqlpatch WHERE status = 'SUCCESS'
SEC-SQL-PAT-002-RC15 Patch delivery and content distribution delays
Condition: row_count > 0 — Review recent patch history to assess distribution timing
Missing Critical Security Patches FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT patch_id, target_version AS version, status, action, action_time FROM dba_registry_sqlpatch WHERE action = 'APPLY' ORDER BY action_time DESC FETCH FIRST 10 ROWS ONLY
SEC-SQL-PRI-001-RC01 Misunderstanding of "Encryption at Rest" (TDE)
Condition: row_count = 0 — No column-level encryption indicates reliance only on tablespace-level TDE without granular data protection
PII Exposed in Clear Text FINNACLEDB_TEST: DETECTED 0 rows FLEXCUBE: DETECTED 0 rows alert: Yes mail: Yes (destructive)
SELECT owner, table_name, column_name, encryption_alg, salt FROM ALL_ENCRYPTED_COLUMNS ORDER BY owner, table_name
SEC-SQL-PRI-001-RC02 Searchability I&O Indexing Requirements
Condition: row_count > 0 — Potentially sensitive indexed columns that may be left unencrypted for searchability
PII Exposed in Clear Text FINNACLEDB_TEST: FAILED FLEXCUBE: FAILED alert: No mail: No
SELECT ic.table_owner, ic.table_name, ic.column_name, ic.index_name, tc.data_type FROM ALL_IND_COLUMNS ic JOIN ALL_TAB_COLUMNS tc ON ic.table_owner = tc.owner AND ic.table_name = tc.table_name AND ic.column_name = tc.column_name WHERE (tc.column_name LIKE '%SSN%' OR (tc.column_name LIKE '%bvn%' OR tc.column_name LIKE '%CARD%' OR tc.column_name LIKE '%ACCOUNT%' OR tc.column_name LIKE '%EMAIL%' OR tc.column_name LIKE '%PHONE%' OR tc.column_name LIKE '%PASSWORD%')) AND ic.table_owner NOT IN ('SYS', 'SYSTEM') ORDER BY ic.table_owner, ic.table_name
SEC-SQL-PRI-001-RC03 Performance Overhead Fears
Condition: row_count > 0 — Permanent tablespaces without encryption may indicate performance concerns blocking adoption
PII Exposed in Clear Text FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 4 rows alert: Yes mail: No
SELECT ts.tablespace_name, ts.encrypted, ts.contents FROM dba_tablespaces ts WHERE ts.contents = 'PERMANENT' AND ts.encrypted = 'NO' AND ts.tablespace_name NOT IN ('SYSTEM', 'SYSAUX', 'UNDO') ORDER BY ts.tablespace_name
SEC-SQL-PRI-001-RC04 Incorrect Data Types
Condition: row_count > 0 — Sensitive columns with non-encryptable data types (LONG, BFILE, ROWID) cannot use TDE column encryption
PII Exposed in Clear Text FINNACLEDB_TEST: FAILED FLEXCUBE: FAILED alert: No mail: No
SELECT tc.owner, tc.table_name, tc.column_name, tc.data_type, tc.data_length FROM ALL_TAB_COLUMNS tc WHERE (tc.column_name LIKE '%SSN%' OR (tc.column_name LIKE '%bvn%' OR tc.column_name LIKE '%CARD%' OR tc.column_name LIKE '%SECRET%')) AND tc.data_type IN ('LONG', 'LONG RAW', 'BFILE', 'ROWID', 'UROWID') AND tc.owner NOT IN ('SYS', 'SYSTEM') ORDER BY tc.owner, tc.table_name
SEC-SQL-PRI-001-RC05 Legacy Schema Constraints
Condition: row_count > 0 — Legacy data types (LONG, LONG RAW) prevent TDE encryption and indicate schema modernization needed

PII Exposed in Clear Text FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 110 rows alert: Yes mail: No
SELECT owner, table_name, column_name, data_type FROM ALL_TAB_COLUMNS WHERE data_type IN ('LONG', 'LONG RAW') AND owner NOT IN ('SYS', 'SYSTEM', 'MDSYS', 'XDB', 'WMSYS') ORDER BY owner, table_name
SEC-SQL-PRI-001-RC06 In-House "Masking" Logic Failure
Condition: row_count > 0 — Custom masking functions indicate in-house masking logic that may have implementation gaps
PII Exposed in Clear Text FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 5 rows alert: Yes mail: No
SELECT owner, object_name, object_type FROM ALL_OBJECTS WHERE object_type IN ('FUNCTION', 'PACKAGE') AND (object_name LIKE '%MASK%' OR object_name LIKE '%ENCRYPT%' OR object_name LIKE '%0BFUSC%' OR object_name LIKE '%REDACT%') AND owner NOT IN ('SYS', 'SYSTEM', 'MDSYS', 'XDB') ORDER BY owner, object_name
SEC-SQL-PRI-001-RC07 ETL & Data Warehousing Stripping
Condition: row_count > 0 — Active or recent Data Pump jobs may export encrypted data in unencrypted dump files
PII Exposed in Clear Text FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 66 rows alert: Yes mail: No
SELECT job_name, operation, job_mode, state, attached_sessions FROM dba_datapump_jobs WHERE state = 'EXECUTING' OR state = 'NOT RUNNING' ORDER BY state, job_name
SEC-SQL-PRI-001-RC08 Third-Party Integration Limitations
Condition: row_count > 0 — Third-party applications connecting may have limitations with encryption features
PII Exposed in Clear Text FINNACLEDB_TEST: DETECTED 5 rows FLEXCUBE: DETECTED 20 rows alert: Yes mail: No
SELECT DISTINCT program, module, COUNT(*) AS sessions FROM v\$session WHERE type = 'USER' AND username IS NOT NULL GROUP BY program, module ORDER BY sessions DESC FETCH FIRST 20 ROWS ONLY
SEC-SQL-PRI-001-RC09 Key Management Complexity
Condition: status != 'OPEN' OR wallet_type = 'UNKNOWN' — Wallet not open or not configured indicates key management complexity blocking encryption
PII Exposed in Clear Text FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT wrl_type, wrl_parameter, status, wallet_type, wallet_order, fully_backed_up FROM v\$encryption_wallet
SEC-SQL-PRI-001-RC10 Lack of Native Masking Features (Old Versions)
Condition: version starts with 11 or earlier — Oracle 11g and earlier lack Data Redaction feature (introduced in 12c)
PII Exposed in Clear Text FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT BANNER_LEGACY, banner_full FROM v\$version WHERE banner_full LIKE '%Oracle%'
SEC-SQL-PRI-001-RC11 Unsecured Backups/Snapshots
Condition: row_count = 0 OR value = 'OFF' — No RMAN encryption configuration means backups are stored unencrypted
PII Exposed in Clear Text FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT conf#, name, value FROM v\$rman_configuration WHERE name LIKE '%ENCRYPTION%' ORDER BY name
SEC-SQL-PRI-001-RC12 Sensitive columns discovered in schema (PII)
Condition: row_count > 0 — Columns with names suggesting PII/sensitive data found in database schema
PII Exposed in Clear Text FINNACLEDB_TEST: FAILED FLEXCUBE: FAILED alert: No mail: No

SELECT owner AS schema_name, table_name, column_name, data_type, data_length, CASE WHEN LOWER(column_name) LIKE '%ssn%' OR LOWER(column_name) LIKE '%bvn%' OR LOWER(column_name) LIKE '%social_security%' THEN 'ssn bvn' WHEN LOWER(column_name) LIKE '%credit_card%' OR LOWER(column_name) LIKE '%card_number%' OR LOWER(column_name) LIKE '%cvv%' THEN 'Credit Card' WHEN LOWER(column_name) LIKE '%password%' OR LOWER(column_name) LIKE '%passwd%' OR LOWER(column_name) LIKE '%secret%' OR LOWER(column_name) LIKE '%api_key%' THEN 'Credential' WHEN LOWER(column_name) LIKE '%email%' THEN 'Email' WHEN LOWER(column_name) LIKE '%phone%' OR LOWER(column_name) LIKE '%mobile%' THEN 'Phone' WHEN LOWER(column_name) LIKE '%birth%' OR LOWER(column_name) LIKE '%dob%' THEN 'Date of Birth' WHEN LOWER(column_name) LIKE '%salary%' OR LOWER(column_name) LIKE '%income%' THEN 'Financial' WHEN LOWER(column_name) LIKE '%bank_account%' OR LOWER(column_name) LIKE '%iban%' THEN 'Bank Account' WHEN LOWER(column_name) LIKE '%national_id%' OR LOWER(column_name) LIKE '%passport%' OR LOWER(column_name) LIKE '%tax_id%' THEN 'Government ID' WHEN LOWER(column_name) LIKE '%medical%' OR LOWER(column_name) LIKE '%diagnosis%' THEN 'Medical' ELSE 'Other Sensitive' END AS pii_category FROM all_tab_columns WHERE owner NOT IN ('SYS', 'SYSTEM', 'DBSNMP', 'OUTLN', 'MDSYS', 'ORDSYS', 'CTXSYS', 'XDB', 'WMSYS') AND (LOWER(column_name) LIKE '%ssn%' OR (LOWER(column_name) LIKE '%bvn%' OR LOWER(column_name) LIKE '%social_security%' OR LOWER(column_name) LIKE '%credit_card%' OR LOWER(column_name) LIKE '%card_number%' OR LOWER(column_name) LIKE '%cvv%' OR LOWER(column_name) LIKE '%password%' OR LOWER(column_name) LIKE '%passwd%' OR LOWER(column_name) LIKE '%secret%' OR LOWER(column_name) LIKE '%api_key%' OR LOWER(column_name) LIKE '%email%' OR LOWER(column_name) LIKE '%phone%' OR LOWER(column_name) LIKE '%mobile%' OR LOWER(column_name) LIKE '%birth%' OR LOWER(column_name) LIKE '%dob%' OR LOWER(column_name) LIKE '%salary%' OR LOWER(column_name) LIKE '%income%' OR LOWER(column_name) LIKE '%bank_account%' OR LOWER(column_name) LIKE '%iban%' OR LOWER(column_name) LIKE '%national_id%' OR LOWER(column_name) LIKE '%passport%' OR LOWER(column_name) LIKE '%tax_id%' OR LOWER(column_name) LIKE '%medical%' OR LOWER(column_name) LIKE '%diagnosis%' OR LOWER(column_name) LIKE '%ip_address%' OR LOWER(column_name) LIKE '%biometric%')) ORDER BY pii_category, owner, table_name
SEC-SQL-PRI-001-RC13 Sensitive columns actively queried (PII in use)
Condition: row_count > 0 — Tables with sensitive columns contain data and are in active use
PII Exposed in Clear Text FINNACLEDB_TEST: FAILED FLEXCUBE: FAILED alert: No mail: No
SELECT c.owner AS schema_name, c.table_name, c.column_name, s.num_rows, s.last_analyzed, CASE WHEN LOWER(c.column_name) LIKE '%ssn%' OR LOWER(c.column_name) LIKE '%bvn%' THEN 'ssn bvn' WHEN LOWER(c.column_name) LIKE '%credit_card%' OR LOWER(c.column_name) LIKE '%card_number%' THEN 'Credit Card' WHEN LOWER(c.column_name) LIKE '%password%' OR LOWER(c.column_name) LIKE '%secret%' THEN 'Credential' WHEN LOWER(c.column_name) LIKE '%email%' THEN 'Email' WHEN LOWER(c.column_name) LIKE '%phone%' OR LOWER(c.column_name) LIKE '%mobile%' THEN 'Phone' WHEN LOWER(c.column_name) LIKE '%salary%' OR LOWER(c.column_name) LIKE '%income%' THEN 'Financial' WHEN LOWER(c.column_name) LIKE '%national_id%' OR LOWER(c.column_name) LIKE '%passport%' THEN 'Government ID' WHEN LOWER(c.column_name) LIKE '%medical%' OR LOWER(c.column_name) LIKE '%diagnosis%' THEN 'Medical' ELSE 'Other Sensitive' END AS pii_category FROM all_tab_columns c JOIN all_tables s ON s.owner = c.owner AND s.table_name = c.table_name WHERE c.owner NOT IN ('SYS', 'SYSTEM', 'DBSNMP', 'OUTLN', 'MDSYS', 'ORDSYS', 'CTXSYS', 'XDB') AND s.num_rows > 0 AND (LOWER(c.column_name) LIKE '%ssn%' OR (LOWER(c.column_name) LIKE '%bvn%' OR LOWER(c.column_name) LIKE '%credit_card%' OR LOWER(c.column_name) LIKE '%password%' OR LOWER(c.column_name) LIKE '%email%' OR LOWER(c.column_name) LIKE '%phone%' OR LOWER(c.column_name) LIKE '%salary%' OR LOWER(c.column_name) LIKE '%bank_account%' OR LOWER(c.column_name) LIKE '%national_id%' OR LOWER(c.column_name) LIKE '%passport%' OR LOWER(c.column_name) LIKE '%tax_id%' OR LOWER(c.column_name) LIKE '%medical%' OR LOWER(c.column_name) LIKE '%diagnosis%' OR LOWER(c.column_name) LIKE '%birth%' OR LOWER(c.column_name) LIKE '%secret%' OR LOWER(c.column_name) LIKE '%api_key%' OR LOWER(c.column_name) LIKE '%iban%' OR LOWER(c.column_name) LIKE '%income%' OR LOWER(c.column_name) LIKE '%biometric%')) ORDER BY s.num_rows DESC
SEC-SQL-PRI-001-RC14 Active transactions reading/writing sensitive columns (PII)
Condition: row_count > 0 — Active queries are reading or writing sensitive columns — live PII data exposure
PII Exposed in Clear Text FINNACLEDB_TEST: clean FLEXCUBE: FAILED alert: Yes mail: No
SELECT s.sid AS session_id, s.username AS login_name, s.machine AS host_name, s.program AS program_name, s.module, c.owner AS schema_name, c.table_name, c.column_name, CASE WHEN LOWER(c.column_name) LIKE '%password%' OR LOWER(c.column_name) LIKE '%pass%' THEN 'Password' WHEN LOWER(c.column_name) LIKE '%secret%' OR LOWER(c.column_name) LIKE '%token%' OR LOWER(c.column_name) LIKE '%key%' THEN 'Credential' WHEN LOWER(c.column_name) LIKE '%credit%' OR LOWER(c.column_name) LIKE '%card%' THEN 'Credit Card' WHEN LOWER(c.column_name) LIKE '%ssn%' OR LOWER(c.column_name) LIKE '%bvn%' THEN 'ssn bvn' WHEN LOWER(c.column_name) LIKE '%email%' THEN 'Email' WHEN LOWER(c.column_name) LIKE '%phone%' THEN 'Phone' WHEN LOWER(c.column_name) LIKE '%address%' THEN 'Address' WHEN LOWER(c.column_name) LIKE '%dob%' OR LOWER(c.column_name) LIKE '%birth%' THEN 'Date of Birth' WHEN LOWER(c.column_name) LIKE '%salary%' THEN 'Salary' ELSE 'Other Sensitive' END AS pii_category, SUBSTR(sq.sql_text, 1, 4000) AS query_text, s.status FROM v\$session s JOIN v\$sql sq ON sq.sql_id = s.sql_id JOIN all_tab_columns c ON UPPER(sq.sql_text) LIKE '%' UPPER(c.table_name) '%' WHERE s.type = 'USER' AND s.status = 'ACTIVE' AND s.sid <> SYS_CONTEXT('USERENV', 'SID') AND c.owner NOT IN ('SYS', 'SYSTEM', 'DBSNMP', 'OUTLN', 'MDSYS', 'ORDSYS', 'CTXSYS', 'XDB', 'WMSYS') AND (LOWER(c.column_name) LIKE '%password%' OR LOWER(c.column_name) LIKE '%pass%' OR LOWER(c.column_name) LIKE '%secret%' OR LOWER(c.column_name) LIKE '%token%' OR LOWER(c.column_name) LIKE '%key%' OR LOWER(c.column_name) LIKE '%credit%' OR LOWER(c.column_name) LIKE '%card%' OR LOWER(c.column_name) LIKE '%ssn%' OR LOWER(c.column_name) LIKE '%bvn%' OR LOWER(c.column_name) LIKE '%email%' OR LOWER(c.column_name) LIKE '%phone%' OR LOWER(c.column_name) LIKE '%address%' OR LOWER(c.column_name) LIKE '%dob%' OR LOWER(c.column_name) LIKE '%birth%' OR LOWER(c.column_name) LIKE '%salary%')) ORDER BY s.sid, c.table_name, c.column_name
SEC-SQL-PRI-001-RC15 Sensitive columns in cached plans / active transactions (PII)
Condition: row_count > 0 — Sensitive columns referenced by cached plans (V\$SQL) or active transactions (V\$TRANSACTION) — historical + uncommitted PII exposure
PII Exposed in Clear Text FINNACLEDB_TEST: clean FLEXCUBE: FAILED alert: Yes mail: No

WITH sensitive_cols AS (SELECT c.owner AS schema_name, c.table_name, c.column_name, CASE WHEN LOWER(c.column_name) LIKE '%password%' OR LOWER(c.column_name) LIKE '%pass%' THEN 'Password' WHEN LOWER(c.column_name) LIKE '%secret%' OR LOWER(c.column_name) LIKE '%token%' OR LOWER(c.column_name) LIKE '%key%' THEN 'Credential' WHEN LOWER(c.column_name) LIKE '%credit%' OR LOWER(c.column_name) LIKE '%card%' THEN 'Credit Card' WHEN LOWER(c.column_name) LIKE '%ssn%' OR LOWER(c.column_name) LIKE '%bvn%' THEN 'ssn bvn' WHEN LOWER(c.column_name) LIKE '%email%' THEN 'Email' WHEN LOWER(c.column_name) LIKE '%phone%' THEN 'Phone' WHEN LOWER(c.column_name) LIKE '%address%' THEN 'Address' WHEN LOWER(c.column_name) LIKE '%dob%' OR LOWER(c.column_name) LIKE '%birth%' THEN 'Date of Birth' WHEN LOWER(c.column_name) LIKE '%salary%' THEN 'Salary' WHEN LOWER(c.column_name) LIKE '%passport%' THEN 'Passport' WHEN LOWER(c.column_name) LIKE '%medical%' OR LOWER(c.column_name) LIKE '%diagnosis%' THEN 'Medical' WHEN LOWER(c.column_name) LIKE '%teudat%' THEN 'Teudat Zehut' ELSE 'Other Sensitive' END AS pii_category FROM all_tab_columns c WHERE c.owner NOT IN ('SYS','SYSTEM','DBSNMP','OUTLN','MDSYS','ORDSYS','CTXSYS','XDB','WMSYS') AND (LOWER(c.column_name) LIKE '%password%' OR LOWER(c.column_name) LIKE '%pass%' OR LOWER(c.column_name) LIKE '%secret%' OR LOWER(c.column_name) LIKE '%token%' OR LOWER(c.column_name) LIKE '%key%' OR LOWER(c.column_name) LIKE '%credit%' OR LOWER(c.column_name) LIKE '%card%' OR LOWER(c.column_name) LIKE '%ssn%' OR LOWER(c.column_name) LIKE '%bvn%' OR LOWER(c.column_name) LIKE '%email%' OR LOWER(c.column_name) LIKE '%phone%' OR LOWER(c.column_name) LIKE '%address%' OR LOWER(c.column_name) LIKE '%dob%' OR LOWER(c.column_name) LIKE '%birth%' OR LOWER(c.column_name) LIKE '%salary%' OR LOWER(c.column_name) LIKE '%passport%' OR LOWER(c.column_name) LIKE '%medical%' OR LOWER(c.column_name) LIKE '%diagnosis%' OR LOWER(c.column_name) LIKE '%teudat%')) SELECT 'cached_plan' AS source, CAST(NULL AS NUMBER) AS session_id, CAST(NULL AS VARCHAR2(50)) AS transaction_id, sql.executions AS execution_count, sql.last_active_time AS last_execution_time, sql.buffer_gets / NULLIF(sql.executions, 0) AS avg_reads, sql.cpu_time / NULLIF(sql.executions, 0) / 1000 AS avg_cpu_ms, CAST(NULL AS DATE) AS transaction_begin_time, sc.schema_name, sc.table_name, sc.column_name, sc.pii_category, SUBSTR(sql.sql_text, 1, 4000) AS query_text, CAST(NULL AS VARCHAR2(128)) AS login_name FROM v\$sql sql JOIN sensitive_cols sc ON UPPER(sql.sql_text) LIKE '%' UPPER(sc.table_name) '%' WHERE sql.sql_t...
SEC-SQL-PRI-001-RC16 All columns discovered in schema
Condition: row_count > 0 — All columns in database schema
PII Exposed in Clear Text FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 5000 rows alert: Yes mail: No
SELECT SYS_CONTEXT('USERENV','DB_NAME') AS db_name, owner AS schema_name, table_name AS table_name, column_name AS column_name, data_type AS data_type, data_length AS max_length, CAST(NULL AS VARCHAR2(50)) AS pii_category FROM all_tab_columns WHERE owner NOT IN ('SYS','SYSTEM','XDB','MDSYS','DBSNMP','OUTLN','APPQOSSYS','ORDSYS','WMSYS','GSMADMIN_INTERNAL','AUDSYS','DVSYS','LBACSYS','OJMSYS','ORDDATA','OLAPSYS') ORDER BY owner, table_name, column_id
SEC-SQL-PRI-002-RC01 Global "Log Everything" Configuration
Condition: row_count > 0 — Audit policies capturing ALL statements may log sensitive data in SQL text
Sensitive Data in Logs FINNACLEDB_TEST: DETECTED 16 rows FLEXCUBE: DETECTED 16 rows alert: Yes mail: No
SELECT policy_name, audit_option, audit_option_type, audit_condition FROM audit_unified_policies WHERE audit_option IN ('ALL', 'ALL STATEMENTS') OR audit_condition IS NULL ORDER BY policy_name
SEC-SQL-PRI-002-RC02 Unparameterized Queries (String Concatenation)
Condition: row_count > 0 — SQL with literal sensitive values in V\$SQL exposes data to anyone with V\$SQL access
Sensitive Data in Logs FINNACLEDB_TEST: DETECTED 20 rows FLEXCUBE: DETECTED 20 rows alert: Yes mail: No
SELECT sql_id, SUBSTR(sql_text, 1, 300) AS sql_preview FROM v\$sql WHERE (sql_text LIKE '%password%=' OR sql_text LIKE '%passwd%=' OR sql_text LIKE '%secret%=' OR sql_text LIKE '%ssn%=' OR sql_text LIKE '%credit_card%=') AND parsing_schema_name NOT IN ('SYS', 'SYSTEM') ORDER BY last_active_time DESC FETCH FIRST 20 ROWS ONLY
SEC-SQL-PRI-002-RC03 Verbose Error Logging
Condition: _trace_files_public = 'TRUE' OR sec_return_server_release_banner = 'TRUE' — Public trace files or version banner exposure increases sensitive data leak risk
Sensitive Data in Logs FINNACLEDB_TEST: DETECTED 3 rows FLEXCUBE: DETECTED 3 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('max_dump_file_size', '_trace_files_public', 'sec_return_server_release_banner', 'diagnostic_dest') ORDER BY name
SEC-SQL-PRI-002-RC04 Audit Trail Over-Configuration
Condition: audit_record_count > :record_threshold — Very large audit trail suggests overly broad auditing that may include sensitive data
Sensitive Data in Logs FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT COUNT(*) AS audit_record_count, MIN(event_timestamp) AS oldest_record, MAX(event_timestamp) AS newest_record FROM unified_audit_trail
SEC-SQL-PRI-002-RC05 Sensitive Functions in Logs
Condition: row_count > 0 — PL/SQL debug or warning settings may expose sensitive function processing in logs
Sensitive Data in Logs FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('plsql_debug', 'plsql_warnings') AND value NOT IN ('DISABLE:ALL', 'FALSE') ORDER BY name

SEC-SQL-PRI-002-RC06 Crash Dumps and Core Files
Condition: max_dump_file_size = 'UNLIMITED' OR _trace_files_public = 'TRUE' — Unlimited dump file size or public trace files increase sensitive data retention risk
Sensitive Data in Logs FINNACLEDB_TEST: DETECTED 5 rows FLEXCUBE: DETECTED 5 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('max_dump_file_size', 'diagnostic_dest', 'core_dump_dest', '_trace_files_public', 'background_dump_dest', 'user_dump_dest') ORDER BY name
SEC-SQL-PRI-002-RC07 Slow Query Logs with Low Thresholds
Condition: row_count > 0 — Trace configuration parameters — check if tracing captures bind values and if trace files are publicly readable
Sensitive Data in Logs FINNACLEDB_TEST: DETECTED 3 rows FLEXCUBE: DETECTED 3 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('sql_trace', 'tracefile_identifier', 'max_dump_file_size', '_trace_files_public') ORDER BY name
SEC-SQL-PRI-002-RC08 Client-Side "Debug" Modes
Condition: row_count > 0 — Sessions with debug events — may be logging sensitive data for troubleshooting
Sensitive Data in Logs FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT sid, serial#, username, program, event, action FROM v\$session WHERE event LIKE '%debug%' OR event LIKE '%trace%' OR action LIKE '%debug%' ORDER BY username
SEC-SQL-PRI-002-RC09 Trace Files left Enabled
Condition: row_count > 0 — Long-running sessions with trace still enabled — likely forgotten after troubleshooting
Sensitive Data in Logs FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT s.sid, s.serial#, s.username, s.program, s.sql_trace, s.logon_time, ROUND(SYSDATE - s.logon_time) AS days_active FROM v\$session s WHERE s.sql_trace = 'ENABLED' AND SYSDATE - s.logon_time > 1 ORDER BY s.logon_time
SEC-SQL-PRI-002-RC11 Insufficient Log Rotation/Permissions
Condition: row_count > 0 — ADR purge policies control trace file retention — verify rotation is configured
Sensitive Data in Logs FINNACLEDB_TEST: DETECTED 12 rows FLEXCUBE: DETECTED 12 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name LIKE '%diagnostic%' OR name LIKE '%dump%' OR name LIKE '%trace%' ORDER BY name
SEC-SQL-PRI-003-RC01 No documented security-level classification
Condition: row_count > 0 — Database lacks PPL_SECURITY_LEVEL property
Database security level not classified FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT SYS_CONTEXT('USERENV','DB_NAME') AS database_name, 'missing PPL_SECURITY_LEVEL parameter' AS finding FROM dual WHERE NOT EXISTS (SELECT 1 FROM database_properties WHERE property_name = 'PPL_SECURITY_LEVEL')
SEC-SQL-PRI-003-RC02 Database meets High-level criteria but configured as Basic/Medium
Condition: row_count > 0 — Tables exceed 100k rows
Database security level not classified FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 297 rows alert: Yes mail: No
SELECT owner, table_name, num_rows FROM all_tables WHERE num_rows >= 100000 AND owner NOT IN ('SYS','SYSTEM','DBSNMP','OUTLN','MDSYS','ORDSYS','CTXSYS','XDB','WMSYS') ORDER BY num_rows DESC
SEC-SQL-PRI-003-RC03 External access enabled without High-level controls
Condition: row_count > 0 — Remote auth allowed
Database security level not classified FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name IN ('remote_login_passwordfile','remote_os_authent') AND value NOT IN ('NONE','FALSE')

SEC-SQL-PRI-004-RC01 TDE / database-level encryption not enabled
Condition: row_count > 0 — User tablespaces without encryption
Personal data not encrypted at rest FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 4 rows alert: Yes mail: No
SELECT tablespace_name, encrypted FROM dba_tablespaces WHERE encrypted = 'NO' AND contents = 'PERMANENT' AND tablespace_name NOT IN ('SYSTEM','SYSAUX')
SEC-SQL-PRI-004-RC02 Sensitive columns stored without column-level encryption
Condition: row_count > 0 — Sensitive columns not in dba_encrypted_columns
Personal data not encrypted at rest FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 1321 rows alert: Yes mail: No
SELECT c.owner, c.table_name, c.column_name, c.data_type FROM all_tab_columns c LEFT JOIN dba_encrypted_columns ec ON ec.owner = c.owner AND ec.table_name = c.table_name AND ec.column_name = c.column_name WHERE ec.column_name IS NULL AND c.owner NOT IN ('SYS','SYSTEM','DBSNMP','OUTLN') AND (LOWER(c.column_name) LIKE '%password%' OR LOWER(c.column_name) LIKE '%ssn%' OR LOWER(c.column_name) LIKE '%bvn%' OR LOWER(c.column_name) LIKE '%credit_card%' OR LOWER(c.column_name) LIKE '%email%' OR LOWER(c.column_name) LIKE '%phone%' OR LOWER(c.column_name) LIKE '%medical%')
SEC-SQL-PRI-004-RC03 Connections not enforced over TLS
Condition: row_count > 0 — Old SQL*Net auth allowed
Personal data not encrypted at rest FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name = 'sqlnet.allowed_logon_version_server' AND value < '12'
SEC-SQL-PRI-005-RC01 No audit specification configured
Condition: row_count > 0 — audit_trail = NONE
Audit trail missing on personal-data tables FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT name, value FROM v\$parameter WHERE name = 'audit_trail' AND LOWER(value) IN ('none','')
SEC-SQL-PRI-005-RC02 Audit configured but does not cover SELECT on PII tables
Condition: row_count > 0 — PII tables without SELECT audit option
Audit trail missing on personal-data tables FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 596 rows alert: Yes mail: No
SELECT DISTINCT c.owner, c.table_name FROM all_tab_columns c WHERE c.owner NOT IN ('SYS','SYSTEM','DBSNMP') AND (LOWER(c.column_name) LIKE '%password%' OR LOWER(c.column_name) LIKE '%ssn%' OR LOWER(c.column_name) LIKE '%bvn%' OR LOWER(c.column_name) LIKE '%credit_card%' OR LOWER(c.column_name) LIKE '%email%') AND NOT EXISTS (SELECT 1 FROM dba_obj_audit_opts a WHERE a.object_name = c.table_name AND a.owner = c.owner AND a.alt = 'A/A')
SEC-SQL-PRI-005-RC03 Audit log retention < 24 months
Condition: row_count > 0 — Audit-purge daily job — verify retention before purge
Audit trail missing on personal-data tables FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT job_name, last_start_date FROM dba_scheduler_jobs WHERE job_name LIKE '%AUDIT%PURGE%' AND repeat_interval LIKE '%FREQ=DAILY%'
SEC-SQL-PRI-006-RC01 Public role has SELECT on PII tables
Condition: row_count > 0 — PUBLIC has SELECT on PII tables
Excessive privileges on personal data FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT p.owner, p.table_name, p.privilege FROM dba_tab_privs p JOIN all_tab_columns c ON c.owner = p.owner AND c.table_name = p.table_name WHERE p.grantee = 'PUBLIC' AND p.privilege = 'SELECT' AND (LOWER(c.column_name) LIKE '%password%' OR LOWER(c.column_name) LIKE '%ssn%' OR LOWER(c.column_name) LIKE '%bvn%' OR LOWER(c.column_name) LIKE '%credit_card%' OR LOWER(c.column_name) LIKE '%email%')
SEC-SQL-PRI-006-RC02 Application accounts hold sysadmin / db_owner

Condition: row_count > 0 — Application accounts hold DBA
Excessive privileges on personal data FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT grantee, granted_role FROM dba_role_privs WHERE granted_role IN ('DBA','SYSDBA') AND (LOWER(grantee) LIKE '%app%' OR LOWER(grantee) LIKE '%svc%' OR LOWER(grantee) LIKE '%service%' OR LOWER(grantee) LIKE '%api%')
SEC-SQL-PRI-006-RC03 Stale logins retain access to personal data
Condition: row_count > 0 — Open accounts inactive for >90 days
Excessive privileges on personal data FINNACLEDB_TEST: DETECTED 7 rows FLEXCUBE: DETECTED 10 rows alert: Yes mail: No
SELECT username, last_login FROM dba_users WHERE account_status = 'OPEN' AND (last_login IS NULL OR last_login < SYSDATE - 90) AND username NOT IN ('SYS','SYSTEM','DBSNMP','XS\$NULL')
SEC-SQL-PRI-007-RC01 SQL logins without password policy
Condition: row_count > 0 — Users on DEFAULT profile (often weak)
Weak authentication / shared accounts FINNACLEDB_TEST: DETECTED 11 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT username, profile FROM dba_users WHERE profile = 'DEFAULT' AND account_status = 'OPEN' AND username NOT IN ('SYS','SYSTEM','DBSNMP')
SEC-SQL-PRI-007-RC02 Shared / generic logins in use
Condition: row_count > 0 — Generic / shared user names active
Weak authentication / shared accounts FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT username FROM dba_users WHERE account_status = 'OPEN' AND LOWER(username) IN ('admin','app','svc','dba','root','test','dbuser')
SEC-SQL-PRI-007-RC03 SA / superuser login enabled and active
Condition: row_count > 0 — SYS or SYSTEM accounts open
Weak authentication / shared accounts FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT username, account_status FROM dba_users WHERE username IN ('SYS','SYSTEM') AND account_status = 'OPEN'
SEC-SQL-PRI-008-RC01 No recent backup of database holding personal data
Condition: row_count > 0 — No RMAN backup in last 7 days
Backup hygiene & encryption FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: clean alert: Yes mail: No
SELECT name AS database_name, MAX(end_time) AS last_backup FROM v\$database, v\$rman_backup_job_details WHERE input_type IN ('DB FULL','DB INCR') GROUP BY name HAVING MAX(end_time) < SYSDATE - 7 OR MAX(end_time) IS NULL
SEC-SQL-PRI-008-RC02 Backups not encrypted
Condition: row_count > 0 — Recent RMAN backups not encrypted
Backup hygiene & encryption FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 31 rows alert: Yes mail: No
SELECT bs_key, completion_time, encrypted FROM v\$backup_set_details WHERE completion_time > SYSDATE - 30 AND encrypted = 'NO'
SEC-SQL-PRI-008-RC03 Backups stored on same volume as data files
Condition: row_count > 0 — Manual review needed — db_recovery_file_dest vs datafile paths
Backup hygiene & encryption FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT name AS data_file, 'check db_recovery_file_dest = data dir' AS finding FROM v\$datafile WHERE ROWNUM = 1
SEC-SQL-PRI-009-RC01 Records older than retention threshold in PII tables

Condition: row_count > 0 — PII tables with date columns — manual age check required
Personal data retained beyond purpose FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 16 rows alert: Yes mail: No
SELECT c.owner, c.table_name, c.column_name FROM all_tab_columns c WHERE c.data_type IN ('DATE','TIMESTAMP') AND (LOWER(c.column_name) LIKE '%created%' OR LOWER(c.column_name) LIKE '%insert%') AND c.owner NOT IN ('SYS','SYSTEM') AND EXISTS (SELECT 1 FROM all_tab_columns c2 WHERE c2.owner = c.owner AND c2.table_name = c.table_name AND (LOWER(c2.column_name) LIKE '%email%' OR LOWER(c2.column_name) LIKE '%phone%'))
SEC-SQL-PRI-009-RC02 Inactive subjects with persistent personal data
Condition: row_count > 0 — Tables with last_active columns
Personal data retained beyond purpose FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT owner, table_name, column_name FROM all_tab_columns WHERE (LOWER(column_name) LIKE '%last_login%' OR LOWER(column_name) LIKE '%last_active%') AND owner NOT IN ('SYS','SYSTEM')
SEC-SQL-PRI-009-RC03 Soft-deleted rows never purged
Condition: row_count > 0 — Tables with soft-delete columns
Personal data retained beyond purpose FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 17 rows alert: Yes mail: No
SELECT owner, table_name, column_name FROM all_tab_columns WHERE (LOWER(column_name) IN ('is_deleted','isdeleted','deleted') OR LOWER(column_name) LIKE '%deleted_at%') AND owner NOT IN ('SYS','SYSTEM')
SEC-SQL-PRI-010-RC01 Failed-login spike from a single source
Condition: row_count > 0 — >20 failed logins in last hour from same user
Security incident detection signals FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT username, os_username, COUNT(*) AS failed_count, MAX(timestamp) AS last_failure FROM dba_audit_session WHERE returncode <> 0 AND timestamp > SYSDATE - 1/24 GROUP BY username, os_username HAVING COUNT(*) > 20
SEC-SQL-PRI-010-RC02 Privilege escalation event detected
Condition: row_count > 0 — Non-system grantees of DBA/SYSDBA
Security incident detection signals FINNACLEDB_TEST: DETECTED 4 rows FLEXCUBE: DETECTED 5 rows alert: Yes mail: No
SELECT grantee, granted_role, default_role, admin_option FROM dba_role_privs WHERE granted_role IN ('DBA','SYSDBA') AND grantee NOT IN ('SYS','SYSTEM','DBA_USERS')
SEC-SQL-PRI-010-RC03 After-hours sensitive-data access
Condition: row_count > 0 — Active sessions outside 06:00–22:00
Security incident detection signals FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT s.sid, s.username, s.machine, s.program FROM v\$session s WHERE s.type = 'USER' AND s.status = 'ACTIVE' AND (TO_NUMBER(TO_CHAR(SYSDATE,'HH24')) < 6 OR TO_NUMBER(TO_CHAR(SYSDATE,'HH24')) >= 22)
SEC-SQL-PRI-011-RC01 Teudat Zehut (Israeli national ID) columns unencrypted
Condition: row_count > 0 — Teudat-Zehut columns not encrypted
Israeli-specific sensitive data exposure FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 27 rows alert: Yes mail: No
SELECT c.owner, c.table_name, c.column_name FROM all_tab_columns c LEFT JOIN dba_encrypted_columns ec ON ec.owner = c.owner AND ec.table_name = c.table_name AND ec.column_name = c.column_name WHERE ec.column_name IS NULL AND c.owner NOT IN ('SYS','SYSTEM') AND (LOWER(c.column_name) LIKE '%teudat%zehut%' OR LOWER(c.column_name) = 'tz' OR LOWER(c.column_name) = 'national_id' OR LOWER(c.column_name) = 'israeli_id')
SEC-SQL-PRI-011-RC02 Medical / health-data columns without classification
Condition: row_count > 0 — Medical/health pattern columns

Israeli-specific sensitive data exposure FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 143 rows alert: Yes mail: No
SELECT owner, table_name, column_name FROM all_tab_columns WHERE owner NOT IN ('SYS', 'SYSTEM') AND (LOWER(column_name) LIKE '%medical%' OR LOWER(column_name) LIKE '%diagnosis%' OR LOWER(column_name) LIKE '%treatment%' OR LOWER(column_name) LIKE '%prescription%' OR LOWER(column_name) LIKE '%health%')
SEC-SQL-PRI-011-RC03 Religion / political-opinion / sexual-orientation columns exposed
Condition: row_count > 0 — Religion/political/orientation columns
Israeli-specific sensitive data exposure FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 9 rows alert: Yes mail: No
SELECT owner, table_name, column_name FROM all_tab_columns WHERE owner NOT IN ('SYS', 'SYSTEM') AND (LOWER(column_name) LIKE '%religion%' OR LOWER(column_name) = 'dat' OR LOWER(column_name) LIKE '%political%' OR LOWER(column_name) LIKE '%orientation%' OR LOWER(column_name) LIKE '%ethnic%')
SEC-SQL-VS-001-RC03 Overly permissive default role assignments
Condition: row_count > 0 — Users with ANY privileges (SELECT ANY TABLE, etc.) — these bypass schema-level isolation
Management UI/Database Privilege Overlap FINNACLEDB_TEST: DETECTED 231 rows FLEXCUBE: DETECTED 547 rows alert: Yes mail: No
SELECT grantee, privilege FROM dba_sys_privs WHERE privilege LIKE '%ANY%' AND grantee NOT IN ('SYS', 'SYSTEM', 'DBA', 'DATAPUMP_EXP_FULL_DATABASE', 'DATAPUMP_IMP_FULL_DATABASE') ORDER BY grantee, privilege
SEC-SQL-VS-001-RC05 Lack of regular privilege audits
Condition: row_count = 0 — No audit policies tracking privilege changes — privilege grants go unmonitored
Management UI/Database Privilege Overlap FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT policy_name, audit_option, audit_option_type FROM audit_unified_policies WHERE audit_option IN ('GRANT', 'REVOKE', 'ALTER USER', 'CREATE USER', 'DROP USER', 'CREATE ROLE', 'ALTER ROLE', 'DROP ROLE') ORDER BY policy_name
SEC-SQL-VS-001-RC06 Insufficient training on dual privilege model
Condition: row_count > 0 — Users with many system privileges — may indicate privilege grants without proper understanding
Management UI/Database Privilege Overlap FINNACLEDB_TEST: DETECTED 12 rows FLEXCUBE: DETECTED 15 rows alert: Yes mail: No
SELECT grantee, COUNT(*) AS priv_count FROM dba_sys_privs WHERE grantee NOT IN ('SYS', 'SYSTEM', 'DBA', 'DATAPUMP_EXP_FULL_DATABASE', 'DATAPUMP_IMP_FULL_DATABASE') GROUP BY grantee HAVING COUNT(*) > 10 ORDER BY priv_count DESC FETCH FIRST 20 ROWS ONLY
SEC-SQL-VS-001-RC07 Shared MC administrator accounts
Condition: row_count > 0 — DBA accounts connected from multiple machines — indicates shared account usage
Management UI/Database Privilege Overlap FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 1 rows alert: Yes mail: Yes (destructive)
SELECT username, COUNT(DISTINCT machine) AS distinct_machines, COUNT(DISTINCT program) AS distinct_programs, COUNT(*) AS active_sessions FROM vsession WHERE username IN (SELECT grantee FROM dba_role_privs WHERE granted_role = 'DBA' AND grantee NOT IN ('SYS', 'SYSTEM')) GROUP BY username HAVING COUNT(DISTINCT machine) > 2 ORDER BY distinct_machines DESC
SEC-SQL-VS-001-RC08 Stale user accounts retaining privileges
Condition: row_count > 0 — Open accounts inactive for extended period that still have privileges — stale account risk
Management UI/Database Privilege Overlap FINNACLEDB_TEST: DETECTED 7 rows FLEXCUBE: DETECTED 6 rows alert: Yes mail: No
SELECT u.username, u.account_status, TO_CHAR(u.last_login, 'YYYY-MM-DD HH24:MI:SS') AS last_login, TO_CHAR(u.created, 'YYYY-MM-DD') AS created, ROUND(SYSDATE - NVL(CAST(u.last_login AS DATE), u.created)) AS days_inactive, (SELECT COUNT(*) FROM dba_sys_privs WHERE grantee = u.username) AS sys_priv_count, (SELECT COUNT(*) FROM dba_role_privs WHERE grantee = u.username) AS role_count FROM dba_users u WHERE u.account_status = 'OPEN' AND NVL(CAST(u.last_login AS DATE), u.created) < SYSDATE - 90 AND u.username NOT IN ('SYS', 'SYSTEM', 'DBSNMP', 'OUTLN', 'XDB', 'AUDSYS', 'GSMADMIN_INTERNAL') ORDER BY days_inactive DESC FETCH FIRST 20 ROWS ONLY
SEC-SQL-VS-001-RC09 Inadequate privilege documentation
Condition: row_count > :depth_threshold — Deep role hierarchy with roles granted to roles — complex structure needing documentation

Management UI/Database Privilege Overlap FINNACLEDB_TEST: DETECTED 60 rows FLEXCUBE: DETECTED 66 rows alert: Yes mail: No
SELECT grantee, granted_role, admin_option FROM dba_role_privs WHERE grantee IN (SELECT role FROM dba_roles) AND grantee NOT IN ('SYS', 'SYSTEM') ORDER BY grantee, granted_role
SEC-SQL-VS-001-RC10 No automated monitoring for privilege changes
Condition: row_count = 0 — GRANT/REVOKE operations not being audited — privilege changes go unmonitored
Management UI/Database Privilege Overlap FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT policy_name, audit_option FROM audit_unified_policies WHERE audit_option IN ('GRANT', 'REVOKE', 'ALTER USER', 'CREATE USER', 'DROP USER') AND policy_name IN (SELECT policy_name FROM audit_unified_enabled_policies) ORDER BY policy_name
SEC-SQL-VS-001-RC11 Inheritance of excessive privileges from mapped accounts
Condition: row_count > 0 — Users inheriting powerful privileges through intermediate role grants
Management UI/Database Privilege Overlap FINNACLEDB_TEST: DETECTED 6 rows FLEXCUBE: DETECTED 30 rows alert: Yes mail: No
SELECT grantee, granted_role, admin_option, default_role FROM dba_role_privs WHERE grantee NOT IN (SELECT role FROM dba_roles) AND grantee NOT IN ('SYS', 'SYSTEM', 'PUBLIC') AND granted_role IN (SELECT grantee FROM dba_role_privs WHERE granted_role IN ('DBA', 'RESOURCE') OR grantee IN (SELECT grantee FROM dba_sys_privs WHERE privilege LIKE '%ANY%')) ORDER BY grantee
SEC-SQL-VS-001-RC12 Use of default administrative credentials
Condition: row_count > 0 — Accounts with default passwords — critical security vulnerability
Management UI/Database Privilege Overlap FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: Yes (destructive)
SELECT username FROM dba_users_with_defpwd WHERE username NOT IN ('X\$NULL') ORDER BY username
SEC-SQL-VS-001-RC13 Insufficient testing of privilege configurations
Condition: row_count = 0 — No privilege captures — Oracle Privilege Analysis has never been used to validate privilege assignments
Management UI/Database Privilege Overlap FINNACLEDB_TEST: FAILED FLEXCUBE: FAILED alert: No mail: No
SELECT name, type, status FROM dba_priv_captures ORDER BY name
SEC-SQL-VS-001-RC14 Complexity in multi-cluster environments
Condition: pdb_count > :threshold — Multiple PDBs — privilege management across containers increases complexity
Management UI/Database Privilege Overlap FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT COUNT(*) AS pdb_count FROM v\$containers WHERE con_id > 2
SEC-SQL-VS-002-RC01 Installation of untrusted procedural languages
Condition: row_count > 0 — Non-system Java classes installed — verify they are from trusted sources
Insecure Extensions Loaded FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 20 rows alert: Yes mail: No
SELECT owner, object_name, object_type, status FROM ALL_OBJECTS WHERE object_type LIKE 'JAVA%' AND owner NOT IN ('SYS', 'SYSTEM', 'PUBLIC', 'ORDSYS', 'MDSYS', 'XDB', 'WMSYS') ORDER BY owner, object_name FETCH FIRST 20 ROWS ONLY
SEC-SQL-VS-002-RC02 Outdated extensions with known vulnerabilities
Condition: days_since_patch > :patch_age_days — Last patch applied more than threshold days ago — database may be missing critical security updates
Insecure Extensions Loaded FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: Yes (destructive)
SELECT TO_CHAR(MAX(action_time), 'YYYY-MM-DD HH24:MI:SS') AS last_patch_date, ROUND(SYSDATE - CAST(MAX(action_time) AS DATE)) AS days_since_patch FROM dba_registry_sqlpatch WHERE action = 'APPLY'
SEC-SQL-VS-002-RC03 Lack of extension vetting before installation

Condition: row_count > 10 — Many database options enabled — each option increases the attack surface
Insecure Extensions Loaded FINNACLEDB_TEST: DETECTED 80 rows FLEXCUBE: DETECTED 80 rows alert: Yes mail: No
SELECT parameter, value FROM v\$option WHERE value = 'TRUE' ORDER BY parameter
SEC-SQL-VS-002-RC05 Extensions can CREATE OR REPLACE existing objects
Condition: row_count > 0 — Non-DBA users can CREATE OR REPLACE procedures in any schema
Insecure Extensions Loaded FINNACLEDB_TEST: DETECTED 13 rows FLEXCUBE: DETECTED 25 rows alert: Yes mail: No
SELECT GRANTEE, PRIVILEGE FROM DBA_SYS_PRIVS WHERE PRIVILEGE IN ('CREATE ANY PROCEDURE', 'CREATE ANY TRIGGER', 'CREATE ANY TYPE', 'ALTER ANY PROCEDURE', 'ALTER ANY TRIGGER') AND GRANTEE NOT IN ('SYS', 'SYSTEM', 'DBA') ORDER BY GRANTEE, PRIVILEGE
SEC-SQL-VS-002-RC06 Insufficient supply chain verification
Condition: row_count > 0 — Non-Oracle schemas contain executable code — verify supply chain
Insecure Extensions Loaded FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 27 rows alert: Yes mail: No
SELECT OWNER, OBJECT_TYPE, COUNT(*) AS object_count FROM ALL_OBJECTS WHERE OBJECT_TYPE IN ('PACKAGE', 'PACKAGE BODY', 'PROCEDURE', 'FUNCTION', 'TRIGGER', 'JAVA CLASS', 'JAVA SOURCE') AND OWNER NOT IN ('SYS', 'SYSTEM', 'MDSYS', 'CTXSYS', 'XDB', 'WMSYS', 'ORDSYS', 'DBSNMP', 'APPQOSSYS', 'GSMADMIN_INTERNAL', 'AUDSYS', 'OUTLN', 'ORDDATA', 'ORDPLUGINS', 'SI_INFORMTN_SCHEMA', 'DVSYS', 'LBACSYS', 'DVF') GROUP BY OWNER, OBJECT_TYPE ORDER BY OWNER, object_count DESC
SEC-SQL-VS-002-RC07 Inadequate extension inventory management
Condition: row_count > :threshold — Many non-Oracle schemas with objects — inventory management may be lacking
Insecure Extensions Loaded FINNACLEDB_TEST: DETECTED 8 rows FLEXCUBE: DETECTED 25 rows alert: Yes mail: No
SELECT u.USERNAME, u.ACCOUNT_STATUS, u.CREATED, u.DEFAULT_TABLESPACE, (SELECT COUNT(*) FROM DBA_OBJECTS o WHERE o.OWNER = u.USERNAME) AS object_count FROM DBA_USERS u WHERE u.USERNAME NOT IN ('SYS', 'SYSTEM', 'MDSYS', 'CTXSYS', 'XDB', 'WMSYS', 'ORDSYS', 'DBSNMP', 'APPQOSSYS', 'GSMADMIN_INTERNAL', 'AUDSYS', 'OUTLN', 'ORDDATA', 'ORDPLUGINS', 'SI_INFORMTN_SCHEMA', 'DVSYS', 'LBACSYS', 'DVF', 'ANONYMOUS', 'XS\$NULL') AND (SELECT COUNT(*) FROM DBA_OBJECTS o WHERE o.OWNER = u.USERNAME) > 0 ORDER BY object_count DESC
SEC-SQL-VS-002-RC08 Extensions granted excessive privileges by default
Condition: row_count > 0 — Non-DBA accounts with many system privileges granted WITH ADMIN OPTION
Insecure Extensions Loaded FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT GRANTEE, COUNT(*) AS priv_count FROM DBA_SYS_PRIVS WHERE GRANTEE NOT IN ('SYS', 'SYSTEM', 'DBA', 'IMP_FULL_DATABASE', 'EXP_FULL_DATABASE', 'DATAPUMP_IMP_FULL_DATABASE', 'DATAPUMP_EXP_FULL_DATABASE') AND ADMIN_OPTION = 'YES' GROUP BY GRANTEE HAVING COUNT(*) > 5 ORDER BY priv_count DESC
SEC-SQL-VS-002-RC09 Lack of runtime sandboxing
Condition: row_count > 0 — Non-system users have Java permissions for file, network, or runtime access
Insecure Extensions Loaded FINNACLEDB_TEST: DETECTED 4 rows FLEXCUBE: DETECTED 14 rows alert: Yes mail: No
SELECT x.grantee, x.type_name, x.name, x.action FROM XMLTABLE('/ROWSET/ROW' PASSING XMLTYPE(NVL(DBMS_XMLGEN.getxml(CASE WHEN (SELECT COUNT(*) FROM all_objects WHERE object_name = 'DBA_JAVA_POLICY' AND object_type IN ('VIEW','SYNONYM')) > 0 THEN q'[SELECT grantee, type_name, name, action FROM dba_java_policy WHERE grantee NOT IN ('SYS', 'SYSTEM', 'PUBLIC') AND type_name IN ('java.io.FilePermission', 'java.net.SocketPermission', 'java.lang.RuntimePermission') AND action NOT LIKE '%read%' ORDER BY grantee, type_name]' ELSE q'[SELECT NULL AS grantee, NULL AS type_name, NULL AS name, NULL AS action FROM dual WHERE 1=0]' END), '<ROWSET/>')) COLUMNS grantee VARCHAR2(128) PATH 'GRANTEE', type_name VARCHAR2(256) PATH 'TYPE_NAME', name VARCHAR2(4000) PATH 'NAME', action VARCHAR2(256) PATH 'ACTION') x
SEC-SQL-VS-002-RC10 No automated vulnerability scanning
Condition: row_count > 0 — Dynamic SQL usage in PL/SQL — potential SQL injection risk without code scanning
Insecure Extensions Loaded FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 20 rows alert: Yes mail: No
SELECT OWNER, NAME, TYPE, TEXT FROM ALL_SOURCE WHERE (UPPER(TEXT) LIKE '%EXECUTE IMMEDIATE%' OR UPPER(TEXT) LIKE '%DBMS_SQL%') AND OWNER NOT IN ('SYS', 'SYSTEM', 'MDSYS', 'CTXSYS', 'XDB', 'WMSYS', 'ORDSYS', 'AUDSYS') ORDER BY OWNER, NAME FETCH FIRST 20 ROWS ONLY

SEC-SQL-VS-002-RC11 Developers installing extensions without security review
Condition: row_count > :threshold — Many recently created code objects — security review may not have been performed
Insecure Extensions Loaded FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 60 rows alert: Yes mail: No
SELECT OWNER, OBJECT_NAME, OBJECT_TYPE, CREATED, STATUS FROM ALL_OBJECTS WHERE OBJECT_TYPE IN ('PACKAGE', 'PACKAGE BODY', 'PROCEDURE', 'FUNCTION', 'TRIGGER', 'JAVA CLASS') AND CREATED > SYSDATE - 30 AND OWNER NOT IN ('SYS', 'SYSTEM', 'MDSYS', 'CTXSYS', 'XDB', 'WMSYS', 'ORDSYS', 'AUDSYS') ORDER BY CREATED DESC
SEC-SQL-VS-002-RC12 Extensions not included in patch management
Condition: row_count > 0 — Schemas with code not modified in a long time — likely unpatched
Insecure Extensions Loaded FINNACLEDB_TEST: DETECTED 5 rows FLEXCUBE: DETECTED 5 rows alert: Yes mail: Yes (destructive)
SELECT OWNER, COUNT(*) AS objects, MAX(LAST_DDL_TIME) AS last_modified FROM ALL_OBJECTS WHERE OBJECT_TYPE IN ('PACKAGE', 'PACKAGE BODY', 'PROCEDURE', 'FUNCTION', 'JAVA CLASS') AND OWNER NOT IN ('SYS', 'SYSTEM', 'MDSYS', 'CTXSYS', 'XDB', 'WMSYS', 'ORDSYS', 'AUDSYS', 'DBSNMP') GROUP BY OWNER HAVING MAX(LAST_DDL_TIME) < SYSDATE - 365 ORDER BY last_modified
SEC-SQL-VS-002-RC13 Malicious extension objects can persist through backups
Condition: row_count > 0 — Non-system database-level triggers found — these persist through backup/restore
Insecure Extensions Loaded FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT OWNER, TRIGGER_NAME, TRIGGER_TYPE, TRIGGERING_EVENT, TABLE_OWNER, BASE_OBJECT_TYPE, STATUS FROM ALL_TRIGGERS WHERE BASE_OBJECT_TYPE = 'DATABASE' AND OWNER NOT IN ('SYS', 'SYSTEM') ORDER BY OWNER, TRIGGER_NAME
SEC-SQL-VS-002-RC14 Inadequate logging of extension-related activities
Condition: row_count = 0 — No audit policies for procedure/trigger DDL — code changes are unaudited
Insecure Extensions Loaded FINNACLEDB_TEST: DETECTED 0 rows FLEXCUBE: DETECTED 0 rows alert: Yes mail: No
SELECT POLICY_NAME, ENABLED_OPTION FROM AUDIT_UNIFIED_ENABLED_POLICIES WHERE POLICY_NAME LIKE '%DDL%' OR ENABLED_OPTION LIKE '%CREATE%' OR ENABLED_OPTION LIKE '%ALTER%' ORDER BY POLICY_NAME
SEC-SQL-VS-002-RC15 Trusted extension mechanism bypasses security controls
Condition: row_count > 0 — Invoker-rights (CURRENT_USER) PL/SQL objects found — review for privilege escalation risks
Insecure Extensions Loaded FINNACLEDB_TEST: DETECTED 51 rows FLEXCUBE: DETECTED 96 rows alert: Yes mail: No
SELECT OWNER, OBJECT_NAME, OBJECT_TYPE, AUTHID FROM ALL_PROCEDURES WHERE AUTHID = 'CURRENT_USER' AND OWNER NOT IN ('SYS', 'SYSTEM', 'MDSYS', 'CTXSYS', 'XDB', 'WMSYS', 'ORDSYS', 'AUDSYS') ORDER BY OWNER, OBJECT_NAME
SEC-SQL-VS-003-RC01 Failure to change default passwords during installation
Condition: row_count > 0 — Accounts with default passwords detected
Default/Factory Passwords Active FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: Yes (destructive)
SELECT USERNAME, ACCOUNT_STATUS FROM DBA_USERS WHERE USERNAME IN (SELECT USERNAME FROM DBA_USERS_WITH_DEFPWD)
SEC-SQL-VS-003-RC02 Databases upgraded from legacy versions preserving insecure defaults
Condition: row_count > 0 — Accounts with legacy 10g password hashes — weaker DES-based hashing
Default/Factory Passwords Active FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 28 rows alert: Yes mail: No
SELECT USERNAME, PASSWORD_VERSIONS FROM DBA_USERS WHERE PASSWORD_VERSIONS LIKE '%10G%' AND ACCOUNT_STATUS = 'OPEN'
SEC-SQL-VS-003-RC03 Applications dependent on default credentials
Condition: row_count > 0 — Open accounts with default passwords — applications may depend on these credentials
Default/Factory Passwords Active FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No

SELECT u.USERNAME, u.ACCOUNT_STATUS, u.PROFILE, u.CREATED FROM DBA_USERS u WHERE u.USERNAME IN (SELECT USERNAME FROM DBA_USERS_WITH_DEFPWD) AND u.ACCOUNT_STATUS = 'OPEN'
SEC-SQL-VS-003-RC04 Lack of automated password policy enforcement
Condition: row_count > 0 — Password profiles captured — check for weak settings
Default/Factory Passwords Active FINNACLEDB_TEST: DETECTED 9 rows FLEXCUBE: DETECTED 36 rows alert: Yes mail: No
SELECT PROFILE, RESOURCE_NAME, LIMIT FROM DBA_PROFILES WHERE RESOURCE_TYPE = 'PASSWORD' AND PROFILE IN (SELECT DISTINCT PROFILE FROM DBA_USERS WHERE ACCOUNT_STATUS = 'OPEN') ORDER BY PROFILE, RESOURCE_NAME
SEC-SQL-VS-003-RC05 Default accounts not properly locked
Condition: row_count > 0 — Default Oracle accounts that are not locked and expired
Default/Factory Passwords Active FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT USERNAME, ACCOUNT_STATUS, LOCK_DATE, EXPIRY_DATE, CREATED, DEFAULT_TABLESPACE FROM DBA_USERS WHERE USERNAME IN ('SCOTT', 'HR', 'OE', 'SH', 'PM', 'IX', 'BI', 'OUTLN', 'MDSYS', 'CTXSYS', 'DBSNMP', 'XDB', 'APEX_PUBLIC_USER', 'FLOWS_FILES', 'WMSYS', 'ORDSYS', 'ORDDATA', 'ORDPLUGINS', 'SI_INFORMTN_SCHEMA', 'DIP', 'MGMT_VIEW', 'SYSMAN') AND ACCOUNT_STATUS NOT LIKE '%LOCKED%' AND ACCOUNT_STATUS NOT LIKE '%EXPIRED%'
SEC-SQL-VS-003-RC06 Inadequate password verification functions
Condition: LIMIT = 'NULL' OR LIMIT IS NULL — No password verification function assigned — no complexity enforcement
Default/Factory Passwords Active FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 4 rows alert: Yes mail: No
SELECT PROFILE, RESOURCE_NAME, LIMIT FROM DBA_PROFILES WHERE RESOURCE_NAME = 'PASSWORD_VERIFY_FUNCTION' AND PROFILE IN (SELECT DISTINCT PROFILE FROM DBA_USERS WHERE ACCOUNT_STATUS = 'OPEN')
SEC-SQL-VS-003-RC07 Fear of breaking database features by changing default passwords
Condition: row_count > 0 — Oracle internal accounts open with default passwords — fear of breaking OEM, Spatial, Text, etc.
Default/Factory Passwords Active FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT u.USERNAME, u.ACCOUNT_STATUS, u.PROFILE FROM DBA_USERS u WHERE u.USERNAME IN (SELECT USERNAME FROM DBA_USERS_WITH_DEFPWD) AND u.ACCOUNT_STATUS = 'OPEN' AND u.USERNAME IN ('DBSNMP', 'SYSMAN', 'APPOSSYS', 'GSMADMIN_INTERNAL', 'XDB', 'MDSYS', 'CTXSYS')
SEC-SQL-VS-003-RC08 No regular password audits
Condition: row_count > :threshold — Many accounts with no password expiry or very long expiry — passwords not being rotated
Default/Factory Passwords Active FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 21 rows alert: Yes mail: Yes (destructive)
SELECT USERNAME, ACCOUNT_STATUS, EXPIRY_DATE, LOCK_DATE, PROFILE FROM DBA_USERS WHERE ACCOUNT_STATUS = 'OPEN' AND (EXPIRY_DATE IS NULL OR EXPIRY_DATE > SYSDATE + 365) ORDER BY USERNAME
SEC-SQL-VS-003-RC09 Insufficient account lifecycle management
Condition: row_count > :account_threshold — Many accounts without password expiration or with far-future expiry dates indicate no lifecycle management
Default/Factory Passwords Active FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 21 rows alert: Yes mail: No
SELECT u.username, u.account_status, u.created, u.expiry_date, u.last_login, u.profile, ROUND(SYSDATE - u.created) AS account_age_days FROM dba_users u WHERE u.oracle_maintained = 'N' AND u.account_status = 'OPEN' AND (u.expiry_date IS NULL OR u.expiry_date > SYSDATE + 365) ORDER BY u.created
SEC-SQL-VS-003-RC10 Weak password hashing in older versions
Condition: row_count > 0 — Users with only 10G or 11G password versions are using weaker hashing algorithms vulnerable to cracking
Default/Factory Passwords Active FINNACLEDB_TEST: clean FLEXCUBE: DETECTED 2 rows alert: Yes mail: Yes (destructive)
SELECT username, password_versions FROM dba_users WHERE oracle_maintained = 'N' AND account_status = 'OPEN' AND password_versions NOT LIKE '%12C%' ORDER BY username
SEC-SQL-VS-003-RC11 Default accounts granted DBA privileges

Condition: row_count > 0 — Oracle-maintained default accounts are not locked — these should be expired and locked after installation
Default/Factory Passwords Active FINNACLEDB_TEST: DETECTED 2 rows FLEXCUBE: DETECTED 2 rows alert: Yes mail: No
SELECT username, account_status, profile, created, authentication_type FROM dba_users WHERE oracle_maintained = 'Y' AND account_status NOT IN ('EXPIRED & LOCKED', 'LOCKED') ORDER BY username
SEC-SQL-VS-003-RC12 Lack of integration with enterprise password management
Condition: any(authentication_type = 'PASSWORD' AND user_count > :password_threshold) — Most users authenticate with local passwords rather than enterprise authentication (EXTERNAL, GLOBAL)
Default/Factory Passwords Active FINNACLEDB_TEST: DETECTED 1 rows FLEXCUBE: DETECTED 1 rows alert: Yes mail: No
SELECT authentication_type, COUNT(*) AS user_count FROM dba_users WHERE oracle_maintained = 'N' AND account_status = 'OPEN' GROUP BY authentication_type ORDER BY user_count DESC
SEC-SQL-VS-003-RC13 Oracle's historical practice of creating numerous default accounts
Condition: row_count > :default_threshold — Oracle installation created many default accounts — each is a potential attack surface if not properly secured
Default/Factory Passwords Active FINNACLEDB_TEST: DETECTED 36 rows FLEXCUBE: DETECTED 35 rows alert: Yes mail: No
SELECT username, account_status, default_tablespace, created, authentication_type FROM dba_users WHERE oracle_maintained = 'Y' ORDER BY account_status, username
SEC-SQL-VS-003-RC14 Inadequate documentation of required default accounts
Condition: row_count > :open_threshold — Multiple default accounts are open with various privileges — documentation should explain why each is required
Default/Factory Passwords Active FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT u.username, u.account_status, u.default_tablespace, u.profile, (SELECT COUNT(*) FROM dba_role_privs rp WHERE rp.grantee = u.username) AS roles, (SELECT COUNT(*) FROM dba_sys_privs sp WHERE sp.grantee = u.username) AS sys_privs, (SELECT COUNT(*) FROM dba_tab_privs tp WHERE tp.grantee = u.username) AS obj_privs FROM dba_users u WHERE u.oracle_maintained = 'Y' AND u.account_status NOT LIKE '%LOCKED%' AND u.username NOT IN ('SYS', 'SYSTEM') ORDER BY u.username
SEC-SQL-VS-003-RC15 Default passwords in configuration files
Condition: row_count > 0 — Open accounts with known default passwords detected — these are immediately exploitable
Default/Factory Passwords Active FINNACLEDB_TEST: clean FLEXCUBE: clean alert: Yes mail: No
SELECT d.username, d.account_status, d.password_versions FROM dba_users d JOIN dba_users_with_defpwd dp ON d.username = dp.username WHERE d.account_status = 'OPEN' ORDER BY d.username